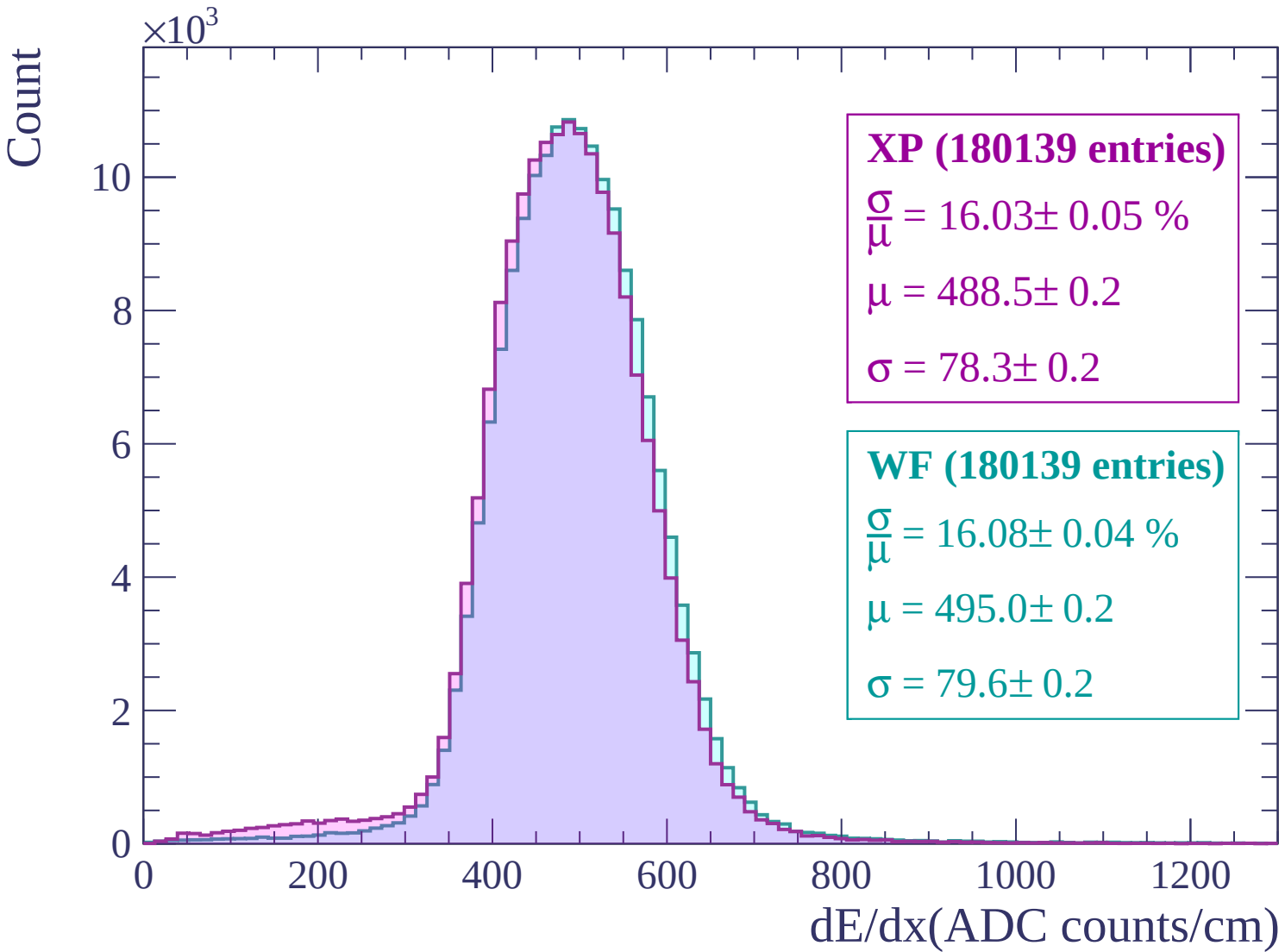
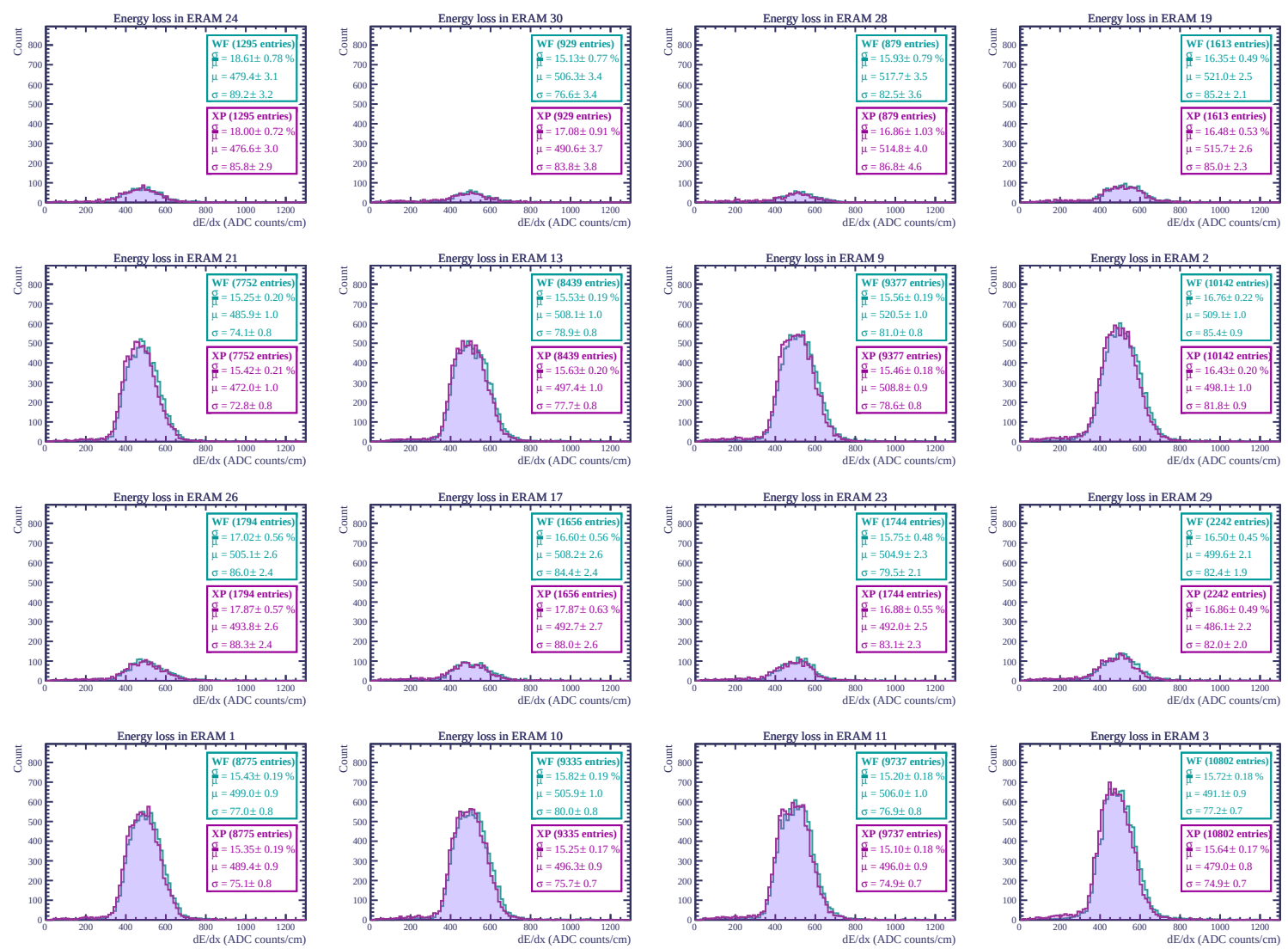
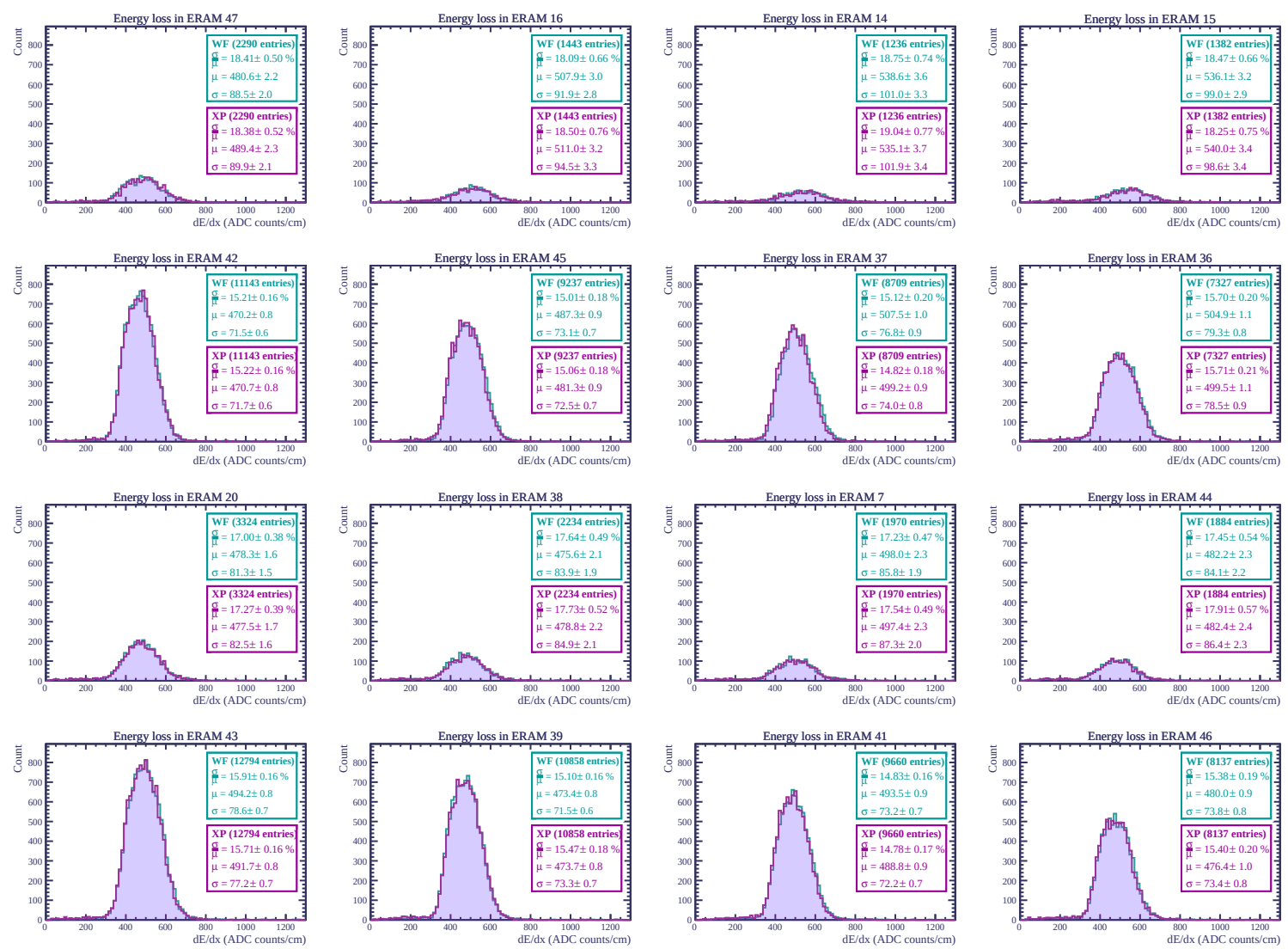
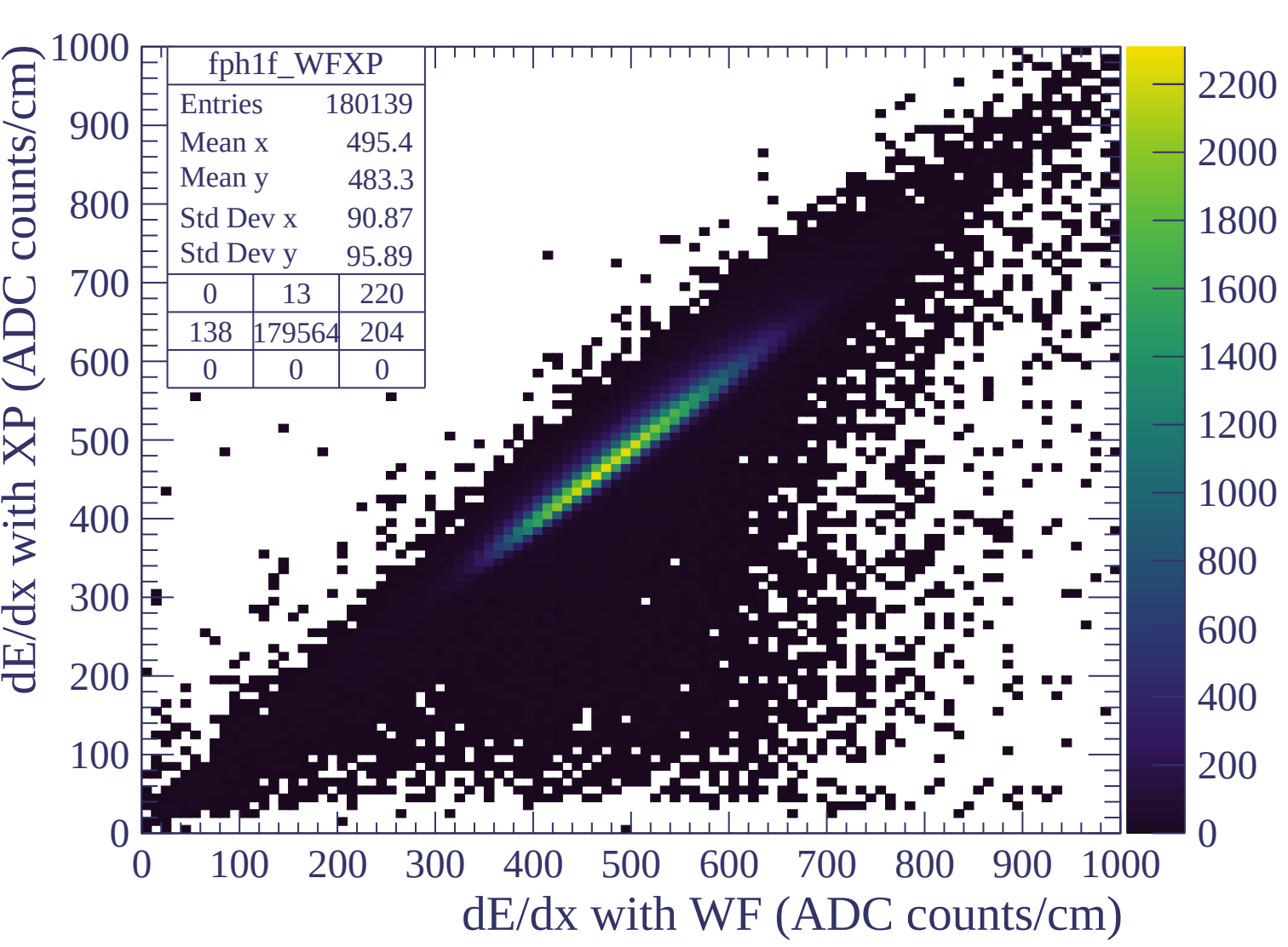
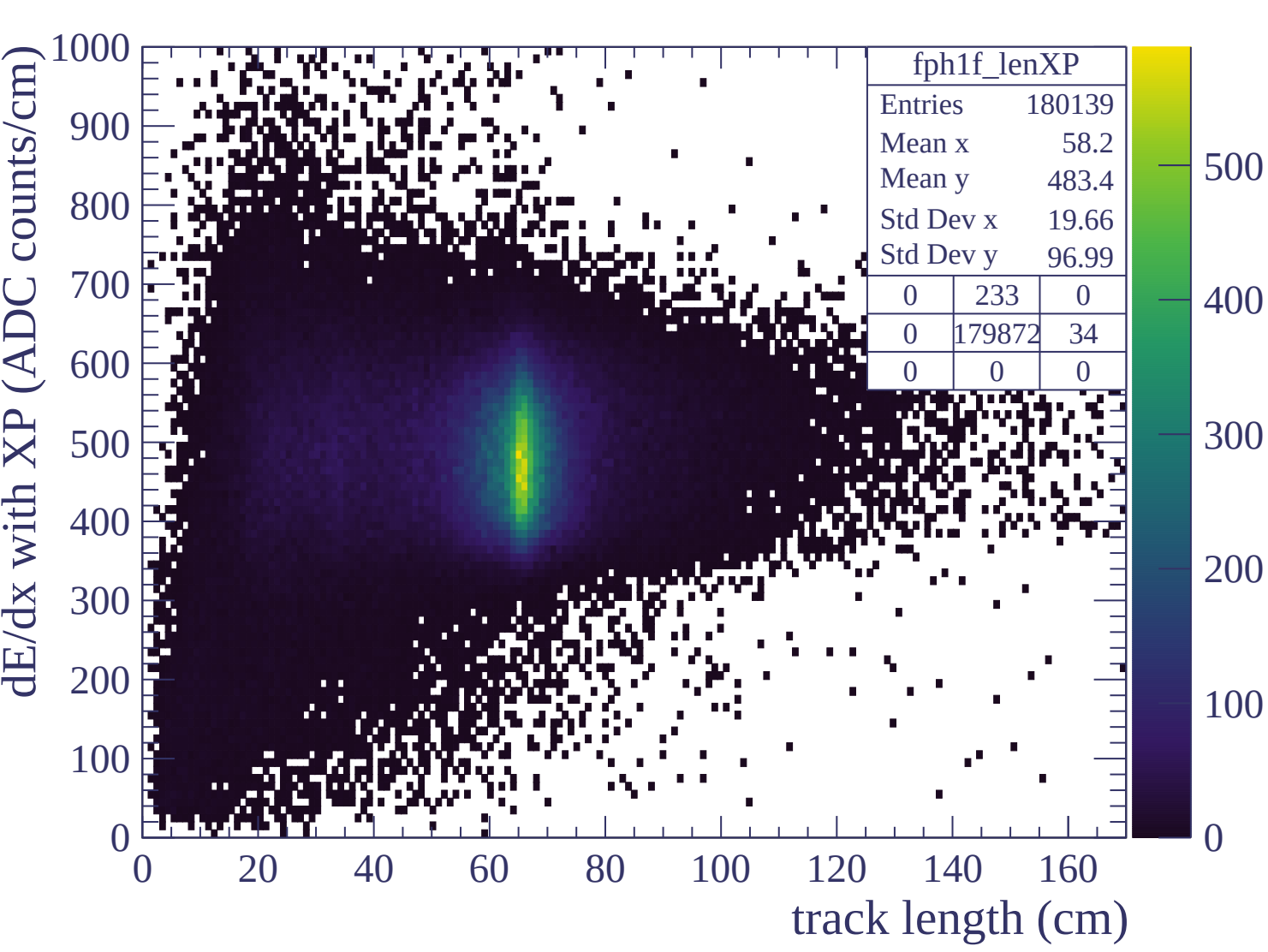


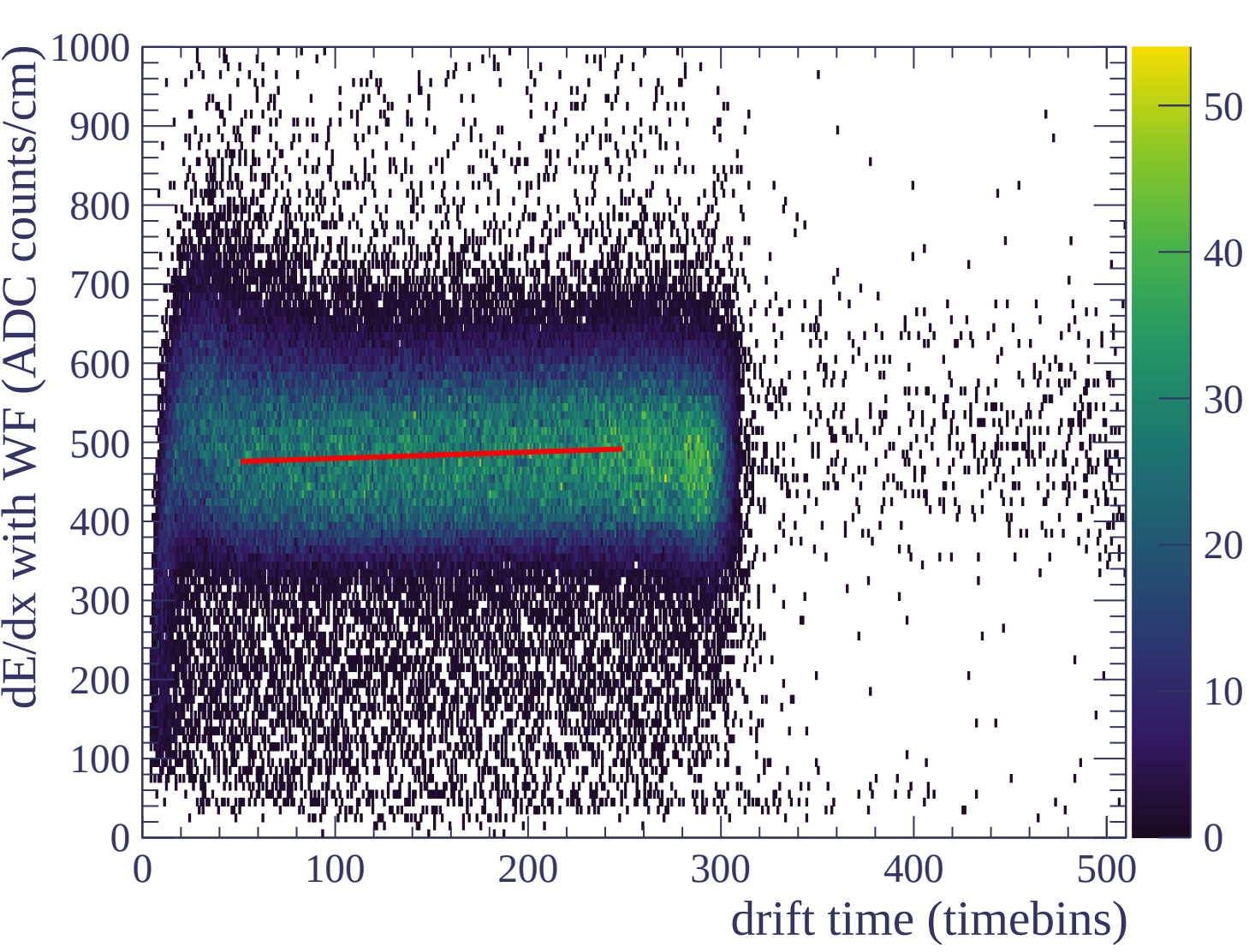


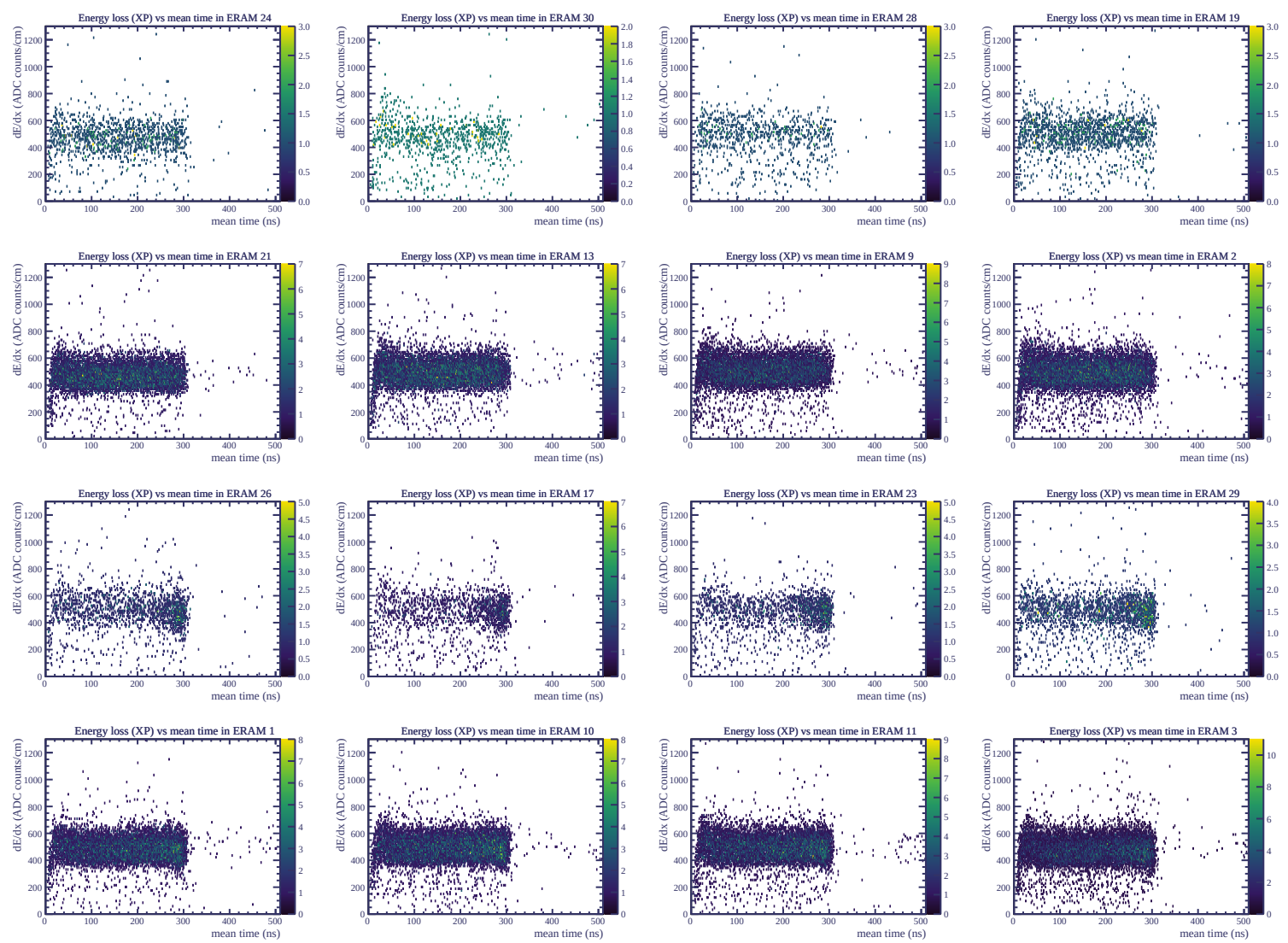


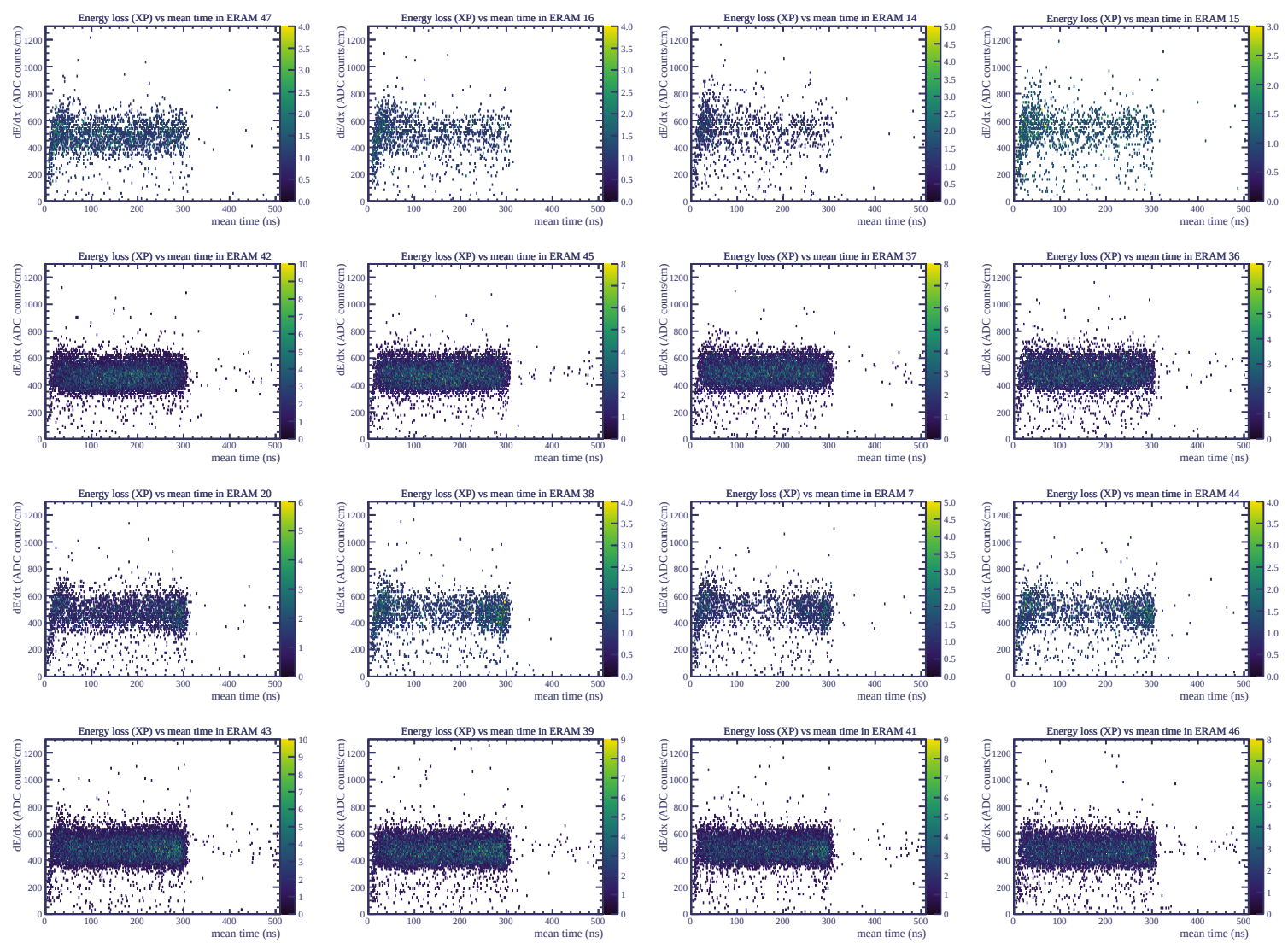




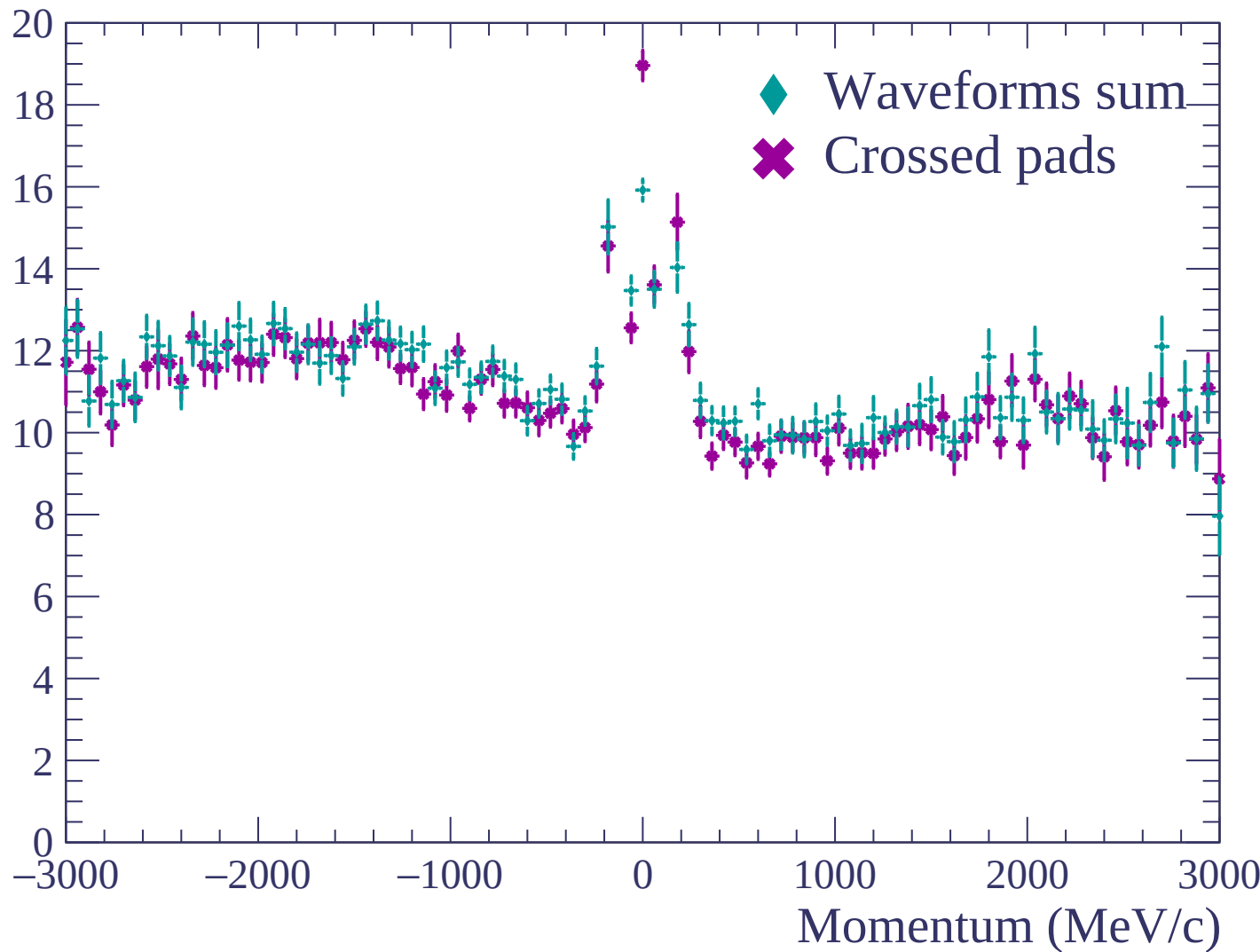


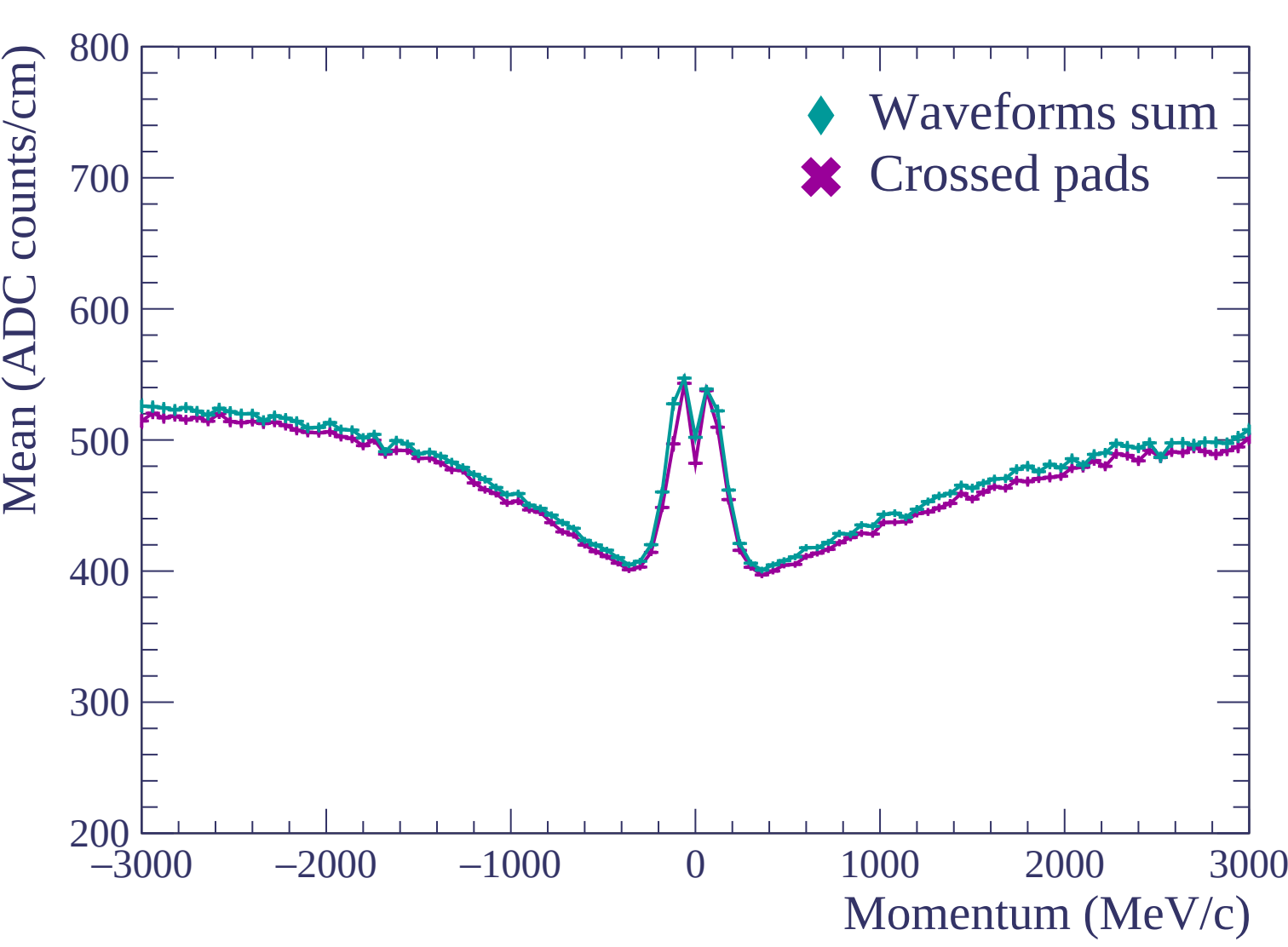


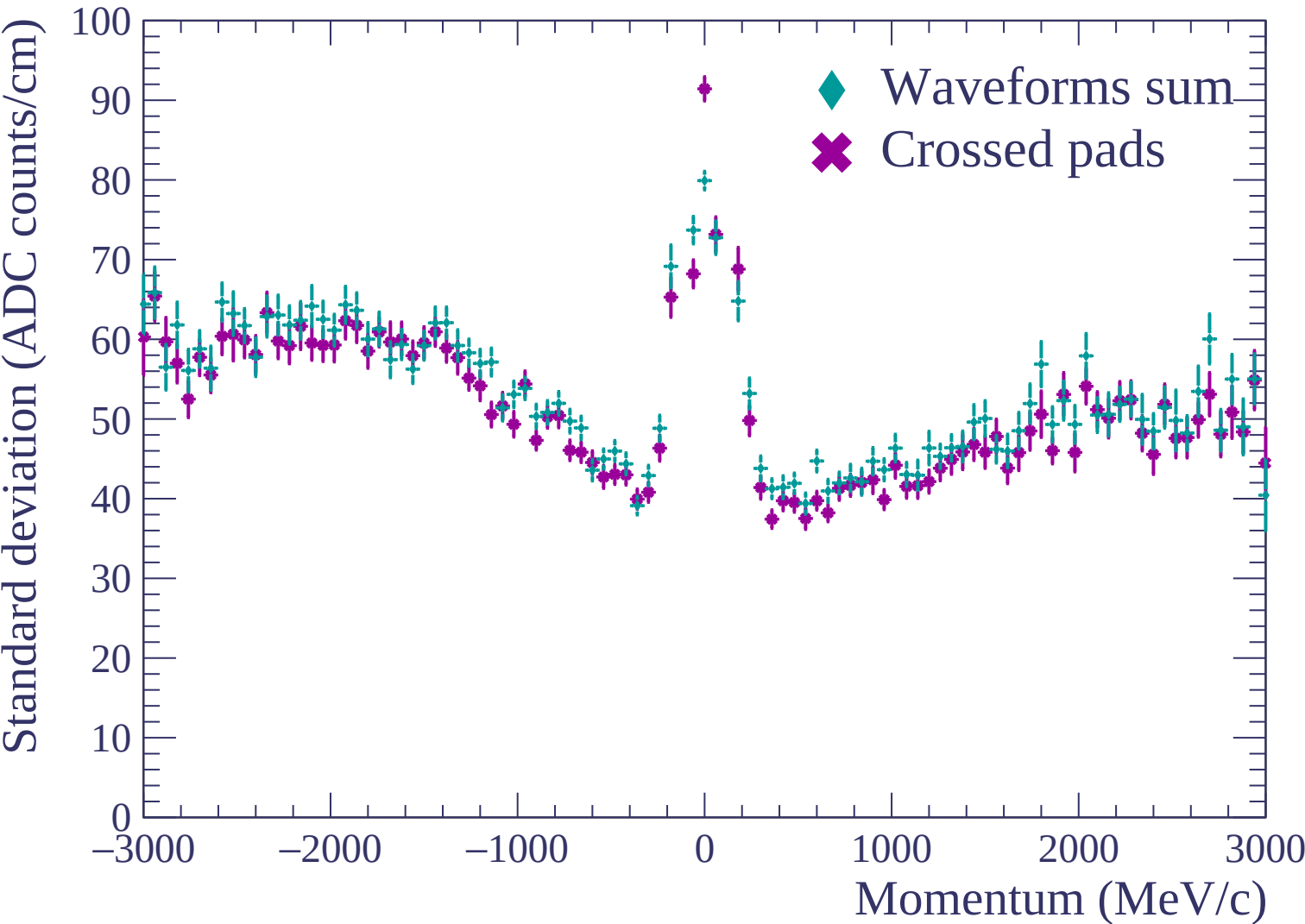


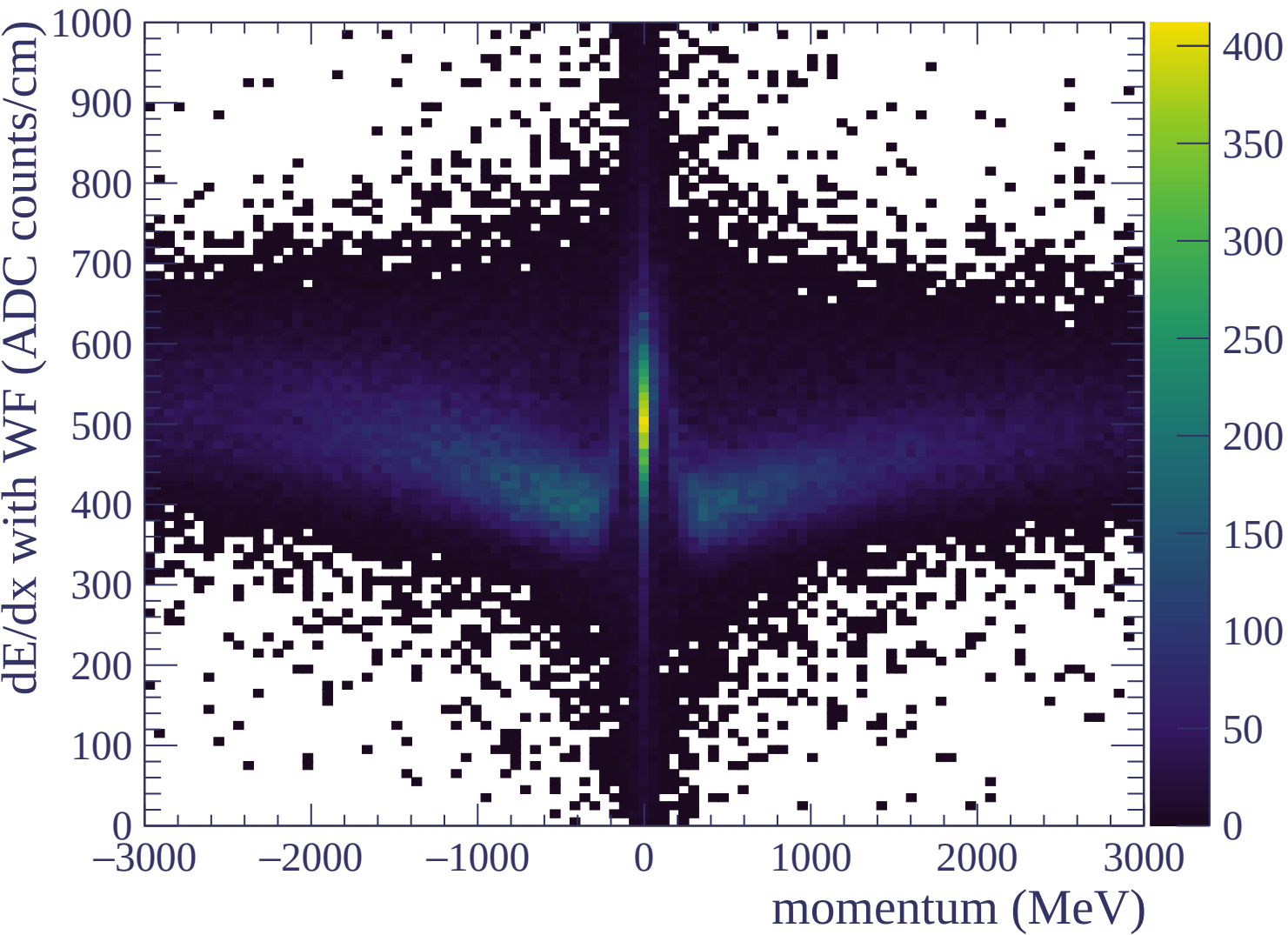


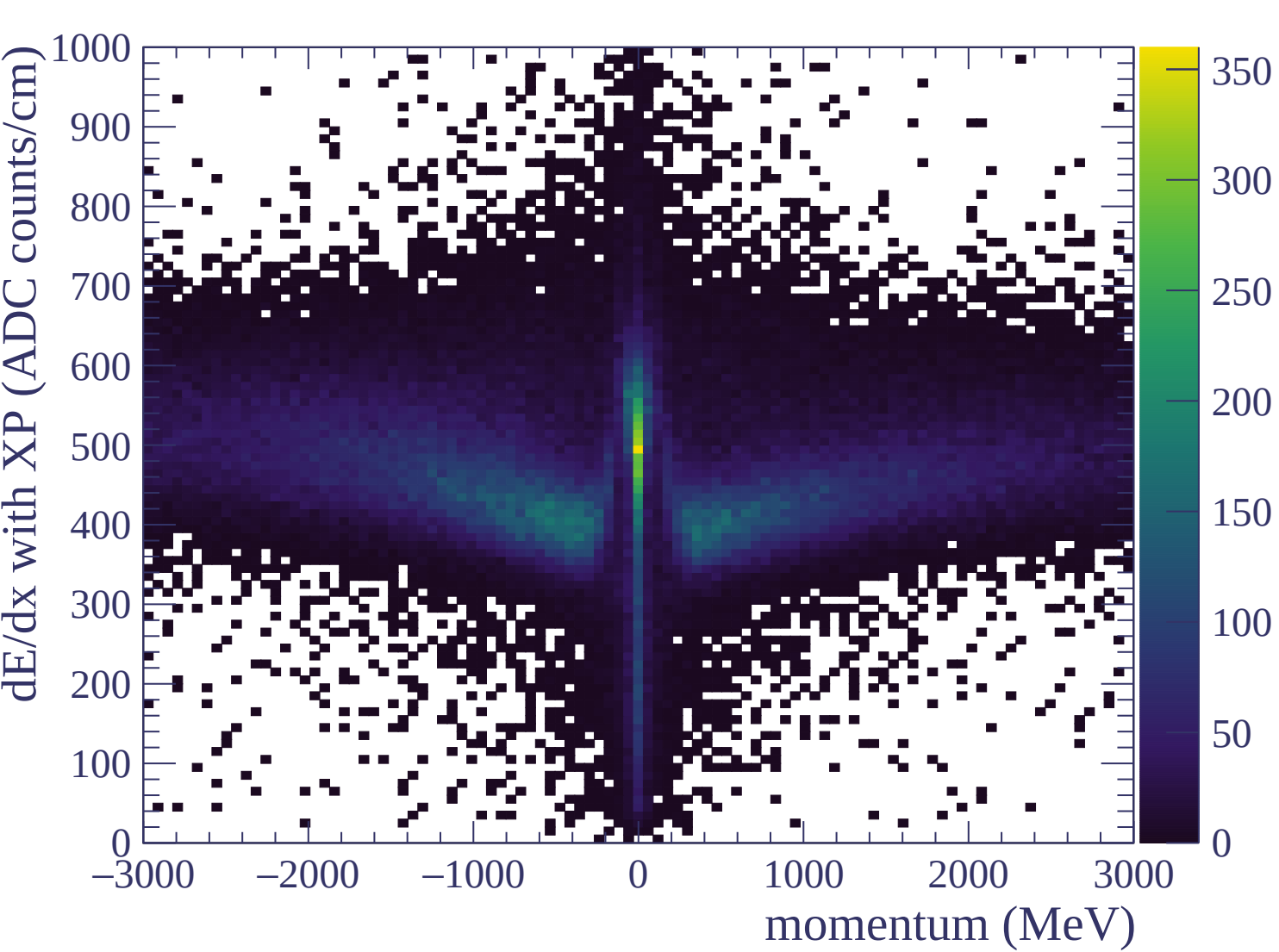
Resolution (%)

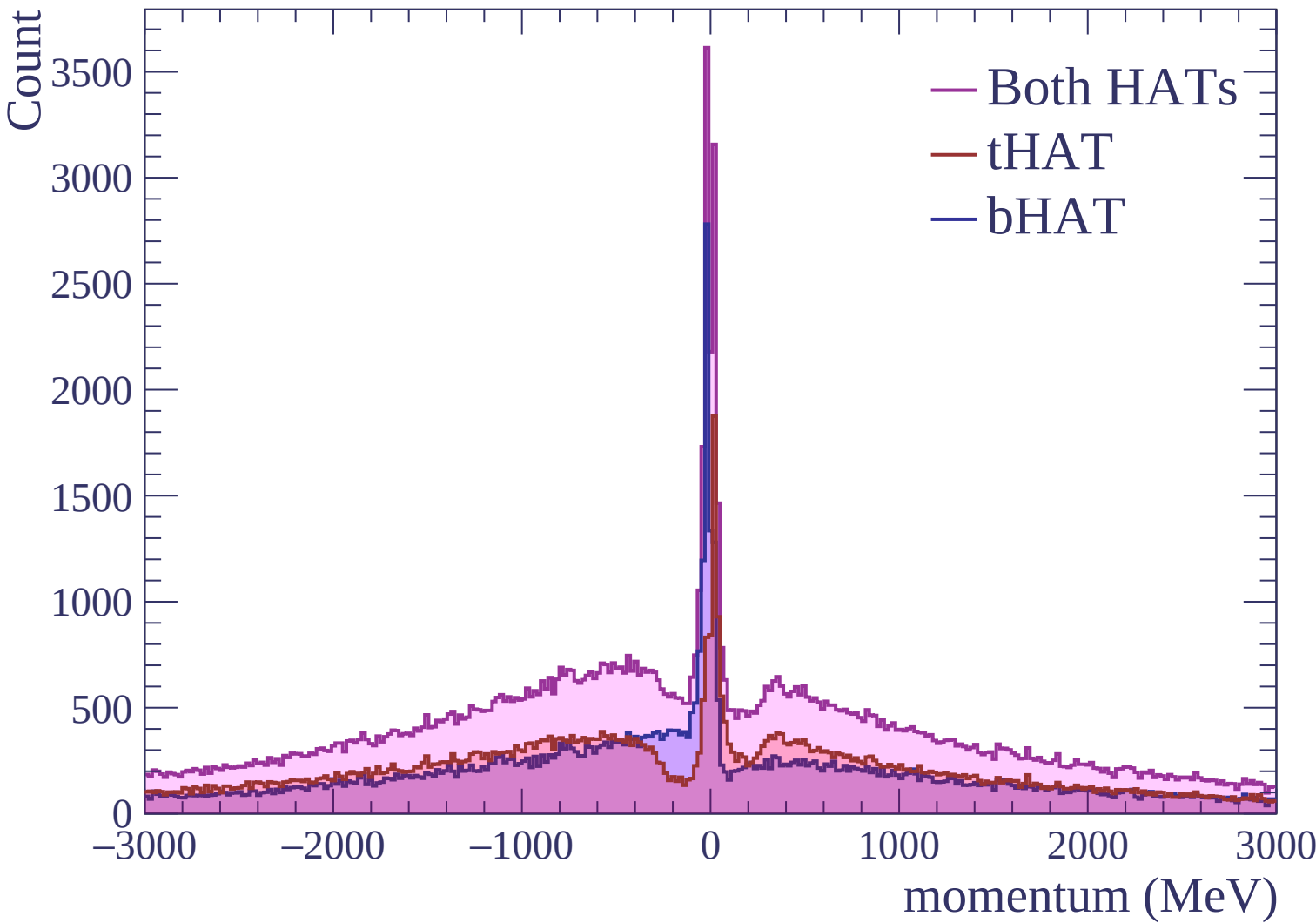


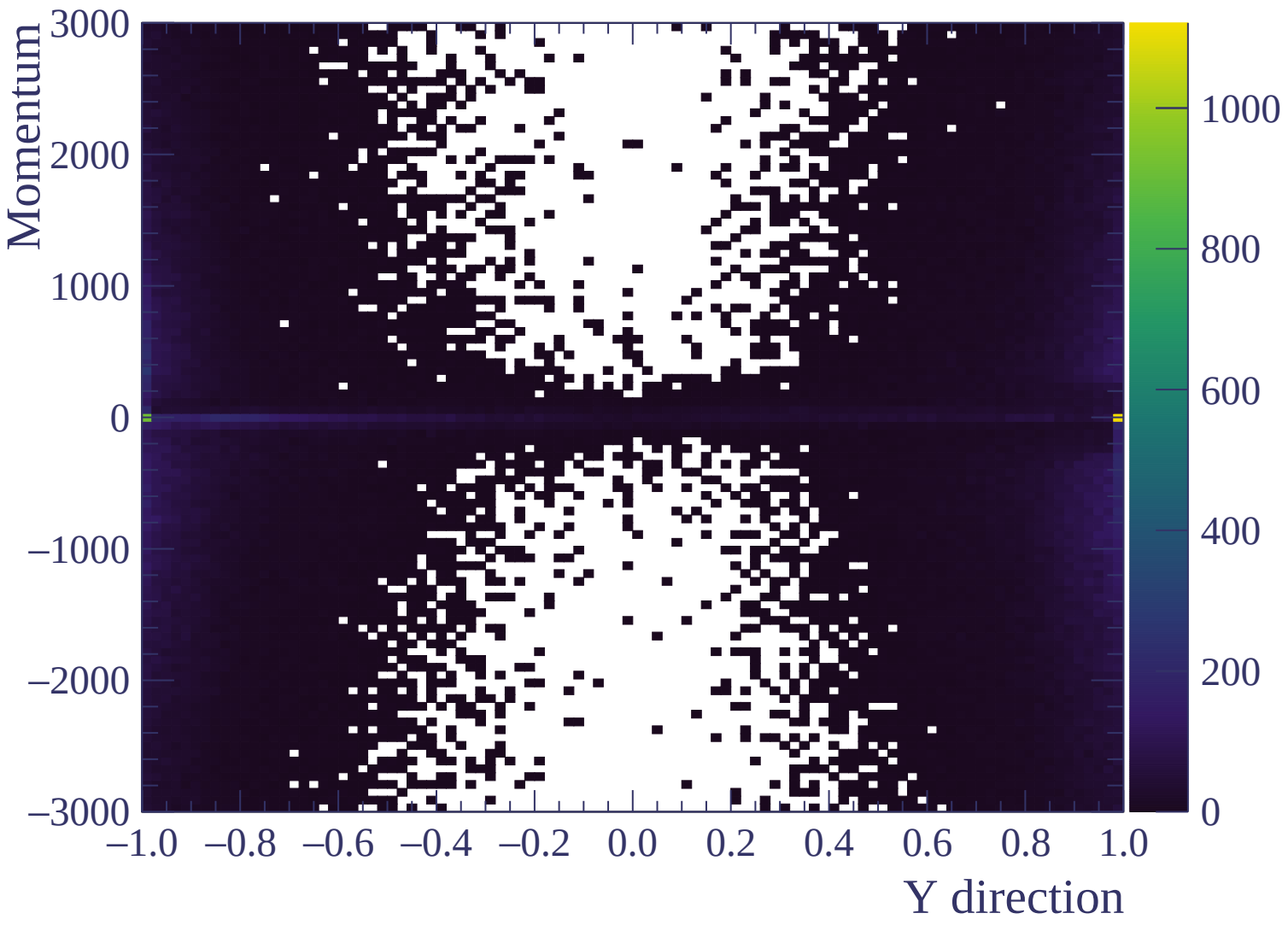


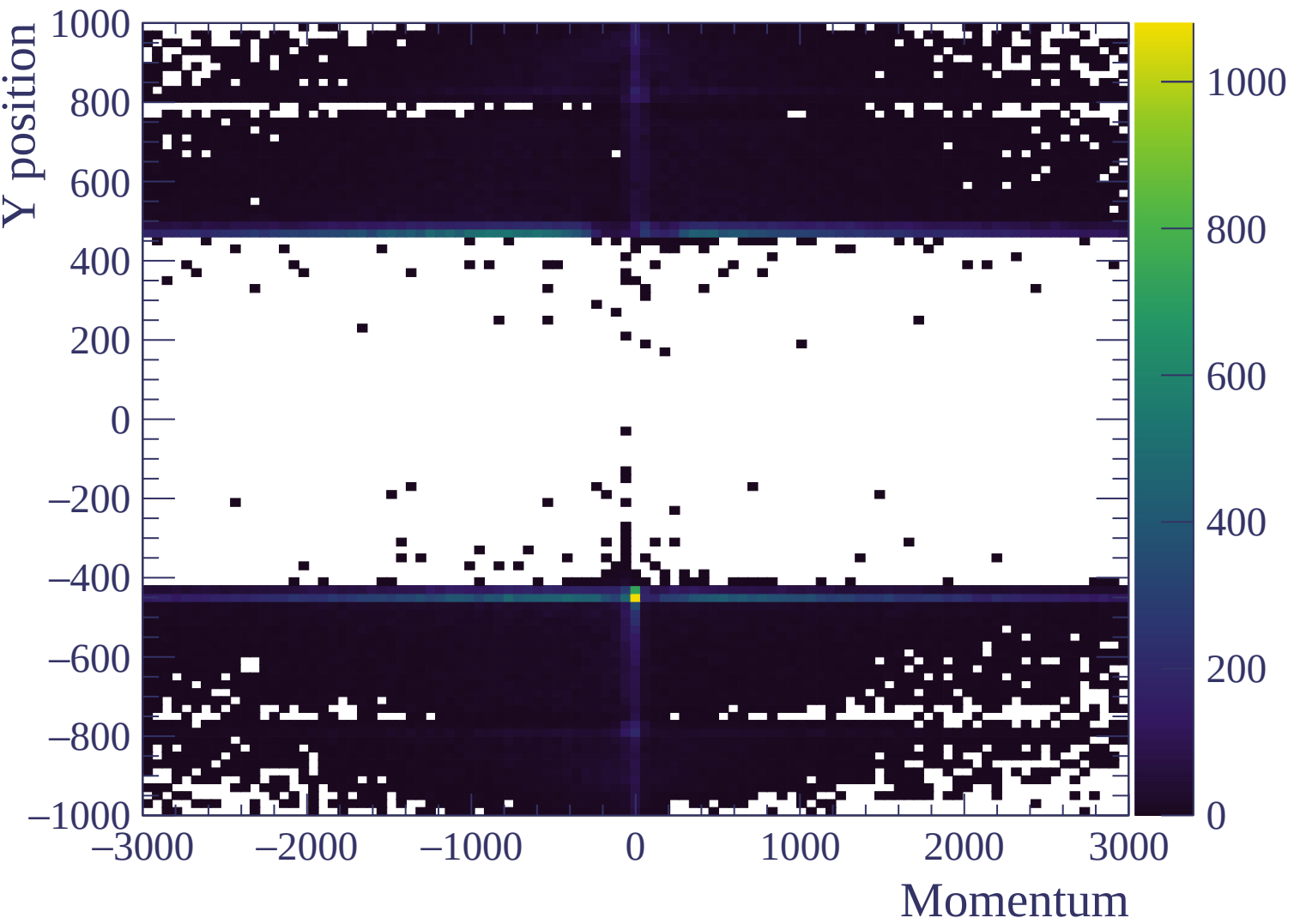


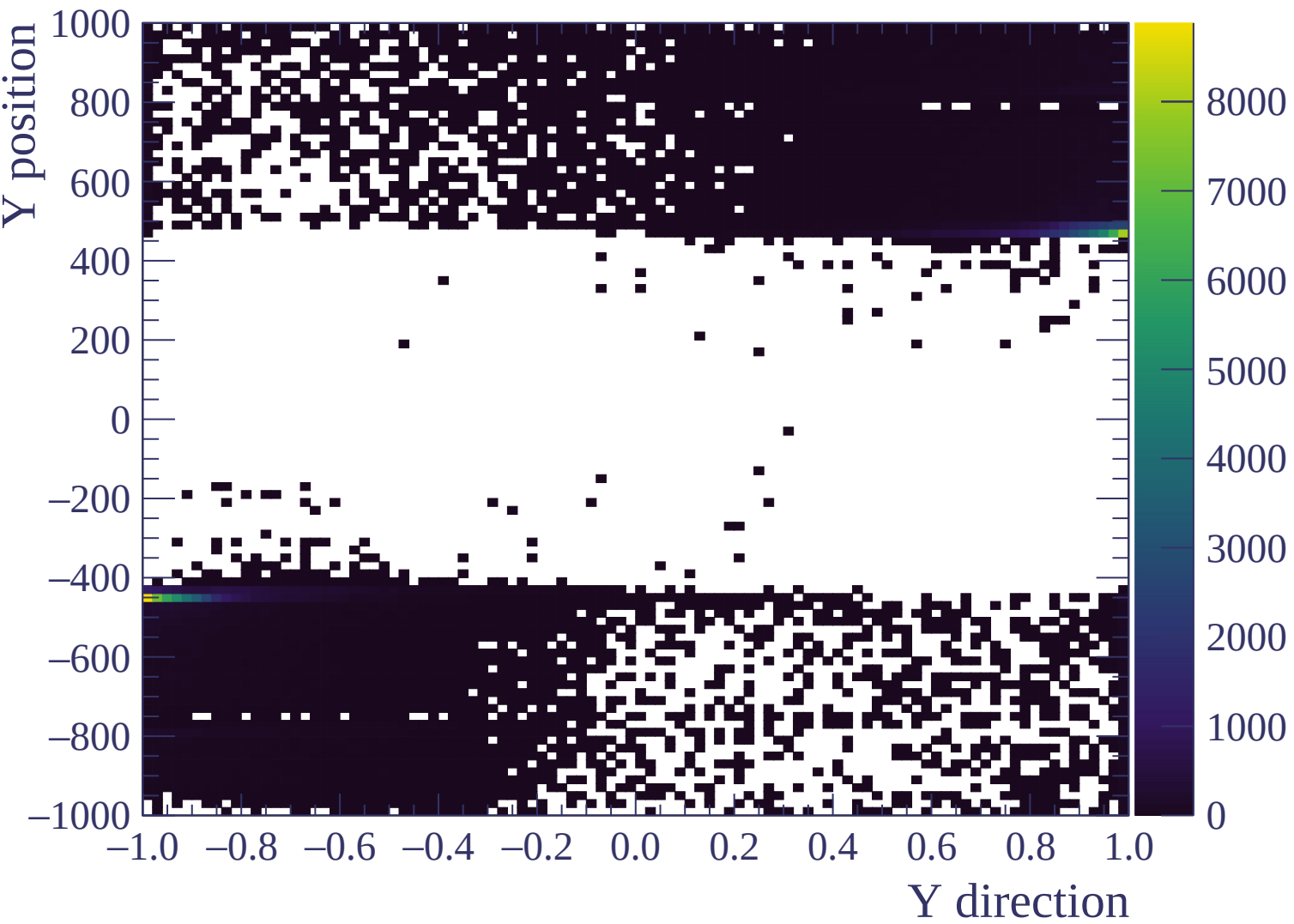






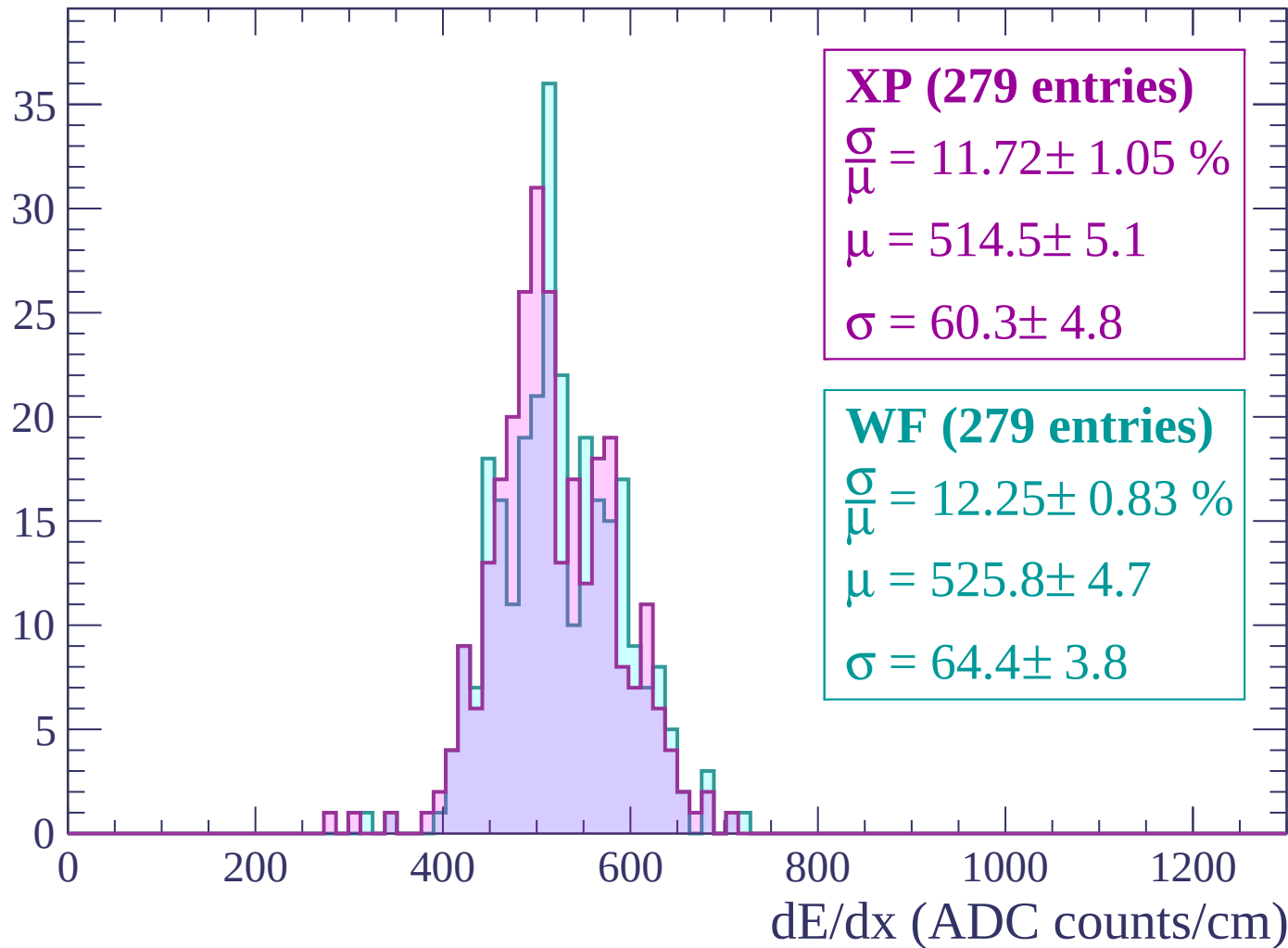






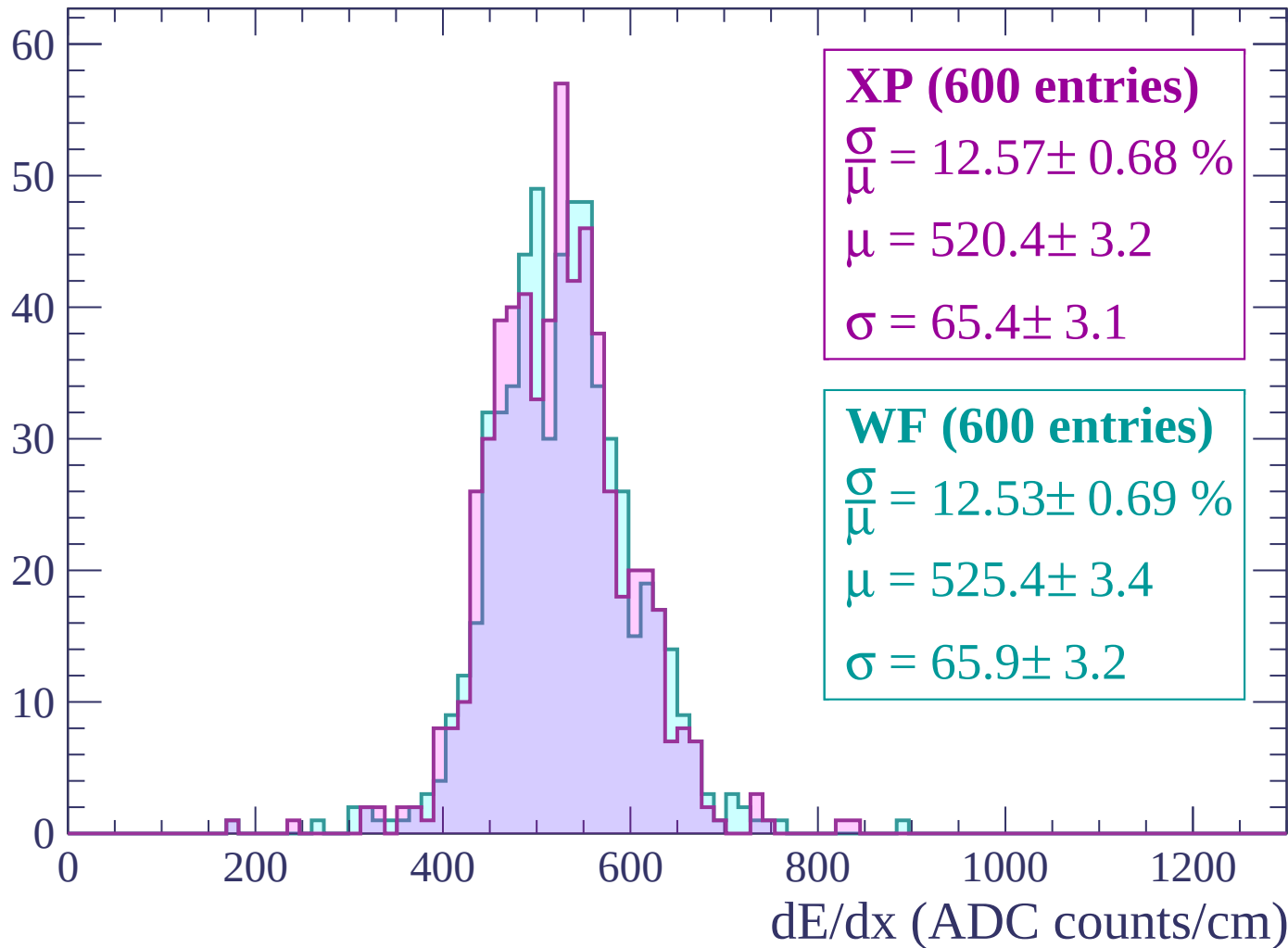
Energy loss | $-3000 < p < -2940$

Count



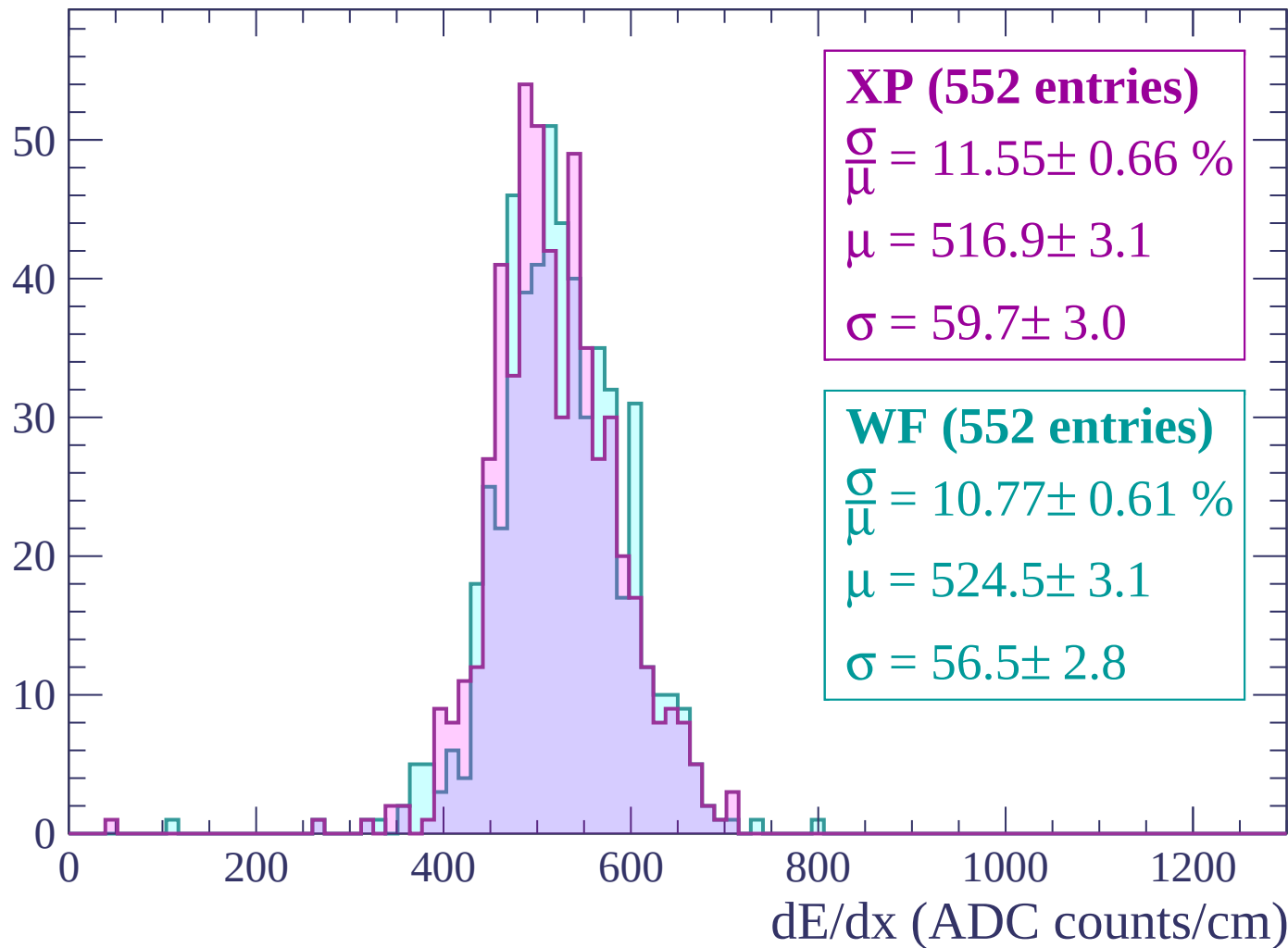
Energy loss | $-2940 < p < -2880$

Count



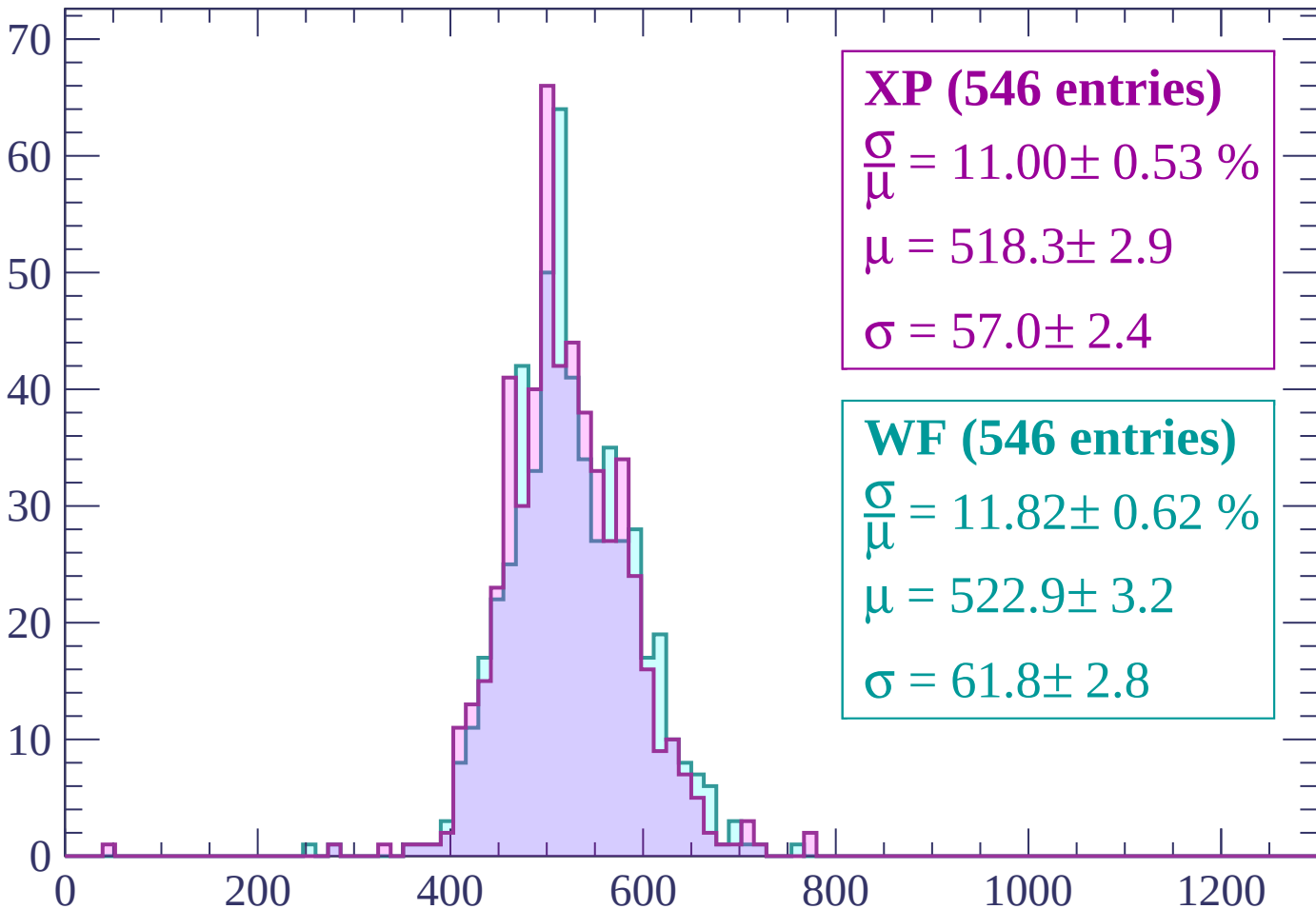
Energy loss | $-2880 < p < -2820$

Count



Energy loss | $-2820 < p < -2760$

Count



XP (546 entries)

$$\frac{\sigma}{\mu} = 11.00 \pm 0.53 \%$$

$$\mu = 518.3 \pm 2.9$$

$$\sigma = 57.0 \pm 2.4$$

WF (546 entries)

$$\frac{\sigma}{\mu} = 11.82 \pm 0.62 \%$$

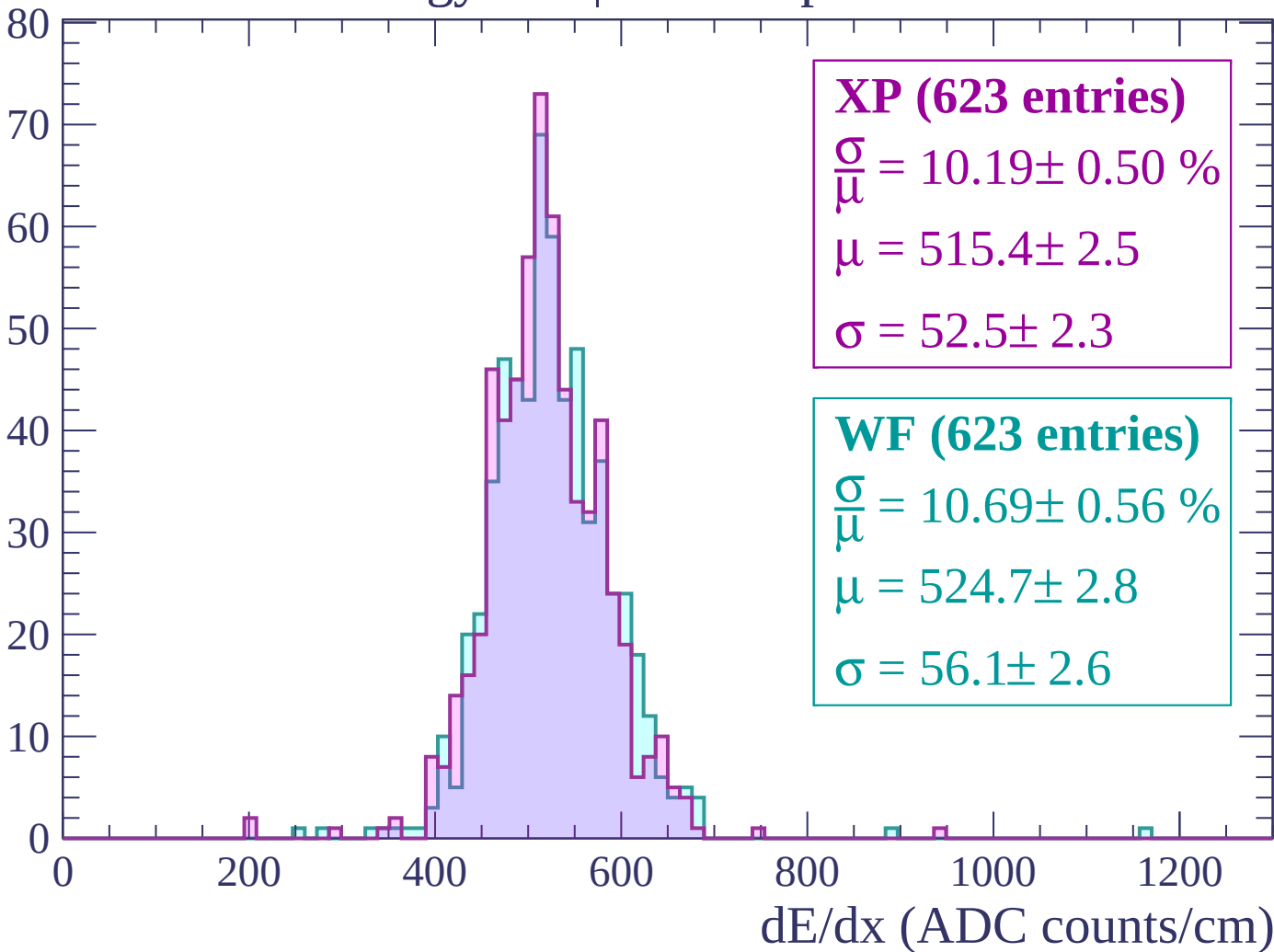
$$\mu = 522.9 \pm 3.2$$

$$\sigma = 61.8 \pm 2.8$$

dE/dx (ADC counts/cm)

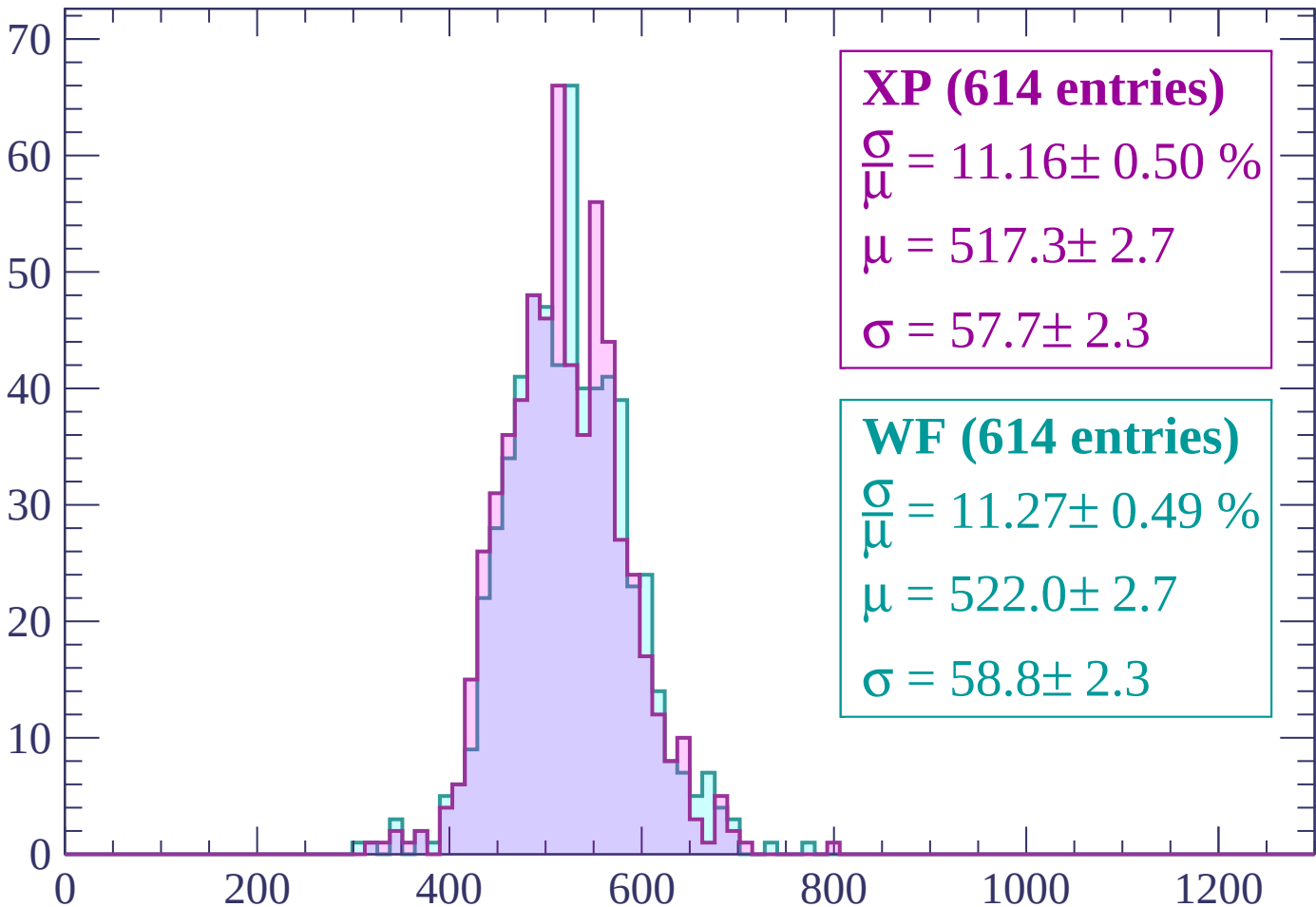
Energy loss | -2760 < p < -2700

Count



Energy loss | $-2700 < p < -2640$

Count



XP (614 entries)

$$\frac{\sigma}{\mu} = 11.16 \pm 0.50 \%$$

$$\mu = 517.3 \pm 2.7$$

$$\sigma = 57.7 \pm 2.3$$

WF (614 entries)

$$\frac{\sigma}{\mu} = 11.27 \pm 0.49 \%$$

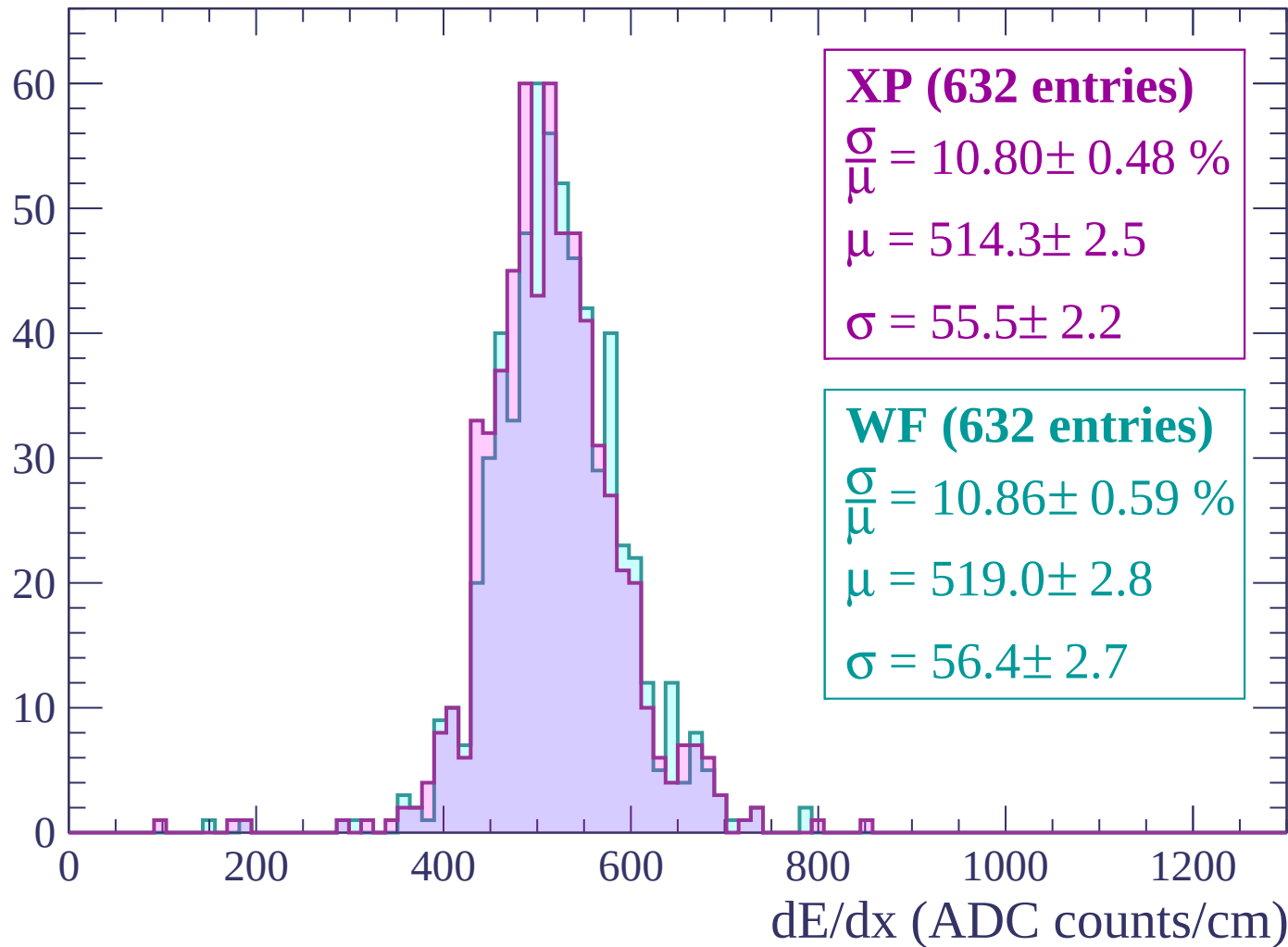
$$\mu = 522.0 \pm 2.7$$

$$\sigma = 58.8 \pm 2.3$$

dE/dx (ADC counts/cm)

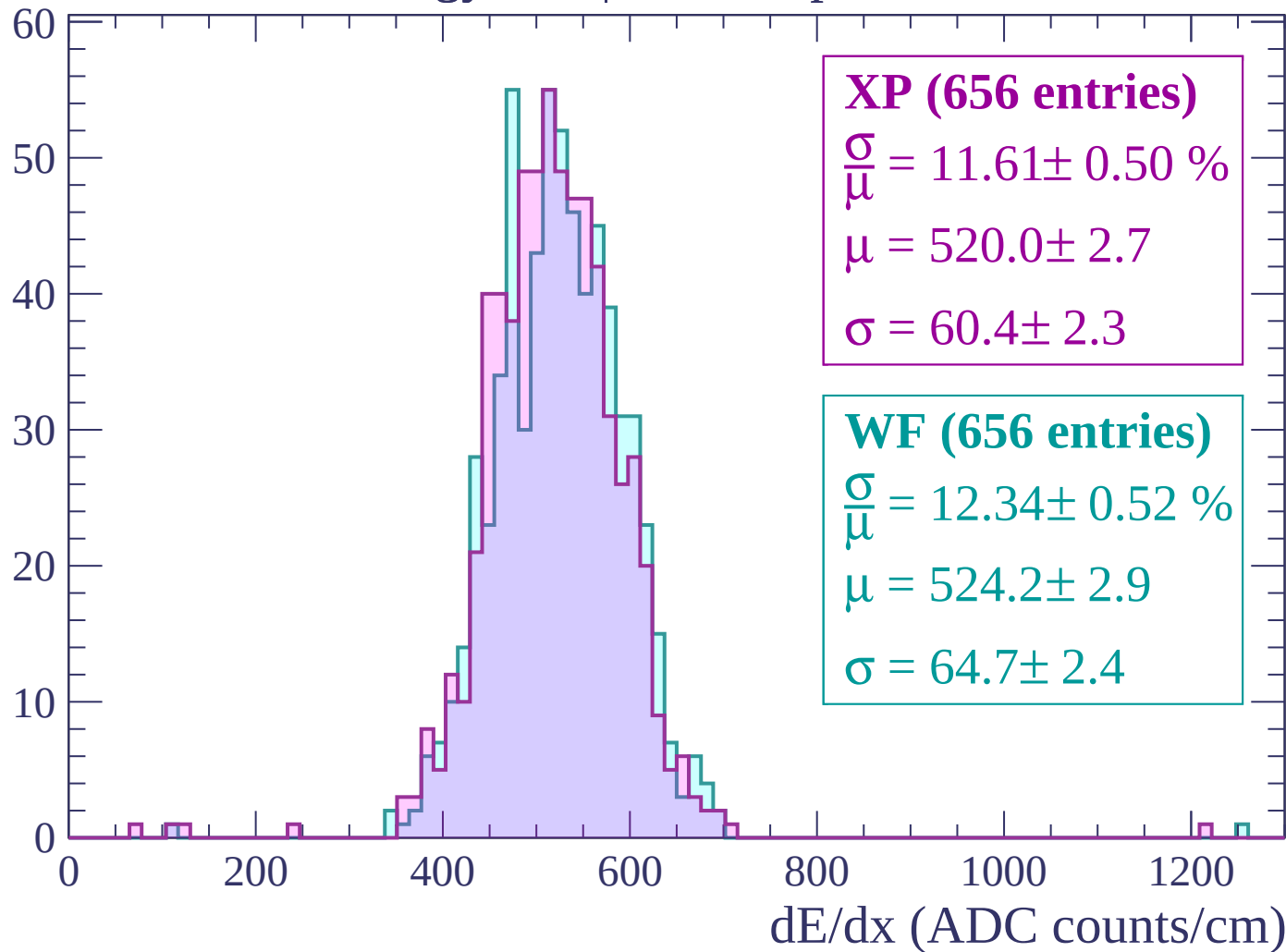
Energy loss | $-2640 < p < -2580$

Count



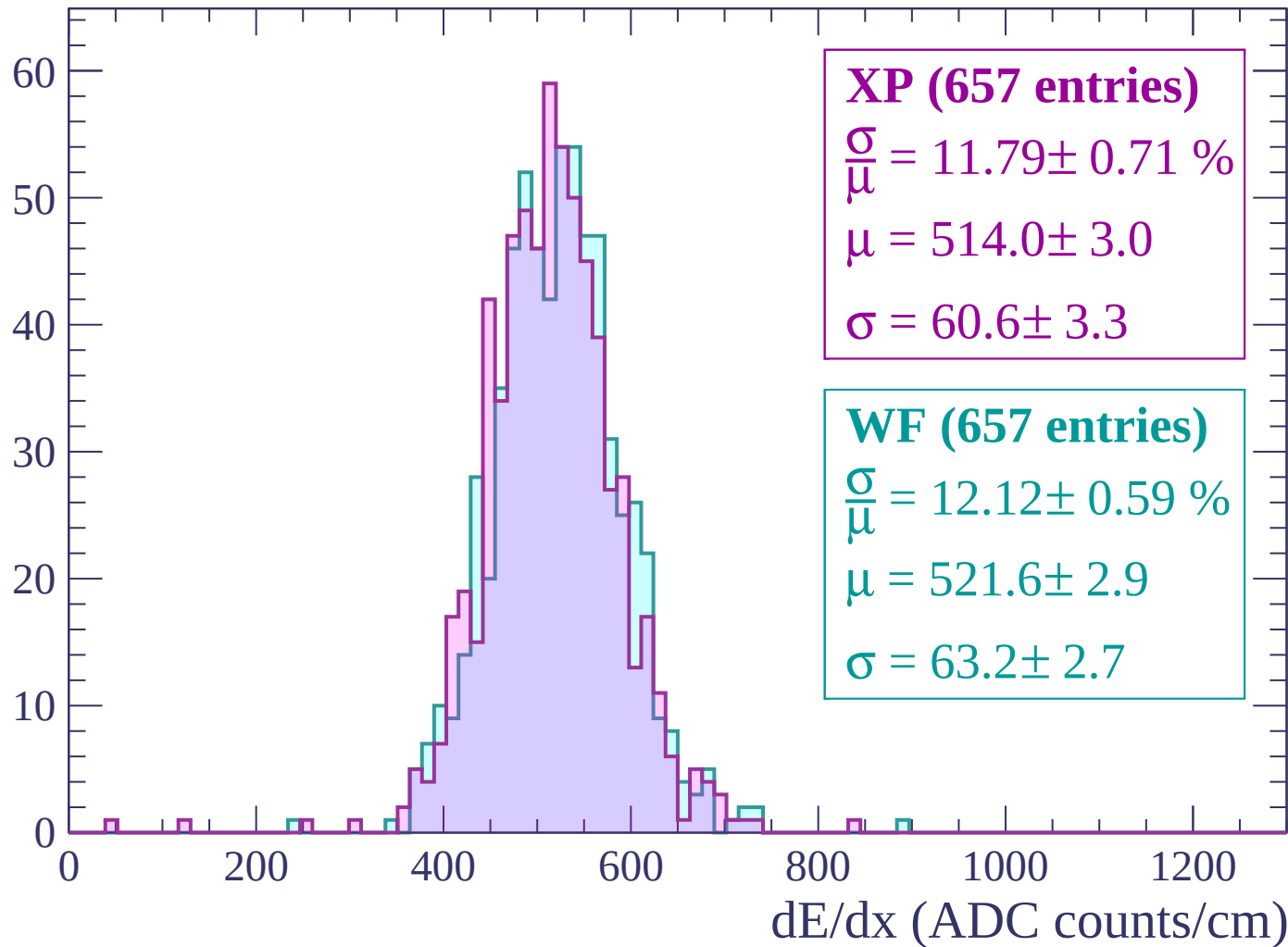
Energy loss | $-2580 < p < -2520$

Count



Energy loss | $-2520 < p < -2460$

Count



Energy loss | $-2460 < p < -2400$

Count

XP (702 entries)

$$\frac{\sigma}{\mu} = 11.68 \pm 0.50 \%$$

$$\mu = 513.2 \pm 2.8$$

$$\sigma = 59.9 \pm 2.2$$

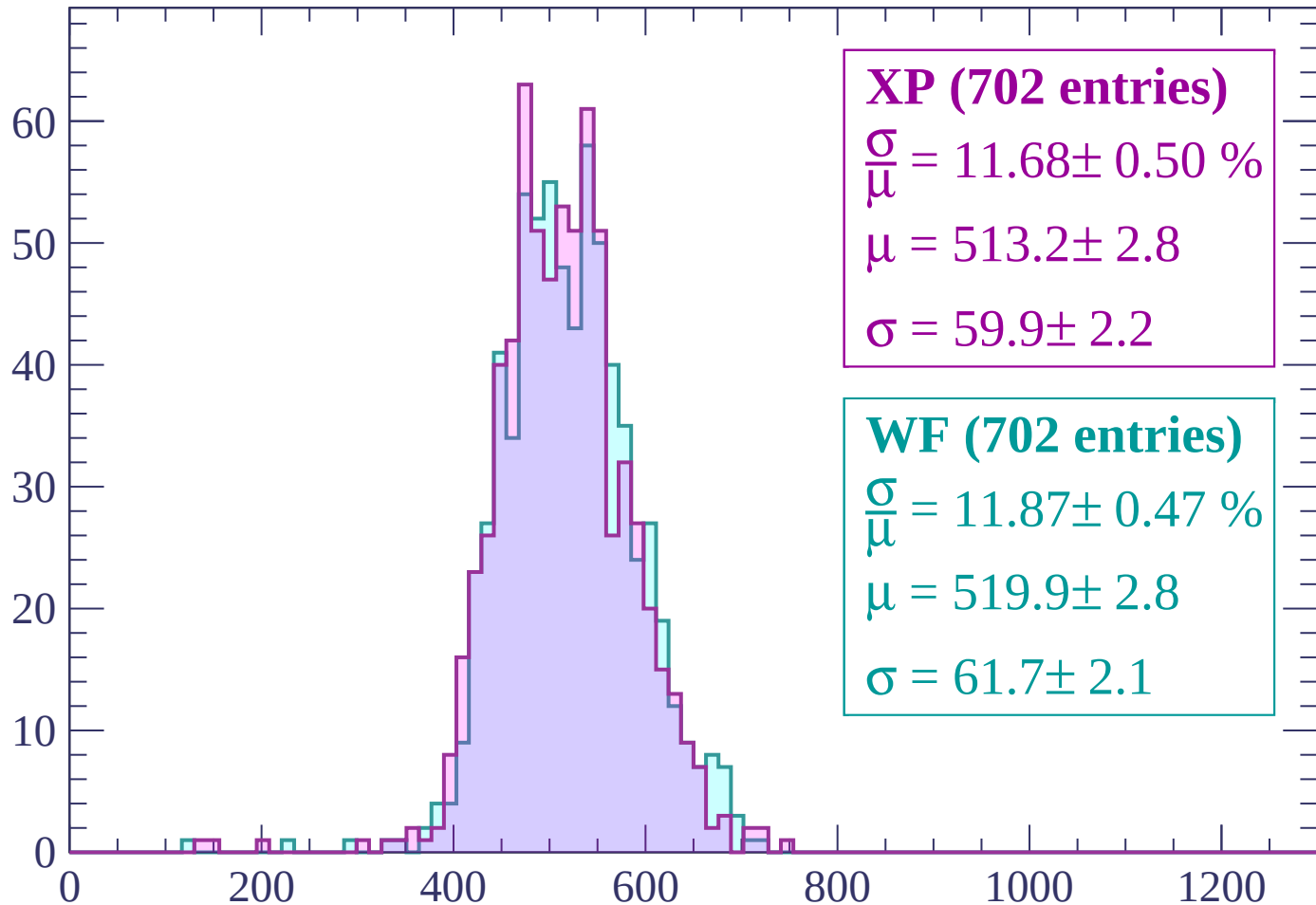
WF (702 entries)

$$\frac{\sigma}{\mu} = 11.87 \pm 0.47 \%$$

$$\mu = 519.9 \pm 2.8$$

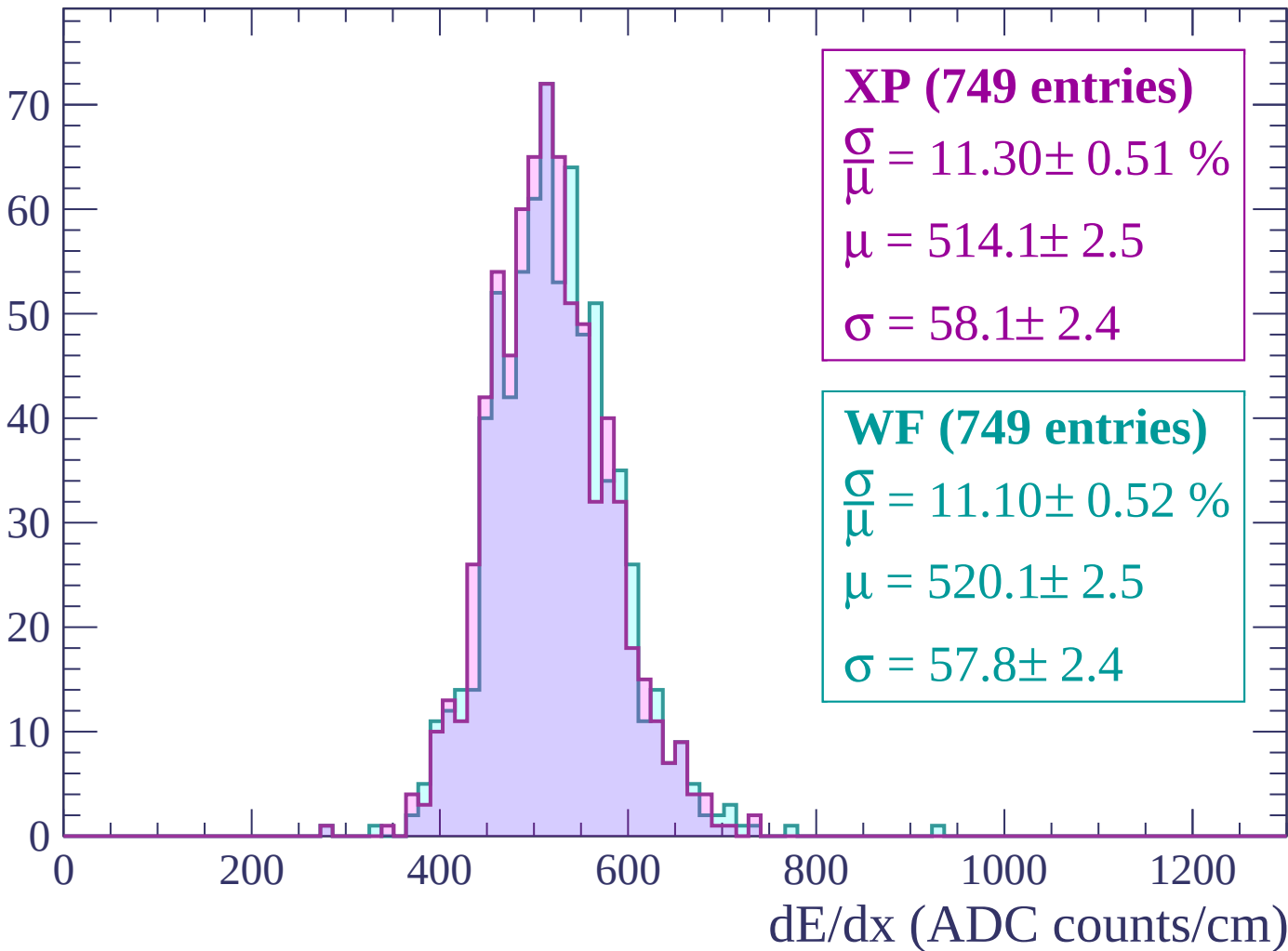
$$\sigma = 61.7 \pm 2.1$$

dE/dx (ADC counts/cm)



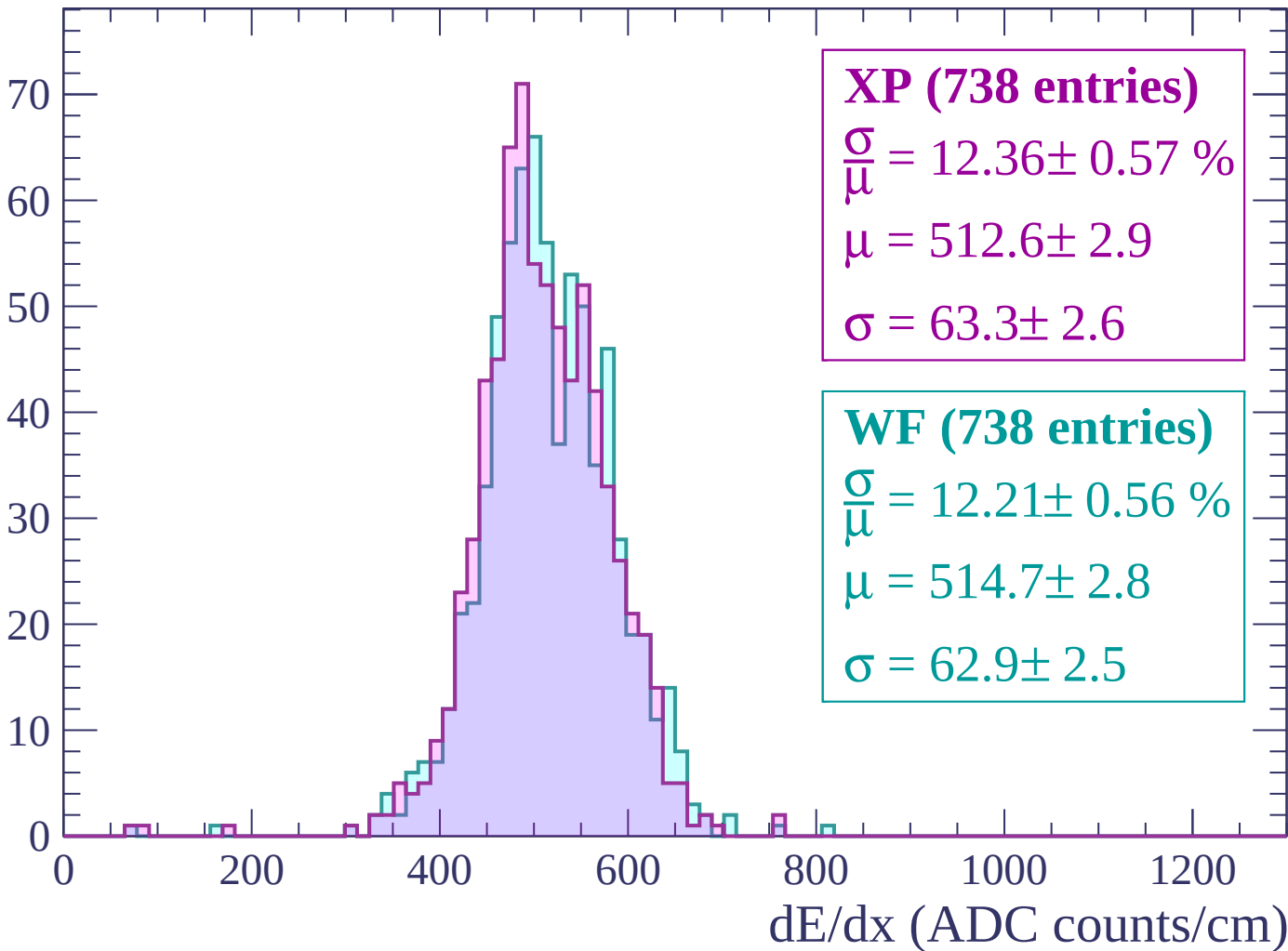
Energy loss | $-2400 < p < -2340$

Count



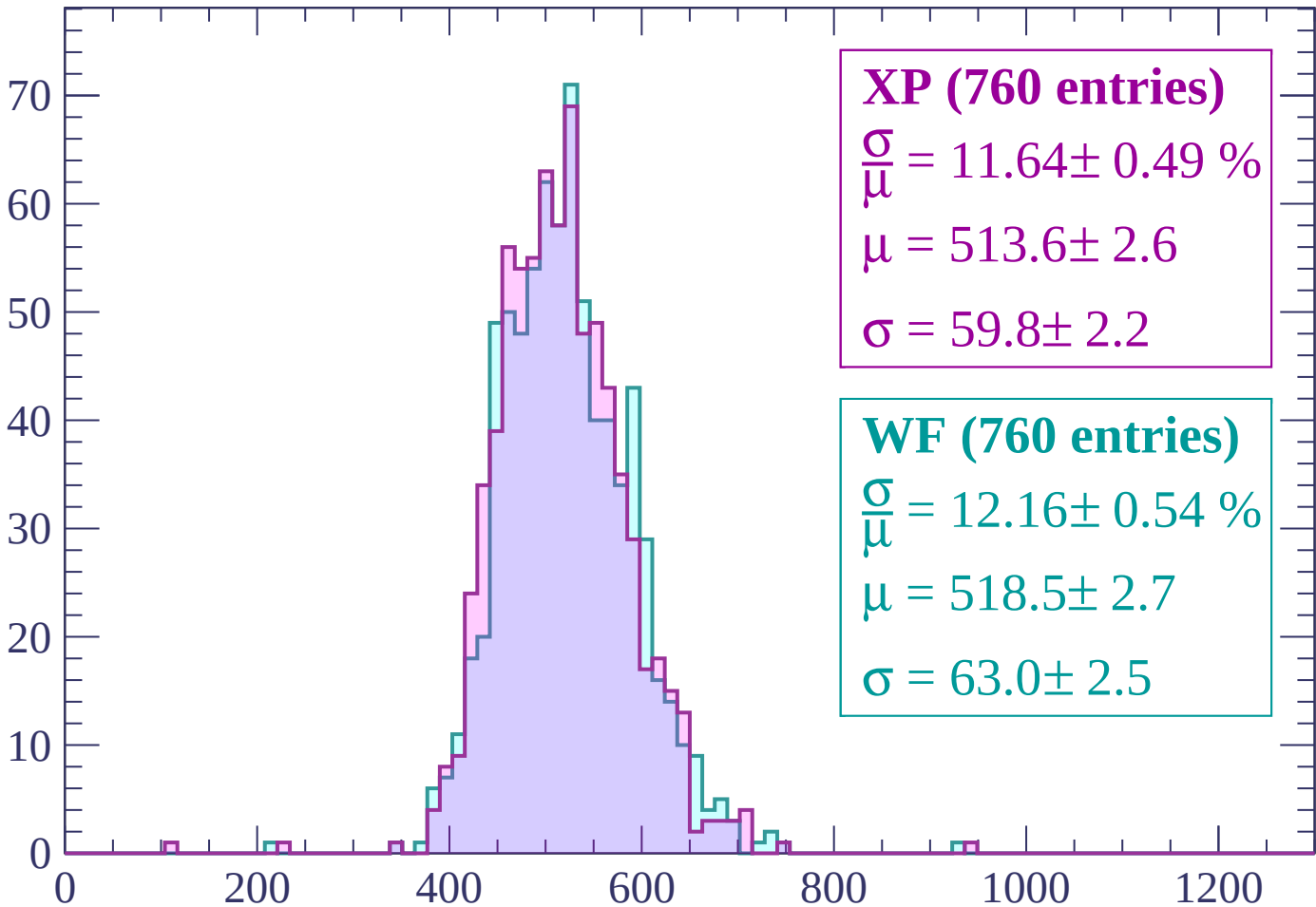
Energy loss | $-2340 < p < -2280$

Count



Energy loss | $-2280 < p < -2220$

Count



XP (760 entries)

$$\frac{\sigma}{\mu} = 11.64 \pm 0.49 \%$$

$$\mu = 513.6 \pm 2.6$$

$$\sigma = 59.8 \pm 2.2$$

WF (760 entries)

$$\frac{\sigma}{\mu} = 12.16 \pm 0.54 \%$$

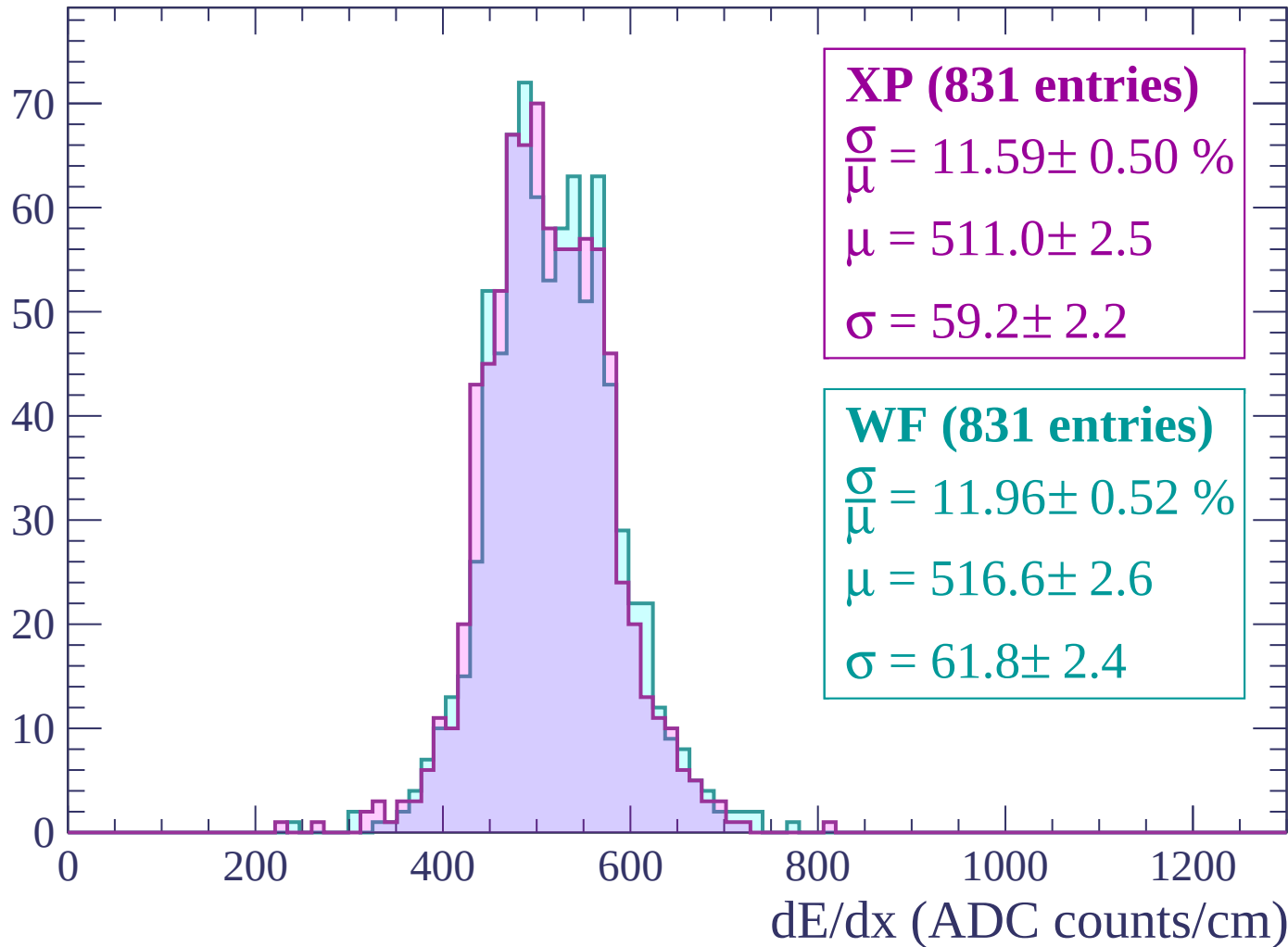
$$\mu = 518.5 \pm 2.7$$

$$\sigma = 63.0 \pm 2.5$$

dE/dx (ADC counts/cm)

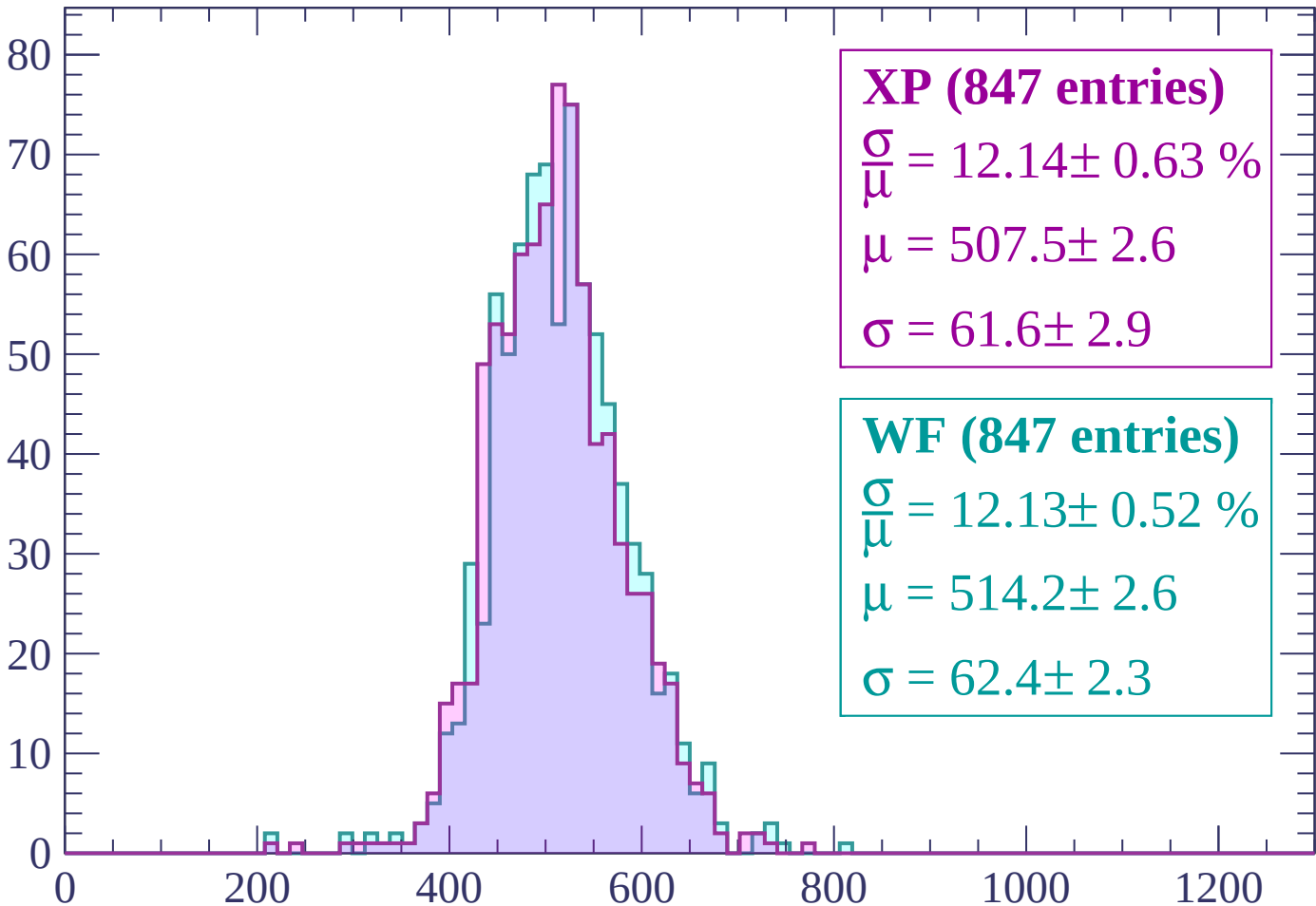
Energy loss | -2220 < p < -2160

Count



Energy loss | -2160 < p < -2100

Count



XP (847 entries)

$$\frac{\sigma}{\mu} = 12.14 \pm 0.63 \%$$

$$\mu = 507.5 \pm 2.6$$

$$\sigma = 61.6 \pm 2.9$$

WF (847 entries)

$$\frac{\sigma}{\mu} = 12.13 \pm 0.52 \%$$

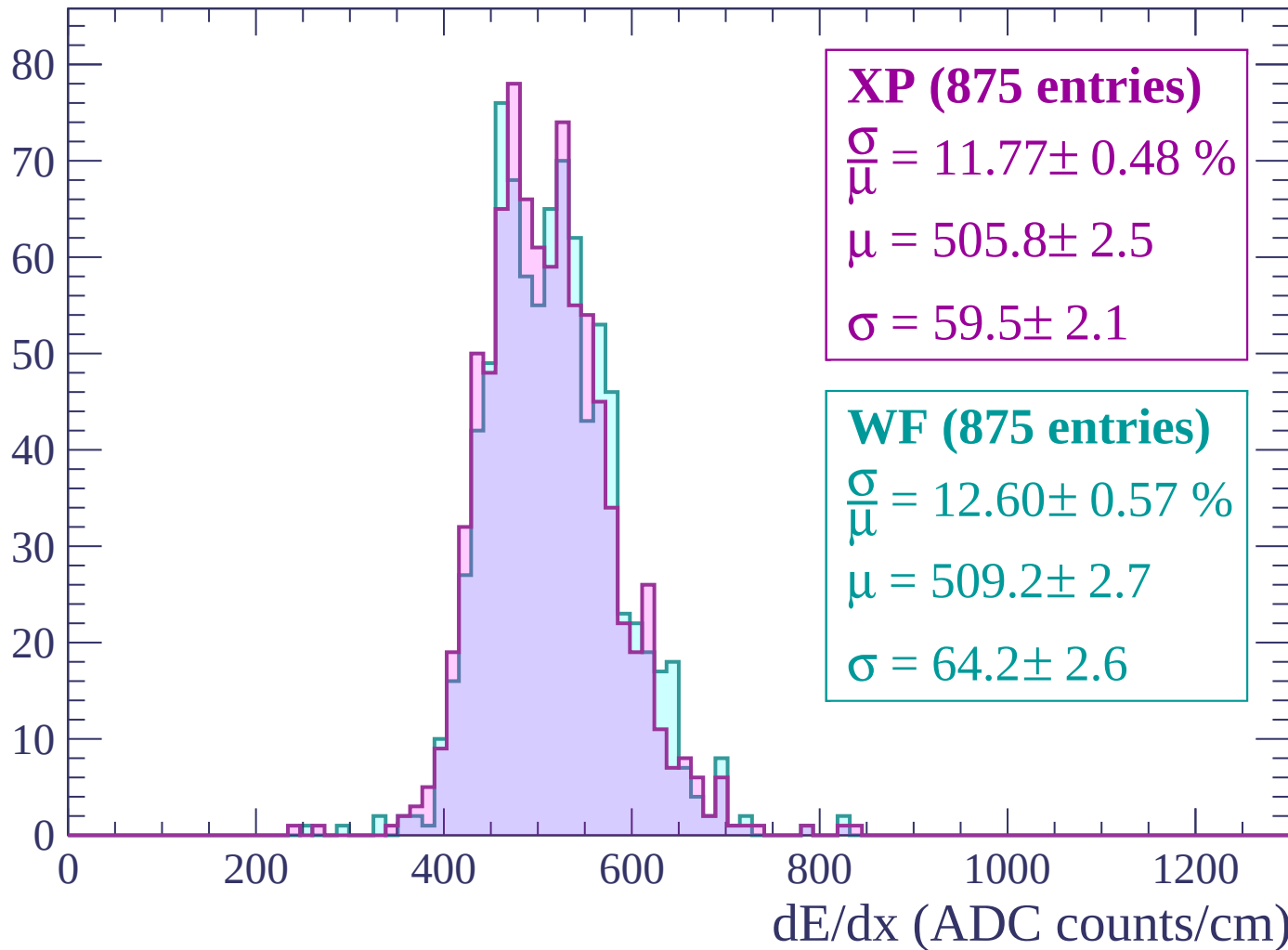
$$\mu = 514.2 \pm 2.6$$

$$\sigma = 62.4 \pm 2.3$$

dE/dx (ADC counts/cm)

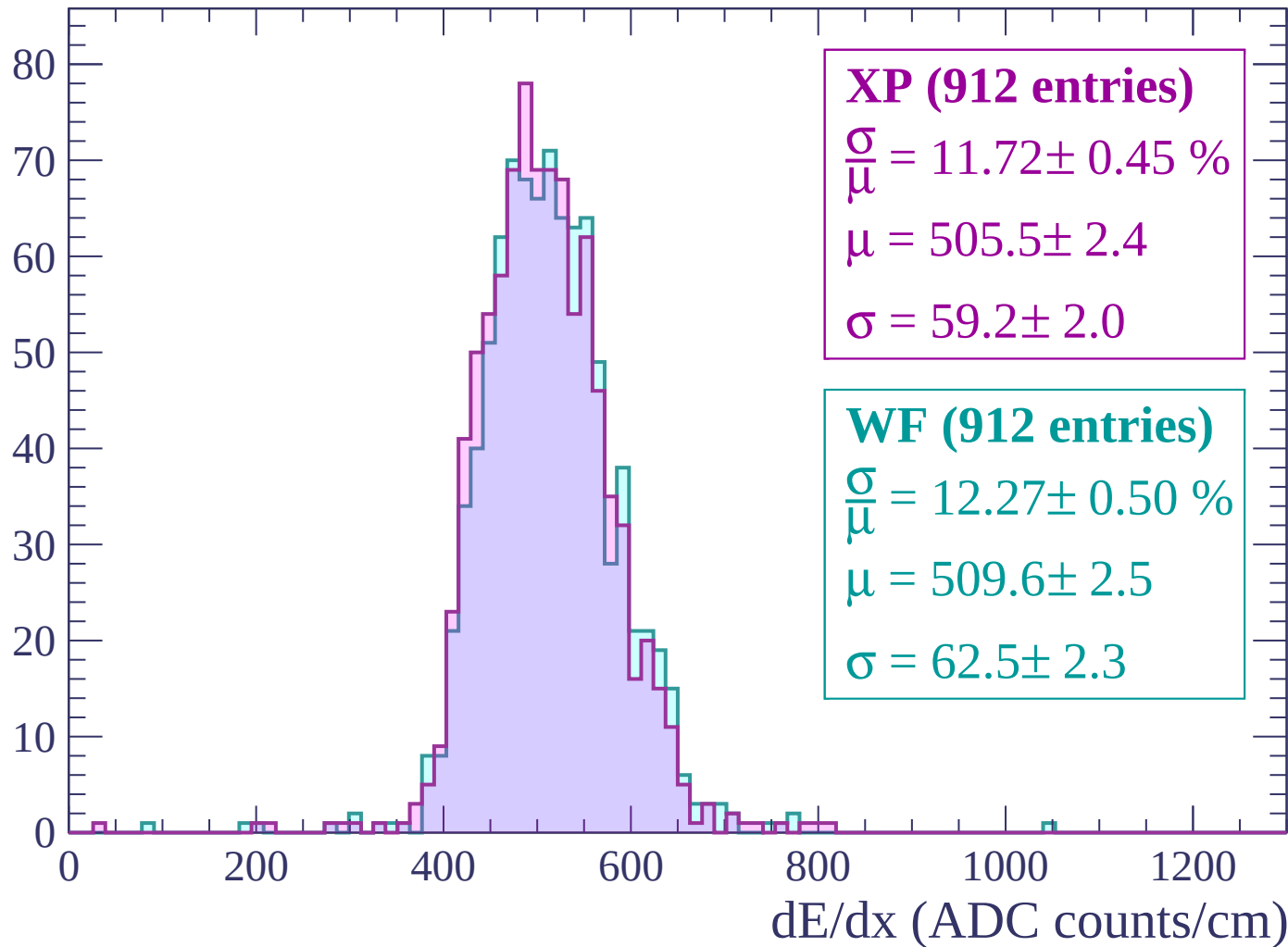
Energy loss | -2100 < p < -2040

Count



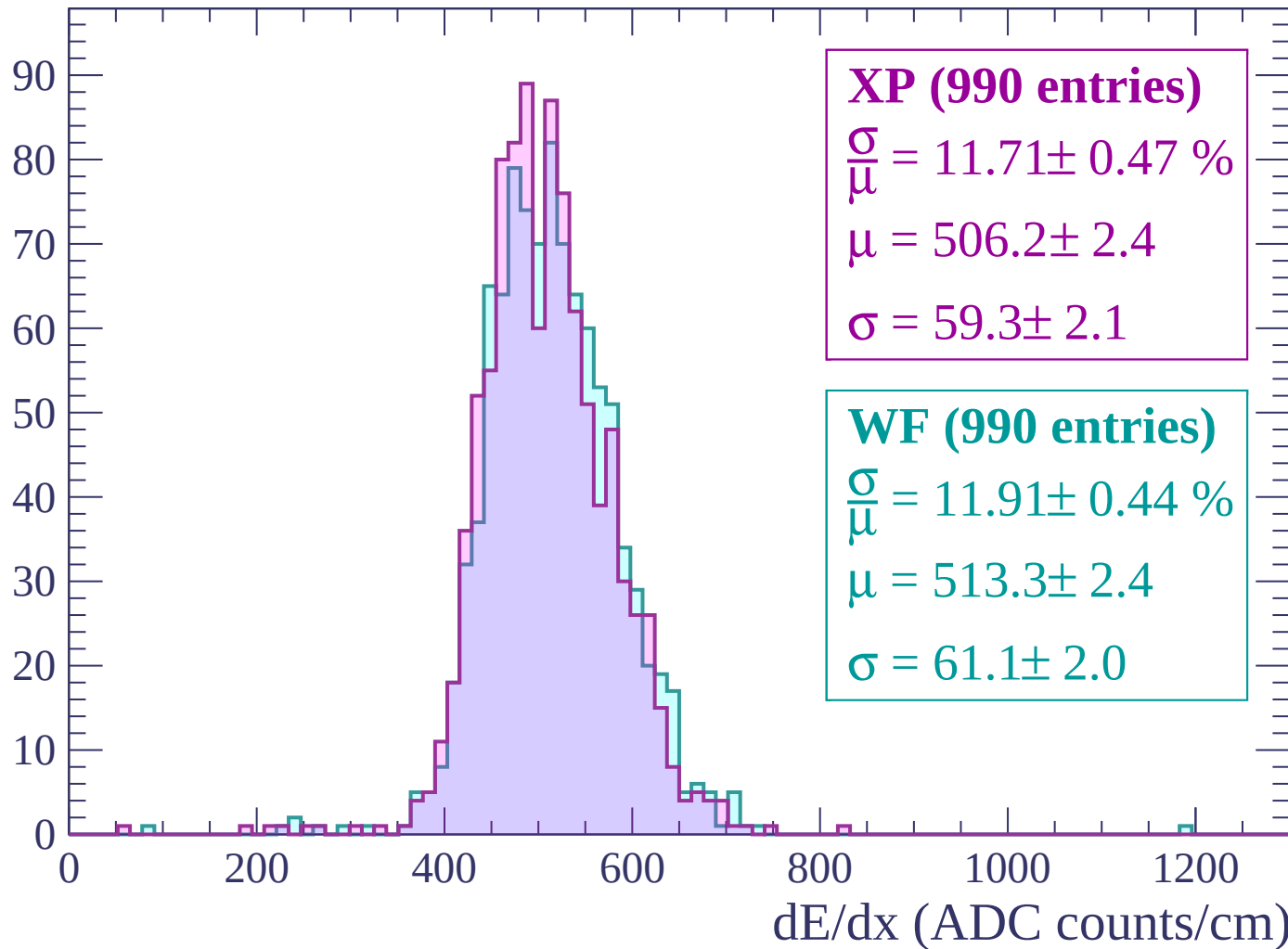
Energy loss | -2040 < p < -1980

Count



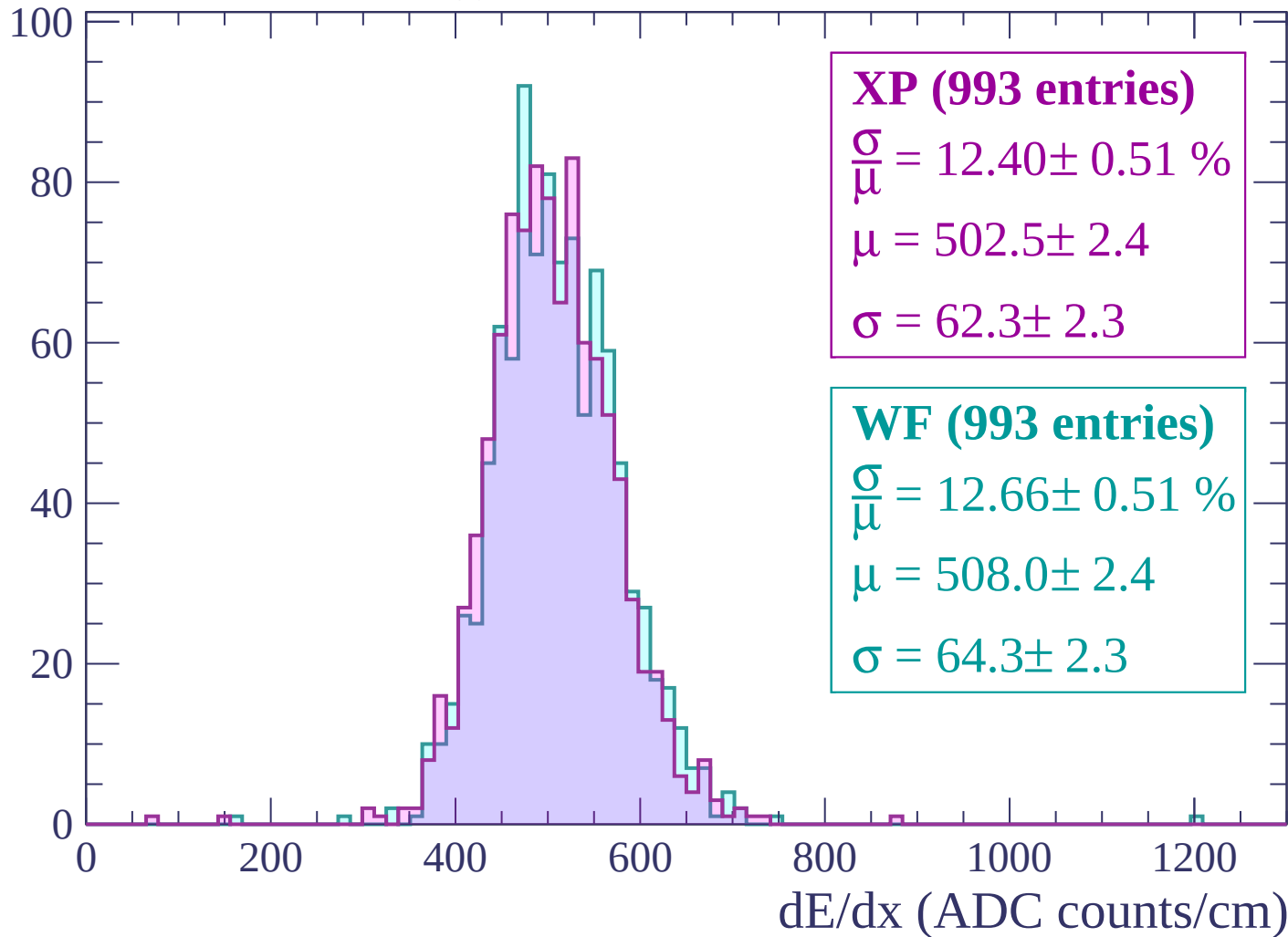
Energy loss | -1980 < p < -1920

Count



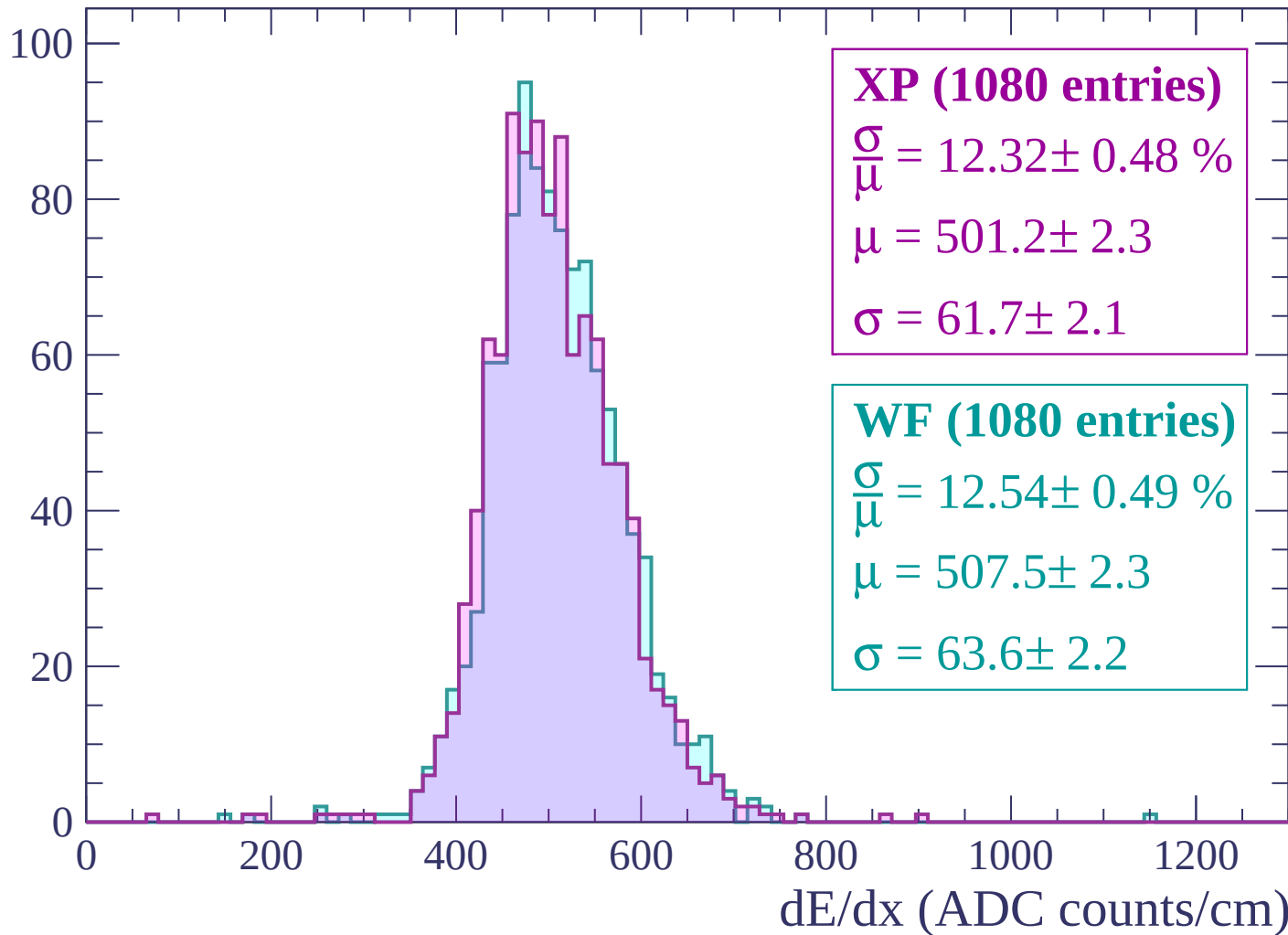
Energy loss | -1920 < p < -1860

Count



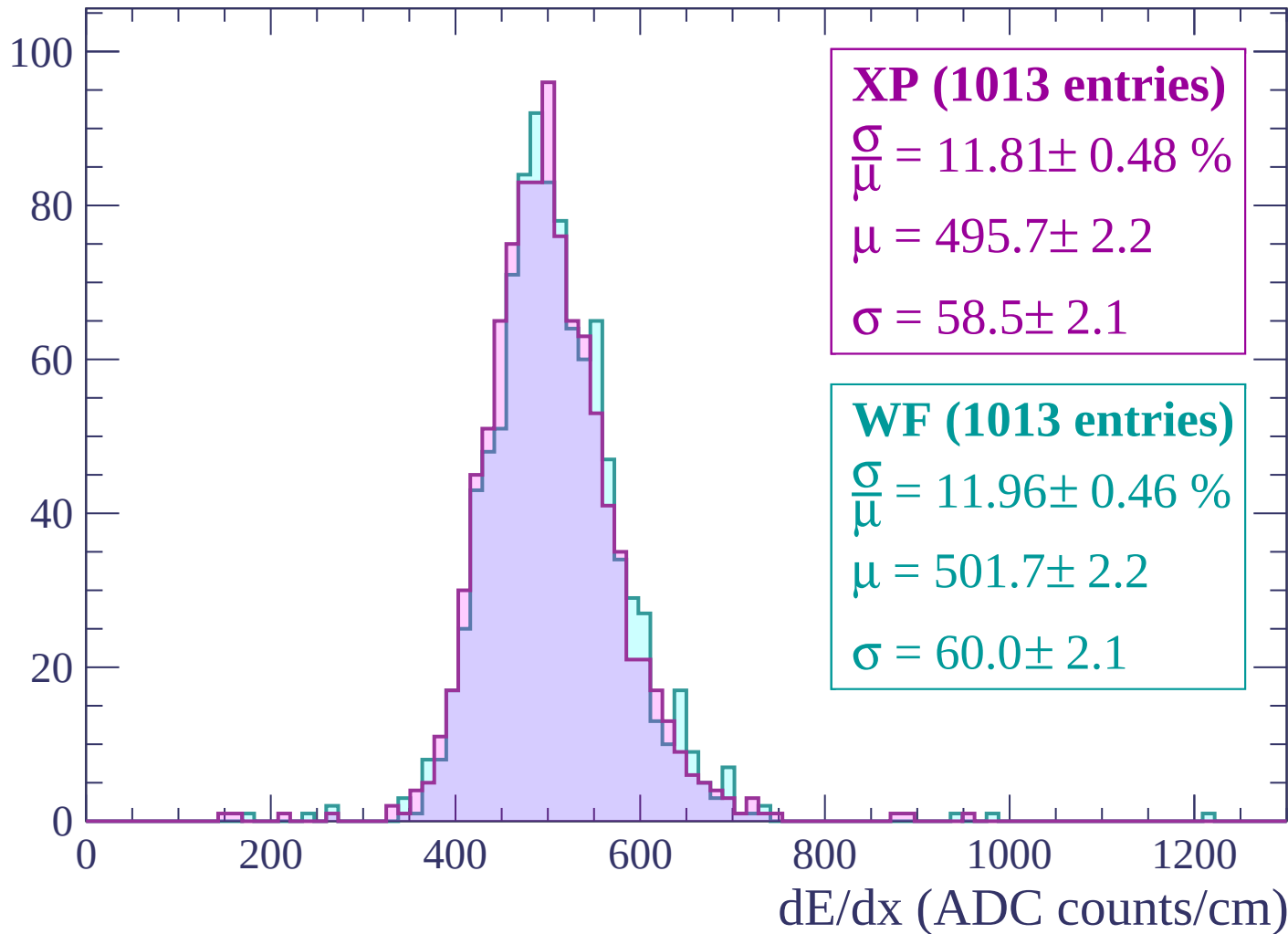
Energy loss | $-1860 < p < -1800$

Count



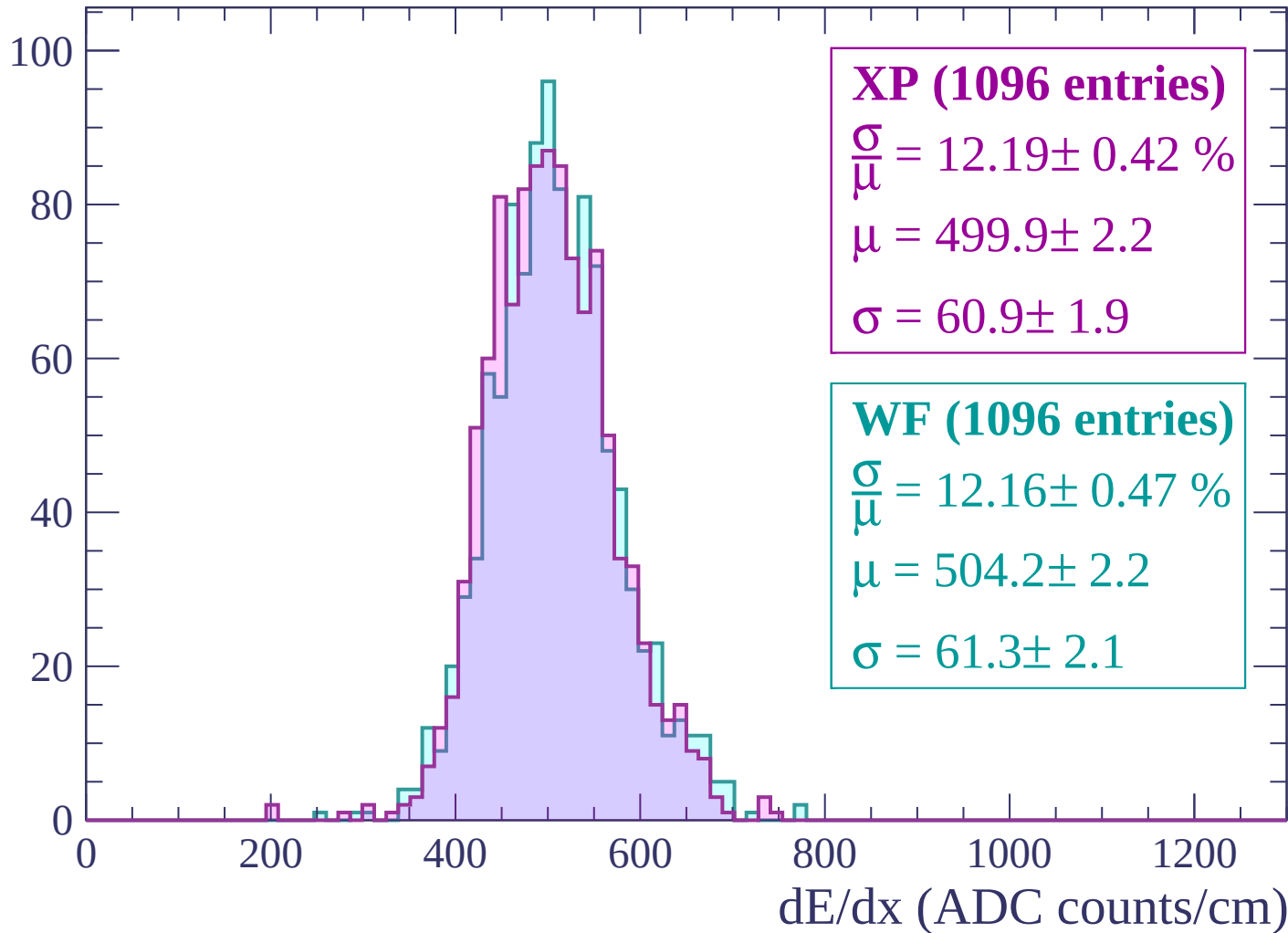
Energy loss | $-1800 < p < -1740$

Count



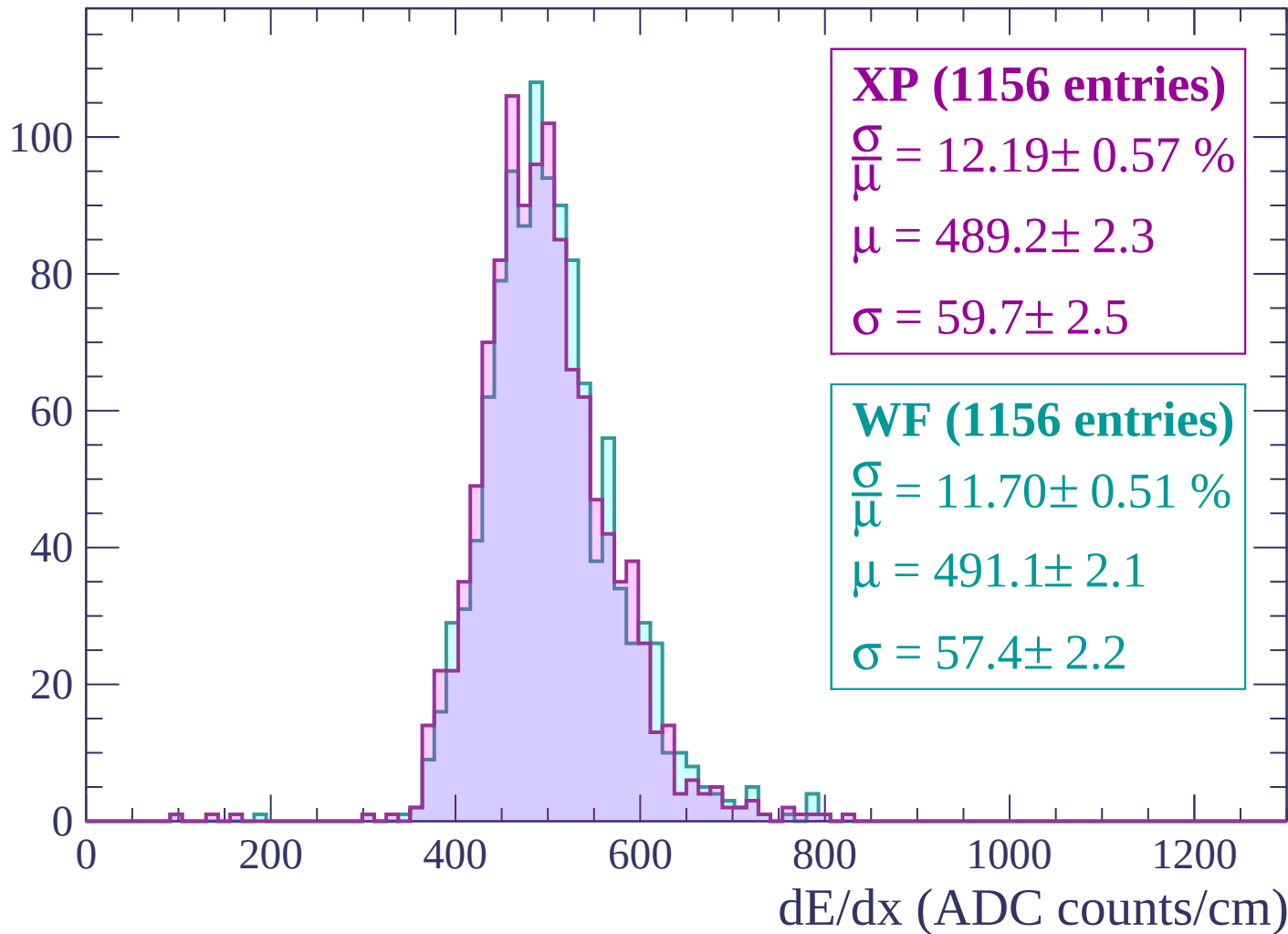
Energy loss | -1740 < p < -1680

Count



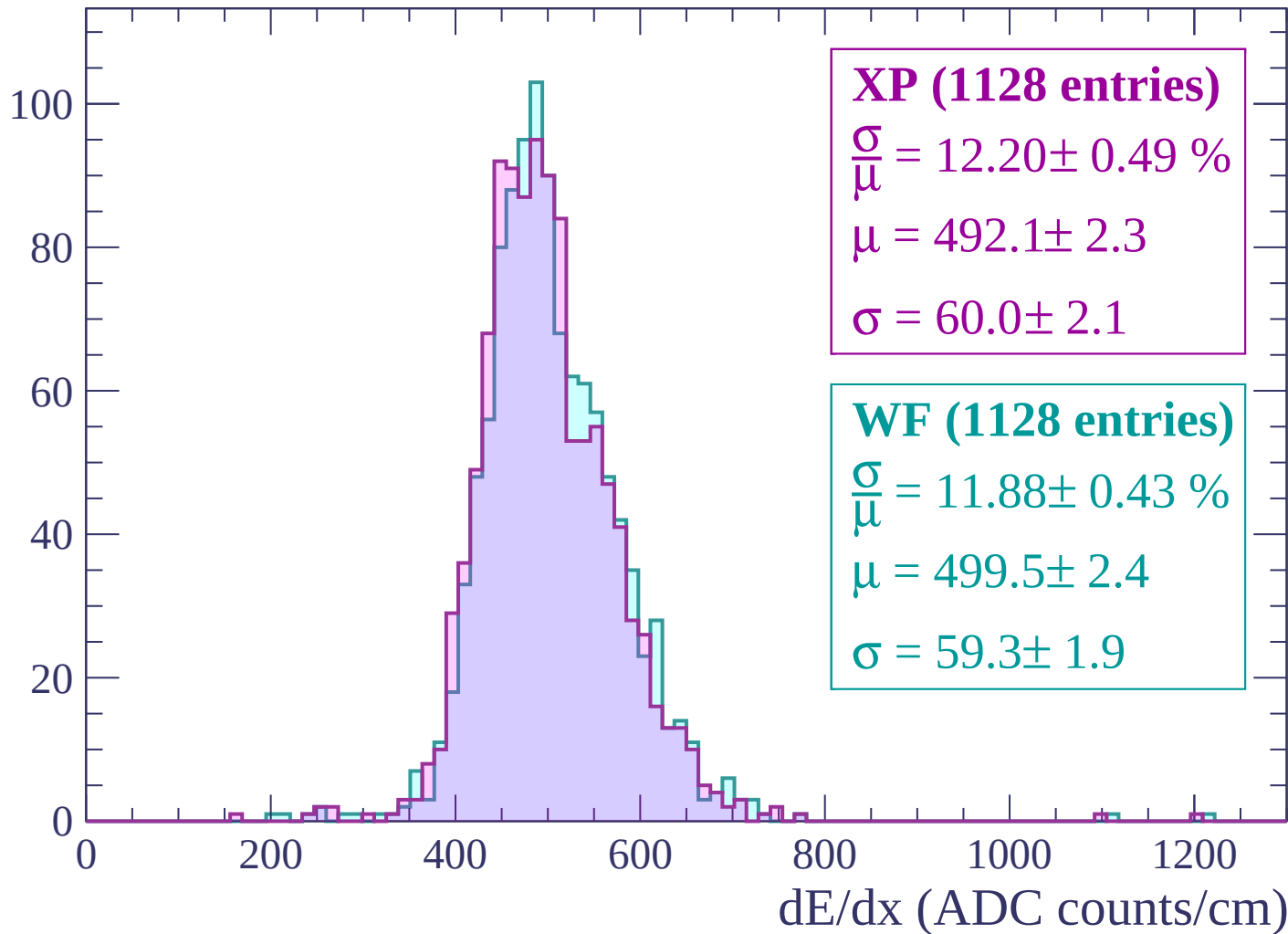
Energy loss | -1680 < p < -1620

Count



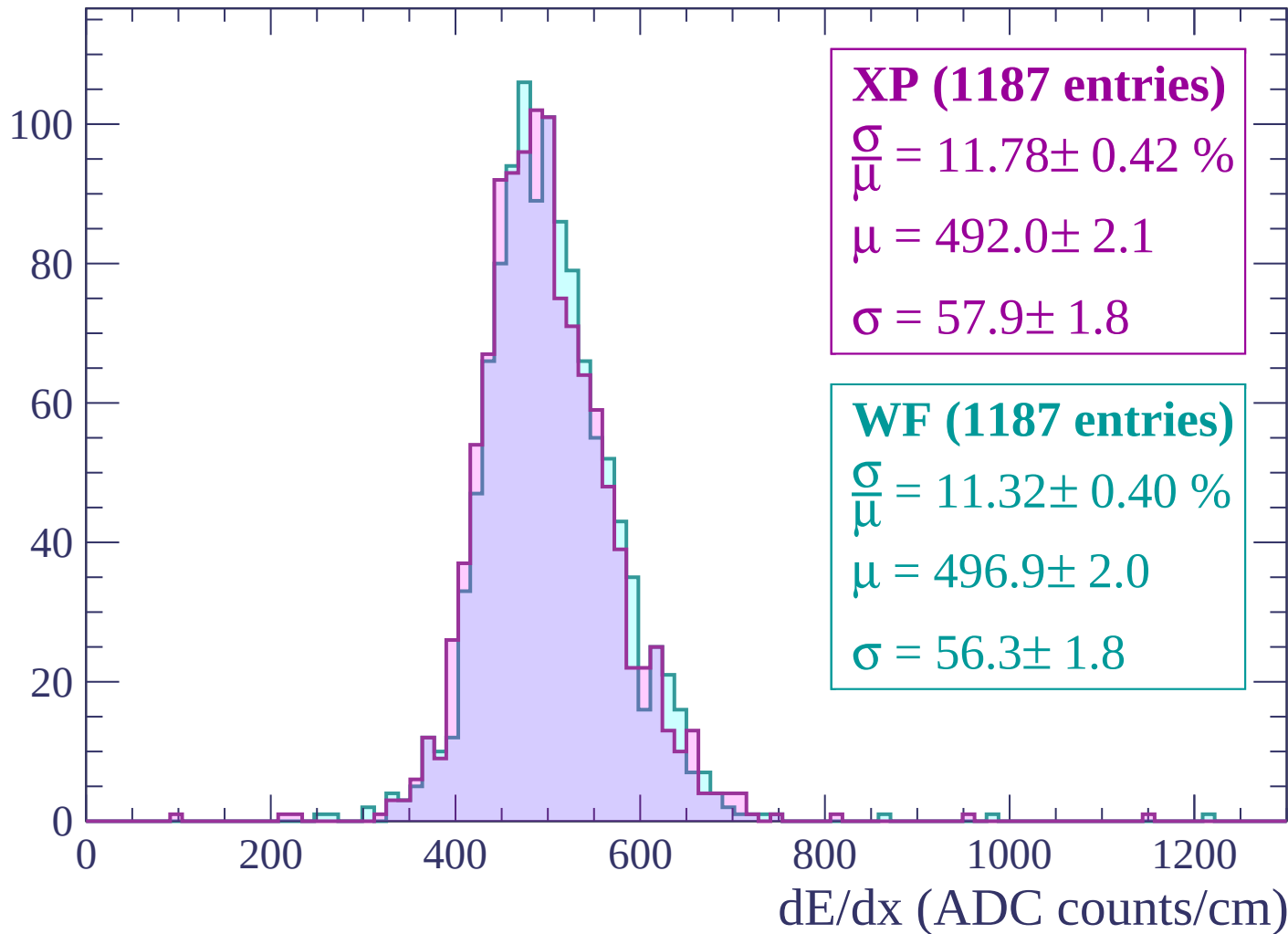
Energy loss | -1620 < p < -1560

Count



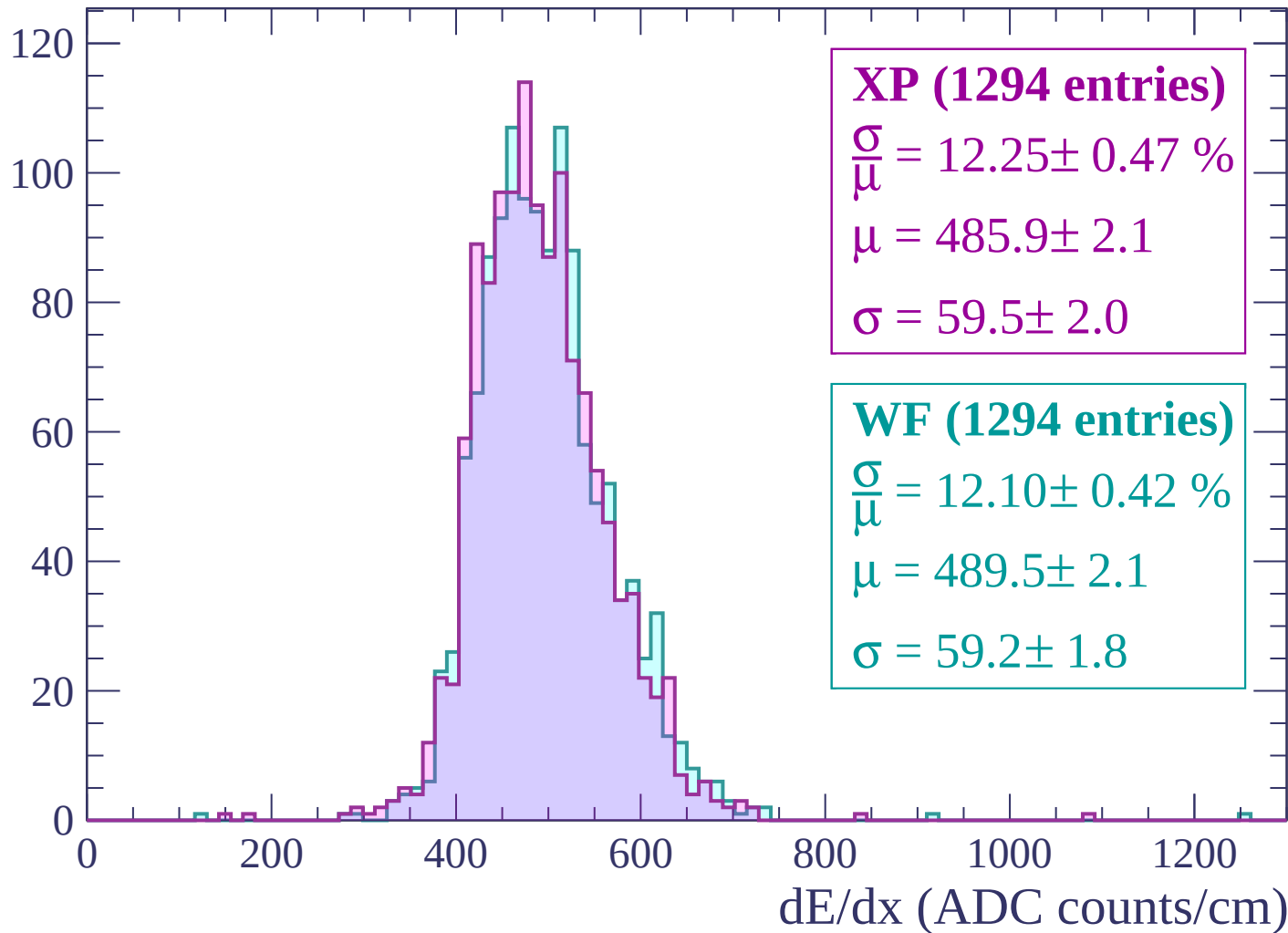
Energy loss | $-1560 < p < -1500$

Count



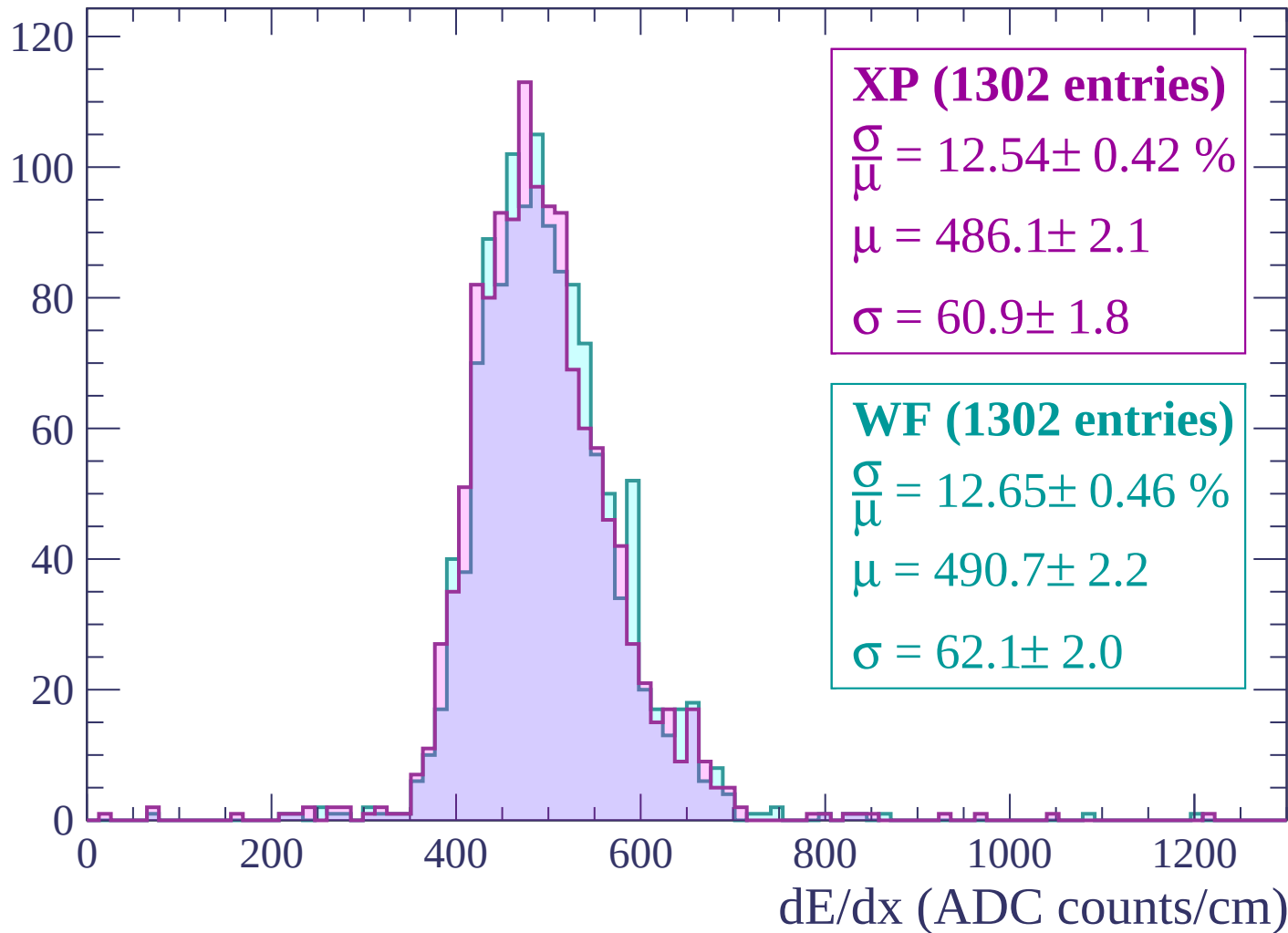
Energy loss | -1500 < p < -1440

Count



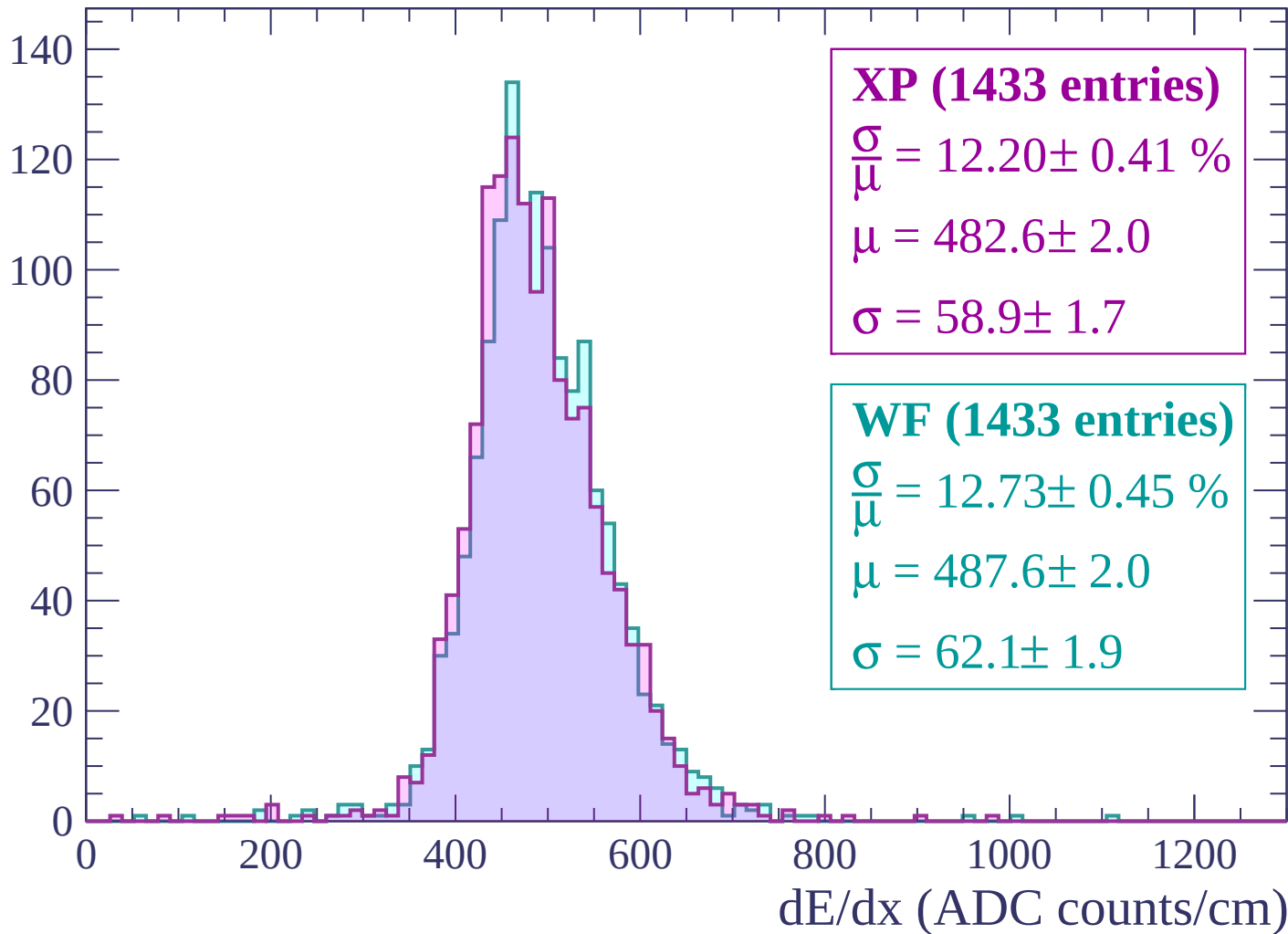
Energy loss | $-1440 < p < -1380$

Count



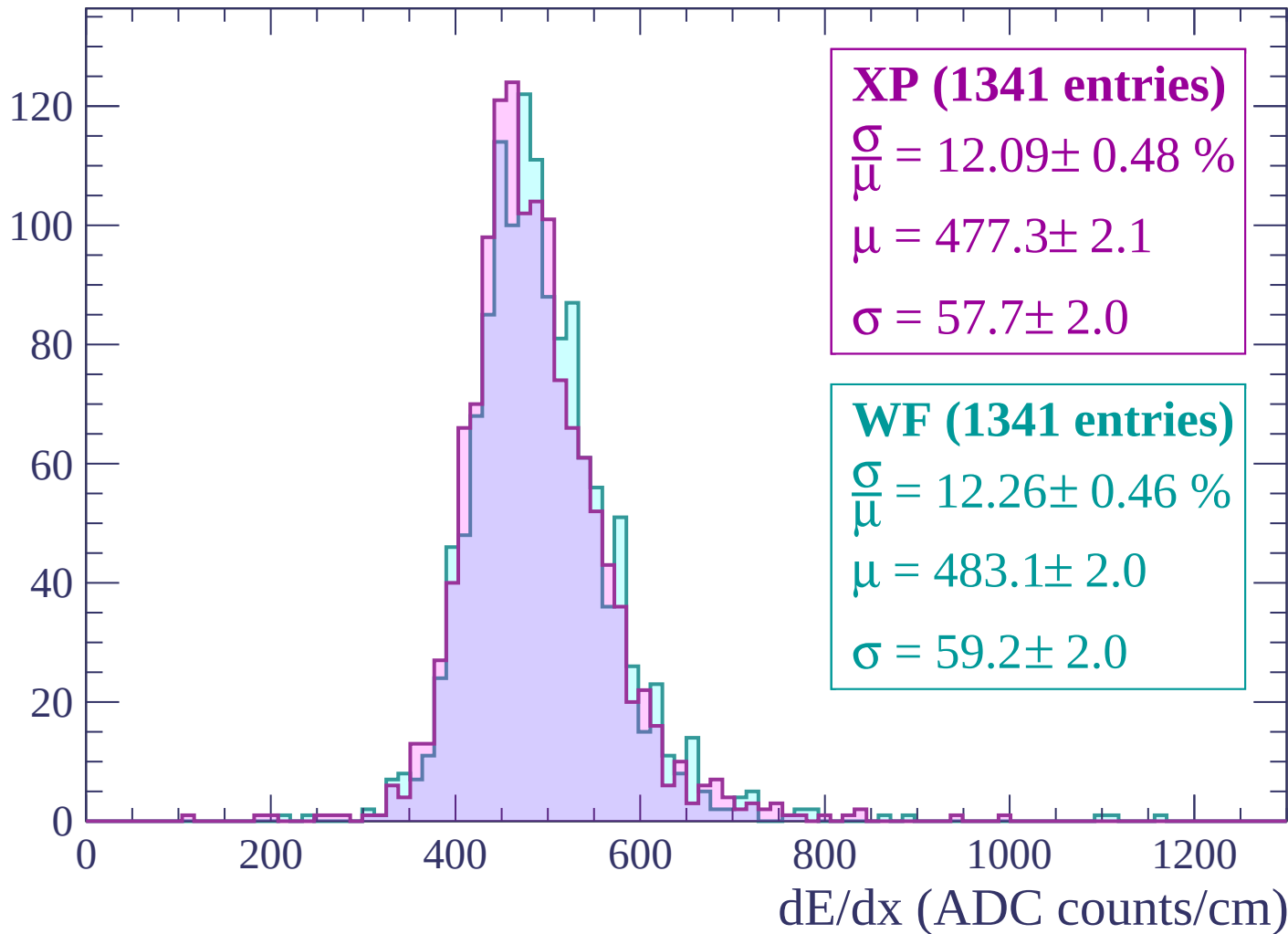
Energy loss | -1380 < p < -1320

Count



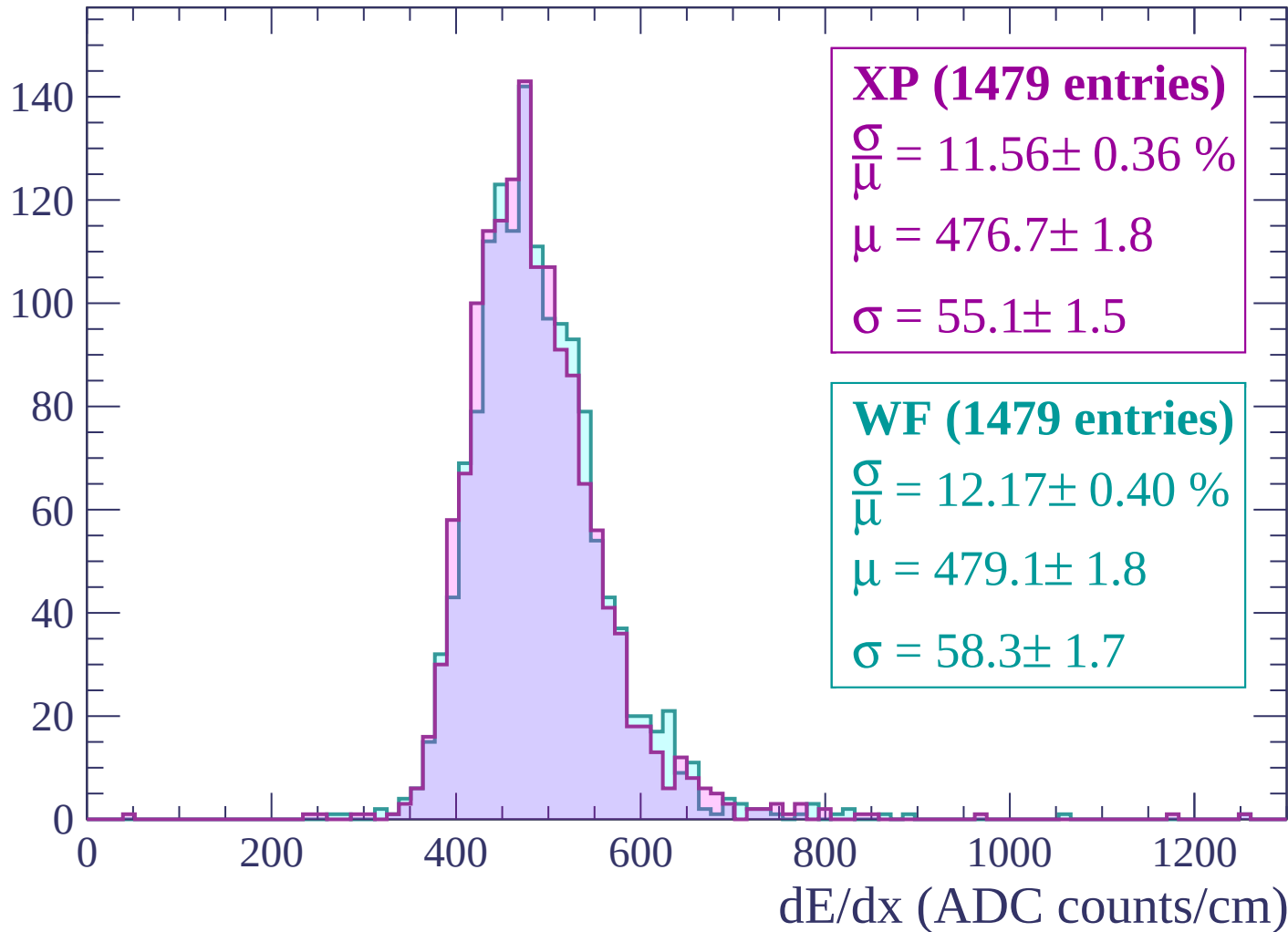
Energy loss | -1320 < p < -1260

Count



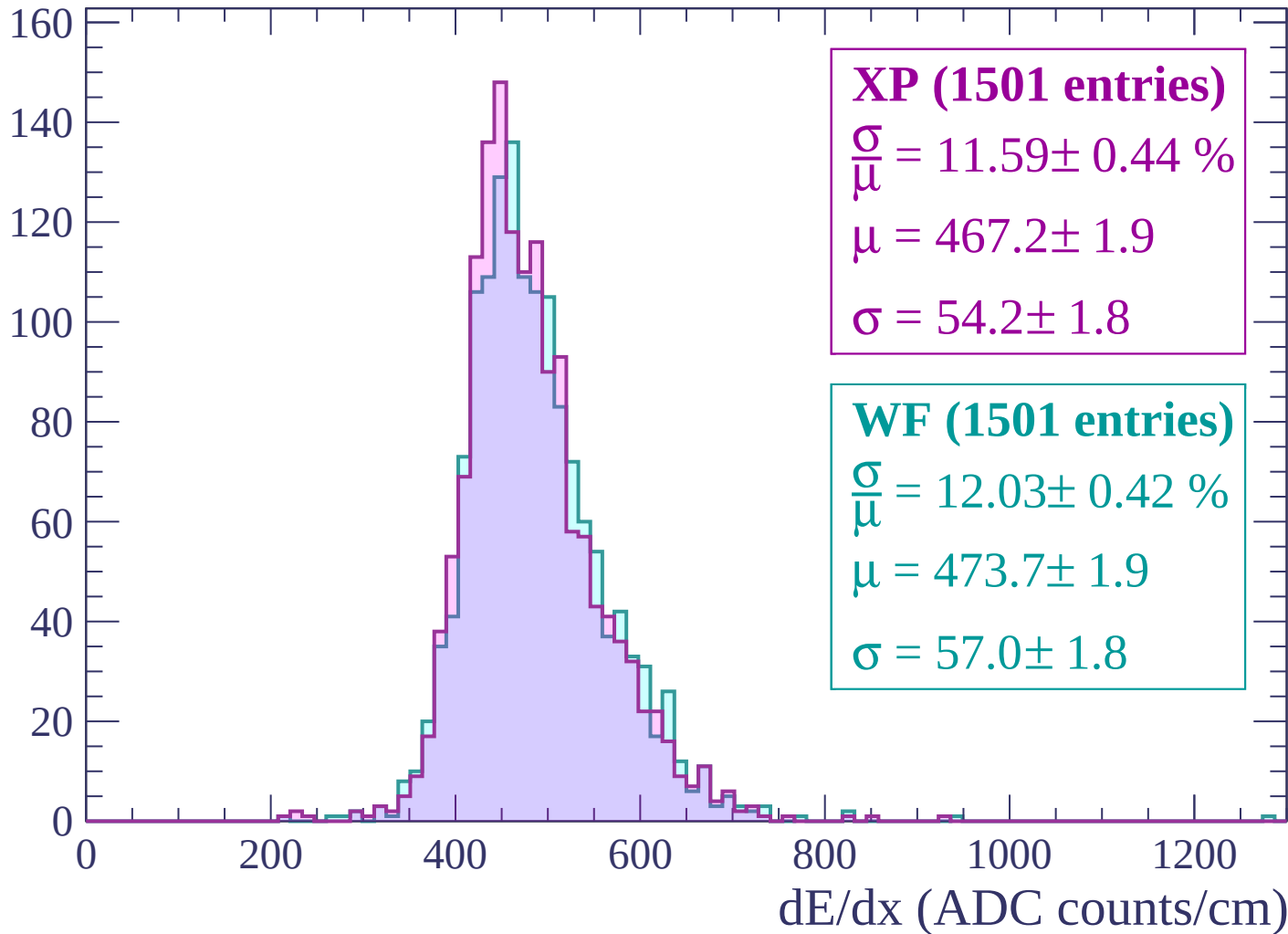
Energy loss | -1260 < p < -1200

Count



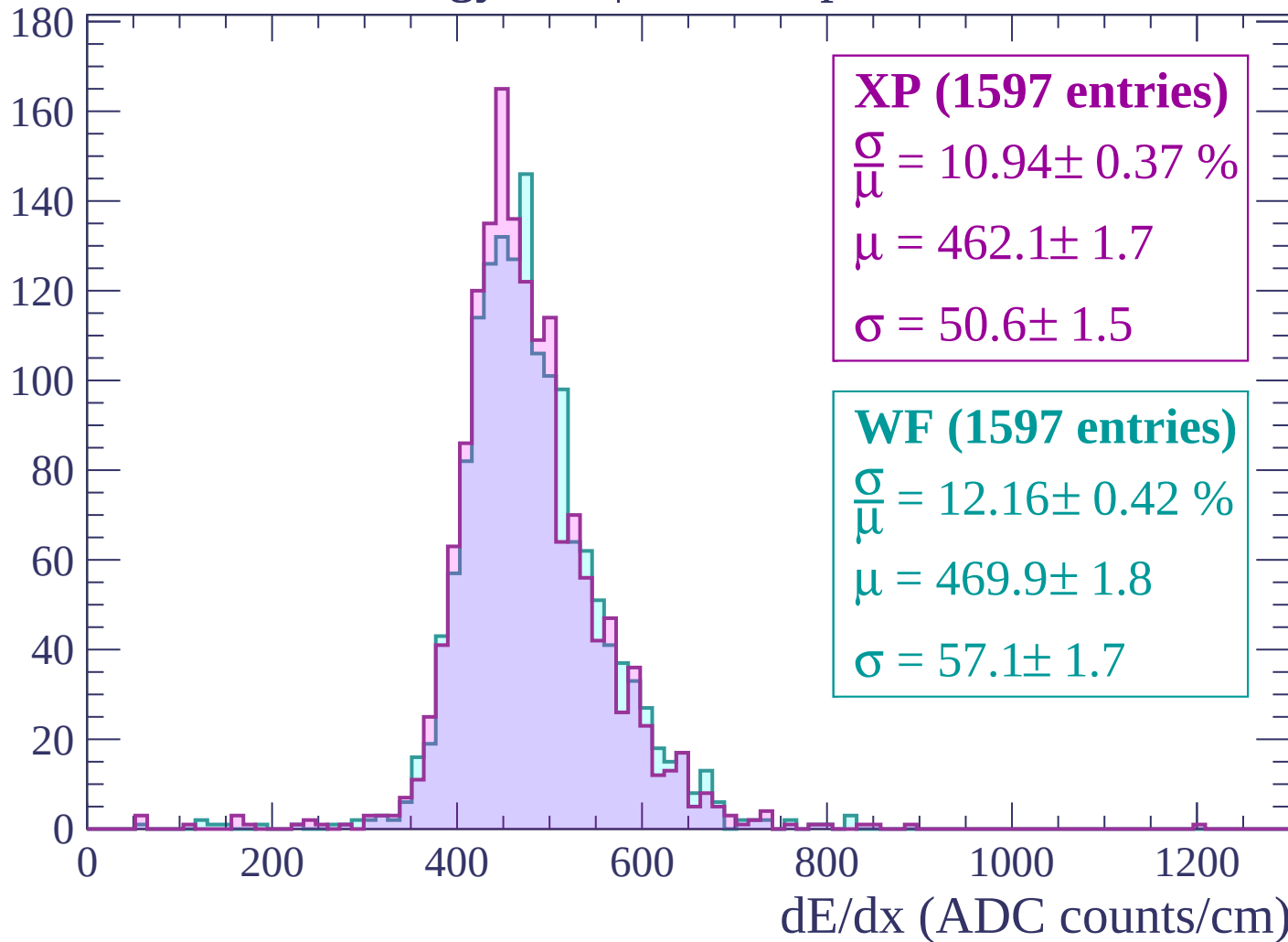
Energy loss | $-1200 < p < -1140$

Count



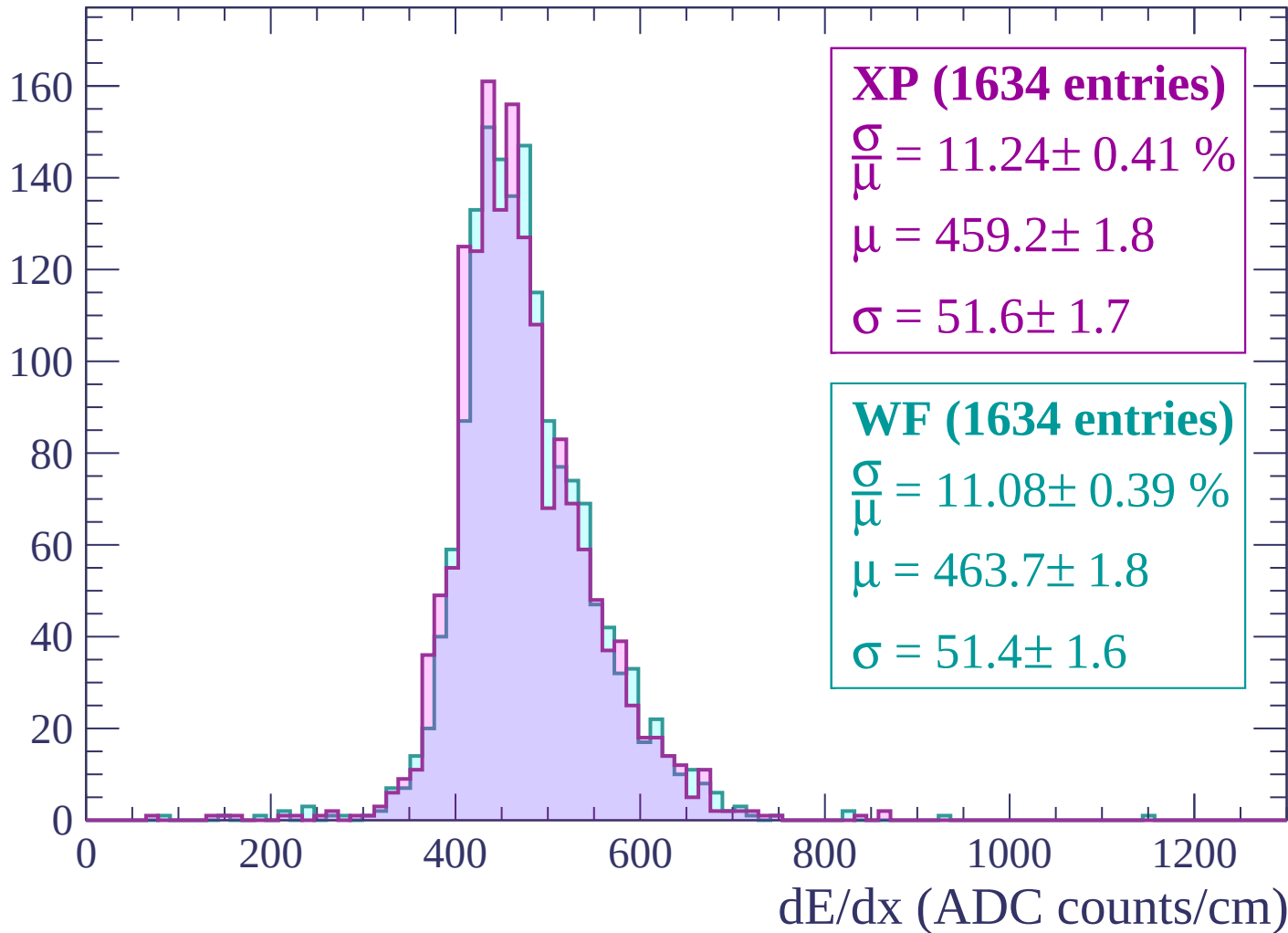
Energy loss | $-1140 < p < -1080$

Count



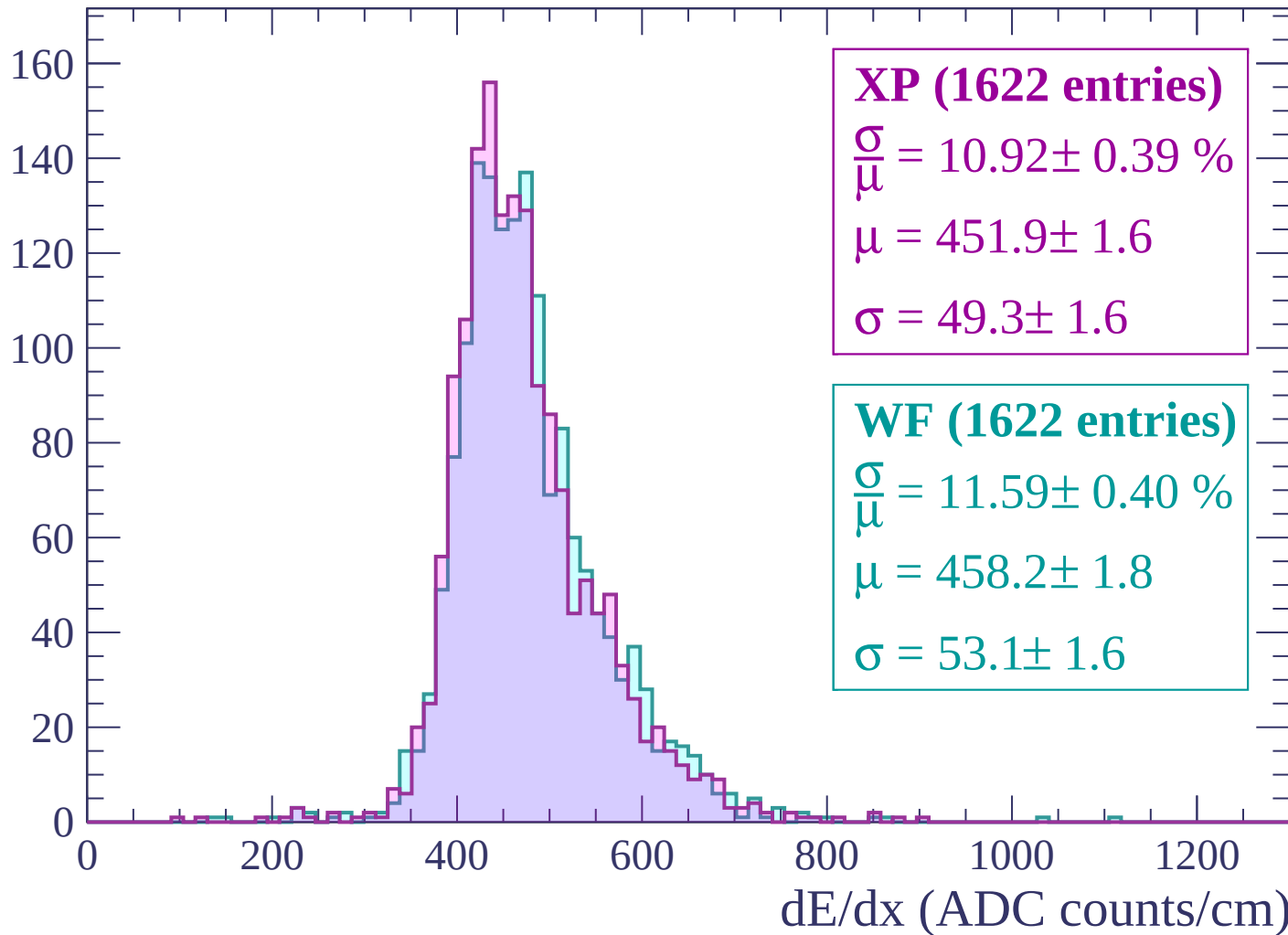
Energy loss | -1080 < p < -1020

Count



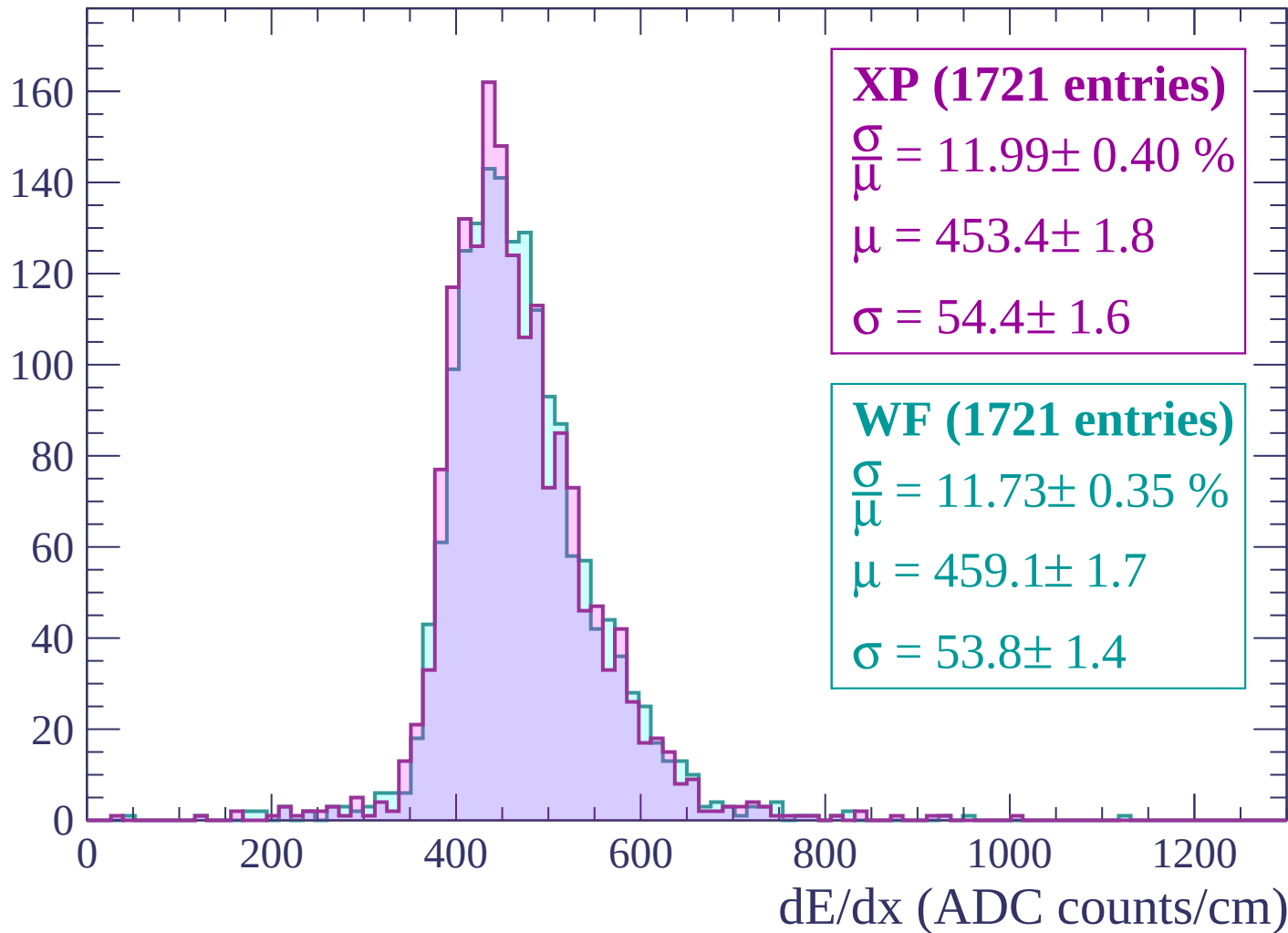
Energy loss | $-1020 < p < -960$

Count



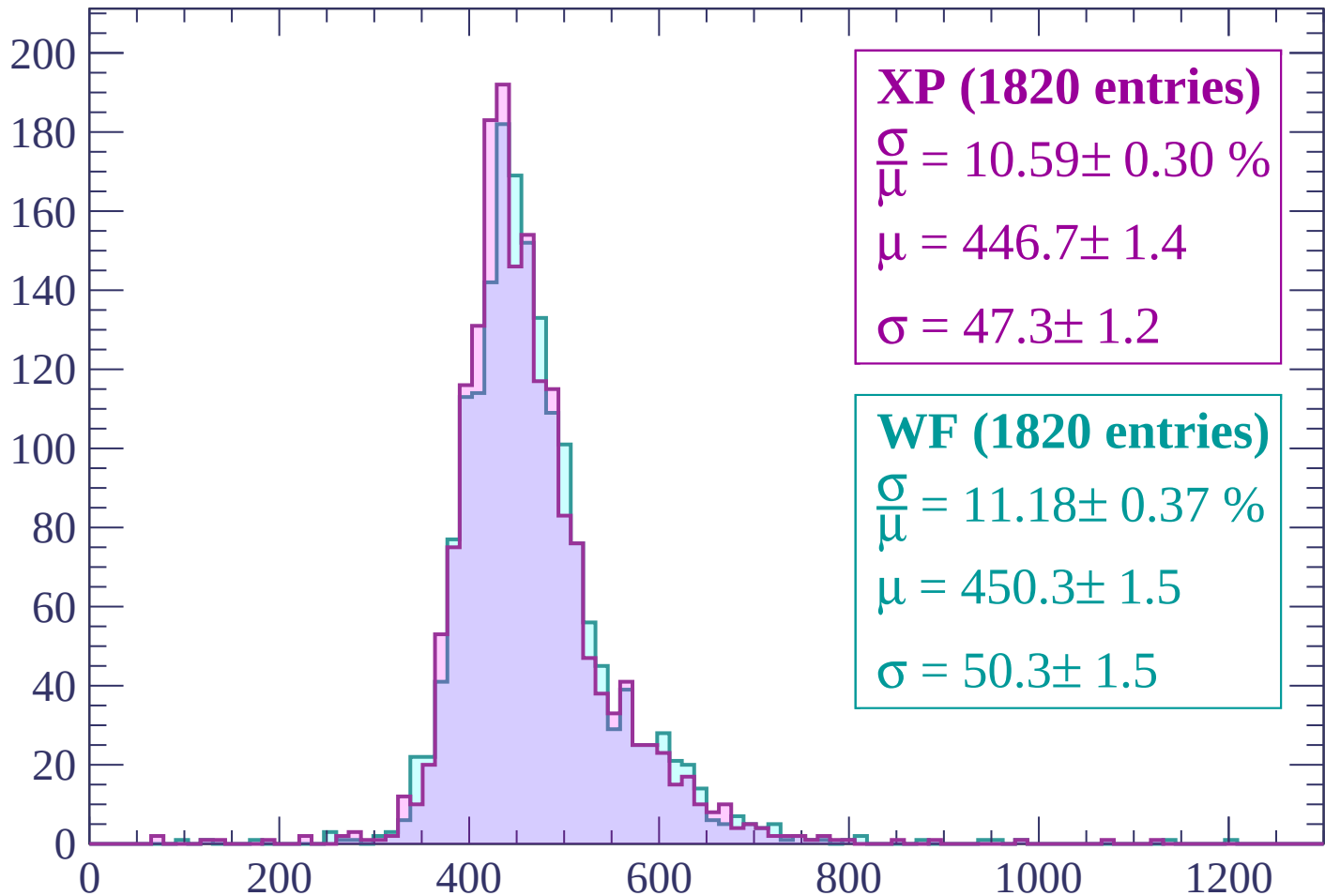
Energy loss | $-960 < p < -900$

Count



Energy loss | $-900 < p < -840$

Count



XP (1820 entries)

$$\frac{\sigma}{\mu} = 10.59 \pm 0.30 \%$$

$$\mu = 446.7 \pm 1.4$$

$$\sigma = 47.3 \pm 1.2$$

WF (1820 entries)

$$\frac{\sigma}{\mu} = 11.18 \pm 0.37 \%$$

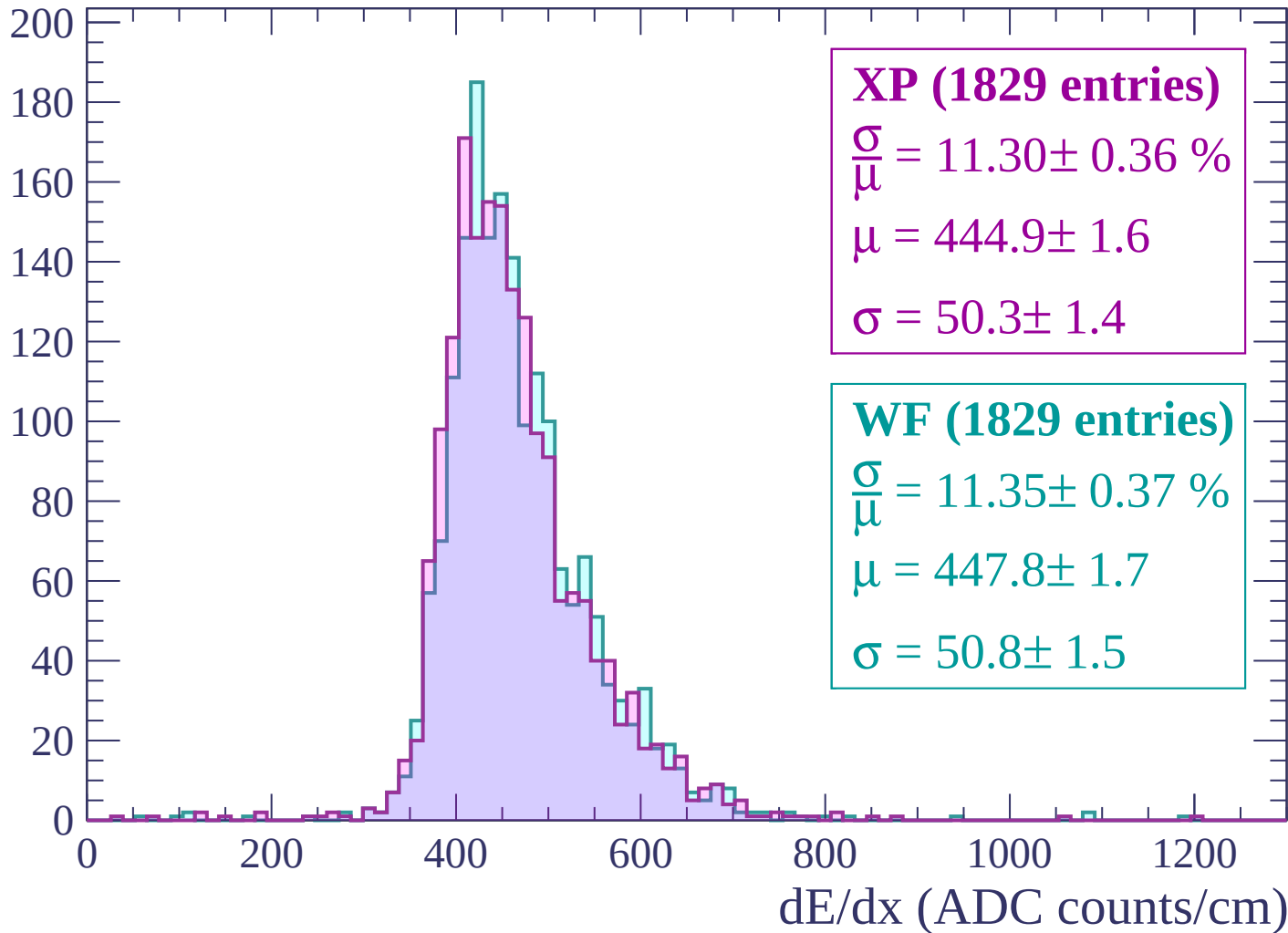
$$\mu = 450.3 \pm 1.5$$

$$\sigma = 50.3 \pm 1.5$$

dE/dx (ADC counts/cm)

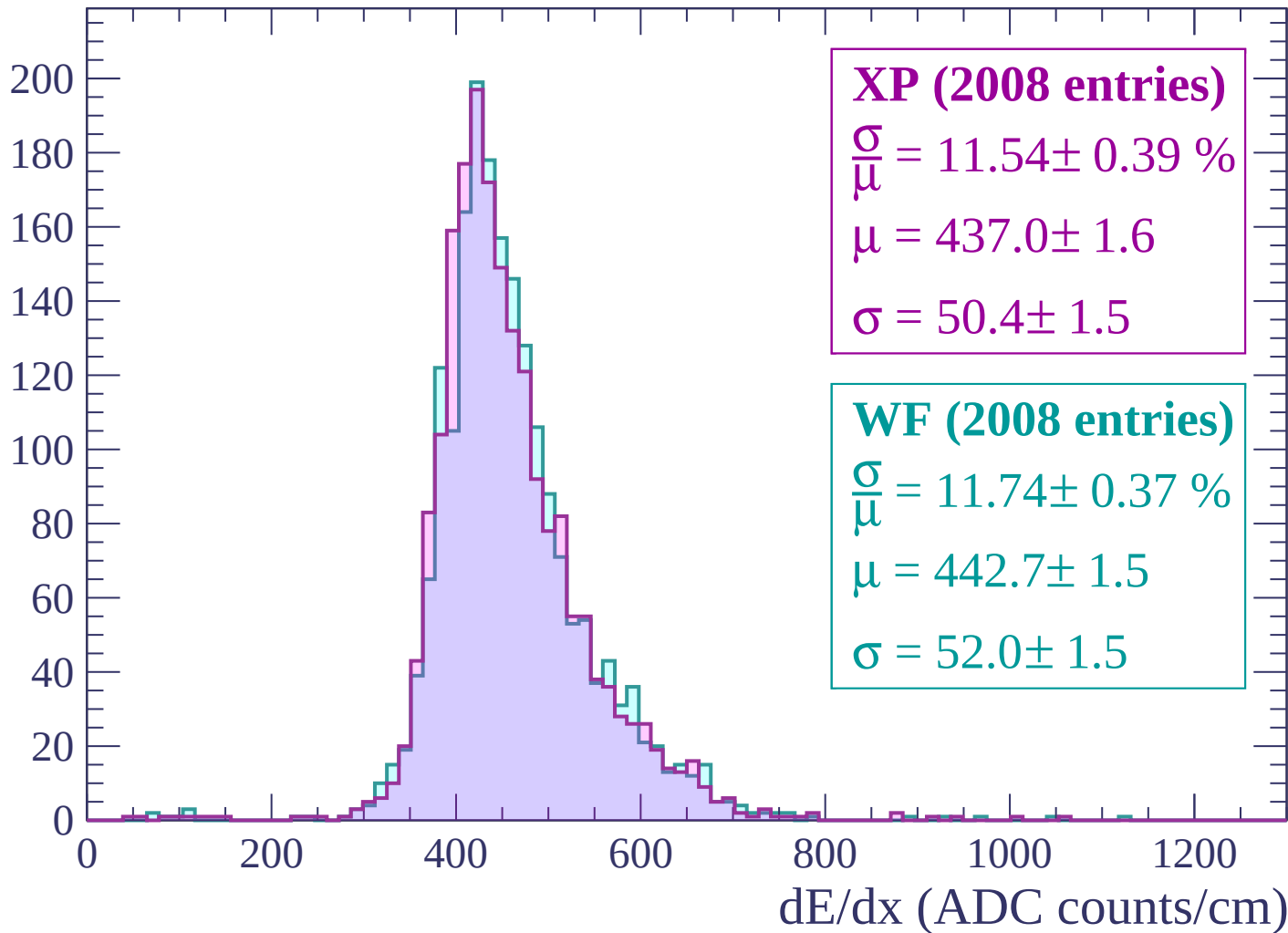
Energy loss | -840 < p < -780

Count



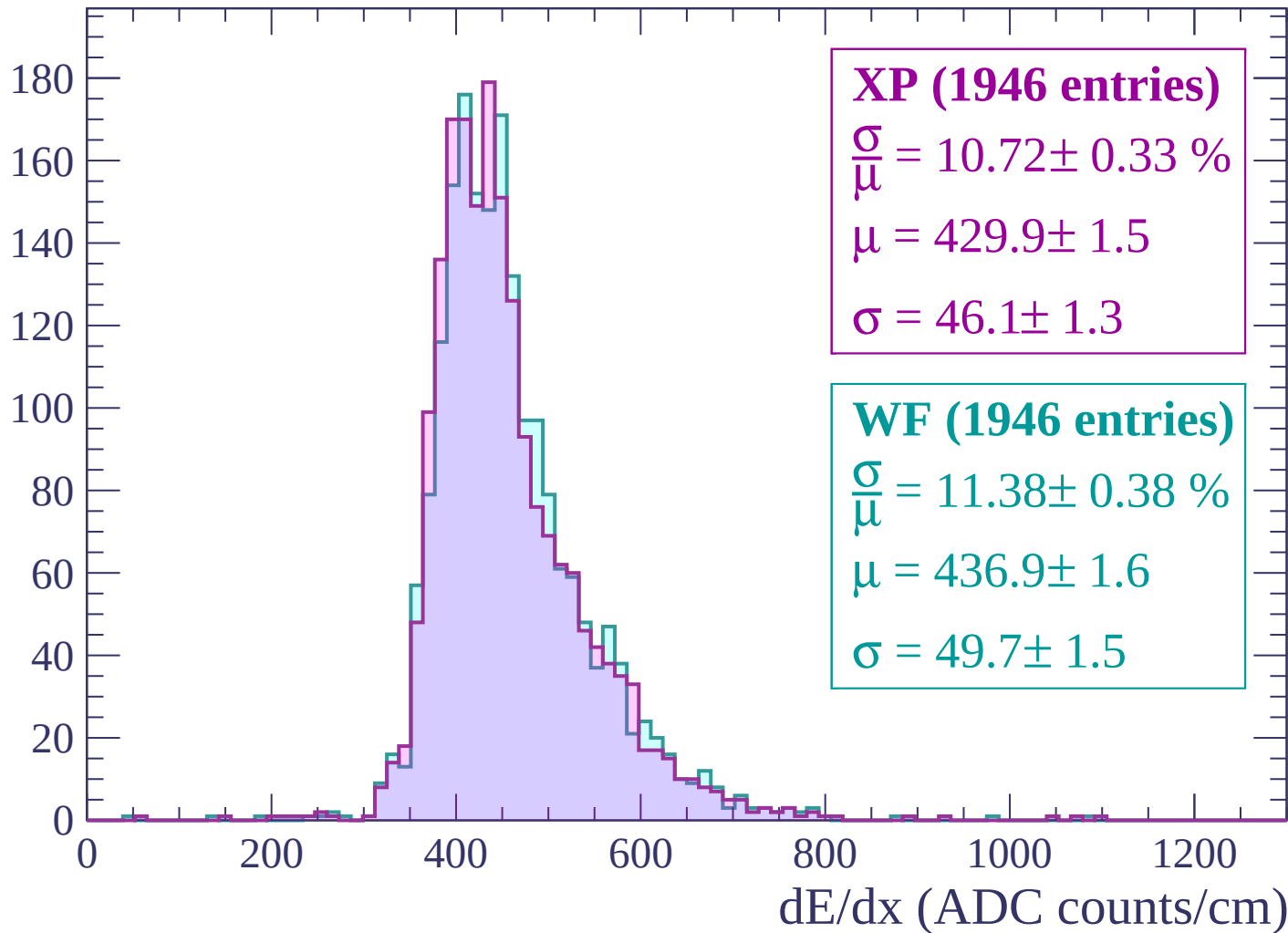
Energy loss | $-780 < p < -720$

Count



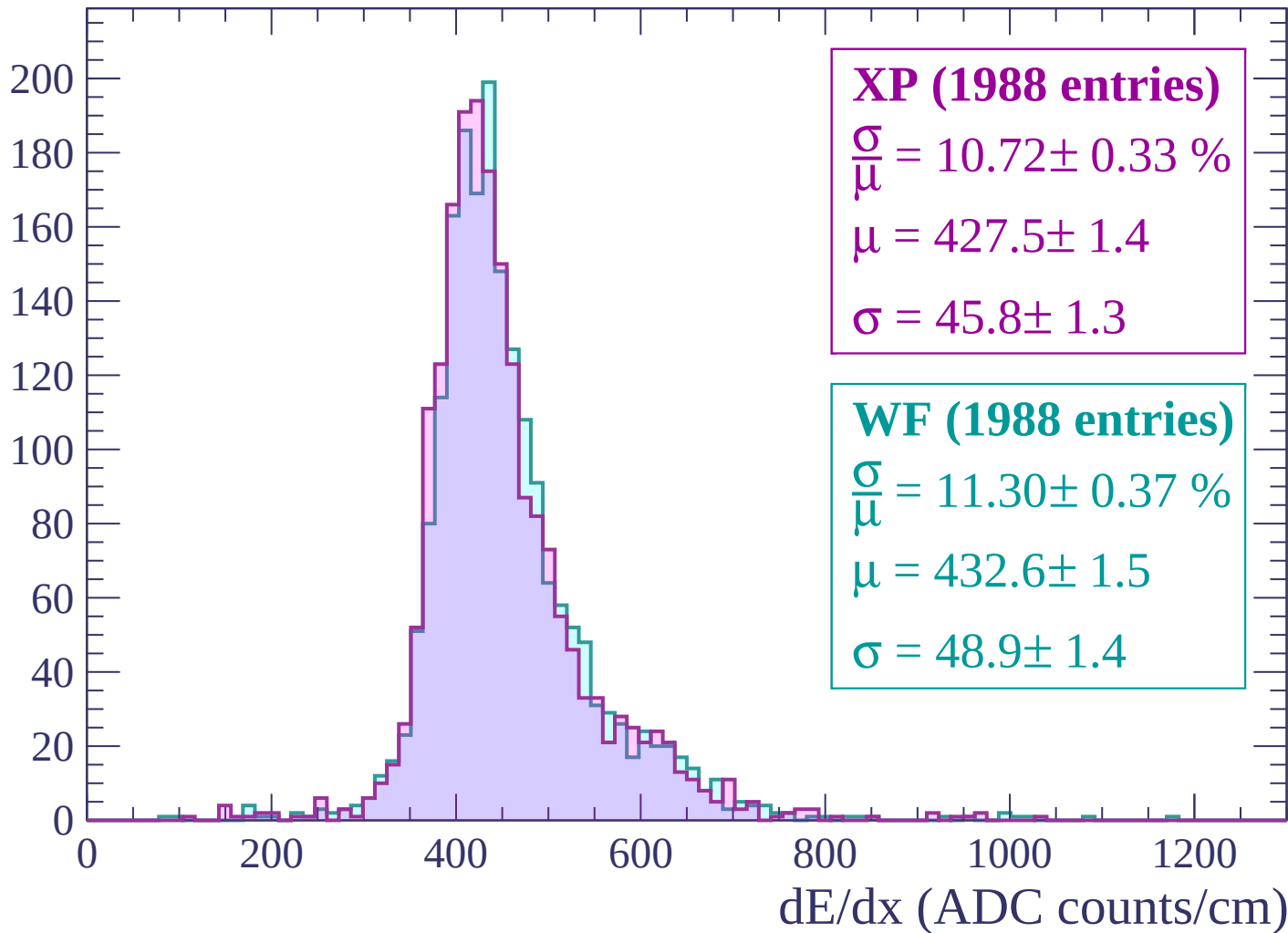
Energy loss | $-720 < p < -660$

Count



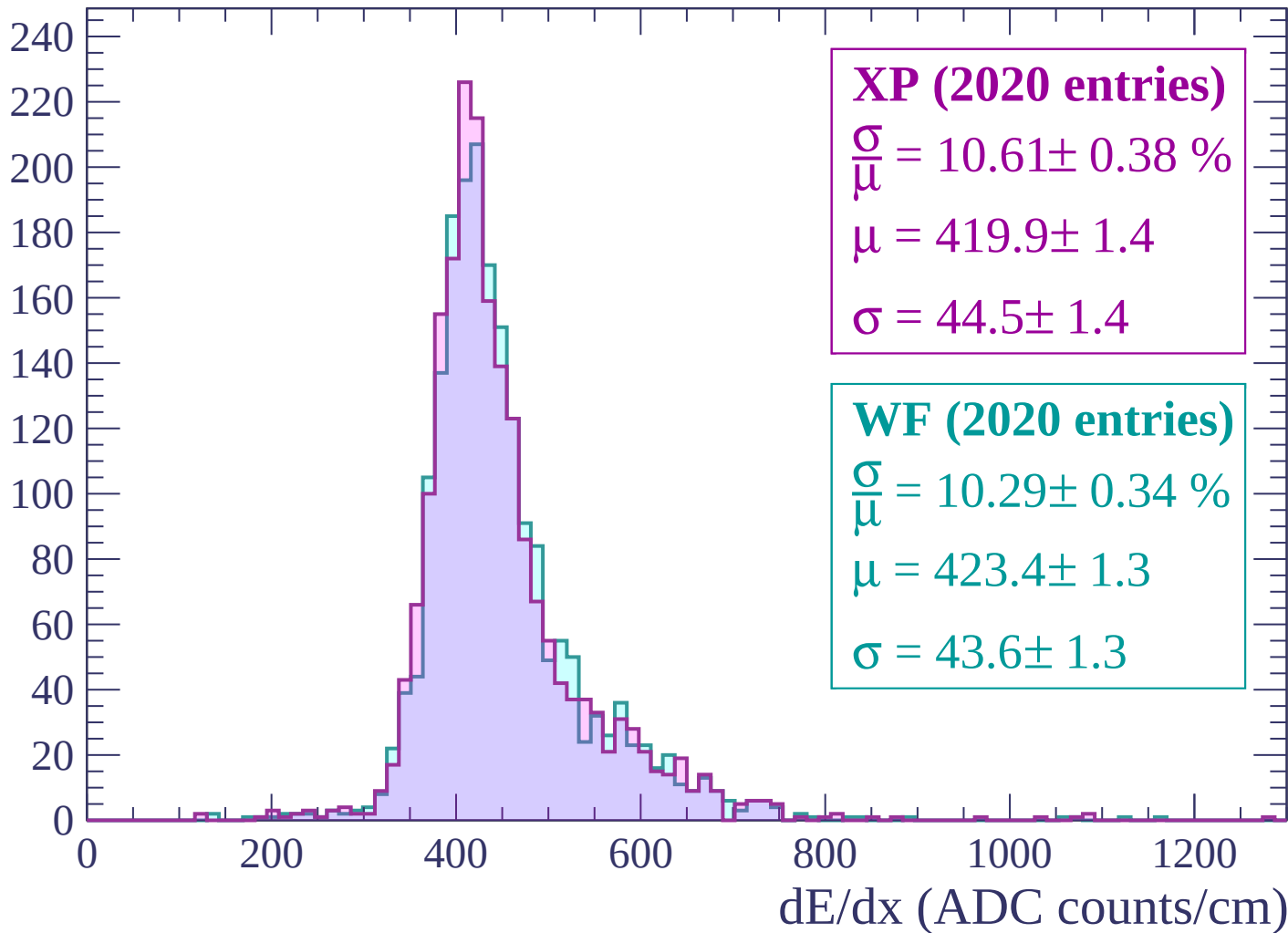
Energy loss | $-660 < p < -600$

Count



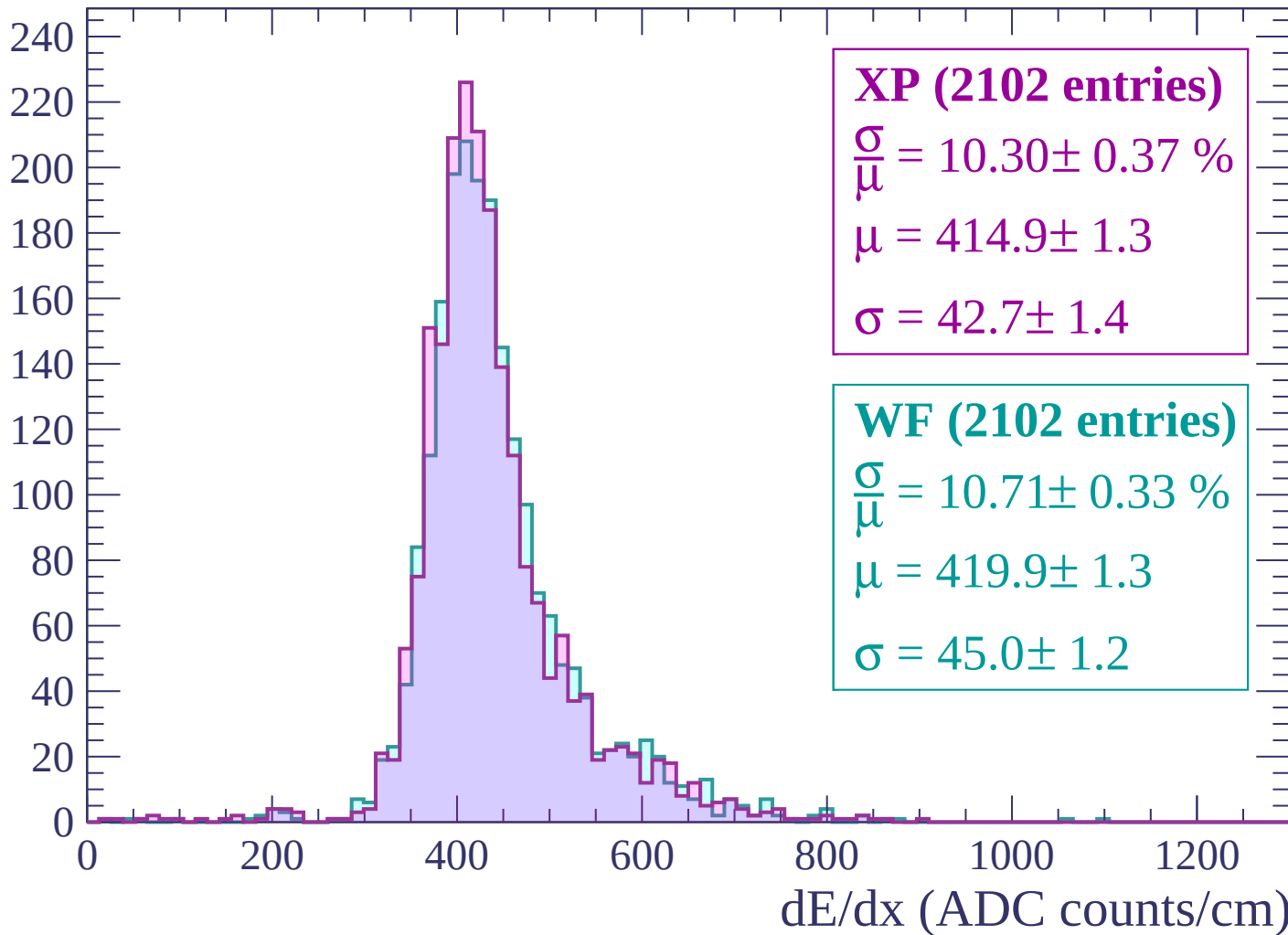
Energy loss | $-600 < p < -540$

Count



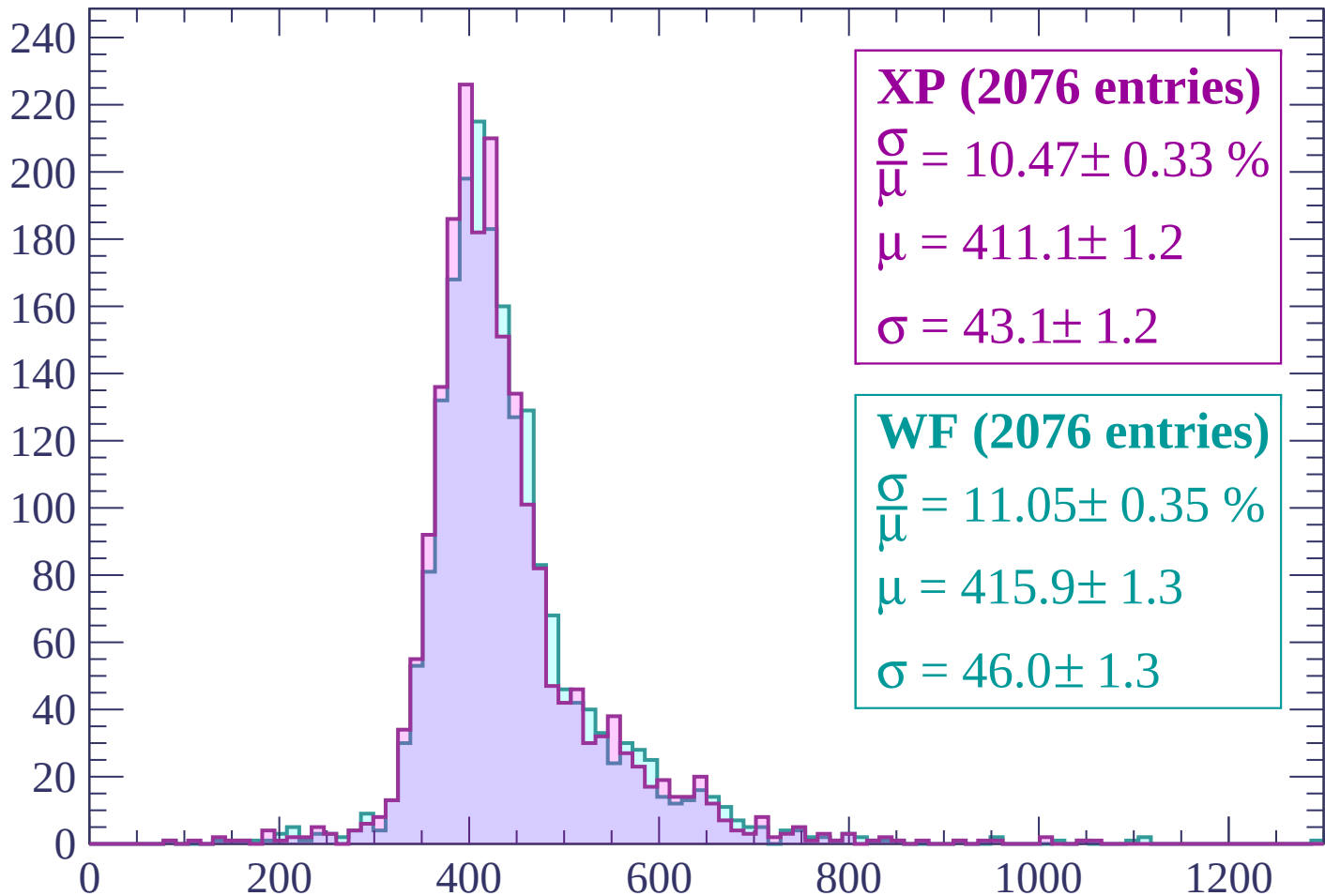
Energy loss | $-540 < p < -480$

Count



Energy loss | $-480 < p < -420$

Count



XP (2076 entries)

$$\frac{\sigma}{\mu} = 10.47 \pm 0.33 \%$$

$$\mu = 411.1 \pm 1.2$$

$$\sigma = 43.1 \pm 1.2$$

WF (2076 entries)

$$\frac{\sigma}{\mu} = 11.05 \pm 0.35 \%$$

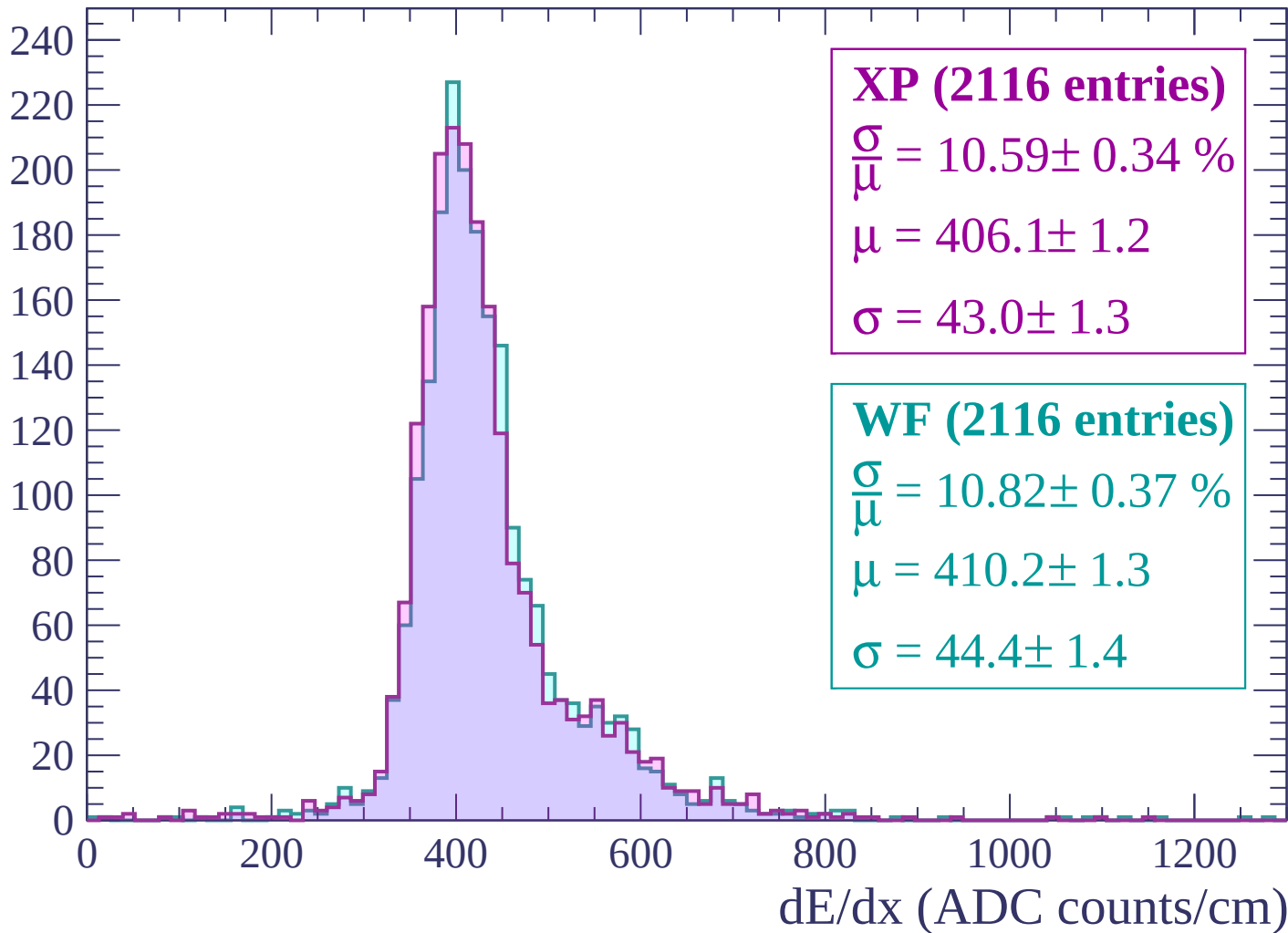
$$\mu = 415.9 \pm 1.3$$

$$\sigma = 46.0 \pm 1.3$$

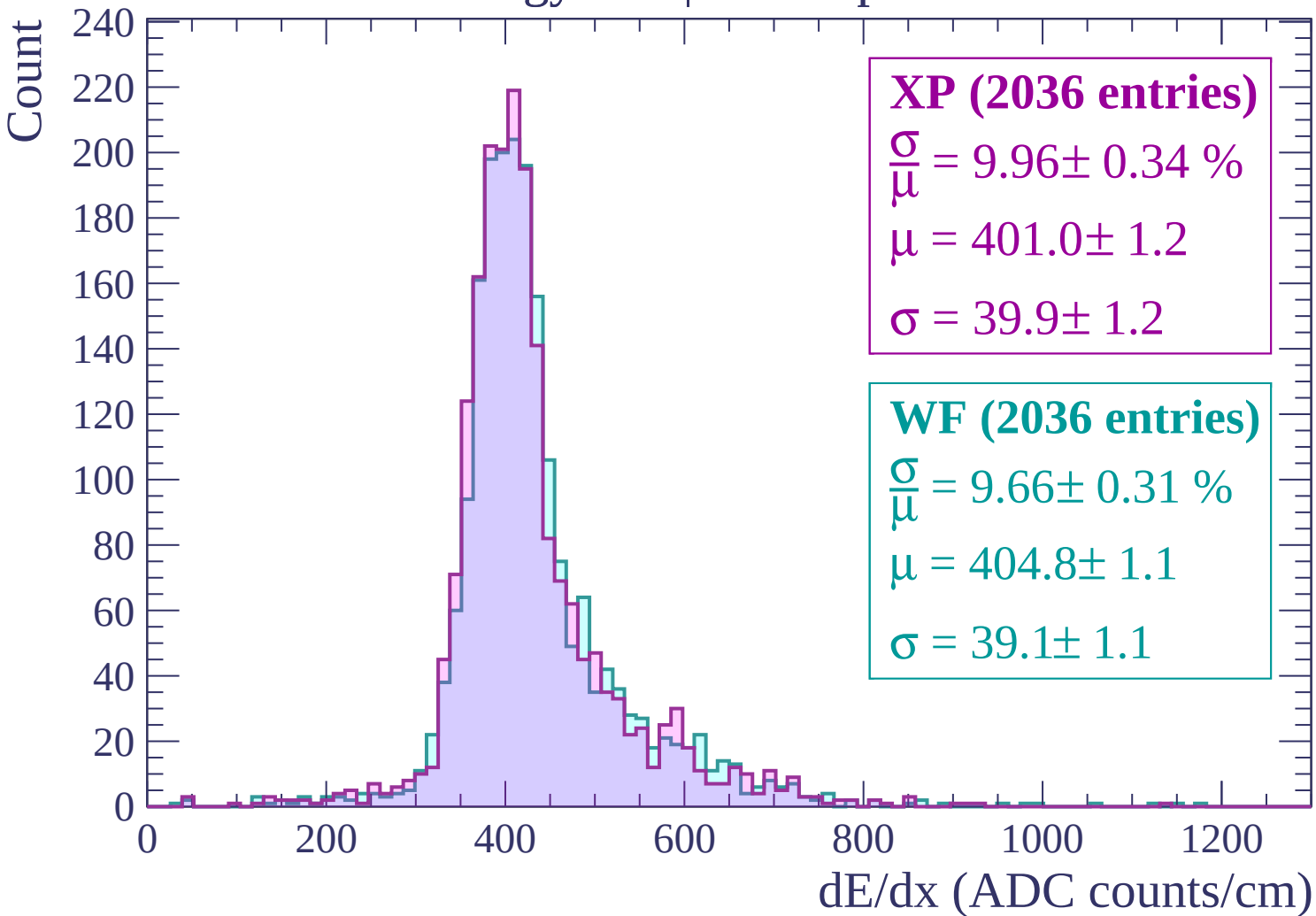
dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count

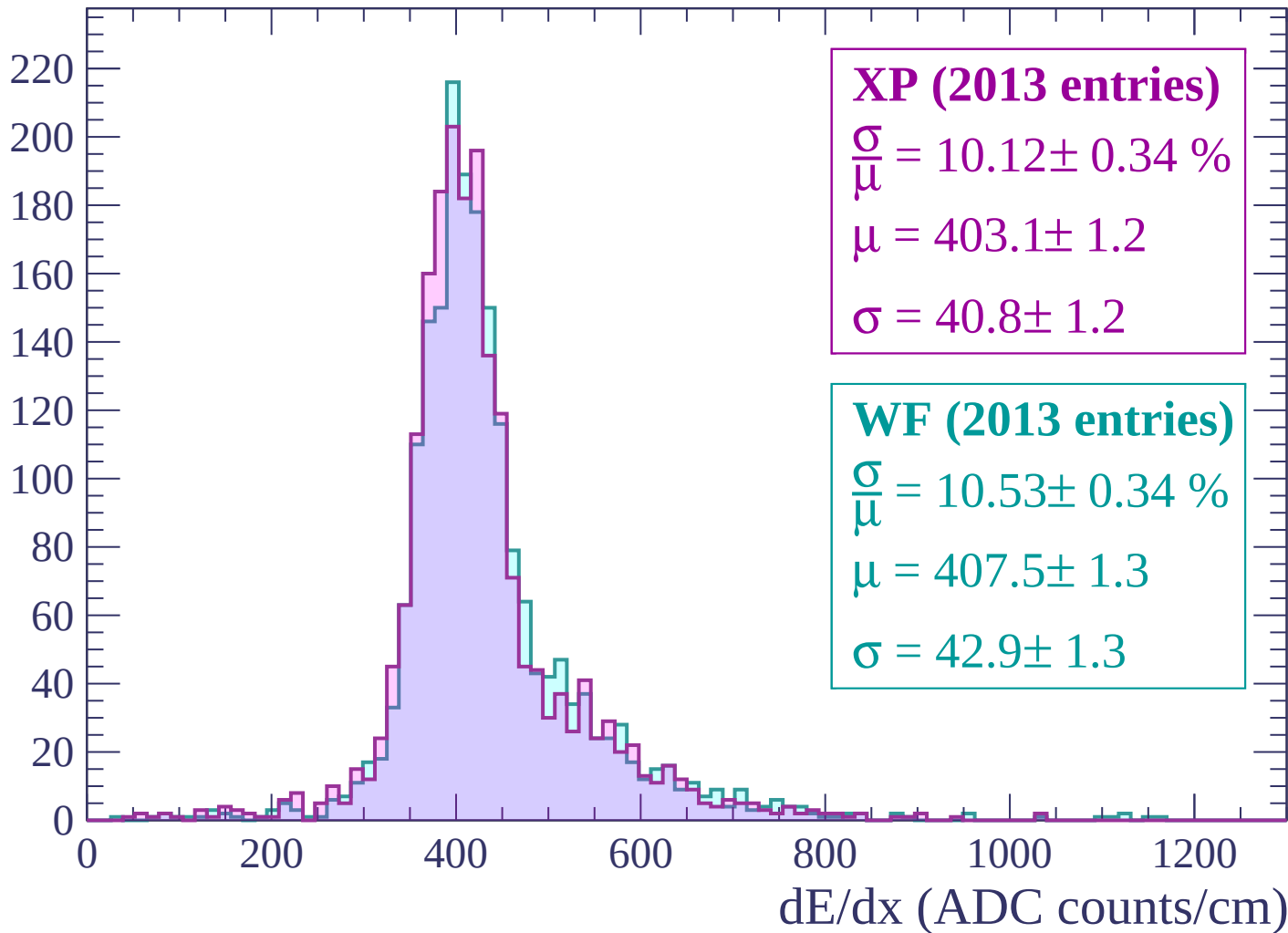


Energy loss | $-360 < p < -300$



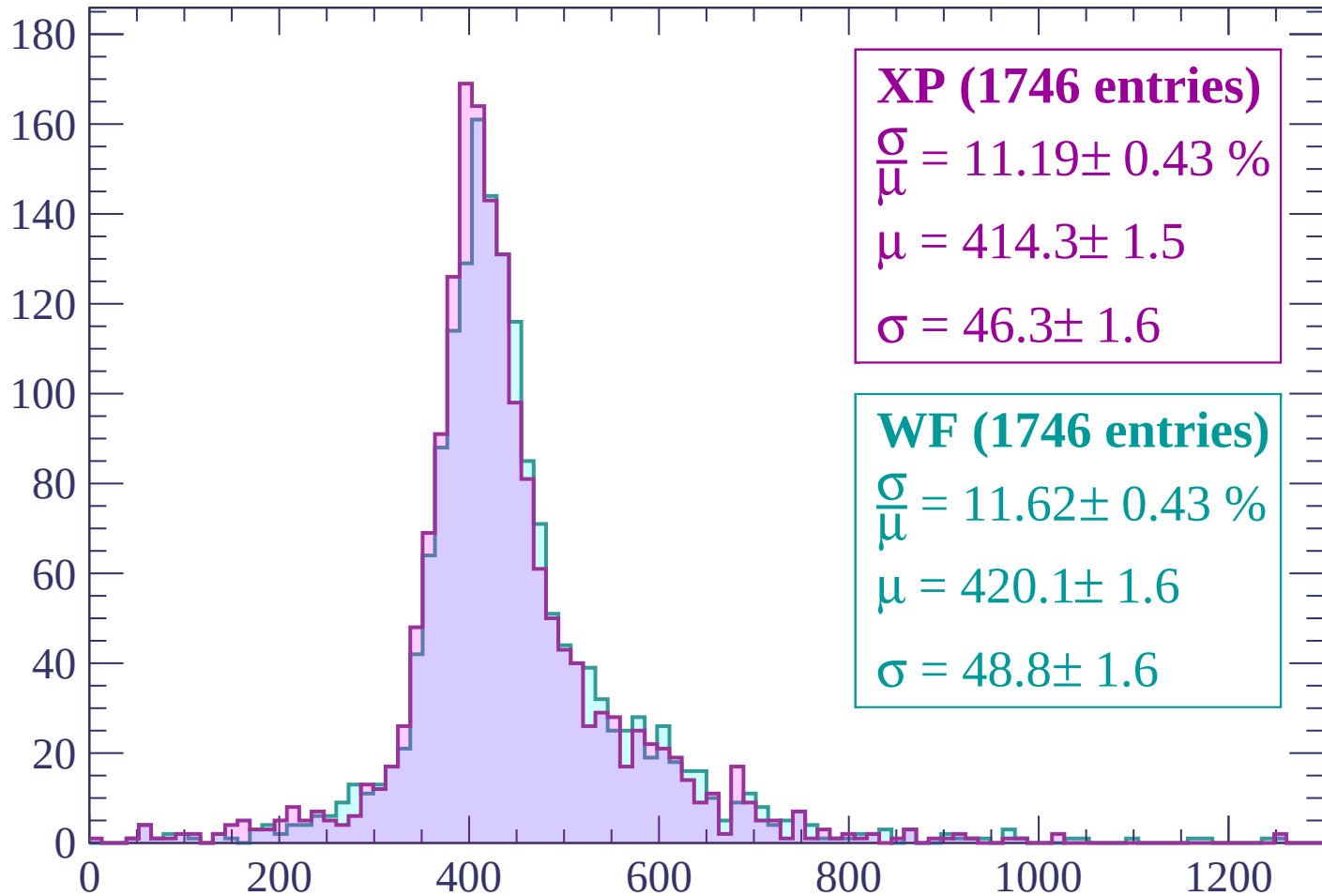
Energy loss | $-300 < p < -240$

Count



Energy loss | $-240 < p < -180$

Count



XP (1746 entries)

$$\frac{\sigma}{\mu} = 11.19 \pm 0.43 \%$$

$$\mu = 414.3 \pm 1.5$$

$$\sigma = 46.3 \pm 1.6$$

WF (1746 entries)

$$\frac{\sigma}{\mu} = 11.62 \pm 0.43 \%$$

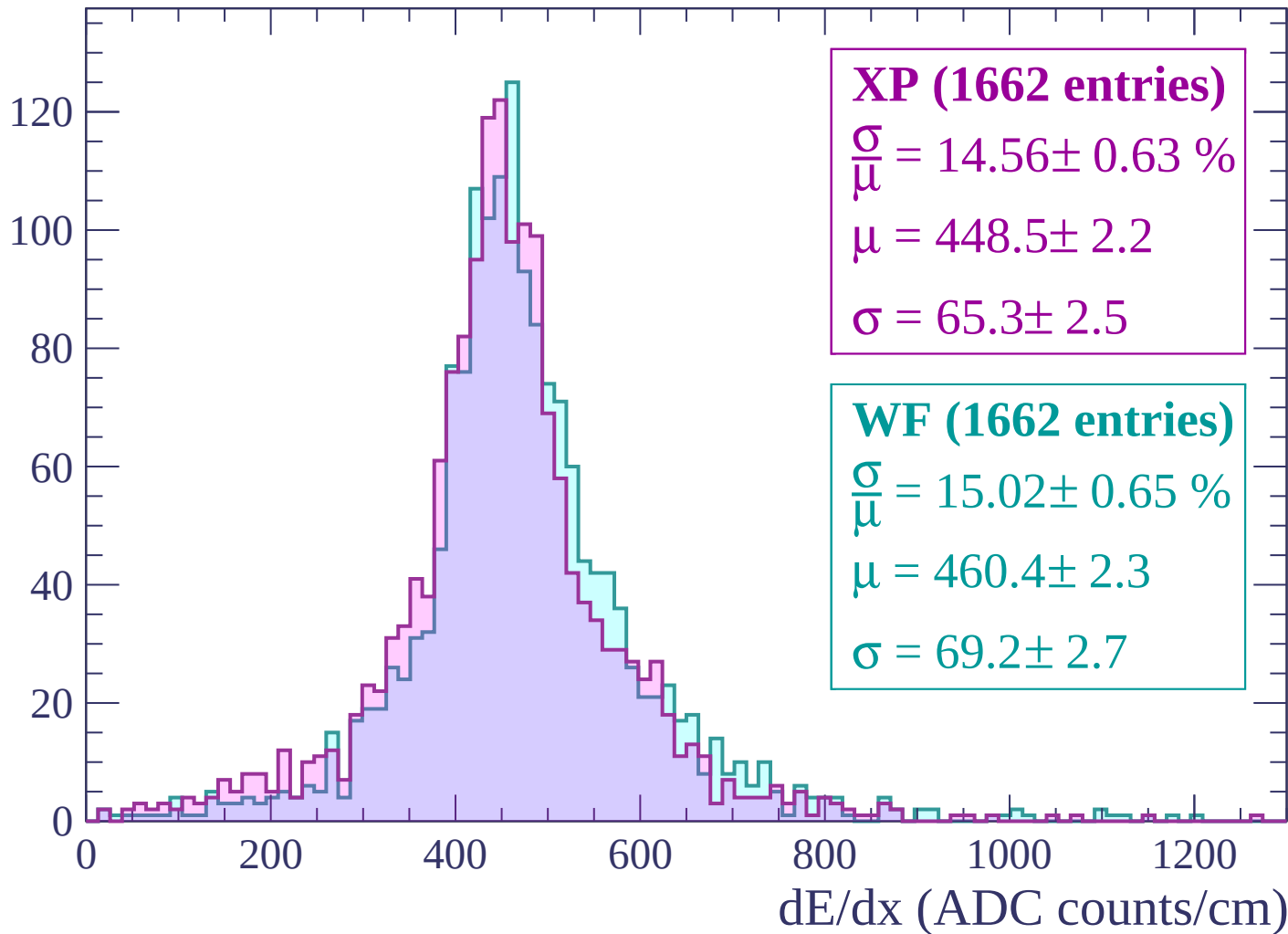
$$\mu = 420.1 \pm 1.6$$

$$\sigma = 48.8 \pm 1.6$$

dE/dx (ADC counts/cm)

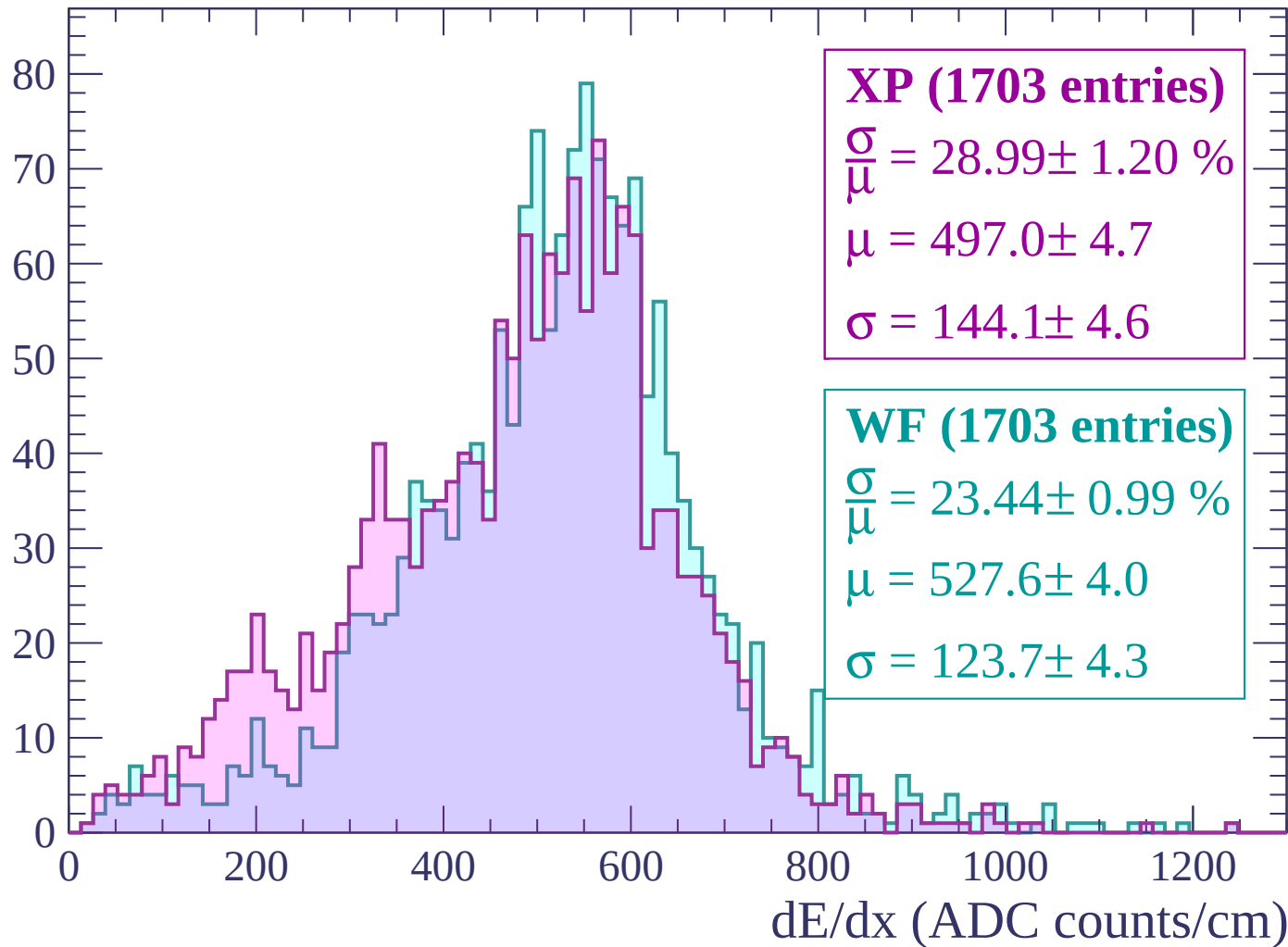
Energy loss | -180 < p < -120

Count



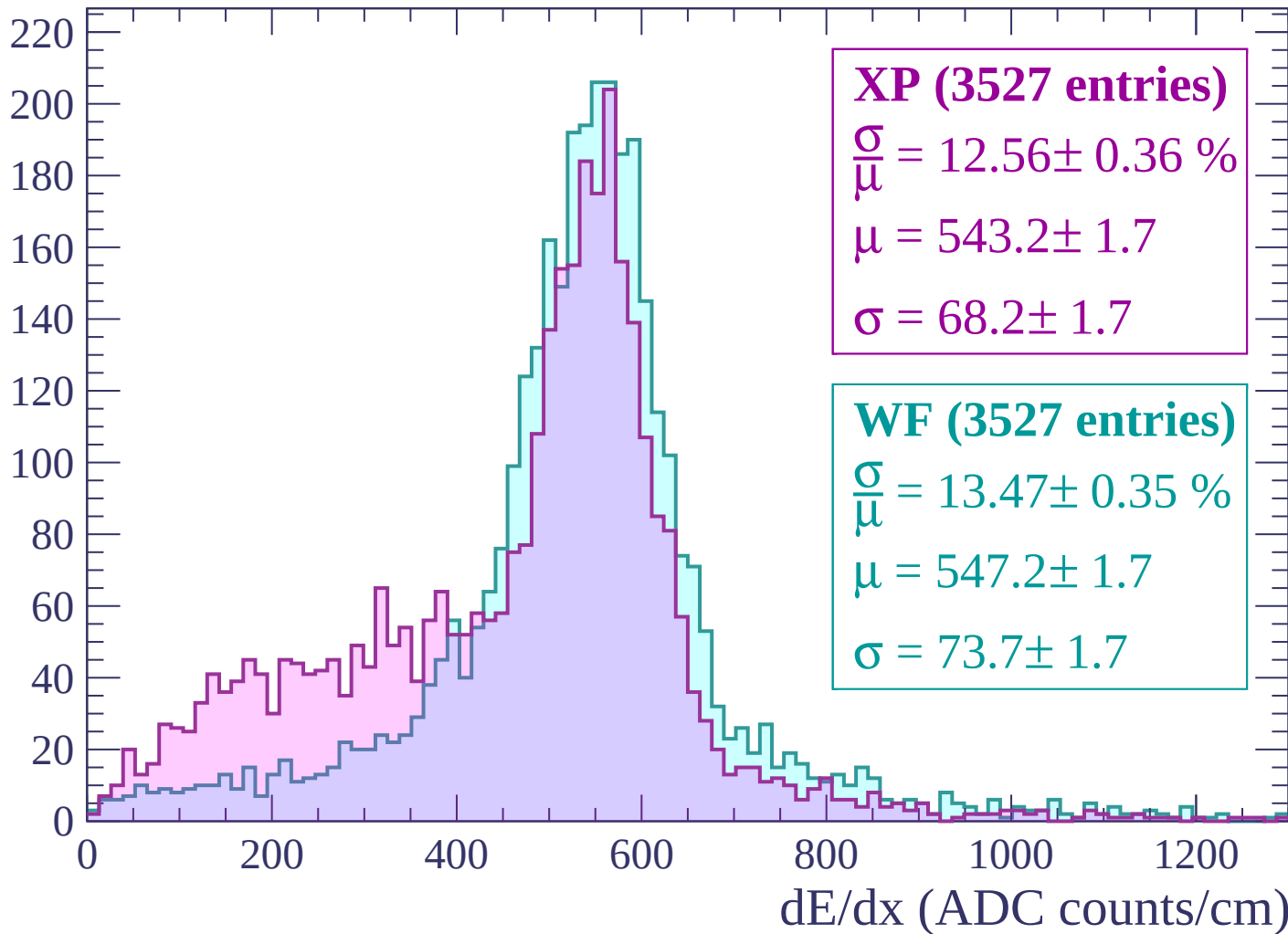
Energy loss | $-120 < p < -60$

Count



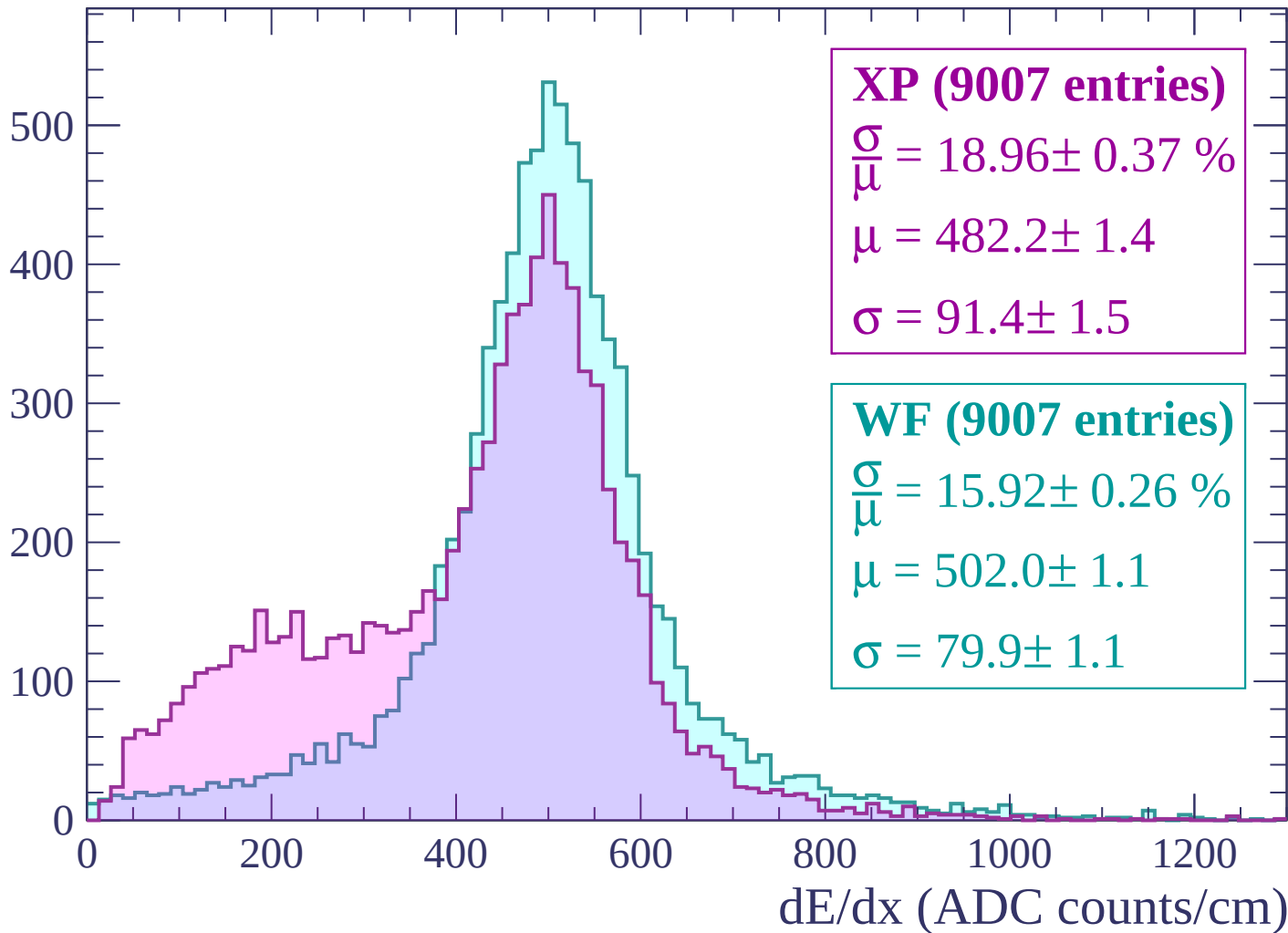
Energy loss | $-60 < p < 0$

Count



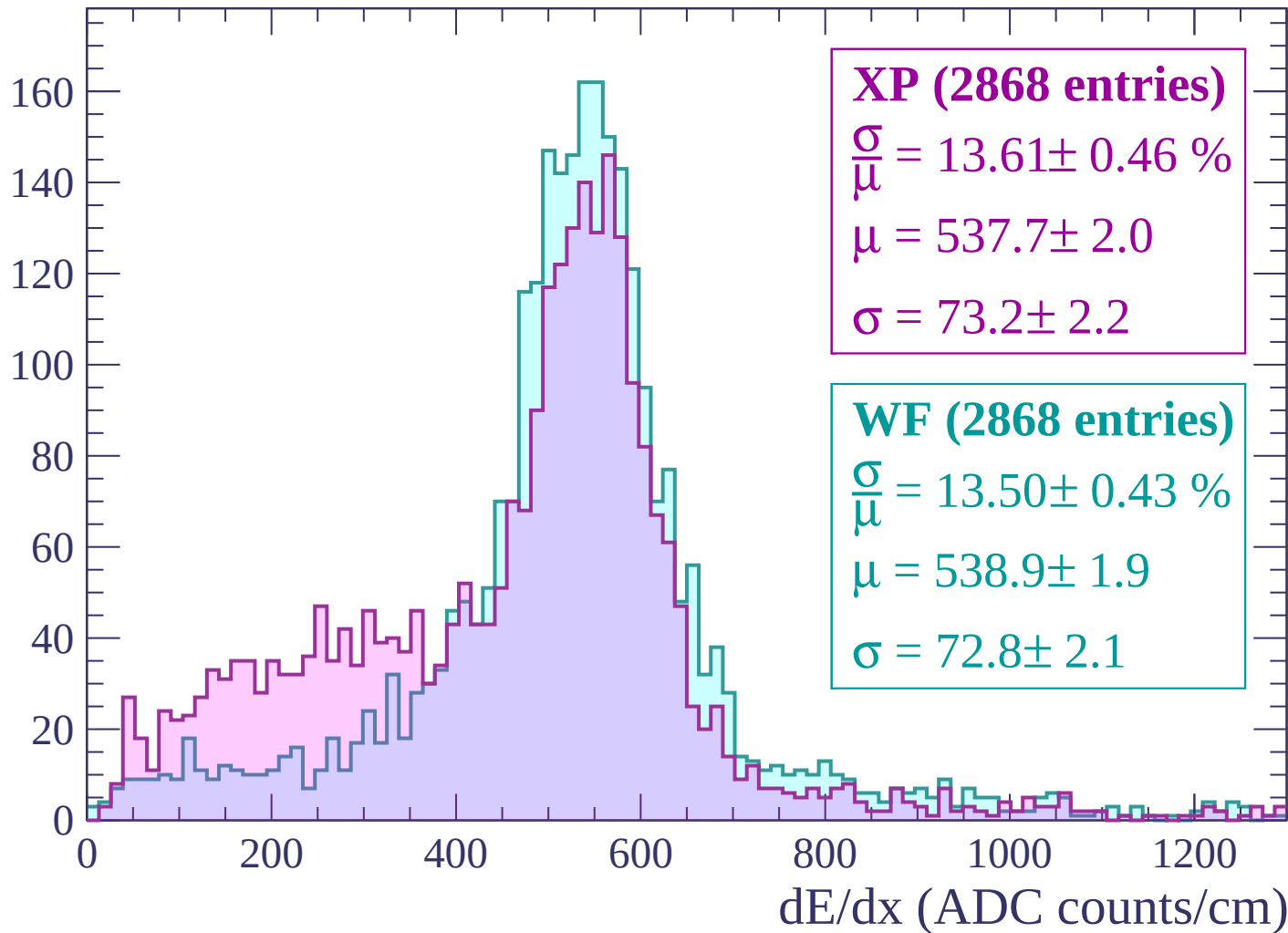
Energy loss | $0 < p < 60$

Count



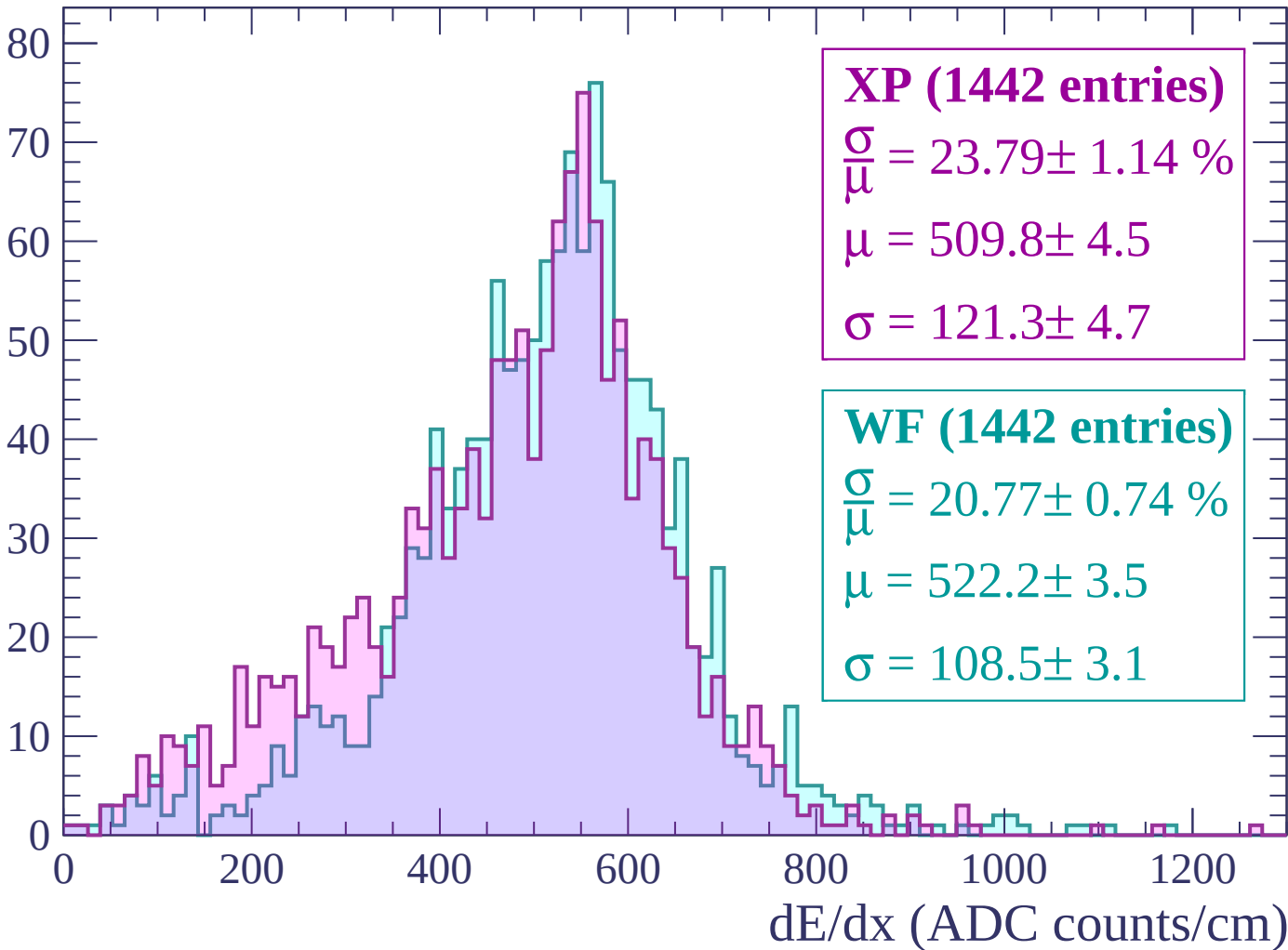
Energy loss | $60 < p < 120$

Count



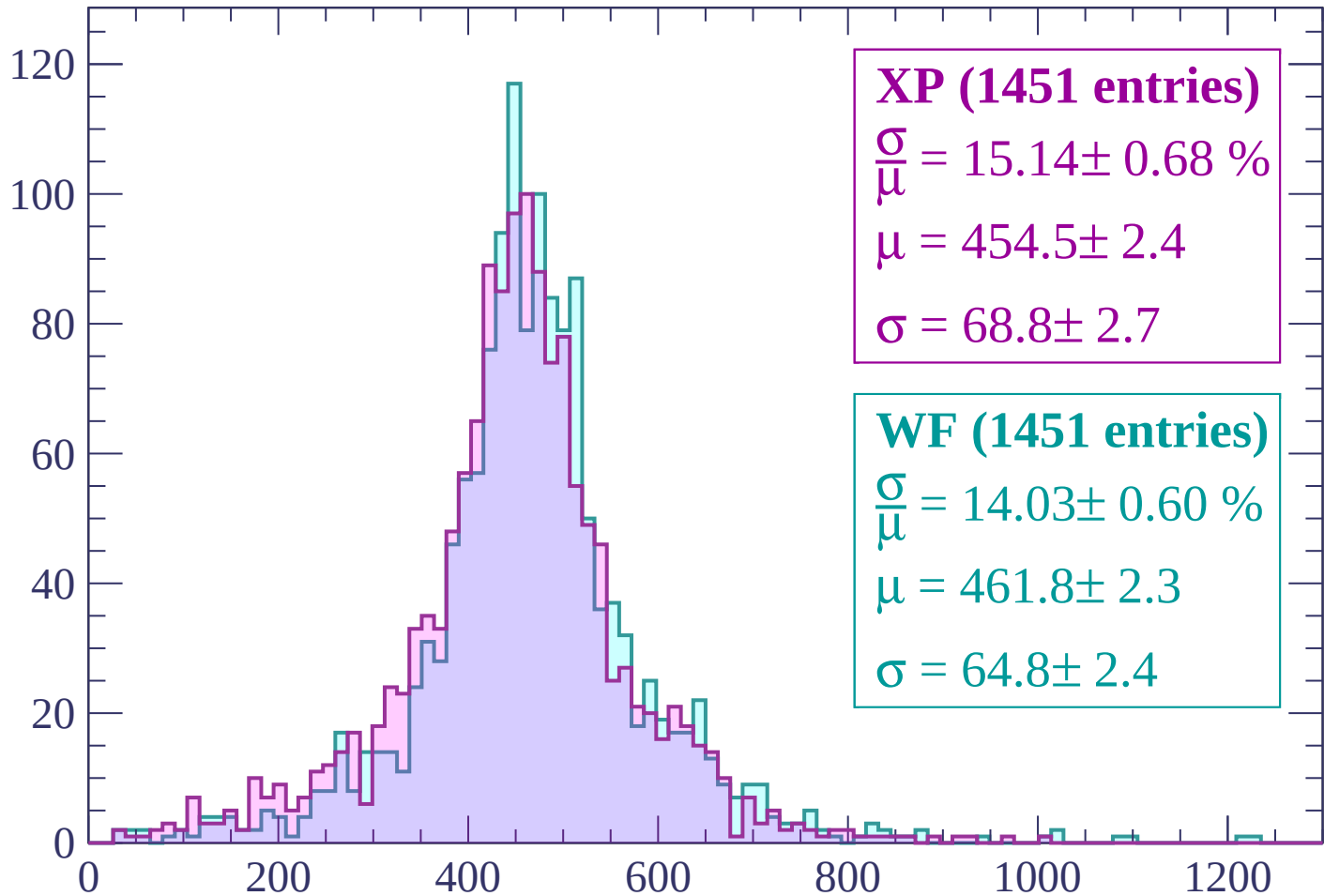
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (1451 entries)

$\frac{\sigma}{\mu} = 15.14 \pm 0.68 \%$

$\mu = 454.5 \pm 2.4$

$\sigma = 68.8 \pm 2.7$

WF (1451 entries)

$\frac{\sigma}{\mu} = 14.03 \pm 0.60 \%$

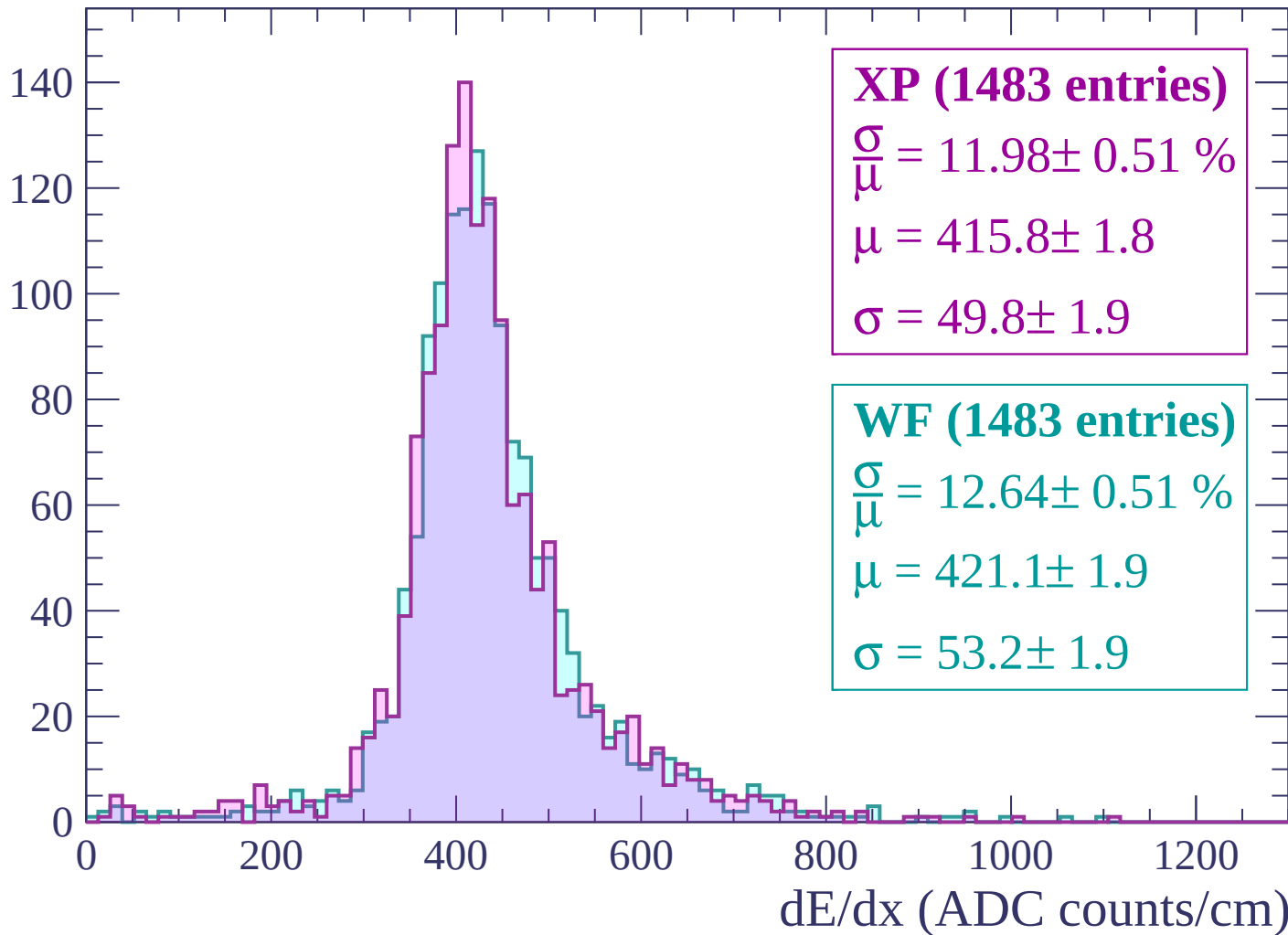
$\mu = 461.8 \pm 2.3$

$\sigma = 64.8 \pm 2.4$

dE/dx (ADC counts/cm)

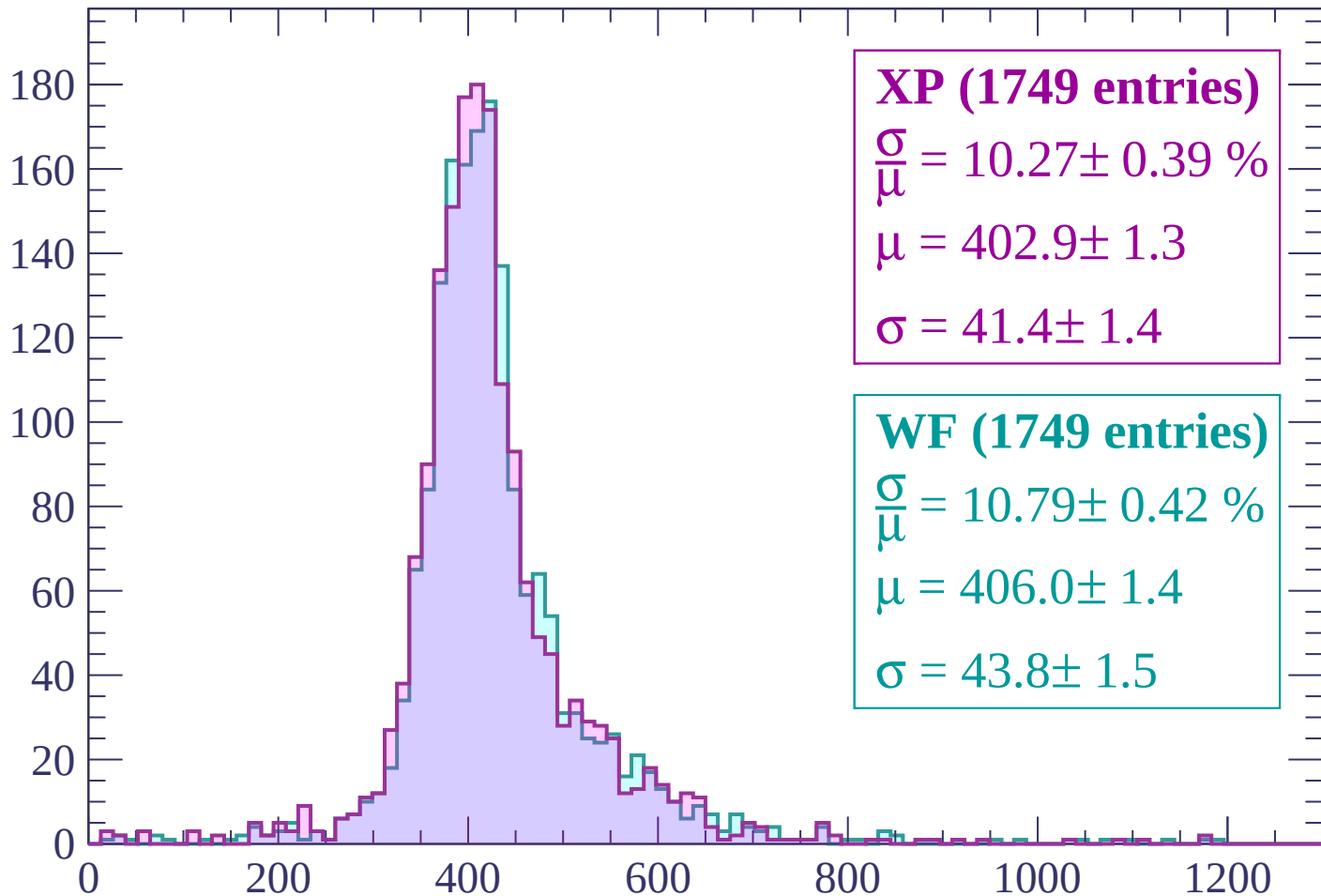
Energy loss | $240 < p < 300$

Count



Energy loss | $300 < p < 360$

Count



XP (1749 entries)

$\frac{\sigma}{\mu} = 10.27 \pm 0.39 \%$

$\mu = 402.9 \pm 1.3$

$\sigma = 41.4 \pm 1.4$

WF (1749 entries)

$\frac{\sigma}{\mu} = 10.79 \pm 0.42 \%$

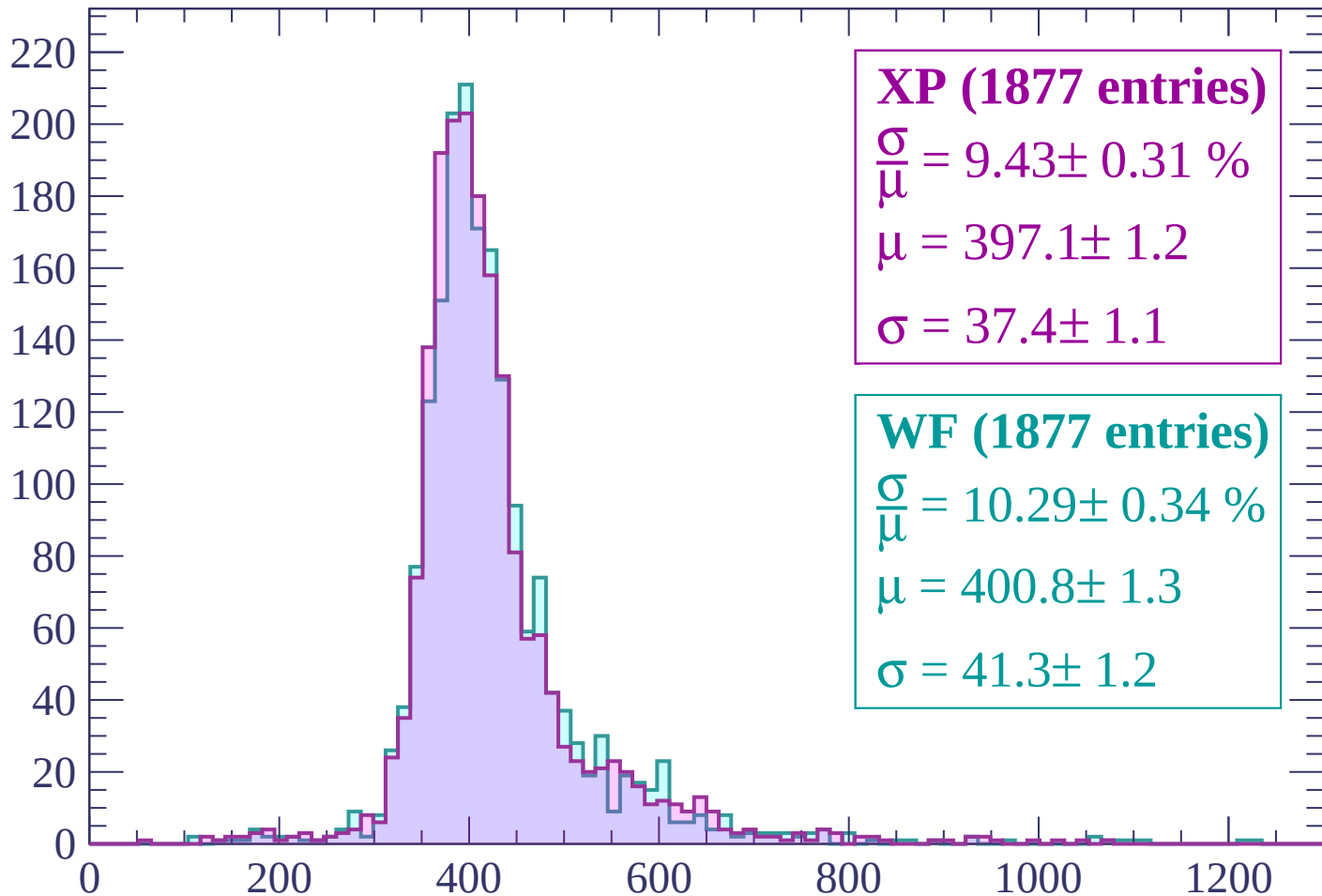
$\mu = 406.0 \pm 1.4$

$\sigma = 43.8 \pm 1.5$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (1877 entries)

$$\frac{\sigma}{\mu} = 9.43 \pm 0.31 \%$$

$$\mu = 397.1 \pm 1.2$$

$$\sigma = 37.4 \pm 1.1$$

WF (1877 entries)

$$\frac{\sigma}{\mu} = 10.29 \pm 0.34 \%$$

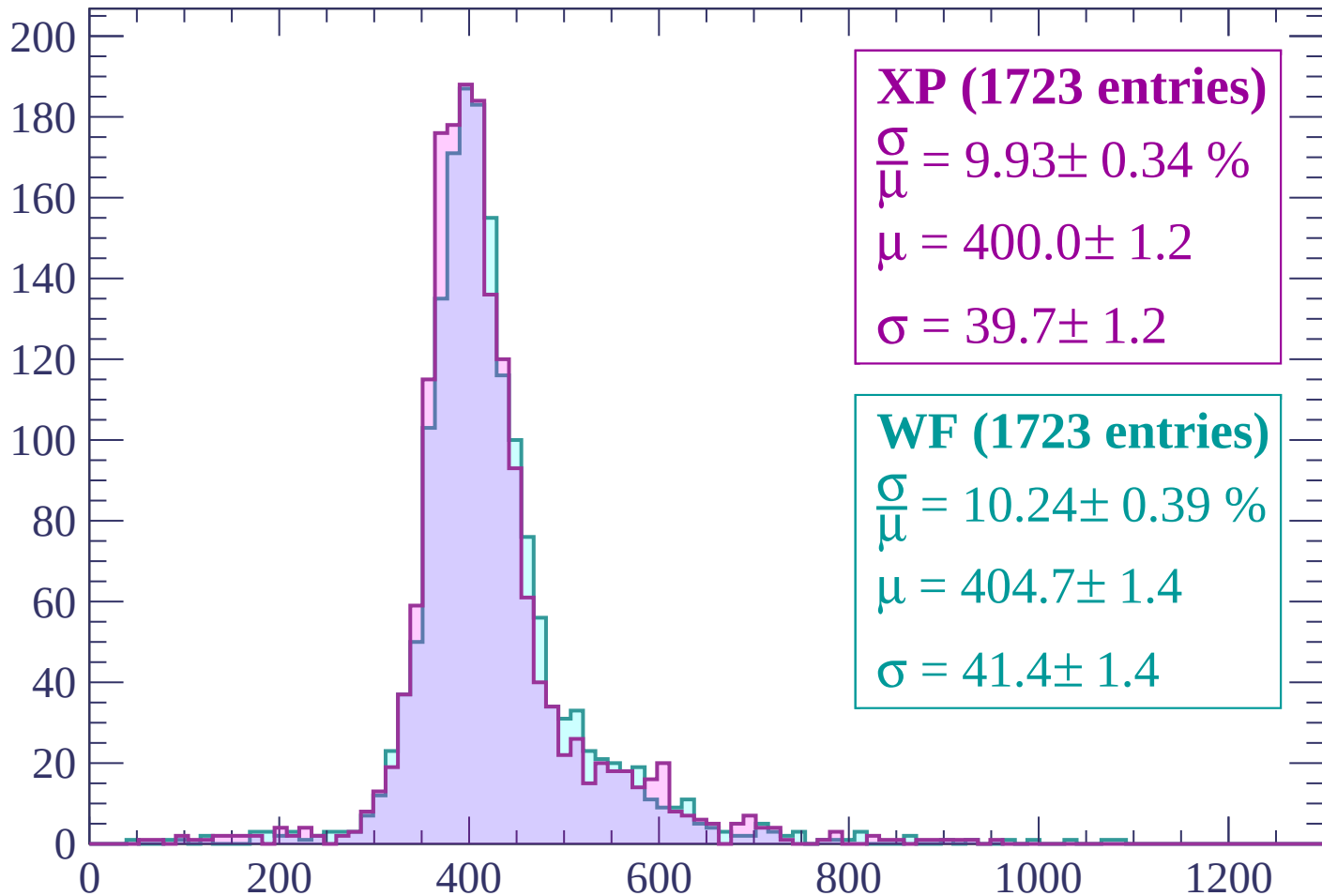
$$\mu = 400.8 \pm 1.3$$

$$\sigma = 41.3 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP (1723 entries)

$$\frac{\sigma}{\mu} = 9.93 \pm 0.34 \%$$

$$\mu = 400.0 \pm 1.2$$

$$\sigma = 39.7 \pm 1.2$$

WF (1723 entries)

$$\frac{\sigma}{\mu} = 10.24 \pm 0.39 \%$$

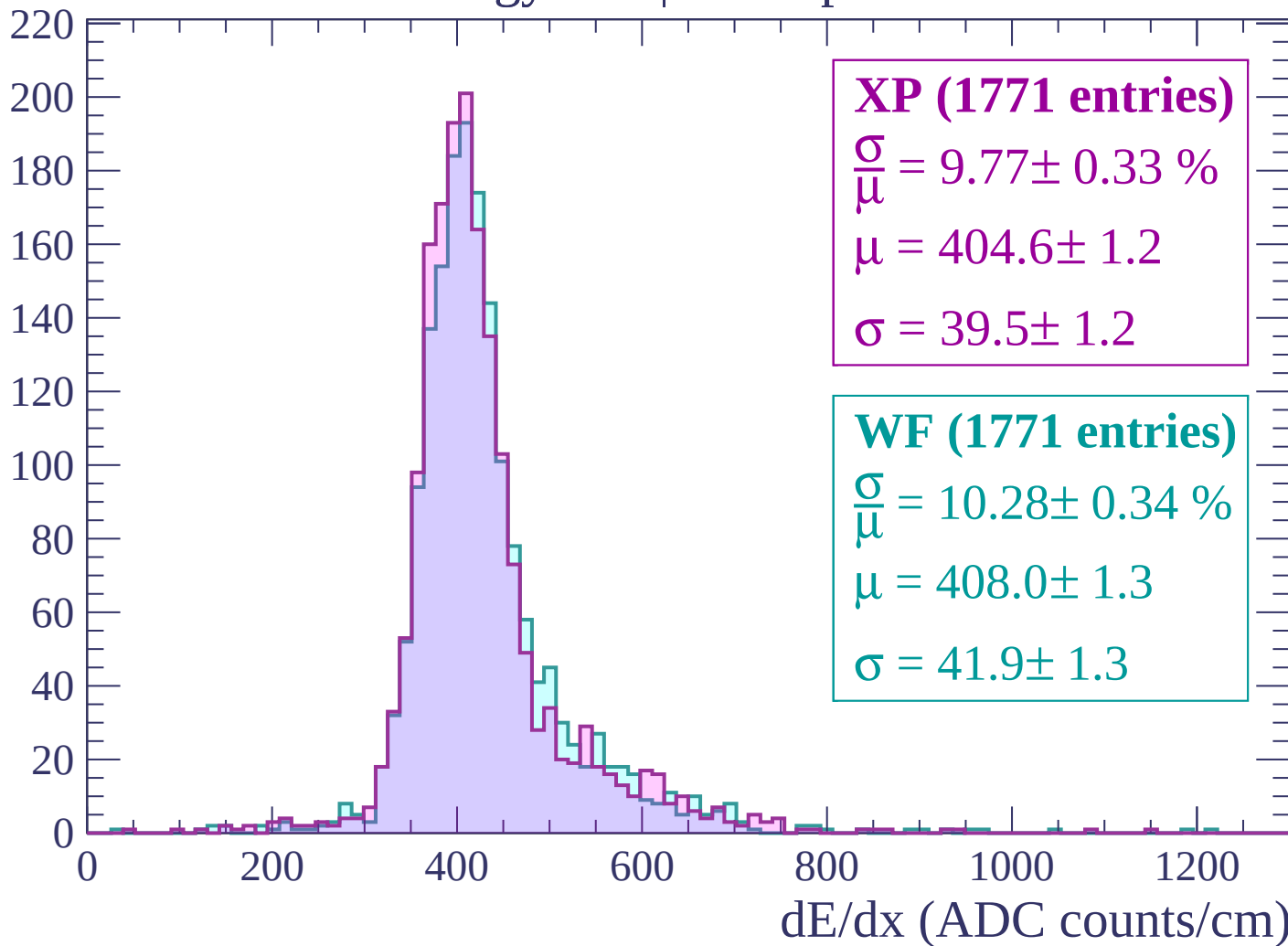
$$\mu = 404.7 \pm 1.4$$

$$\sigma = 41.4 \pm 1.4$$

dE/dx (ADC counts/cm)

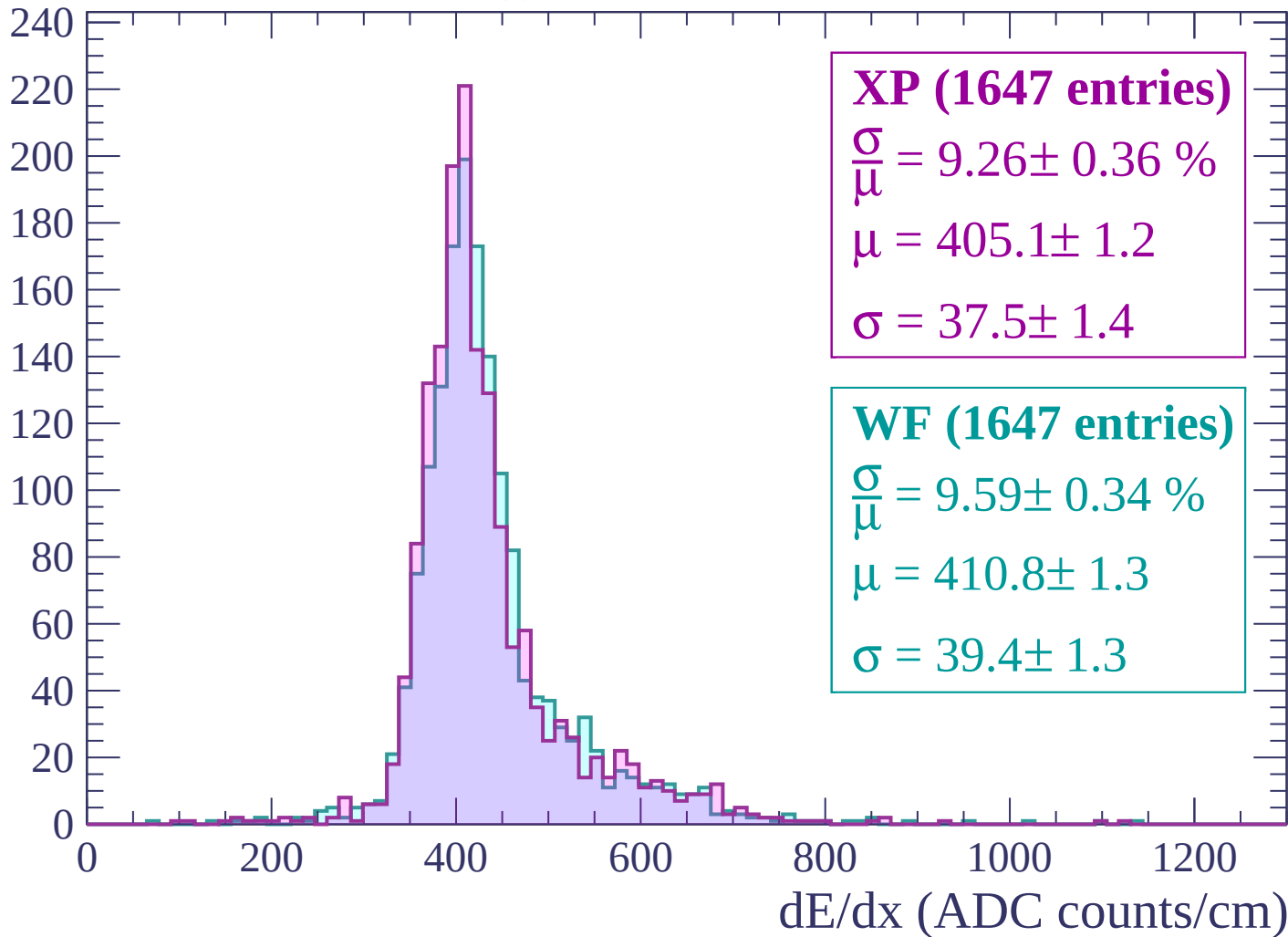
Energy loss | $480 < p < 540$

Count



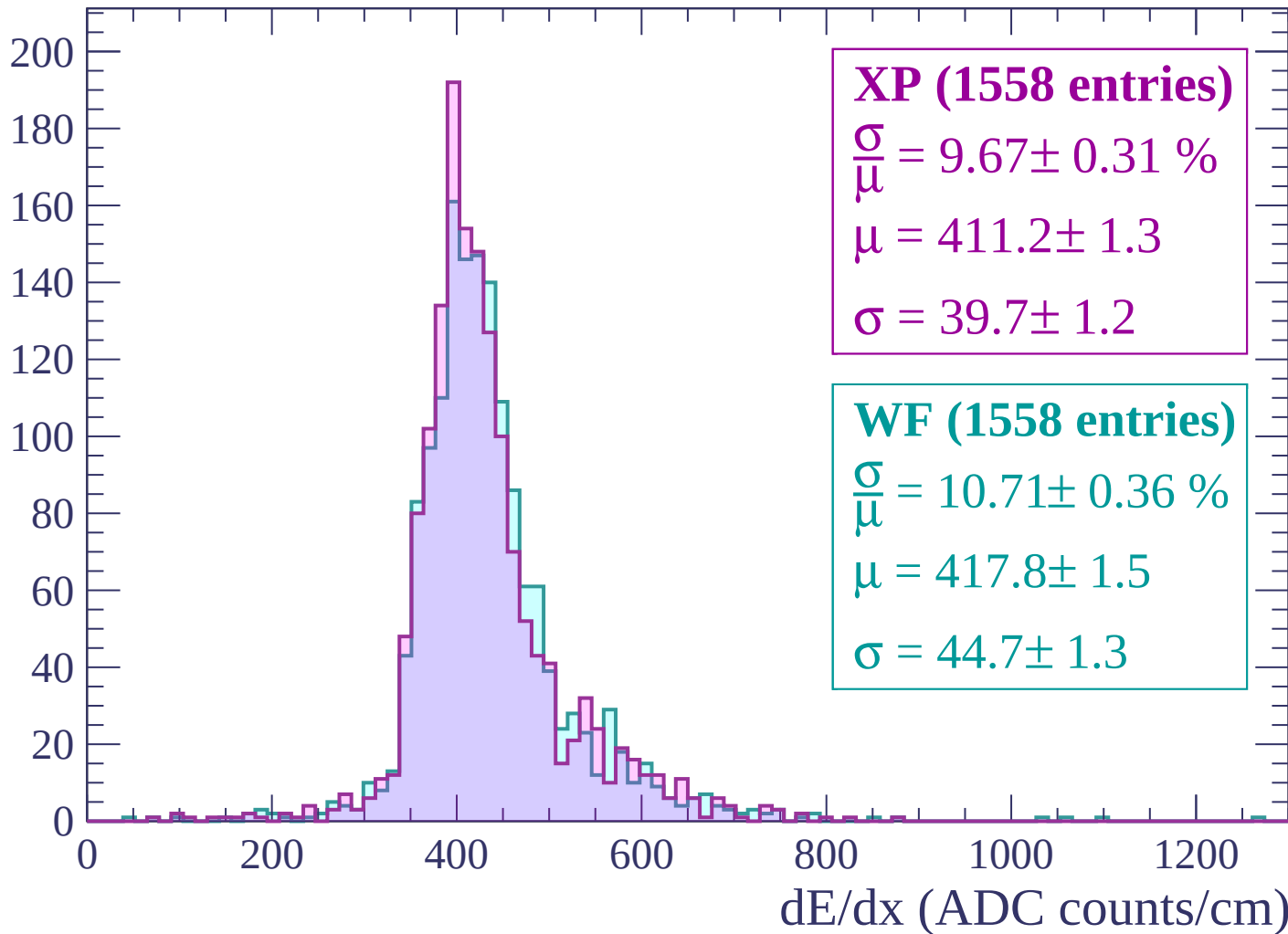
Energy loss | $540 < p < 600$

Count



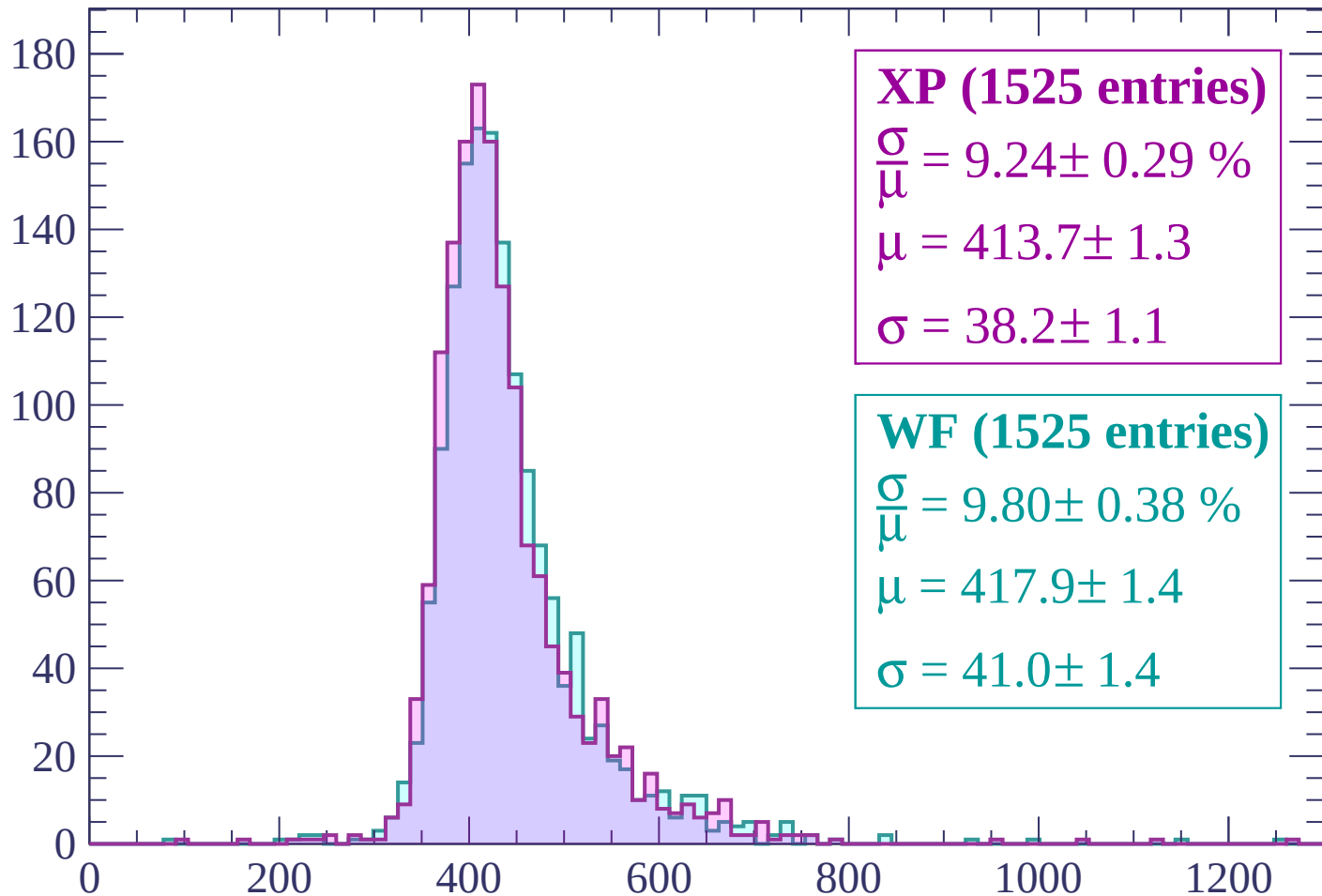
Energy loss | $600 < p < 660$

Count



Energy loss | $660 < p < 720$

Count



XP (1525 entries)

$$\frac{\sigma}{\mu} = 9.24 \pm 0.29 \%$$

$$\mu = 413.7 \pm 1.3$$

$$\sigma = 38.2 \pm 1.1$$

WF (1525 entries)

$$\frac{\sigma}{\mu} = 9.80 \pm 0.38 \%$$

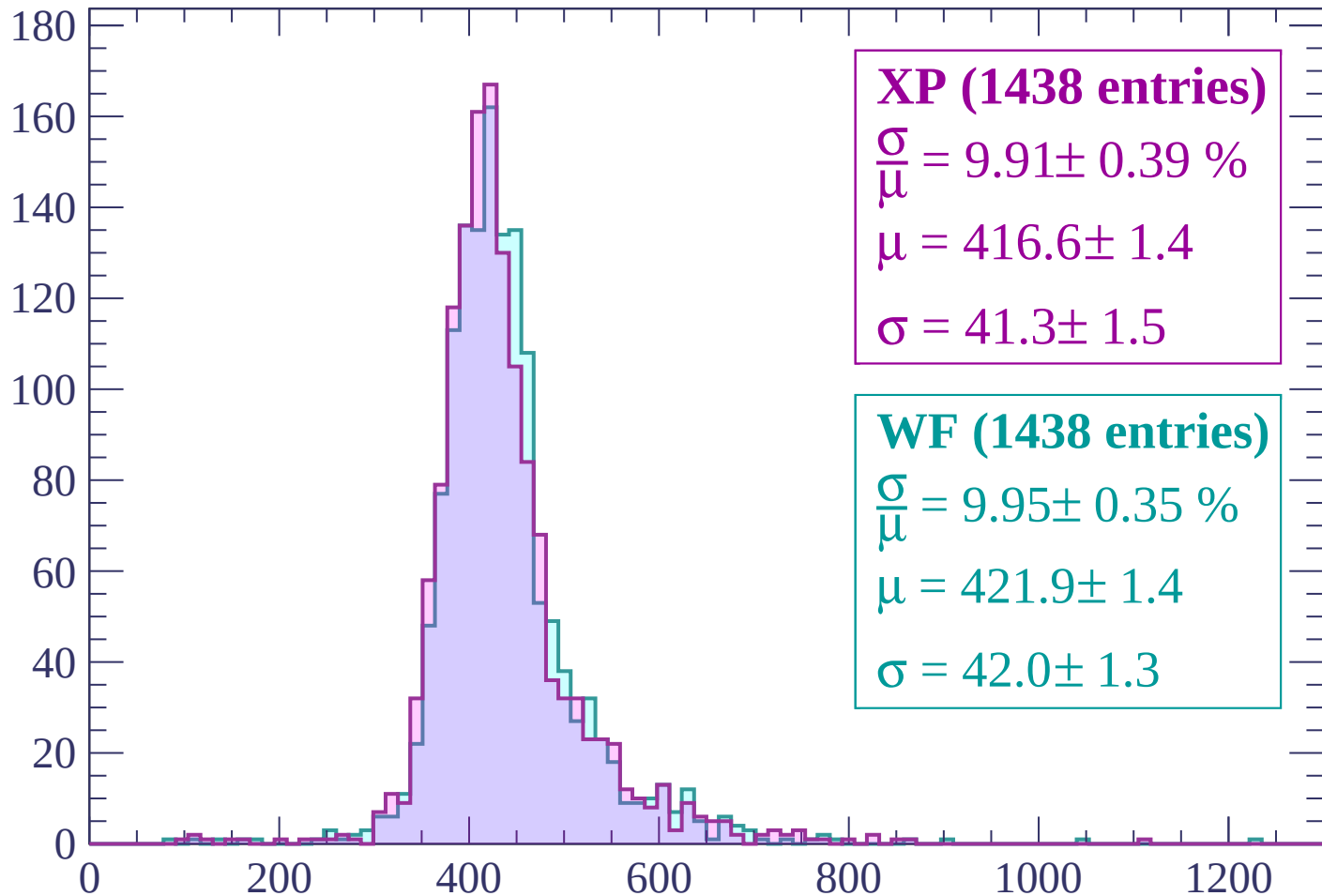
$$\mu = 417.9 \pm 1.4$$

$$\sigma = 41.0 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP (1438 entries)

$$\frac{\sigma}{\mu} = 9.91 \pm 0.39 \%$$

$$\mu = 416.6 \pm 1.4$$

$$\sigma = 41.3 \pm 1.5$$

WF (1438 entries)

$$\frac{\sigma}{\mu} = 9.95 \pm 0.35 \%$$

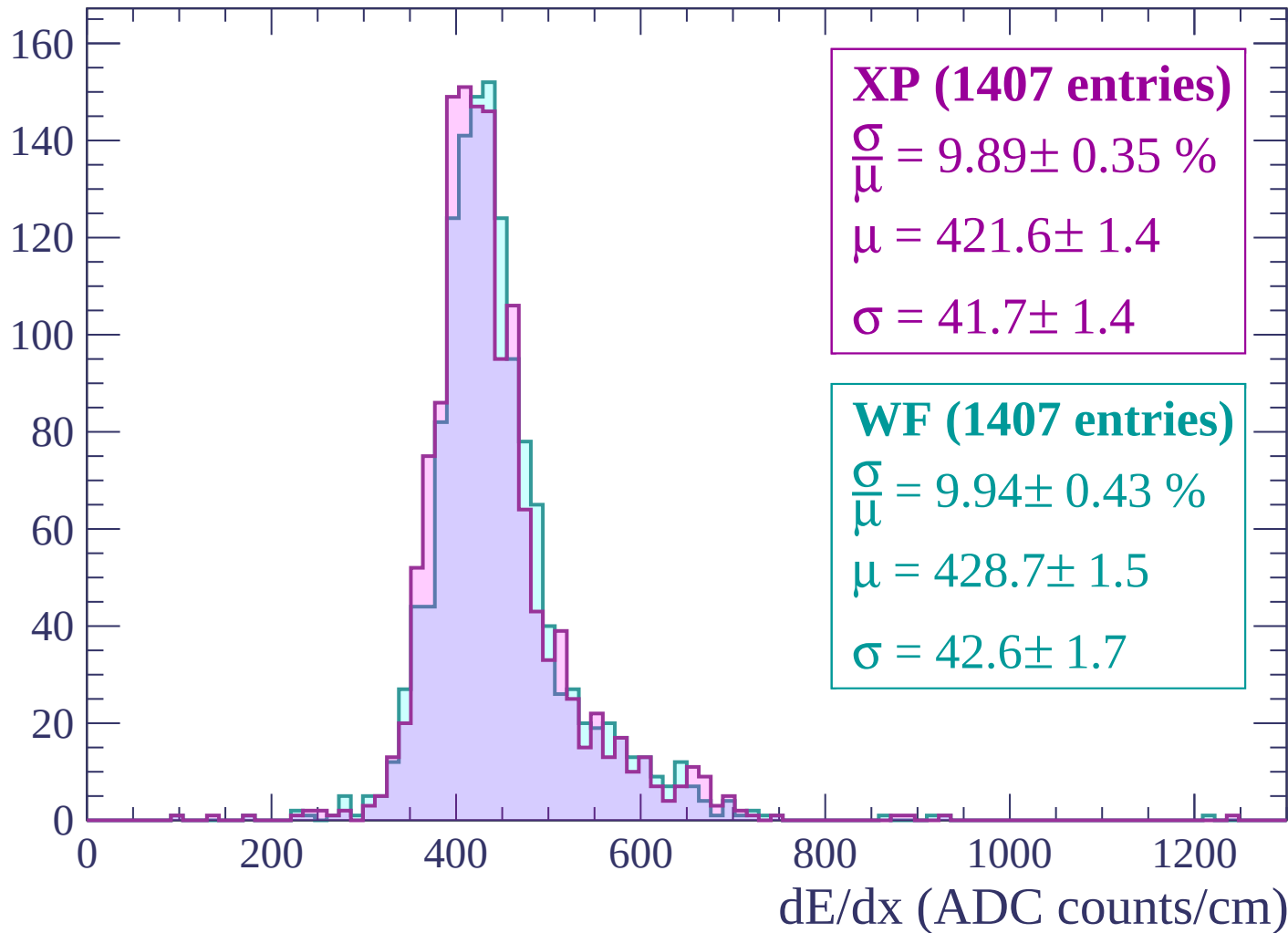
$$\mu = 421.9 \pm 1.4$$

$$\sigma = 42.0 \pm 1.3$$

dE/dx (ADC counts/cm)

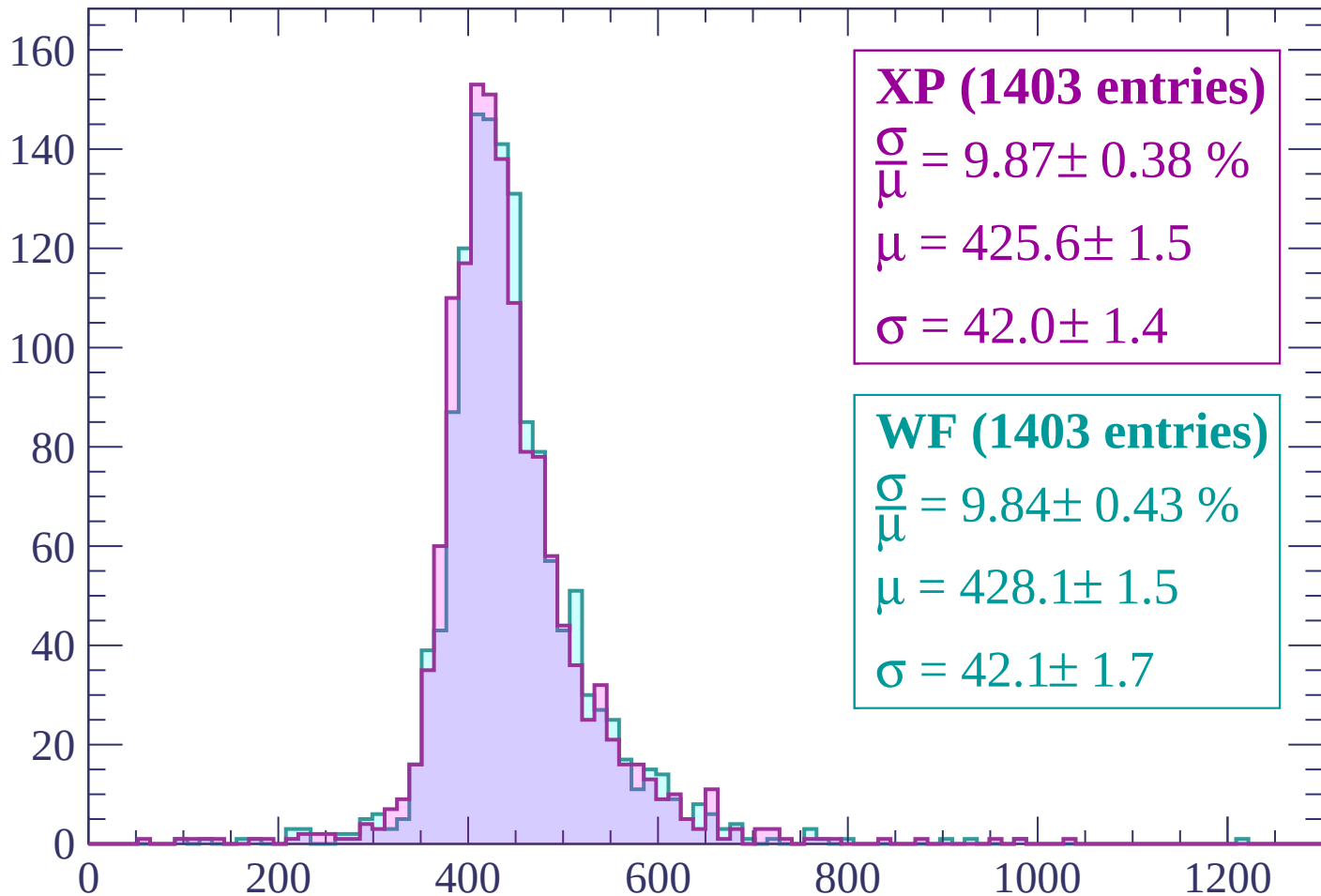
Energy loss | $780 < p < 840$

Count



Energy loss | $840 < p < 900$

Count



XP (1403 entries)

$$\frac{\sigma}{\mu} = 9.87 \pm 0.38 \%$$

$$\mu = 425.6 \pm 1.5$$

$$\sigma = 42.0 \pm 1.4$$

WF (1403 entries)

$$\frac{\sigma}{\mu} = 9.84 \pm 0.43 \%$$

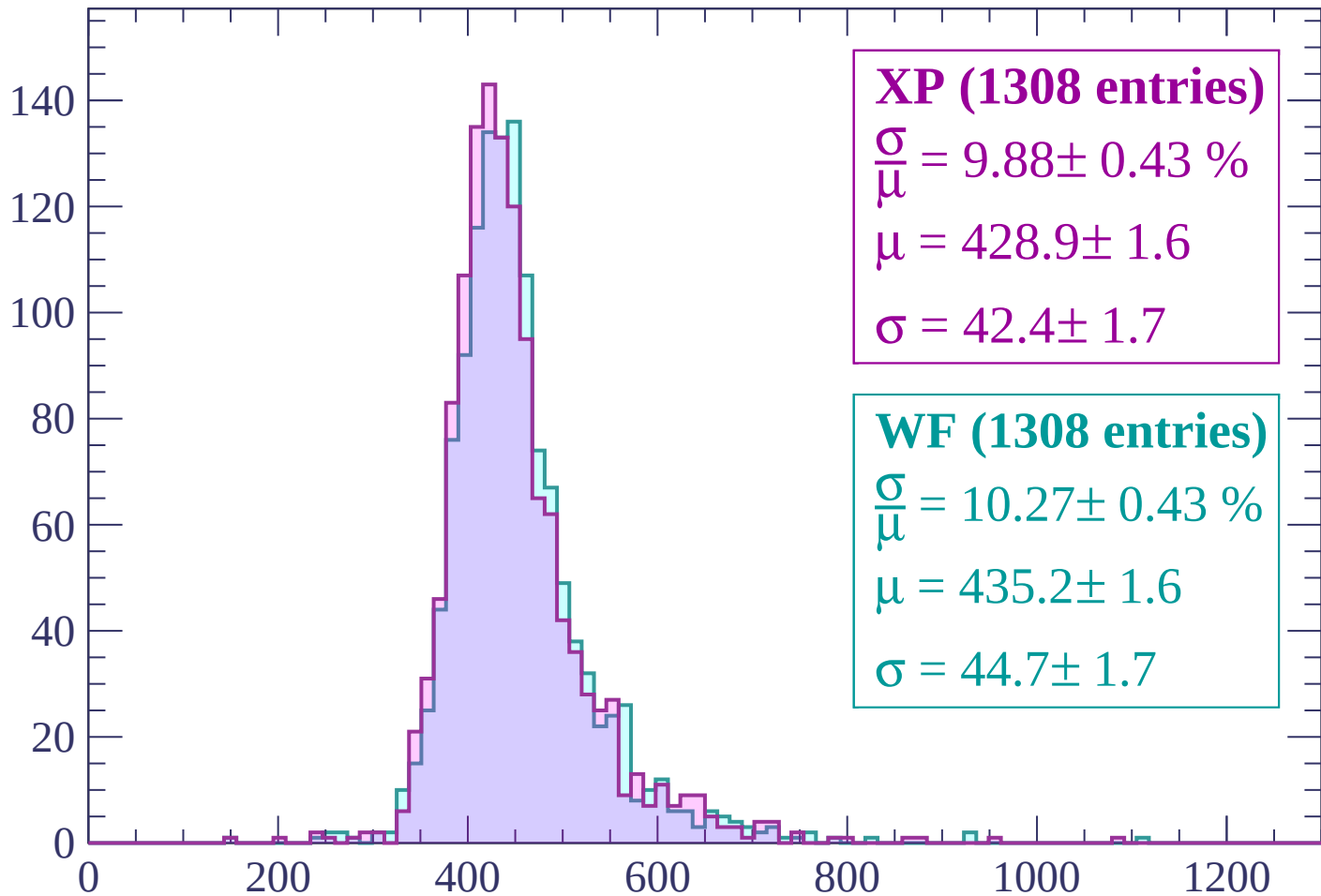
$$\mu = 428.1 \pm 1.5$$

$$\sigma = 42.1 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (1308 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.43 \%$$

$$\mu = 428.9 \pm 1.6$$

$$\sigma = 42.4 \pm 1.7$$

WF (1308 entries)

$$\frac{\sigma}{\mu} = 10.27 \pm 0.43 \%$$

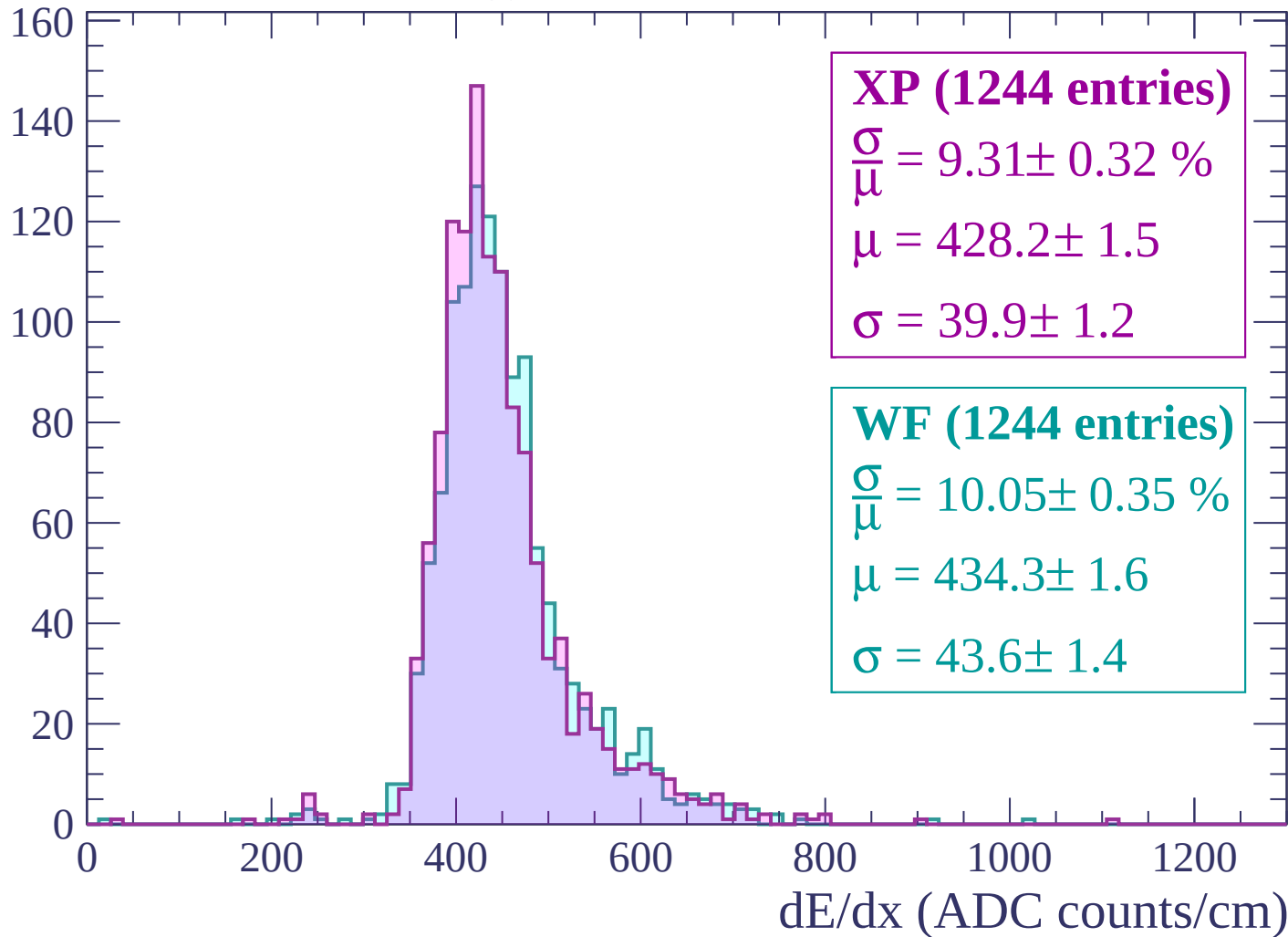
$$\mu = 435.2 \pm 1.6$$

$$\sigma = 44.7 \pm 1.7$$

dE/dx (ADC counts/cm)

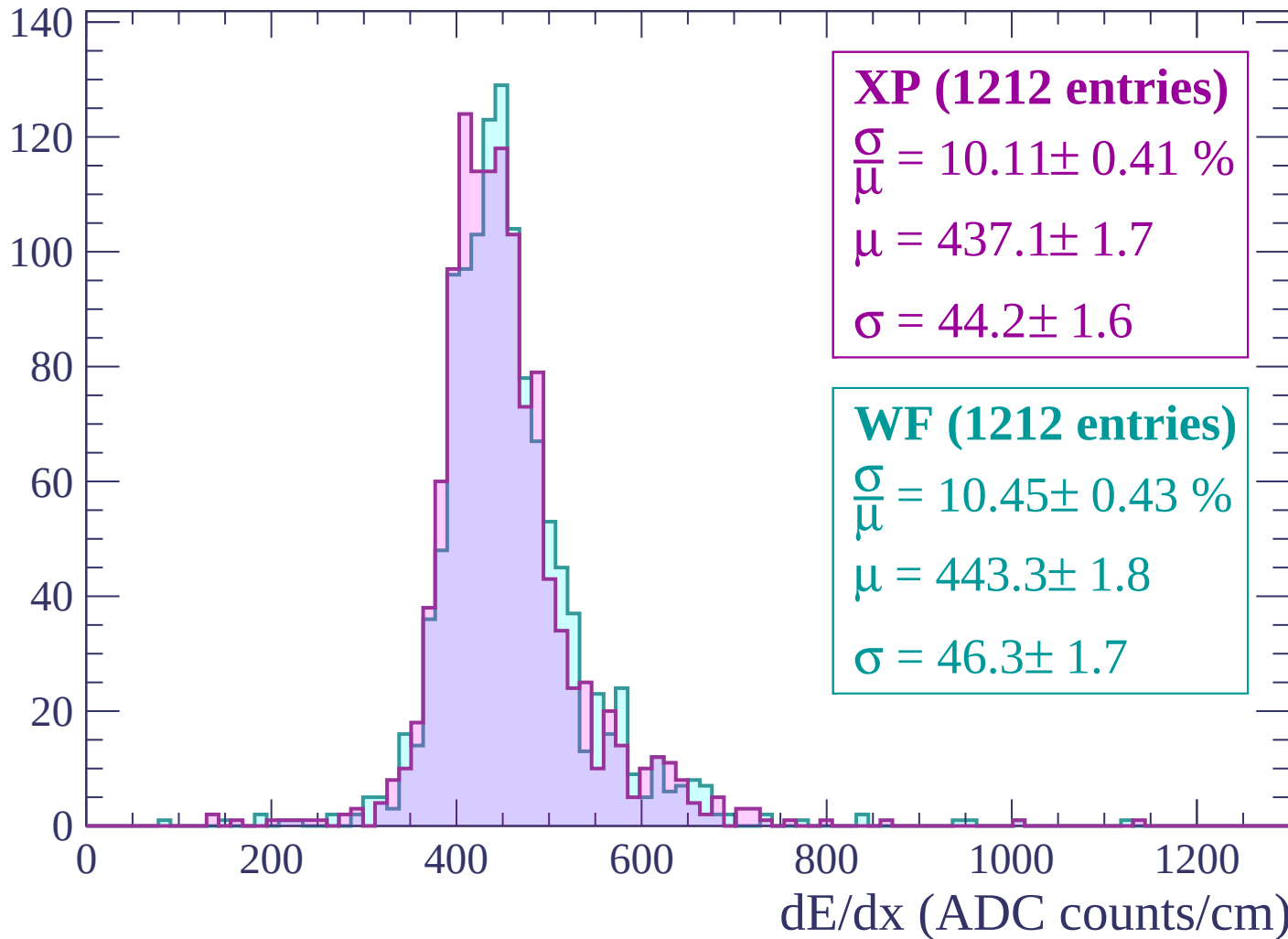
Energy loss | $960 < p < 1020$

Count



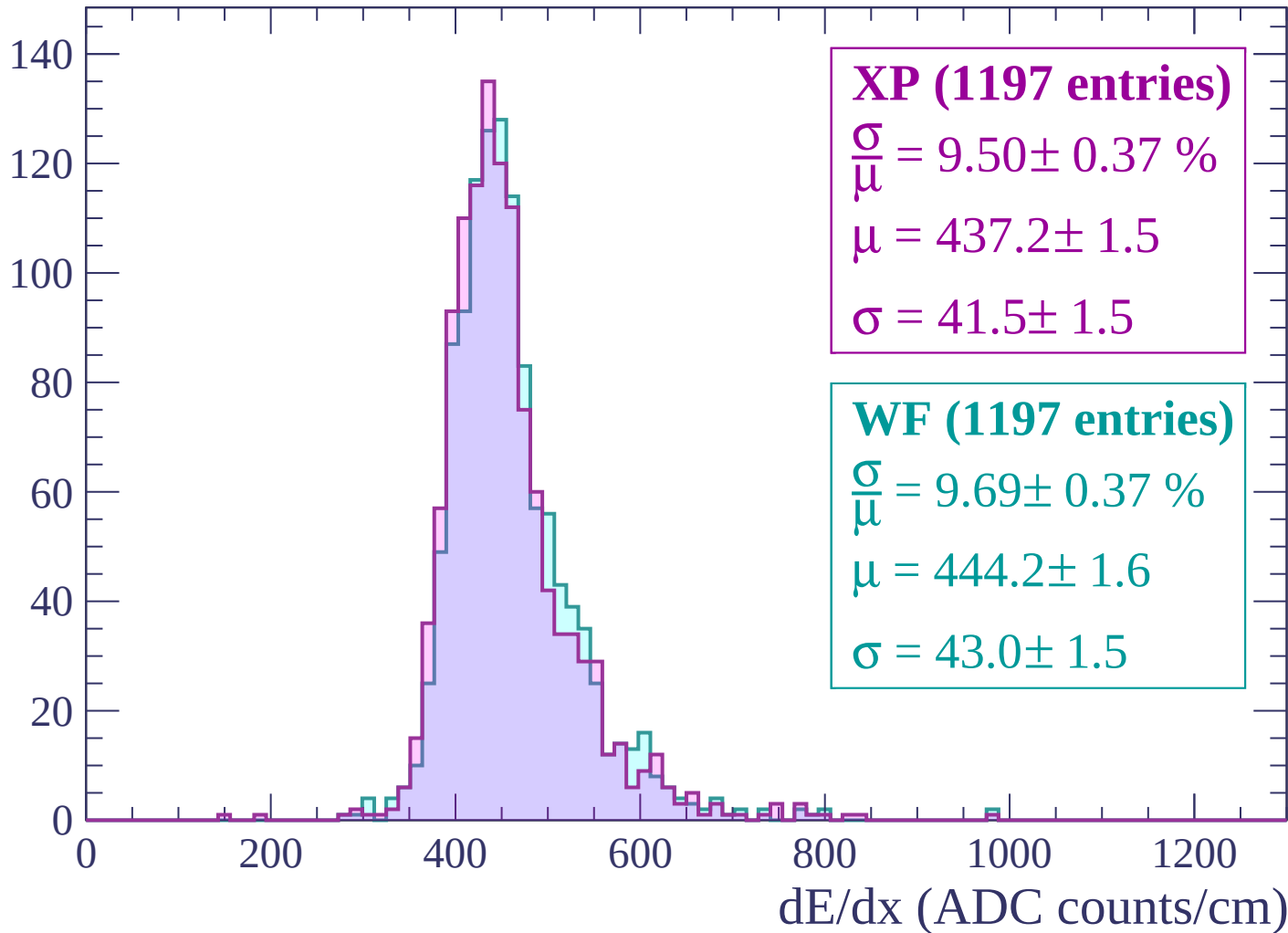
Energy loss | $1020 < p < 1080$

Count



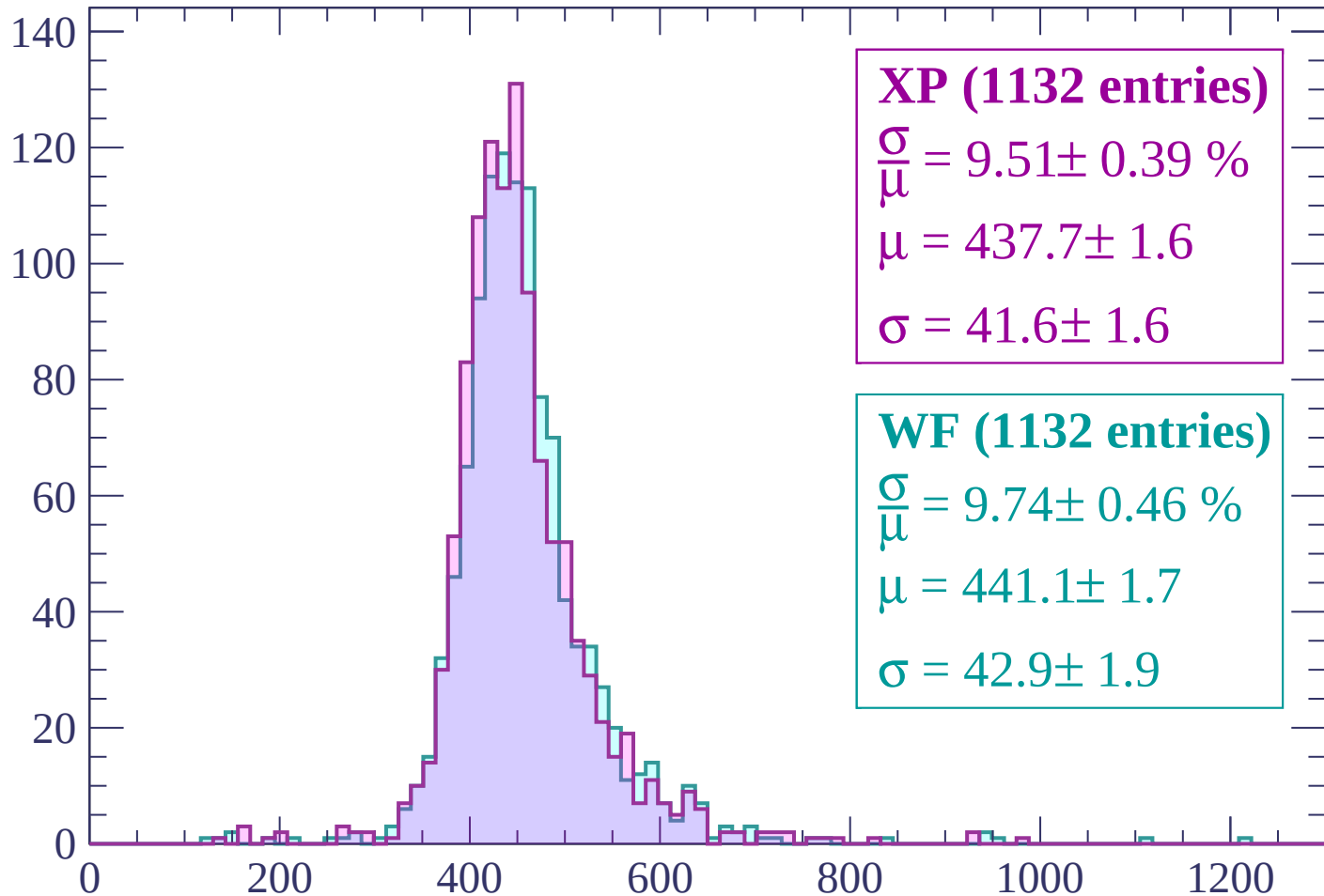
Energy loss | $1080 < p < 1140$

Count



Energy loss | $1140 < p < 1200$

Count



XP (1132 entries)

$$\frac{\sigma}{\mu} = 9.51 \pm 0.39 \%$$

$$\mu = 437.7 \pm 1.6$$

$$\sigma = 41.6 \pm 1.6$$

WF (1132 entries)

$$\frac{\sigma}{\mu} = 9.74 \pm 0.46 \%$$

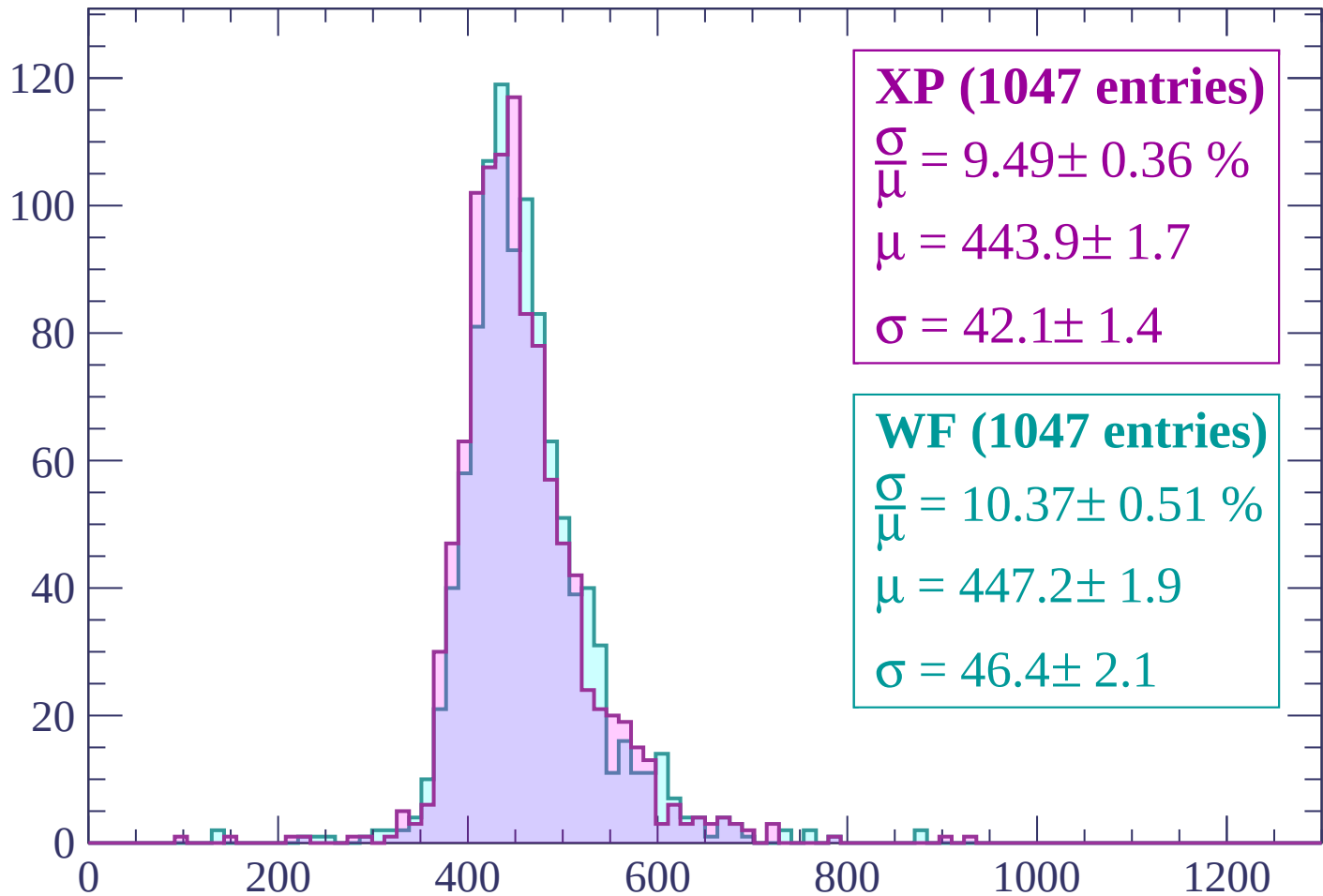
$$\mu = 441.1 \pm 1.7$$

$$\sigma = 42.9 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$

Count



XP (1047 entries)

$$\frac{\sigma}{\mu} = 9.49 \pm 0.36 \%$$

$$\mu = 443.9 \pm 1.7$$

$$\sigma = 42.1 \pm 1.4$$

WF (1047 entries)

$$\frac{\sigma}{\mu} = 10.37 \pm 0.51 \%$$

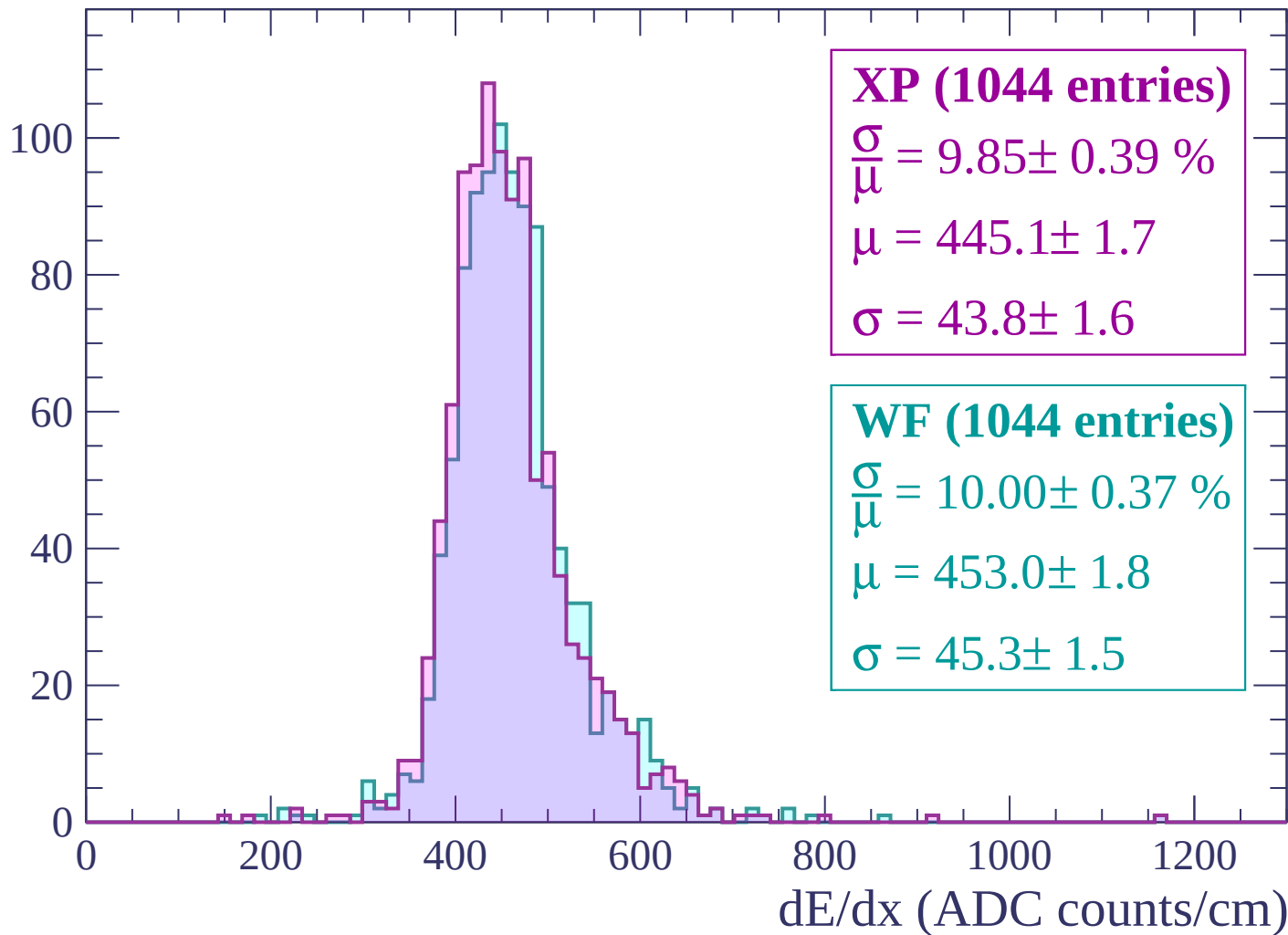
$$\mu = 447.2 \pm 1.9$$

$$\sigma = 46.4 \pm 2.1$$

dE/dx (ADC counts/cm)

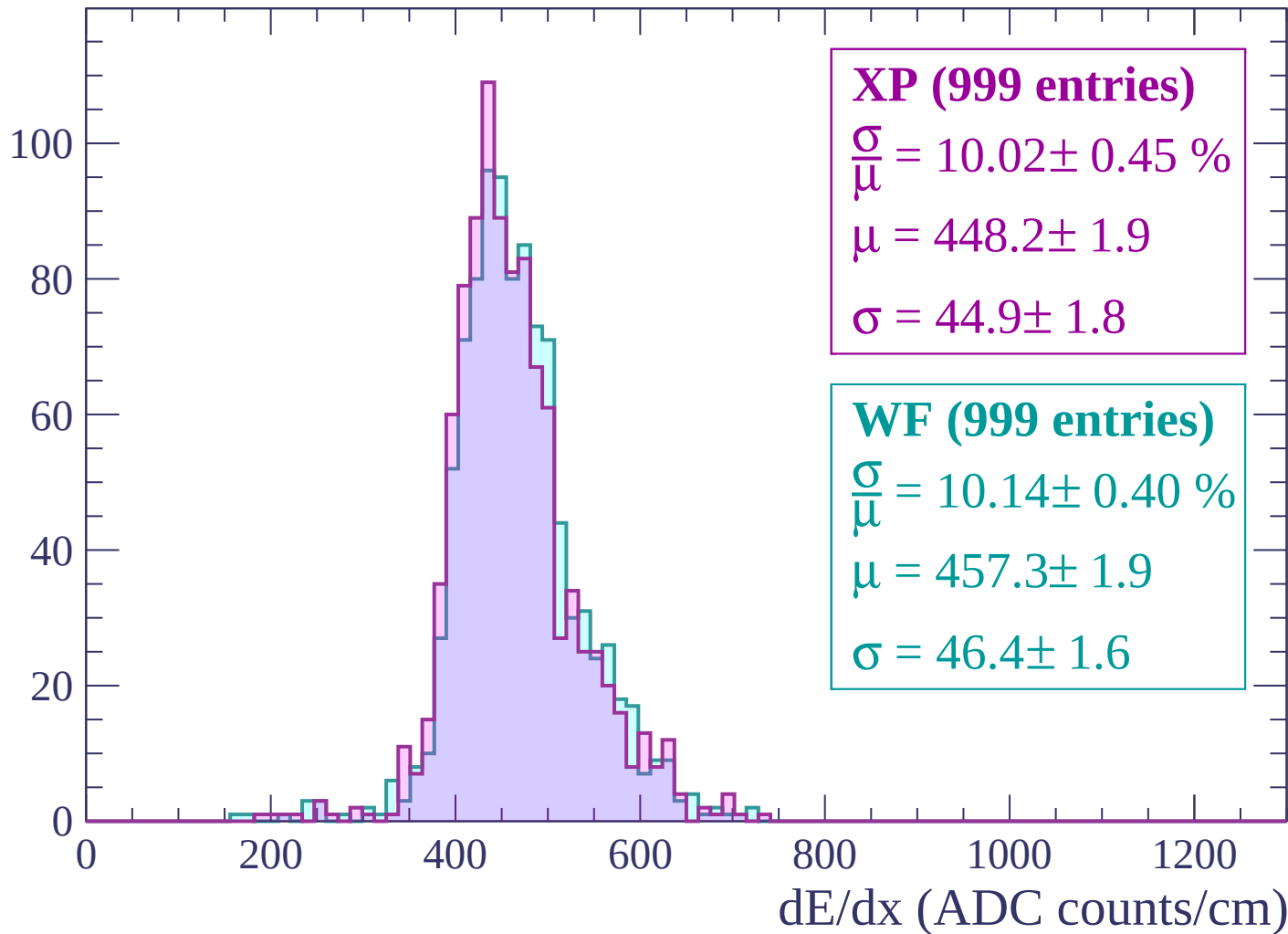
Energy loss | $1260 < p < 1320$

Count



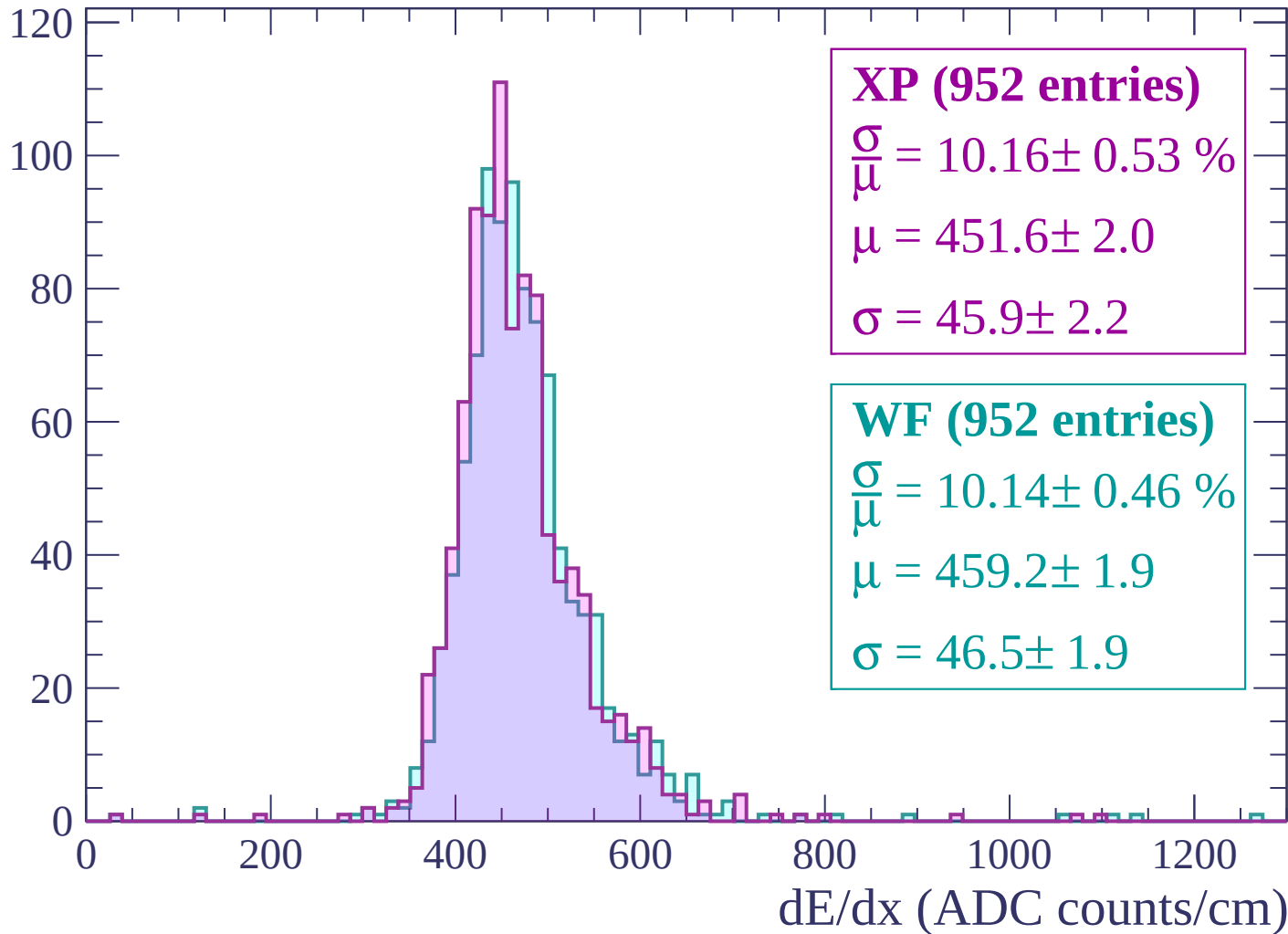
Energy loss | $1320 < p < 1380$

Count



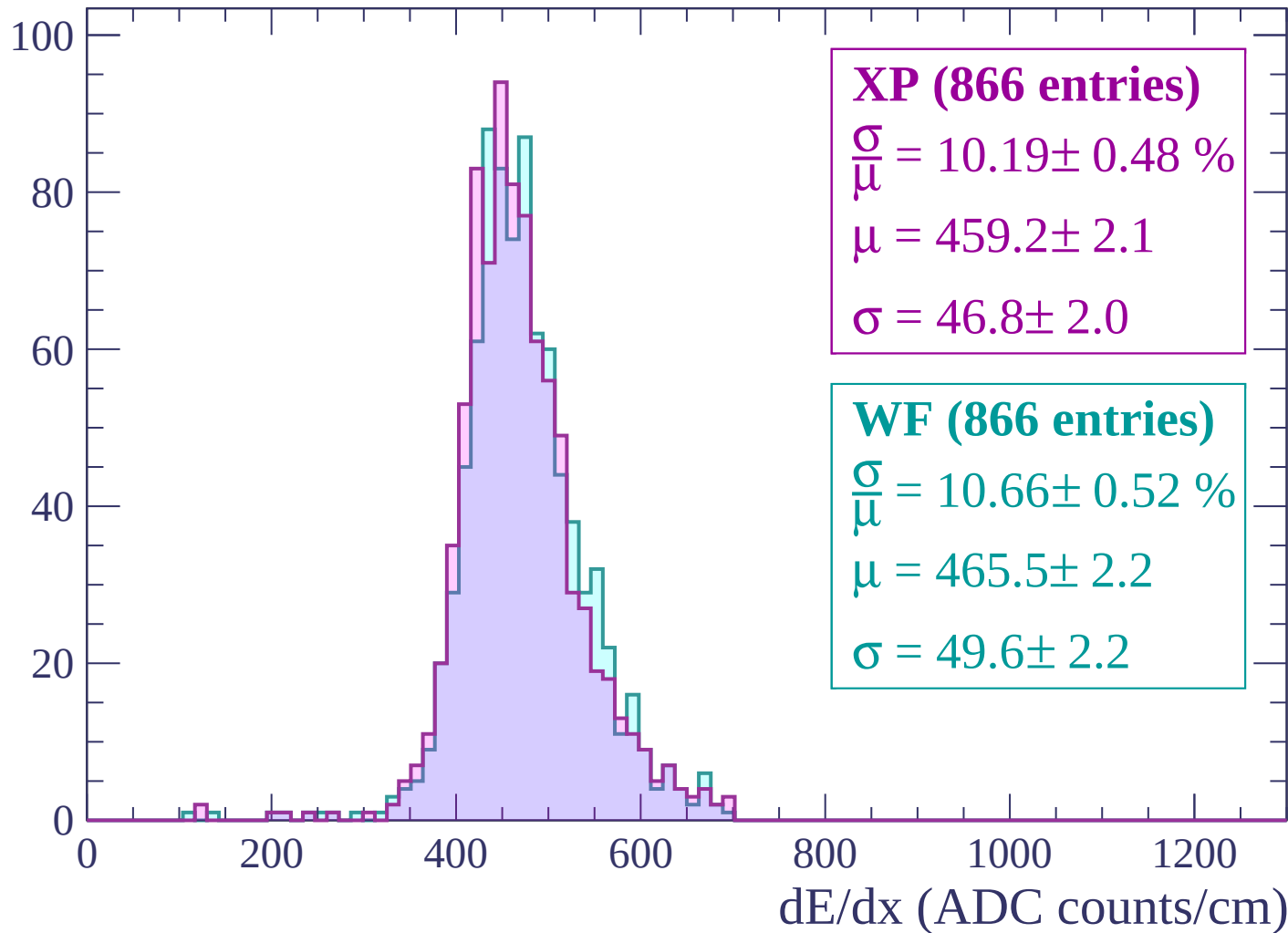
Energy loss | $1380 < p < 1440$

Count



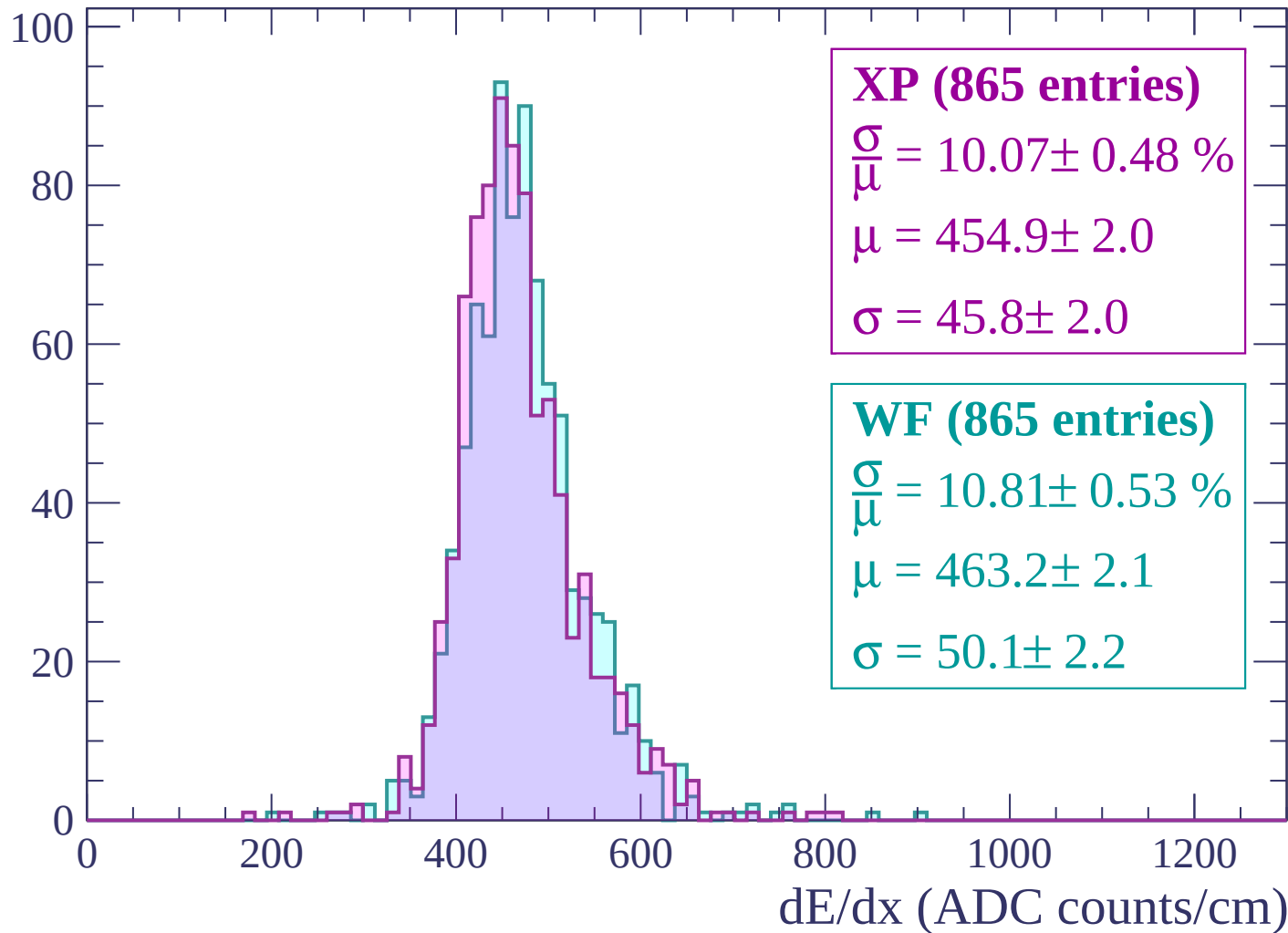
Energy loss | $1440 < p < 1500$

Count



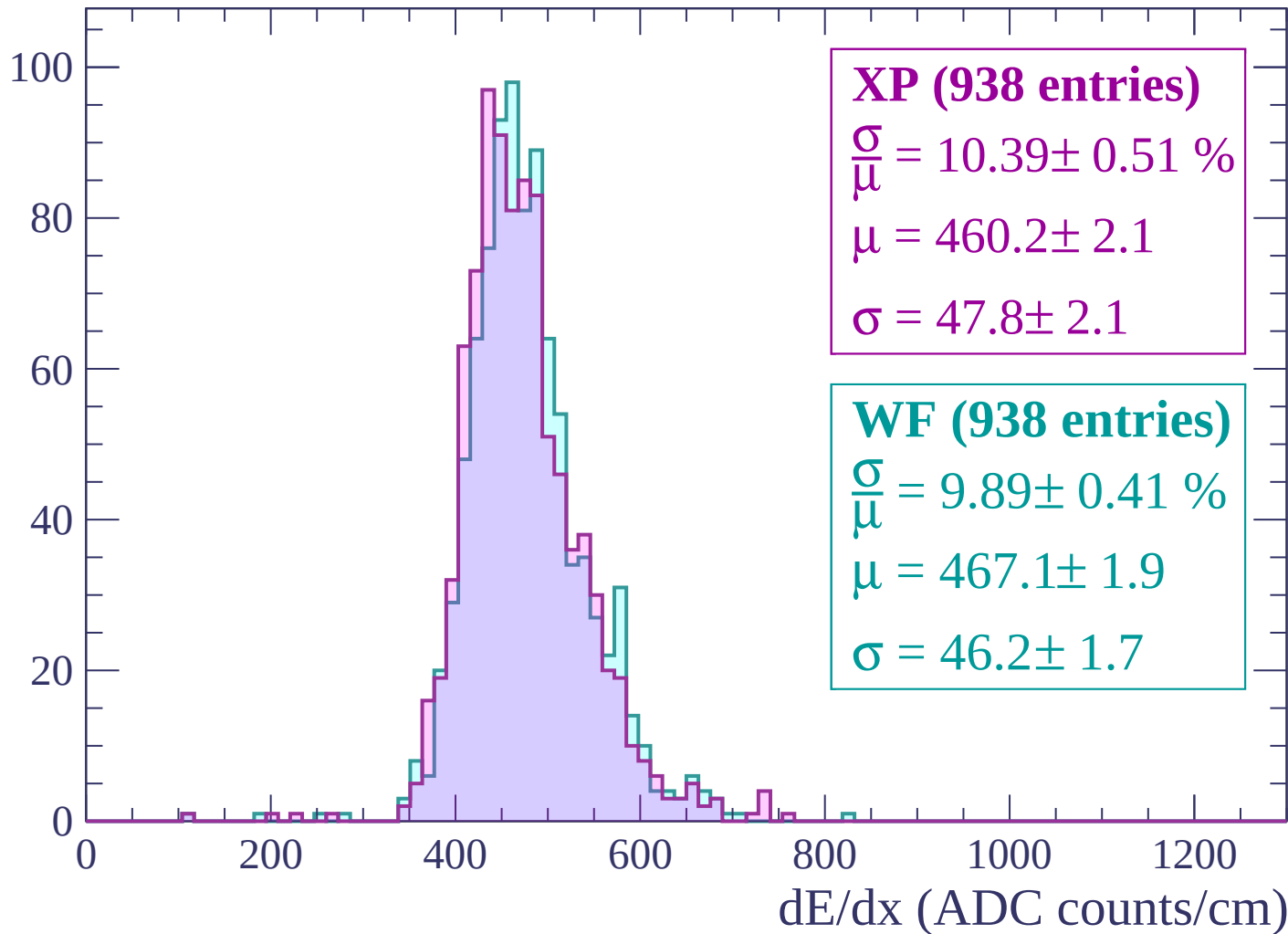
Energy loss | $1500 < p < 1560$

Count



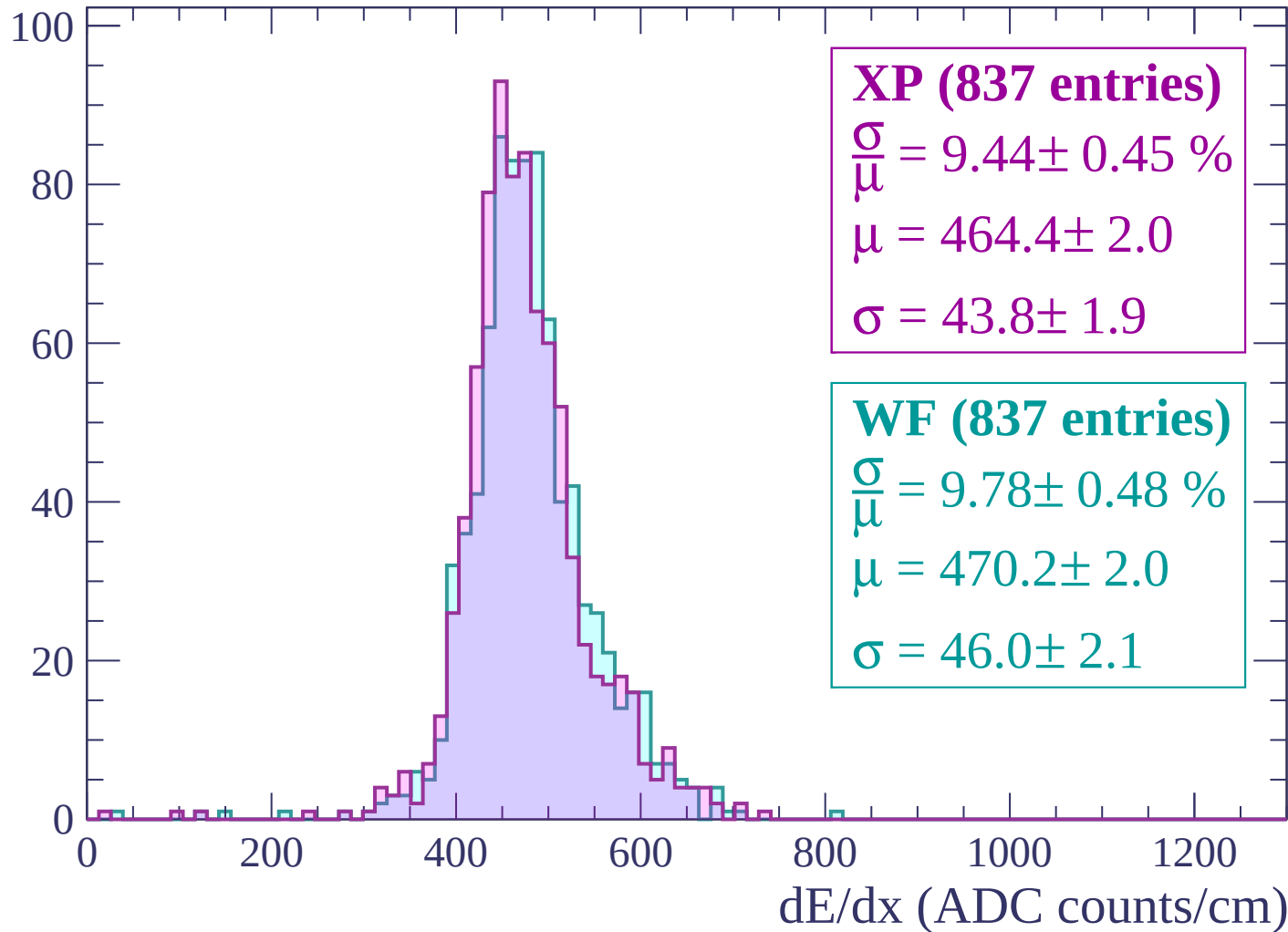
Energy loss | $1560 < p < 1620$

Count



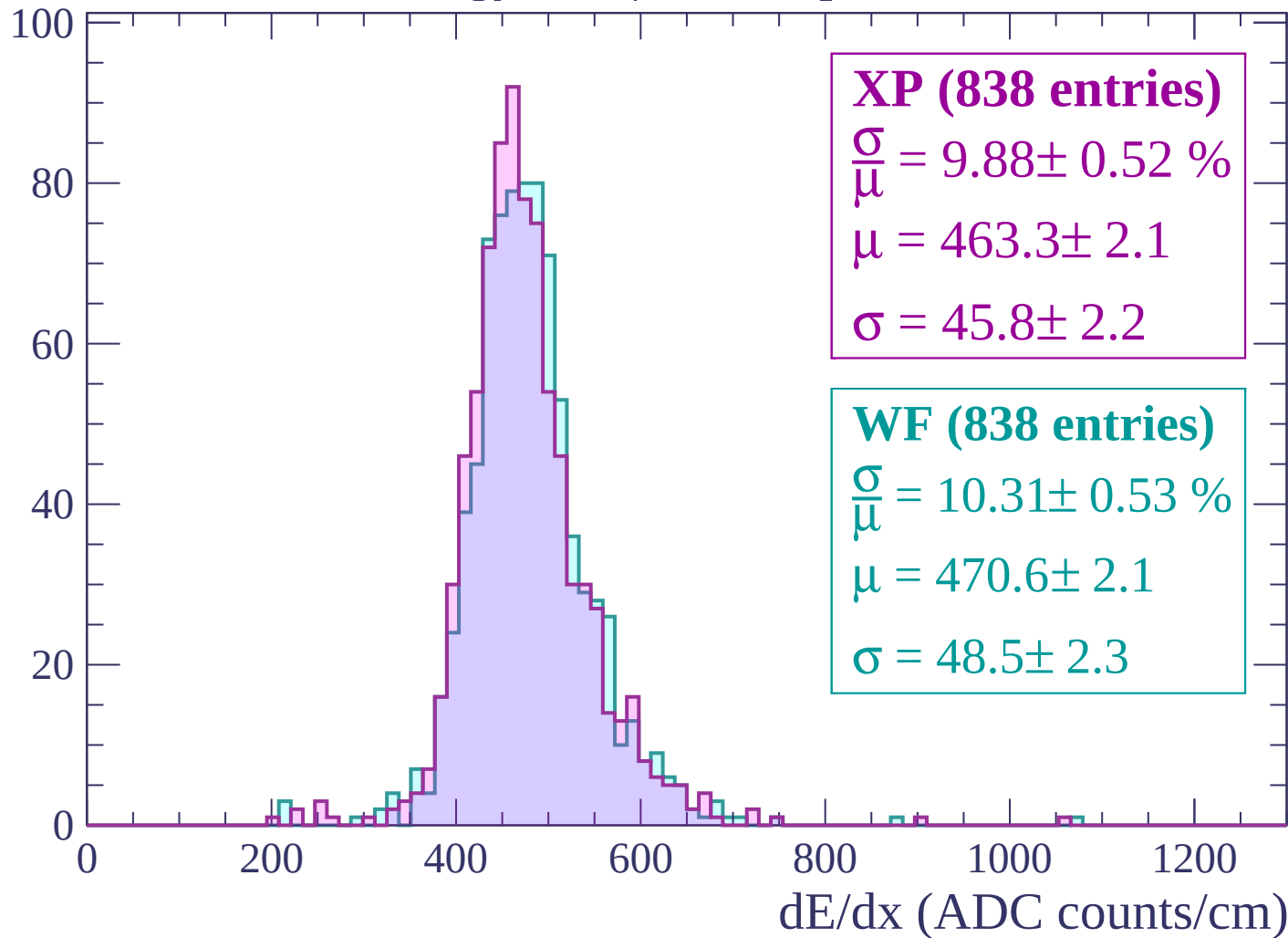
Energy loss | $1620 < p < 1680$

Count



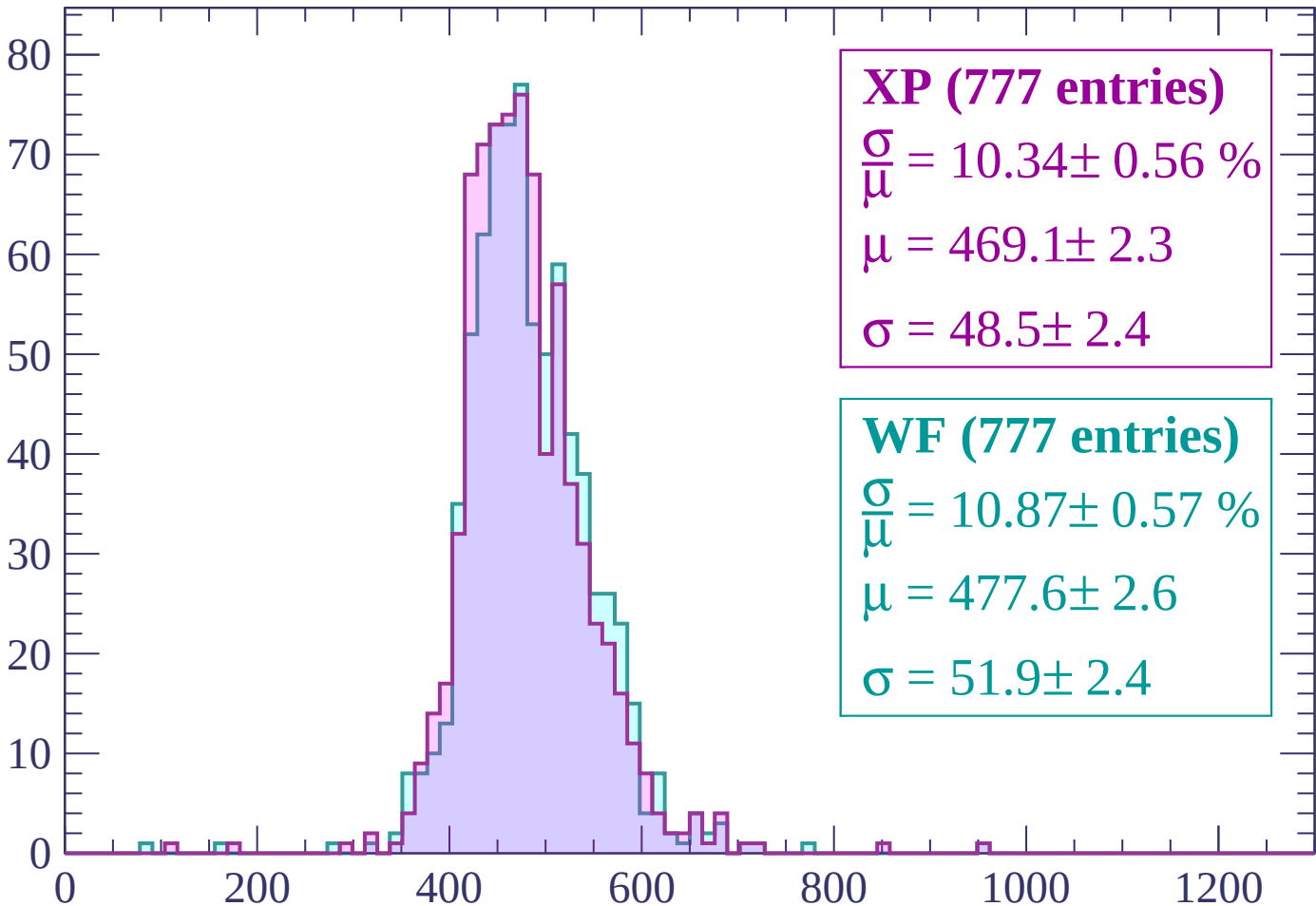
Energy loss | $1680 < p < 1740$

Count



Energy loss | $1740 < p < 1800$

Count



XP (777 entries)

$$\frac{\sigma}{\mu} = 10.34 \pm 0.56 \%$$

$$\mu = 469.1 \pm 2.3$$

$$\sigma = 48.5 \pm 2.4$$

WF (777 entries)

$$\frac{\sigma}{\mu} = 10.87 \pm 0.57 \%$$

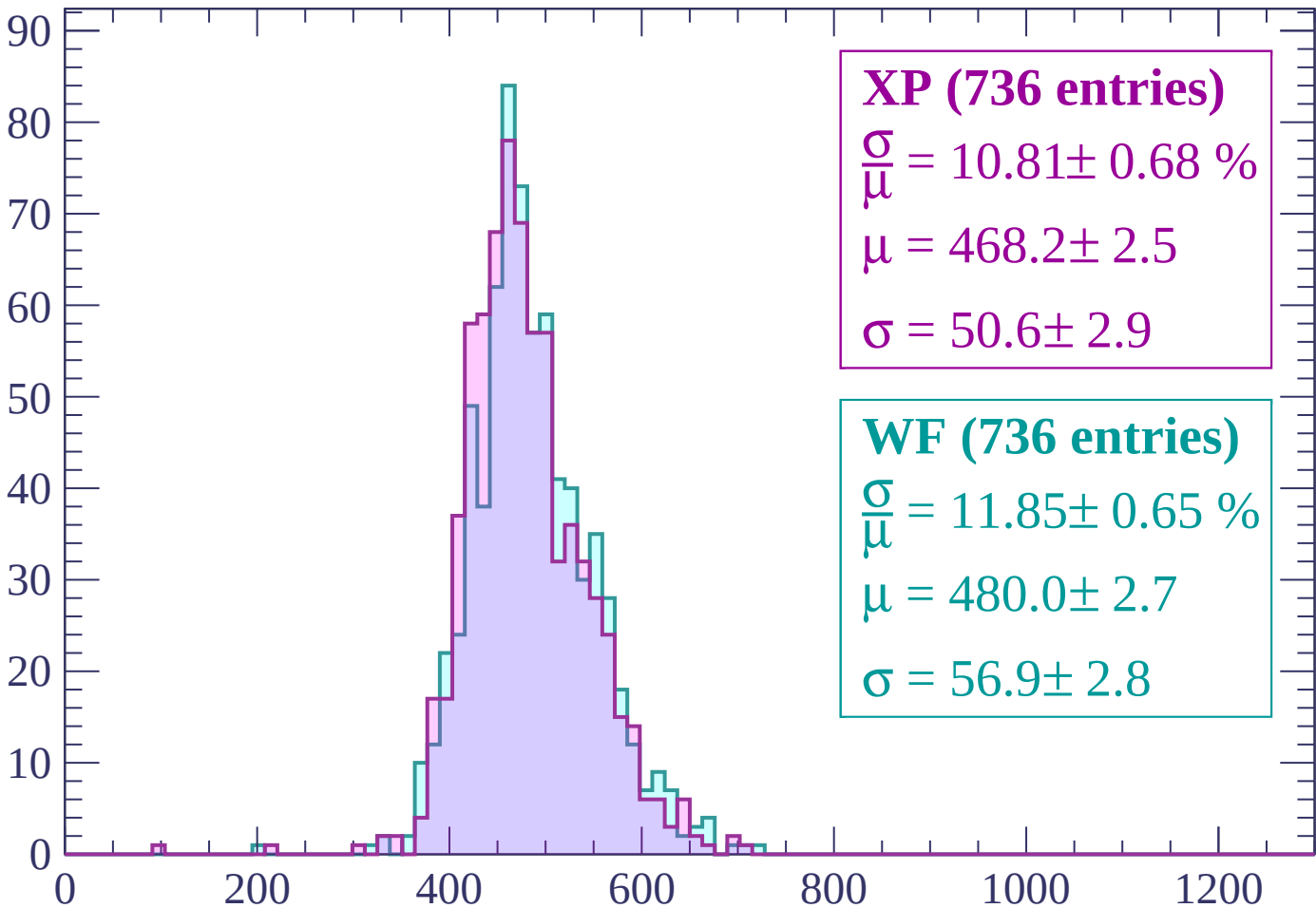
$$\mu = 477.6 \pm 2.6$$

$$\sigma = 51.9 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1860$

Count



XP (736 entries)

$$\frac{\sigma}{\mu} = 10.81 \pm 0.68 \%$$

$$\mu = 468.2 \pm 2.5$$

$$\sigma = 50.6 \pm 2.9$$

WF (736 entries)

$$\frac{\sigma}{\mu} = 11.85 \pm 0.65 \%$$

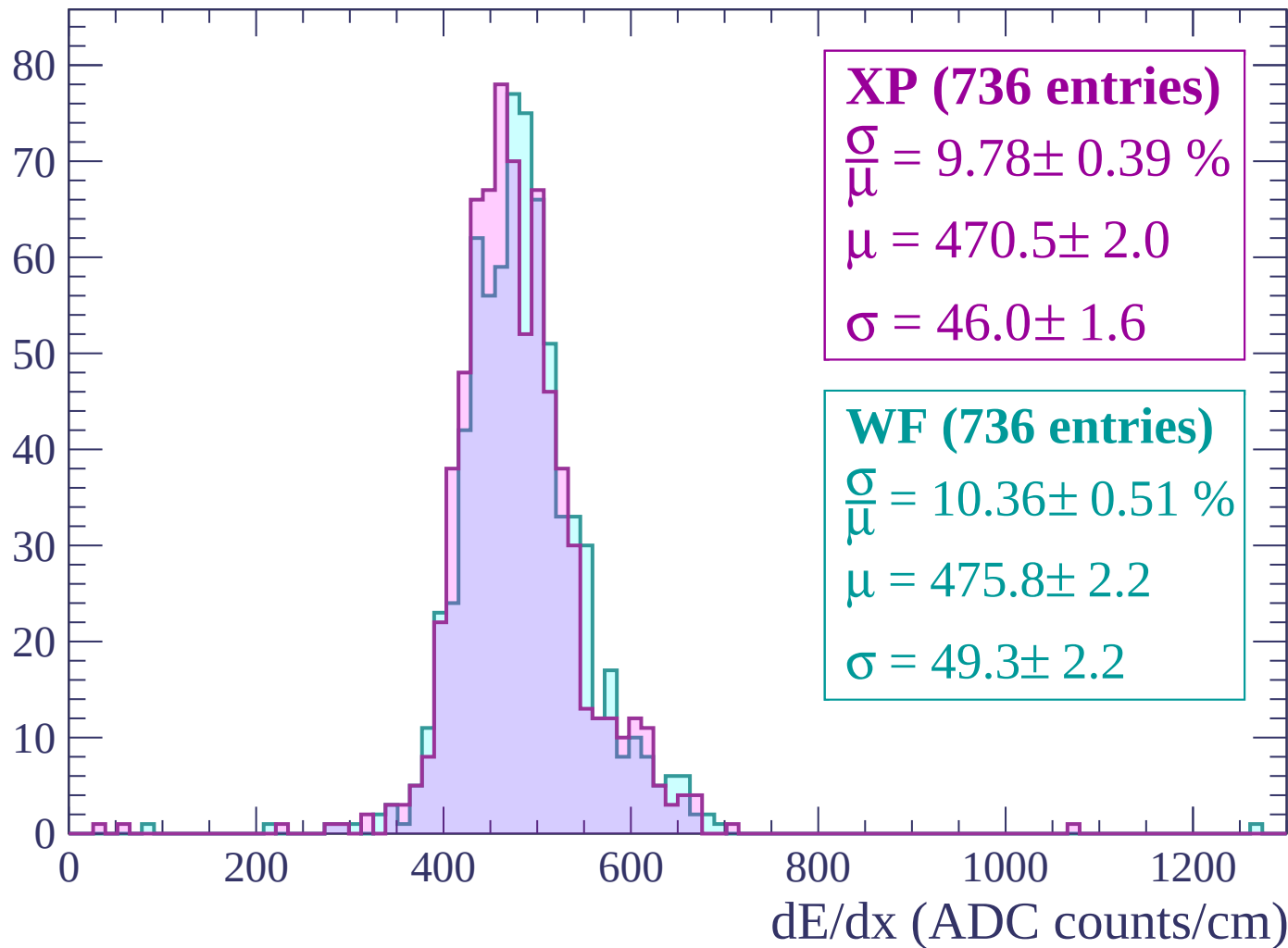
$$\mu = 480.0 \pm 2.7$$

$$\sigma = 56.9 \pm 2.8$$

dE/dx (ADC counts/cm)

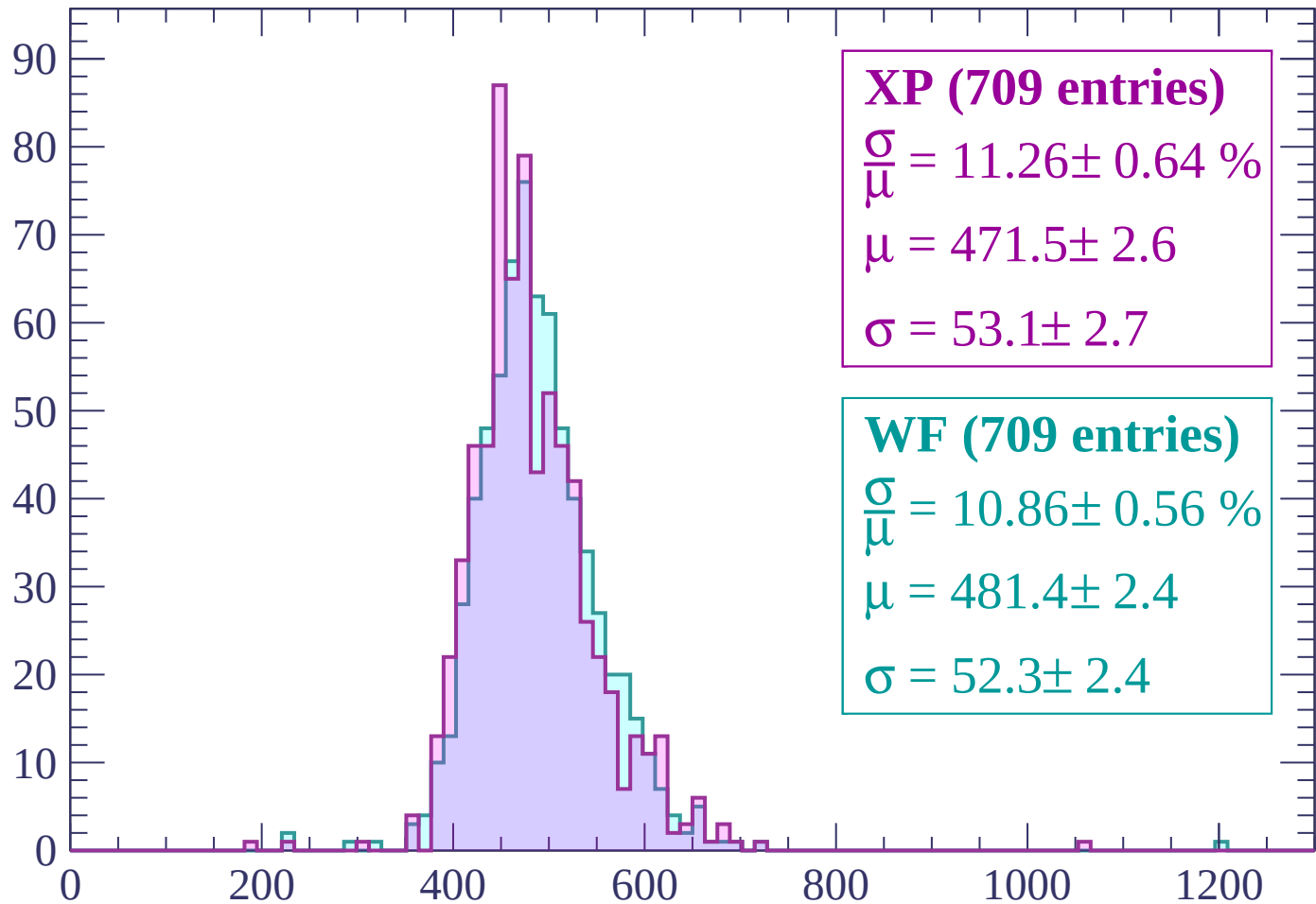
Energy loss | $1860 < p < 1920$

Count



Energy loss | $1920 < p < 1980$

Count



XP (709 entries)

$$\frac{\sigma}{\mu} = 11.26 \pm 0.64 \%$$

$$\mu = 471.5 \pm 2.6$$

$$\sigma = 53.1 \pm 2.7$$

WF (709 entries)

$$\frac{\sigma}{\mu} = 10.86 \pm 0.56 \%$$

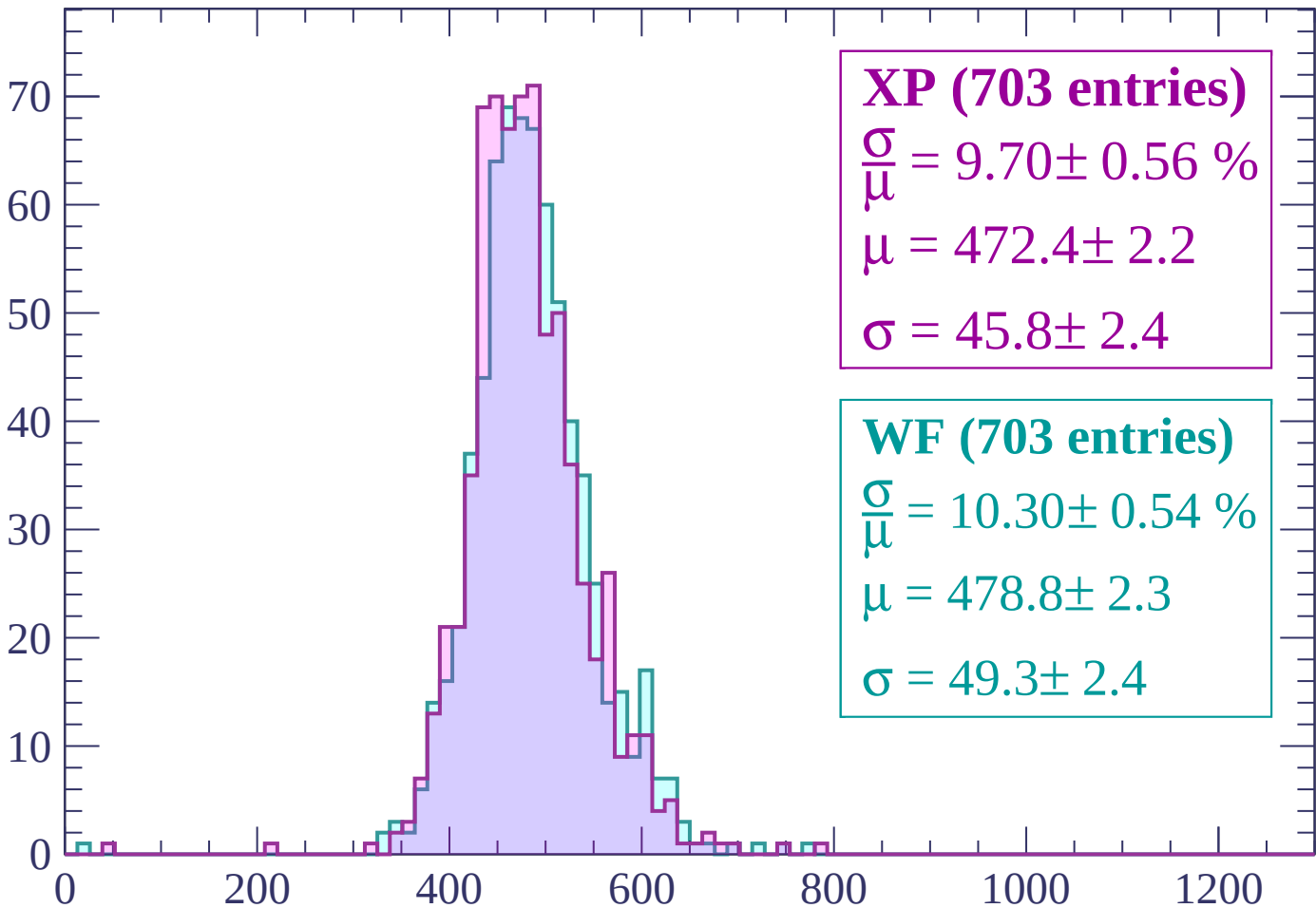
$$\mu = 481.4 \pm 2.4$$

$$\sigma = 52.3 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $1980 < p < 2040$

Count



XP (703 entries)

$$\frac{\sigma}{\mu} = 9.70 \pm 0.56 \%$$

$$\mu = 472.4 \pm 2.2$$

$$\sigma = 45.8 \pm 2.4$$

WF (703 entries)

$$\frac{\sigma}{\mu} = 10.30 \pm 0.54 \%$$

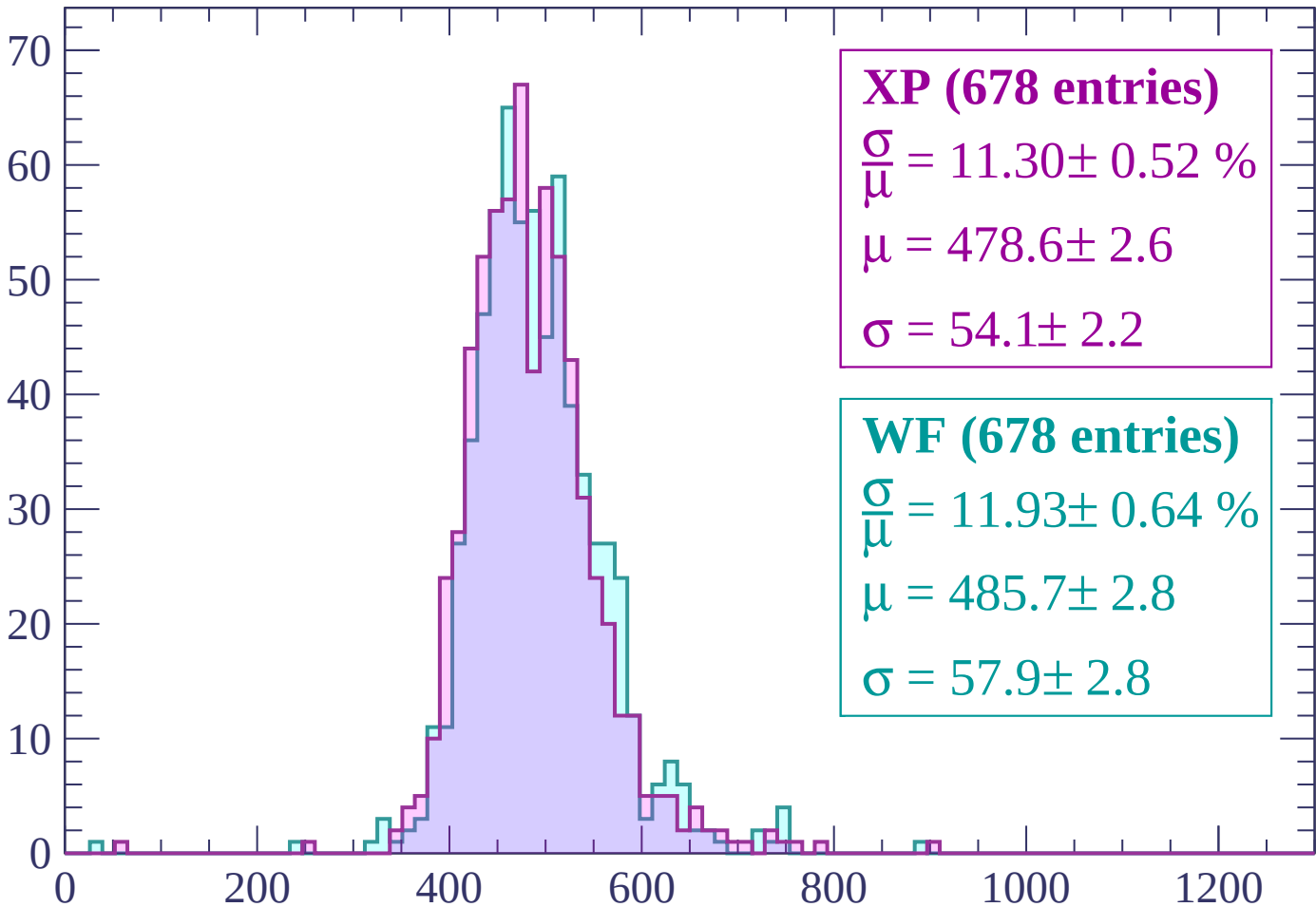
$$\mu = 478.8 \pm 2.3$$

$$\sigma = 49.3 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $2040 < p < 2100$

Count



XP (678 entries)

$\frac{\sigma}{\mu} = 11.30 \pm 0.52 \%$

$\mu = 478.6 \pm 2.6$

$\sigma = 54.1 \pm 2.2$

WF (678 entries)

$\frac{\sigma}{\mu} = 11.93 \pm 0.64 \%$

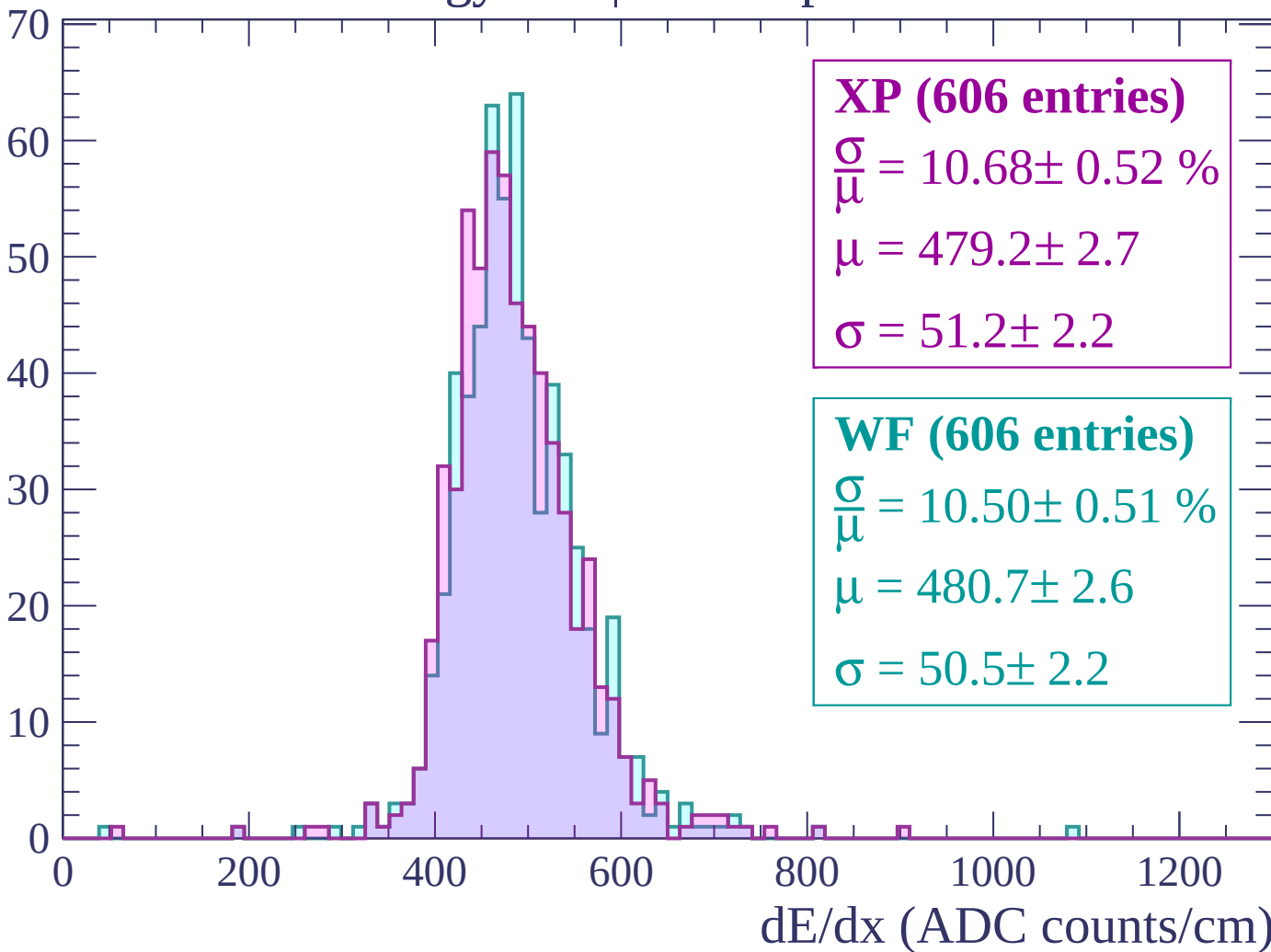
$\mu = 485.7 \pm 2.8$

$\sigma = 57.9 \pm 2.8$

dE/dx (ADC counts/cm)

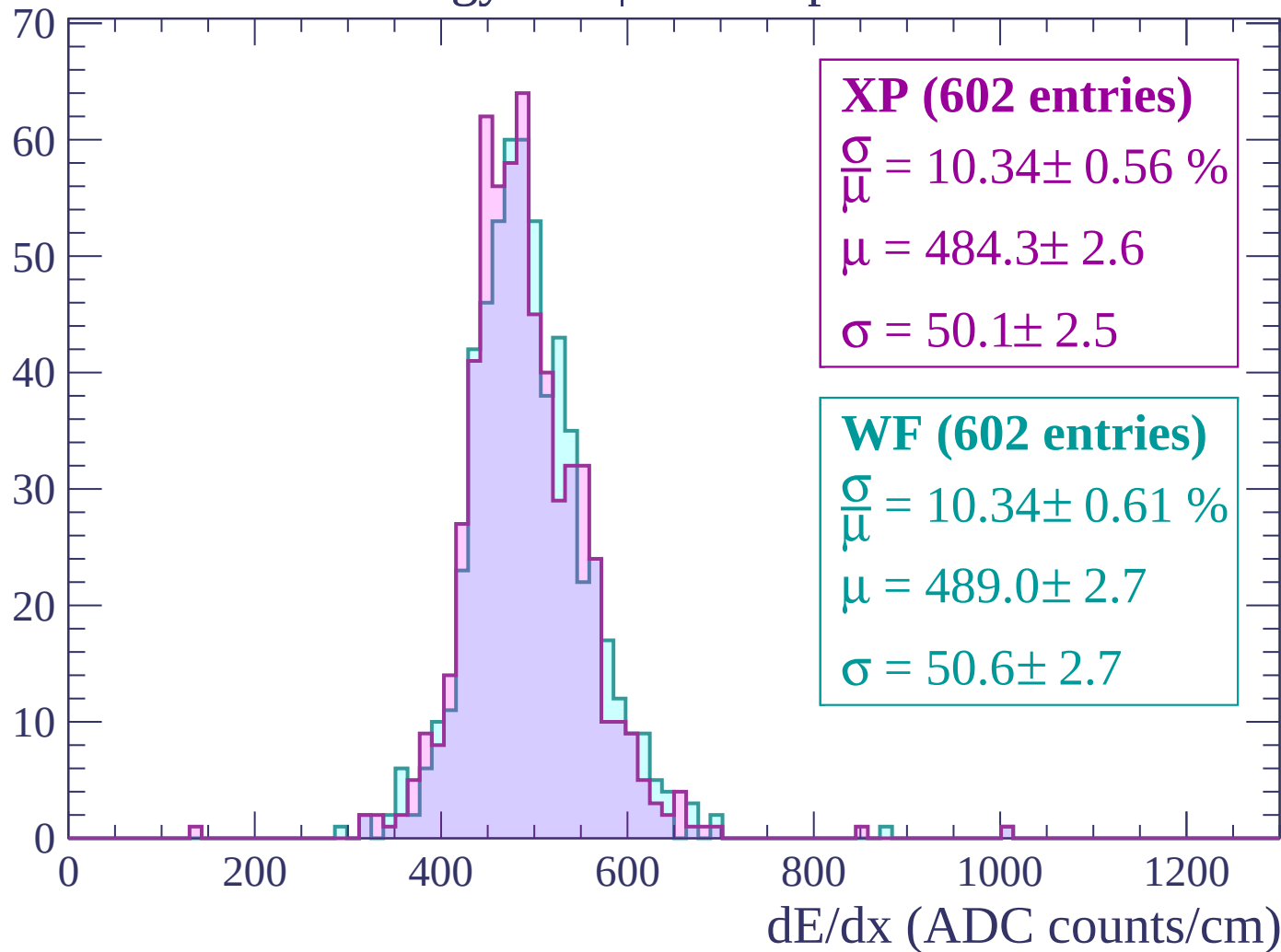
Energy loss | $2100 < p < 2160$

Count



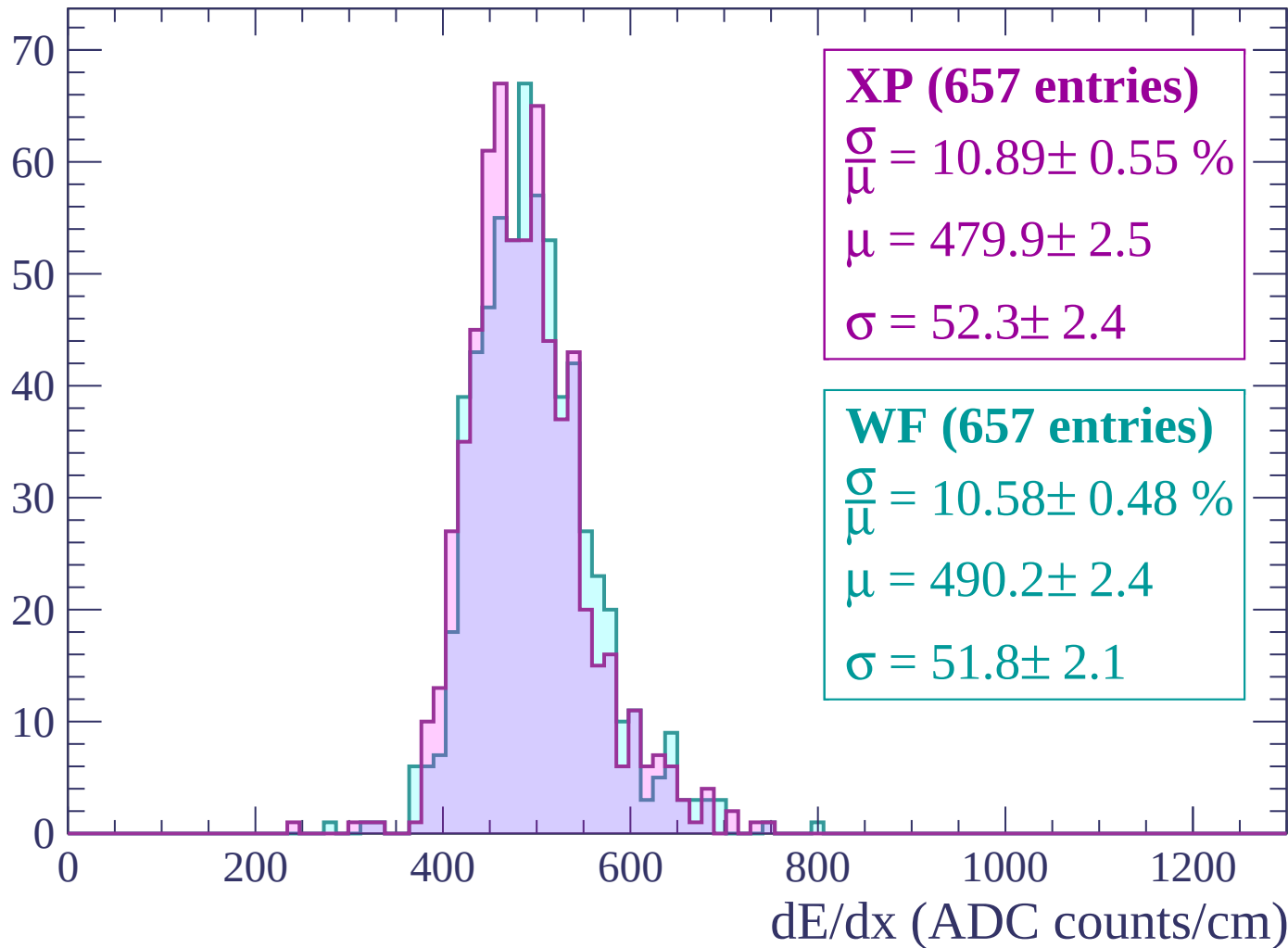
Energy loss | $2160 < p < 2220$

Count



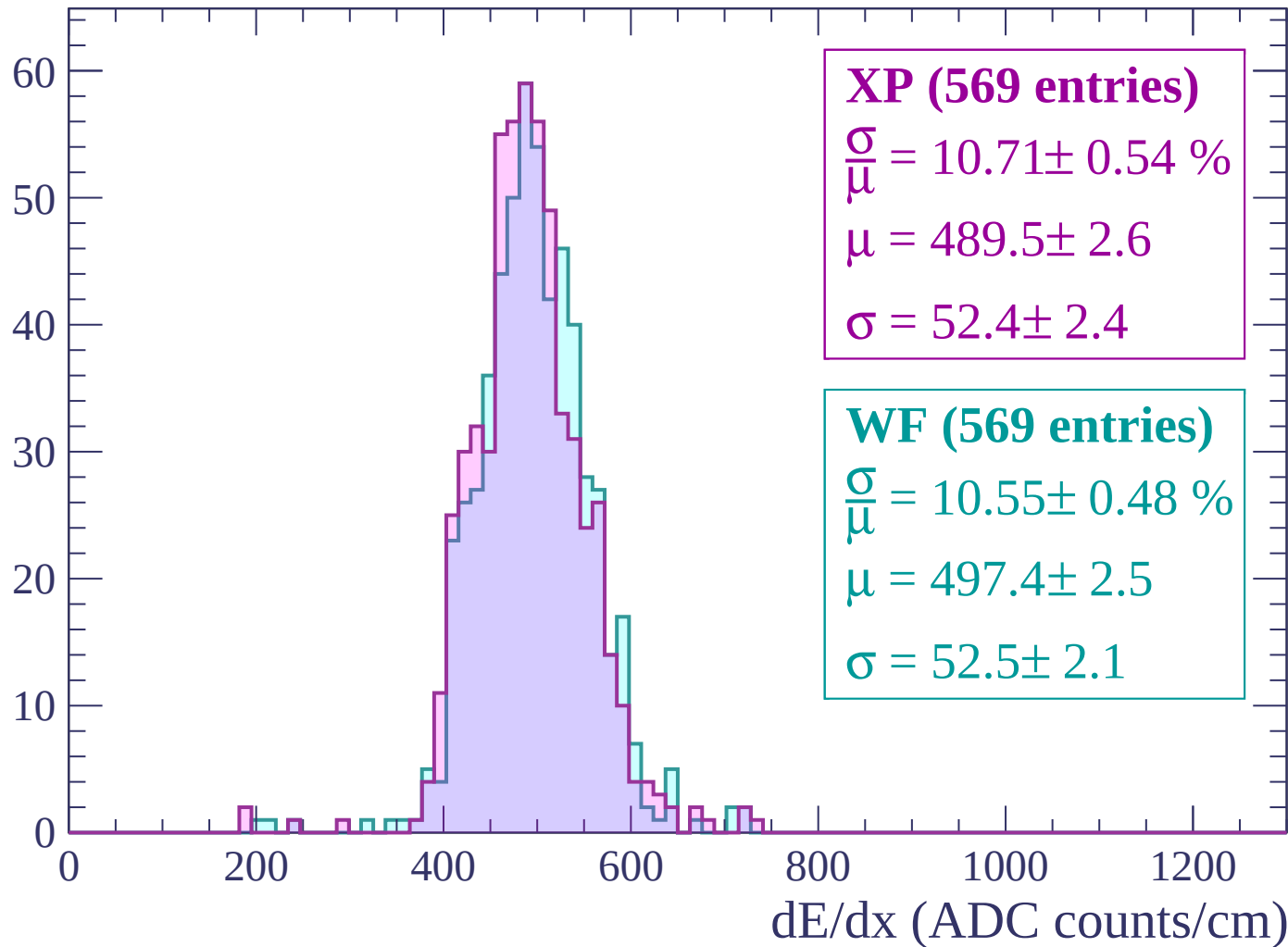
Energy loss | $2220 < p < 2280$

Count



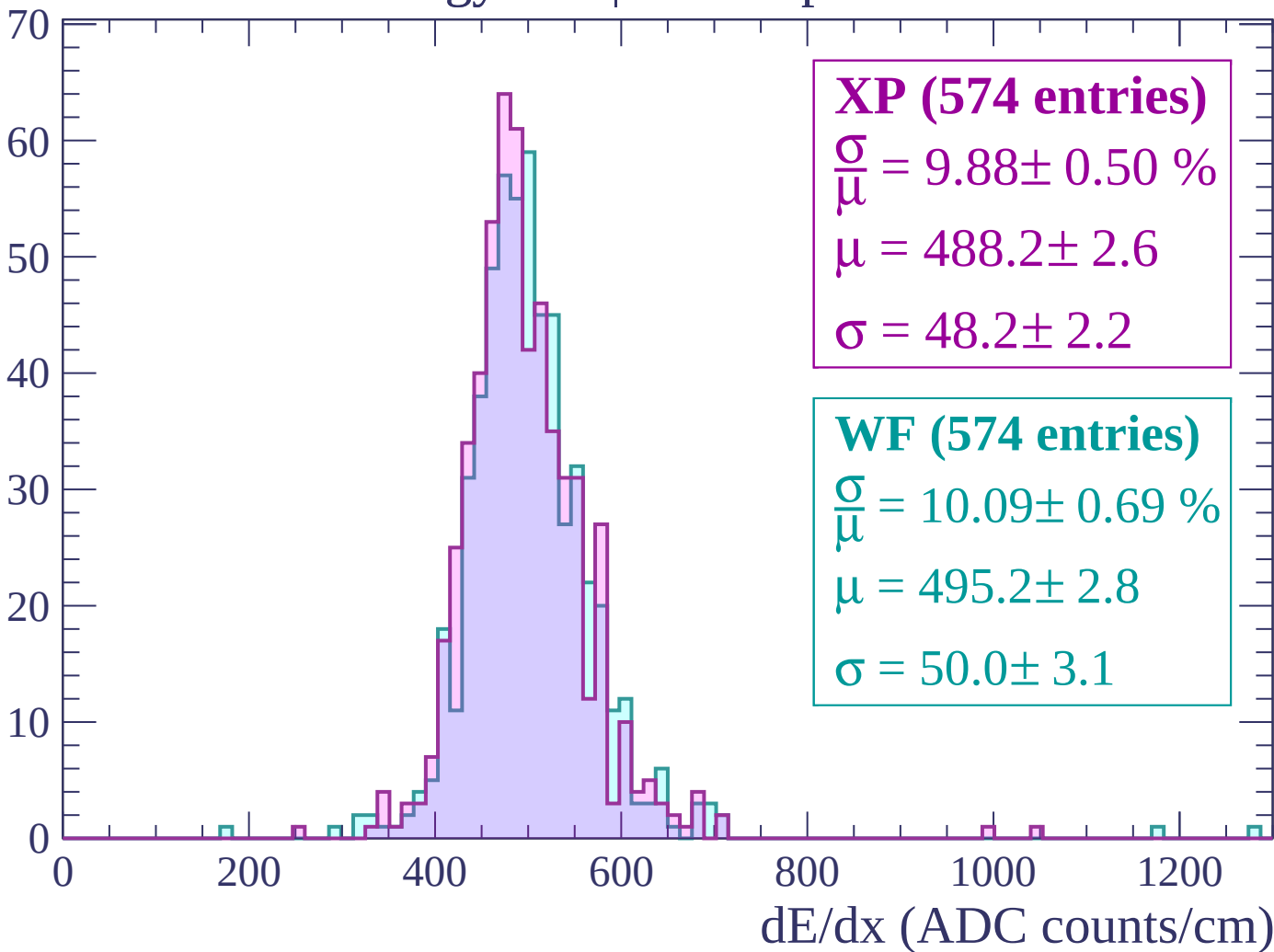
Energy loss | $2280 < p < 2340$

Count



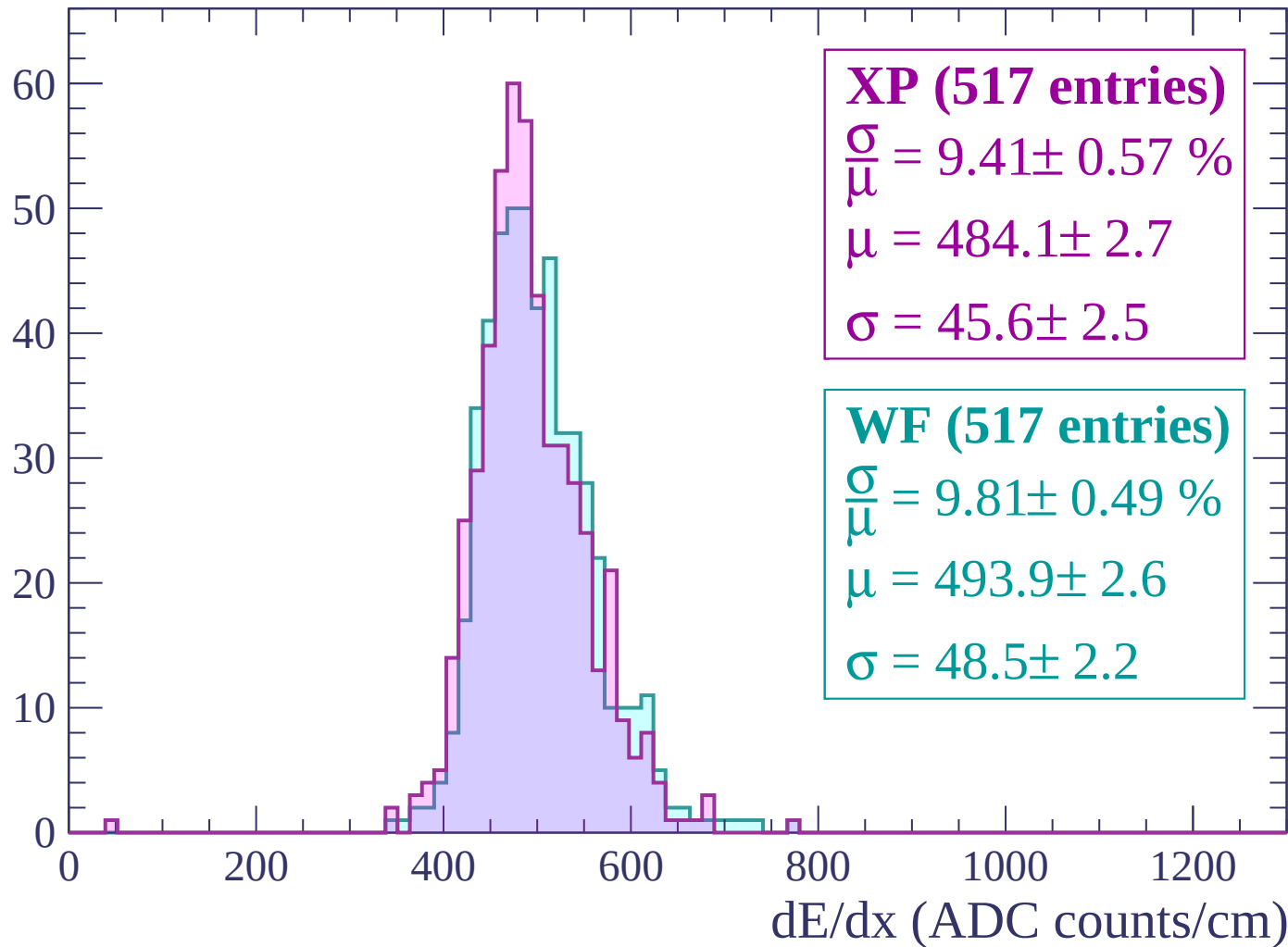
Energy loss | $2340 < p < 2400$

Count



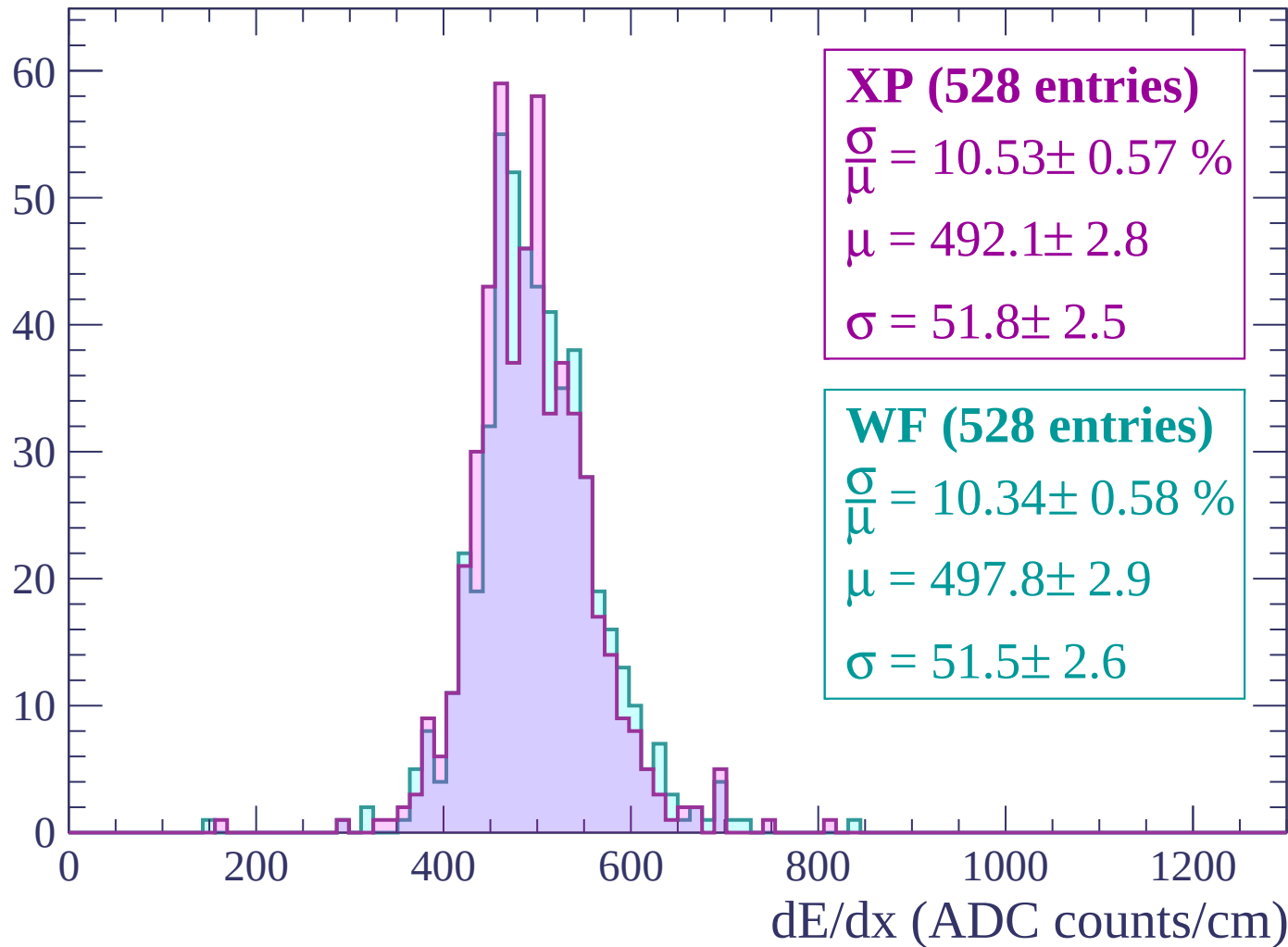
Energy loss | $2400 < p < 2460$

Count



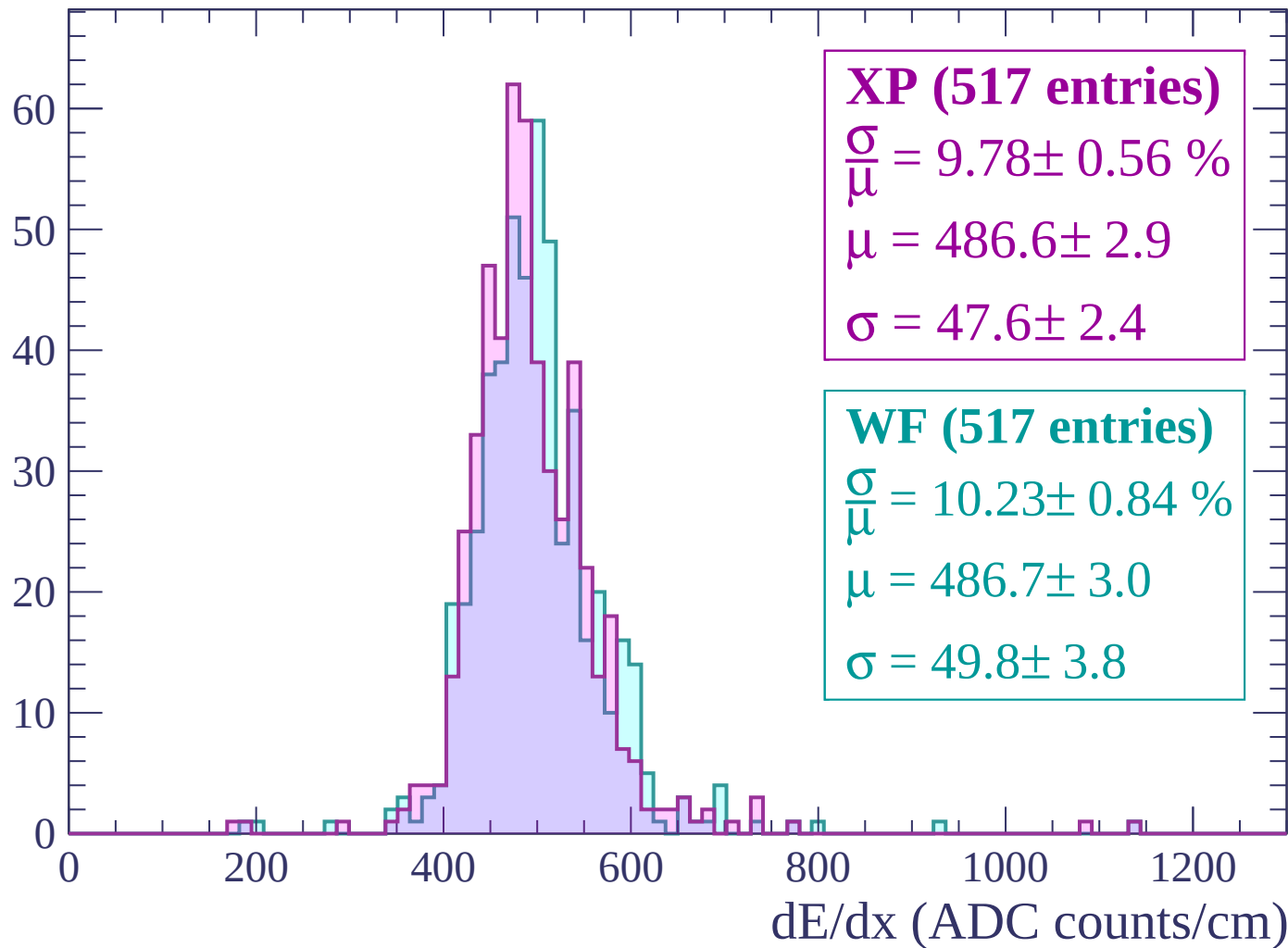
Energy loss | $2460 < p < 2520$

Count



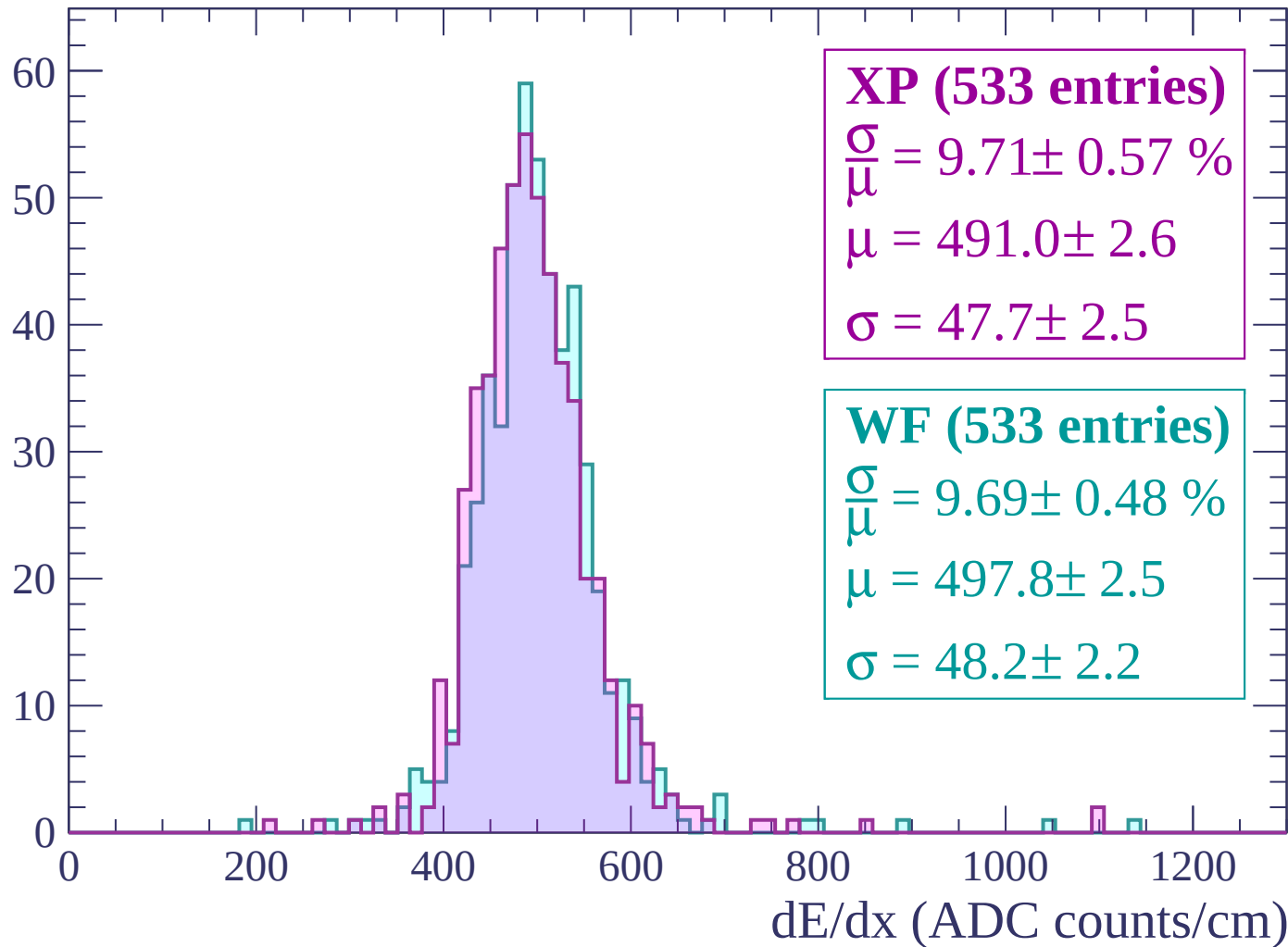
Energy loss | $2520 < p < 2580$

Count



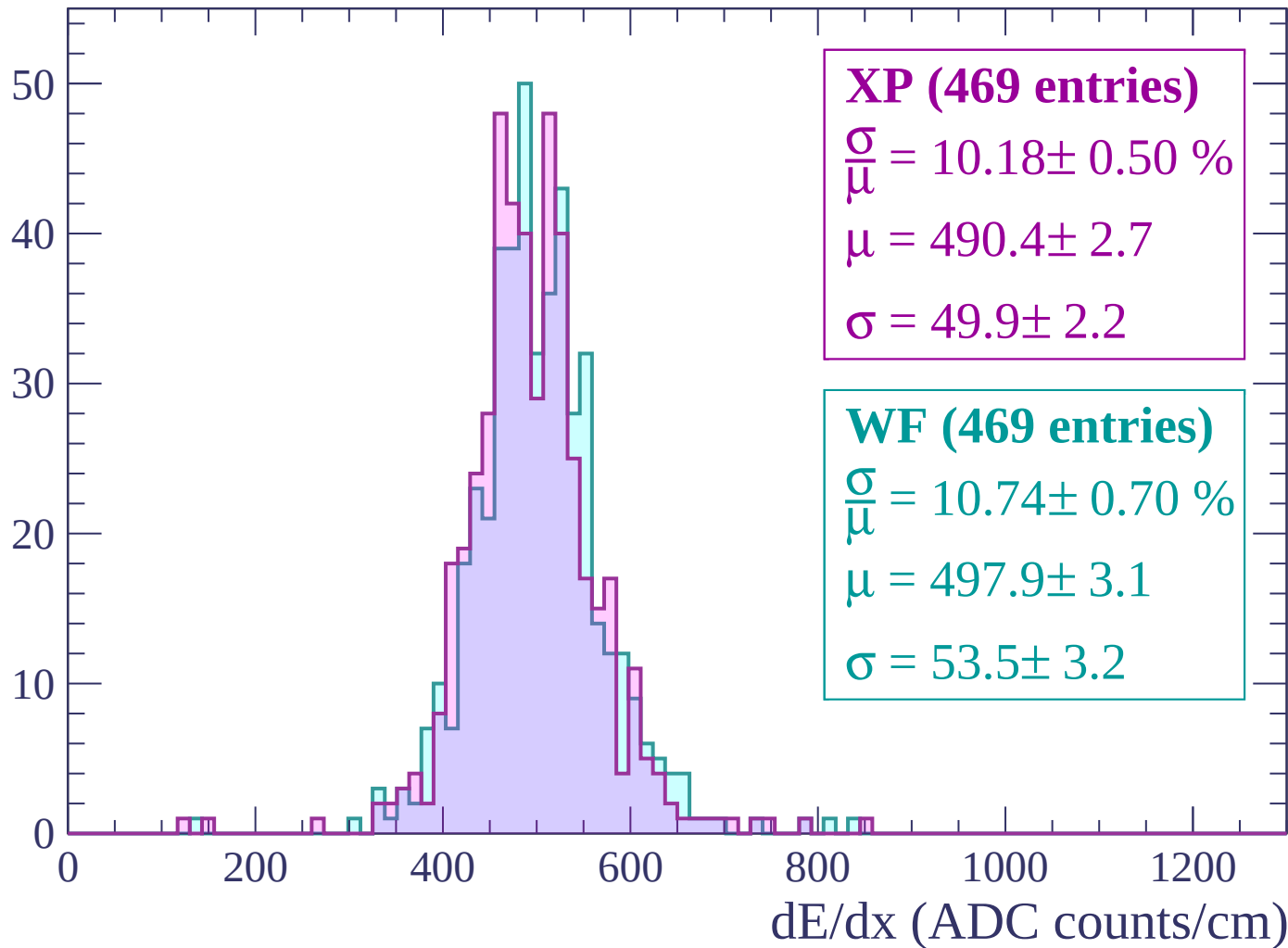
Energy loss | $2580 < p < 2640$

Count



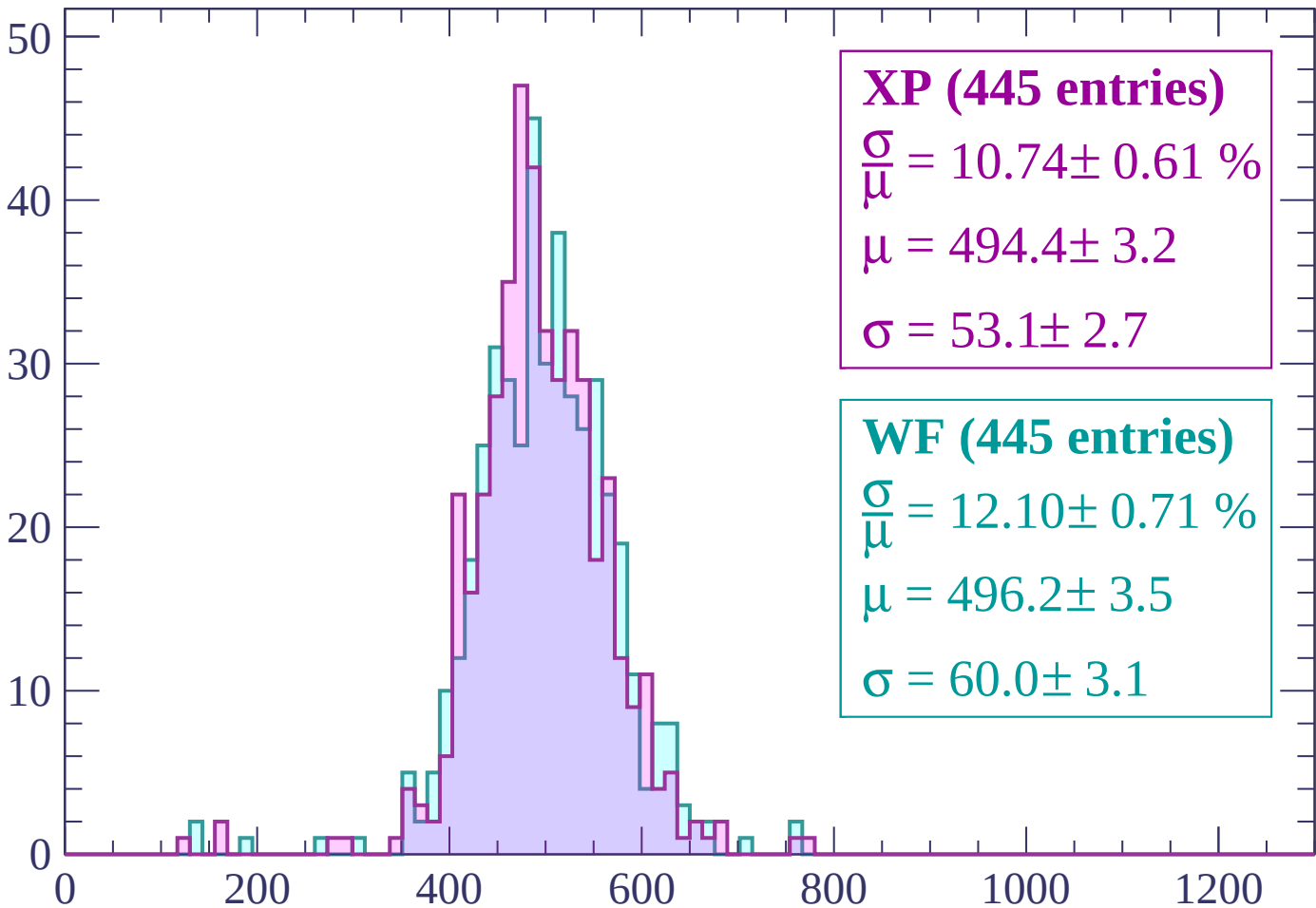
Energy loss | $2640 < p < 2700$

Count



Energy loss | $2700 < p < 2760$

Count



XP (445 entries)

$$\frac{\sigma}{\mu} = 10.74 \pm 0.61 \%$$

$$\mu = 494.4 \pm 3.2$$

$$\sigma = 53.1 \pm 2.7$$

WF (445 entries)

$$\frac{\sigma}{\mu} = 12.10 \pm 0.71 \%$$

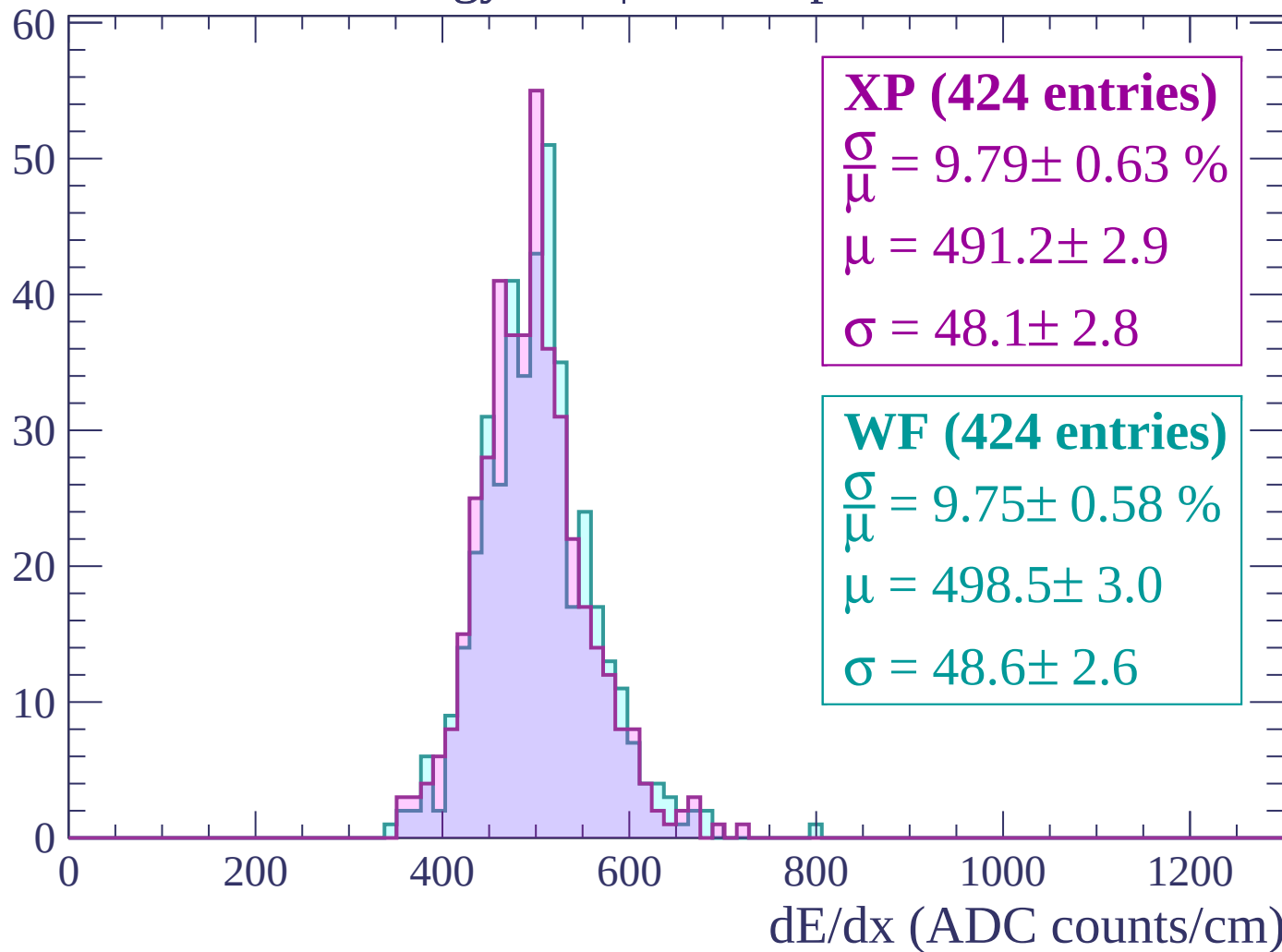
$$\mu = 496.2 \pm 3.5$$

$$\sigma = 60.0 \pm 3.1$$

dE/dx (ADC counts/cm)

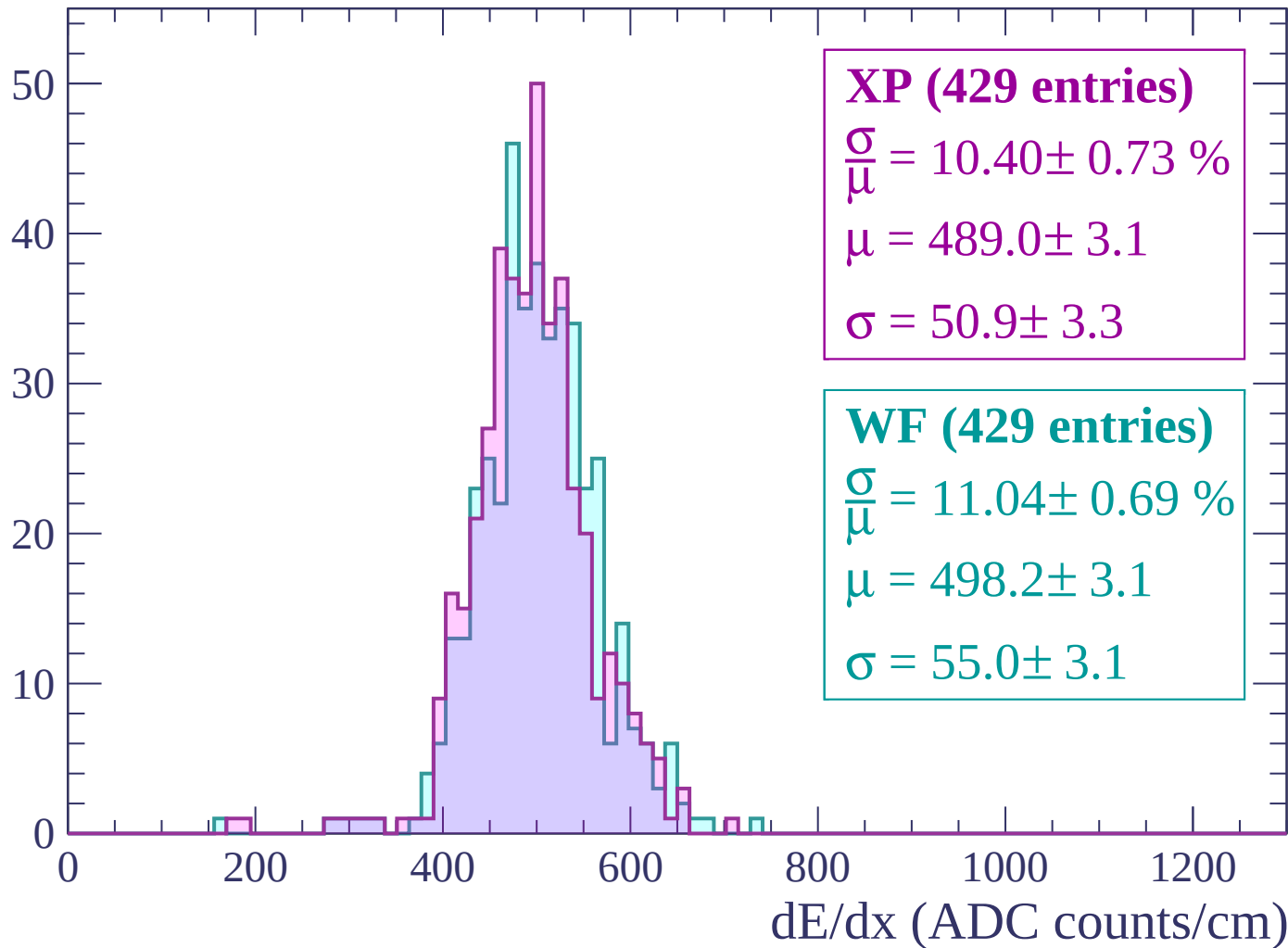
Energy loss | $2760 < p < 2820$

Count



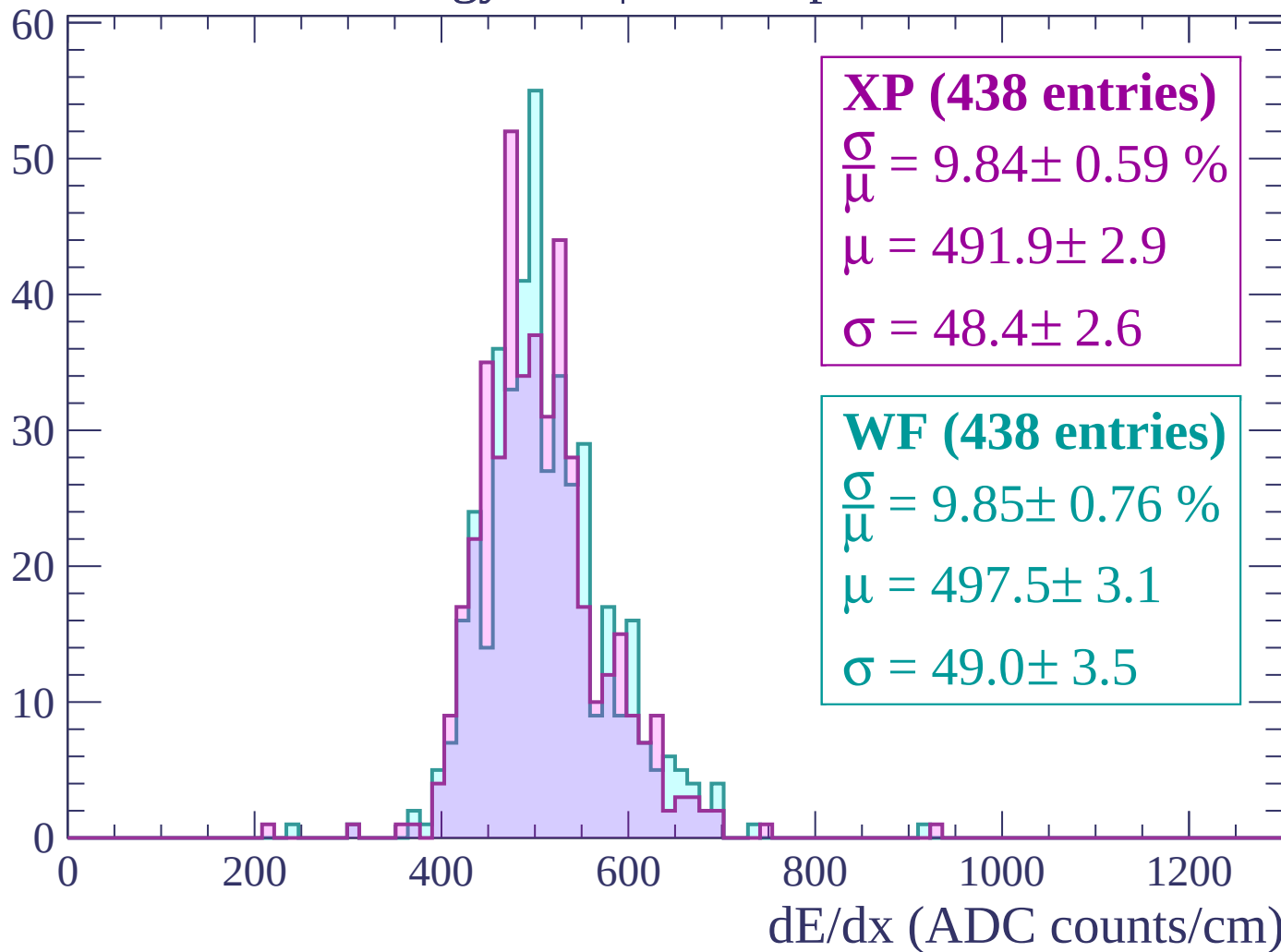
Energy loss | $2820 < p < 2880$

Count



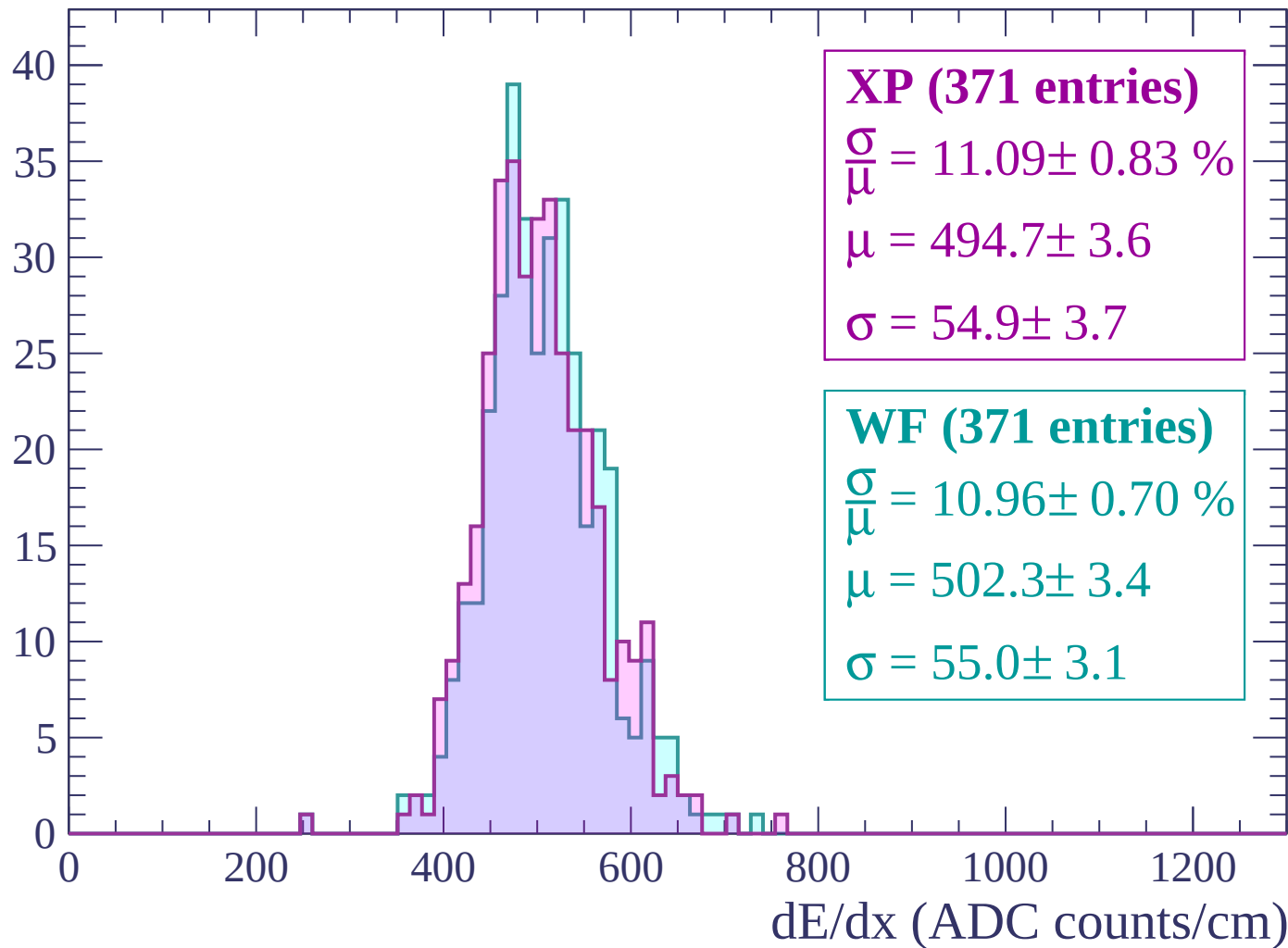
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count

